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IMX586 Application Note

IMX586
Software Reference Manual

Ver.1.0.3
2019/03/14

Sony Semiconductor Solutions Corporation

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Revision History

Version	Date	Description
0.1.0	2018.07.13	First release.
0.1.1	2018.08.31	Revise: “Figure 4-5 Frame structure (Normal)” Revise: “Figure 4-6 Full pixel output mode image data structure” Add: “4.6.2.STATS data control” Revise: “Table 4-8 Data Identifier” Revise: “Table 4-9 Image pixel area and data type” Revise: “Table 5-2 PLL pre divider & multiplier setting (PLL dual mode)” Revise: “Figure 5-3 Image size parameter definition” Revise: “Table 5-10 Typical image output of main capture modes (1)” Revise: “Table 5-16 QBC HDR capture mode setting” Revise: “Table 7-3 List of registers to be set only in SW-standby as mode transition.” Revise: “Table 7-4 List of corrupted frame causing registers” Add: “9.1. Outline of QBC HDR” Add: “9.2. QBC HDR control” Add: “9.3. LSC control” Add: “9.4. AE control” Add: “9.5. STATS (statistic) data control” Add: “9.6. STATS data in image” Revise: “Table A-3 EBD LINE0”
0.1.2	2018.09.04	Revise: “Table 5-11 Typical image output of main capture modes (2)” Revise: “Figure 8-14 Difference between XVS signal and V-Sync signal”
0.1.3	2018.09.28	Revise: “9.5.2. AE-HIST” Revise: “Figure 9-12 MIPI data format for AE HIST”
1.0.0	2018.11.02	Revise: “Glossary” Revise: “4.4. Readout start position” Revise: “4.6.2. STATS data control” Revise: “Table 4-5 Constraints of LINE_LENGTH_PCK” Revise: “Table 5-14 Support list of operation mode of capture mode” Revise: “Table 5-16 QBC HDR capture mode setting” Add: “Table 5-22 Constraints of COARSE_INTEG_TIME” Revise: “Table 5-23 Integration time setting (In case of FRM_LENGTH_CTL=0)” Revise: “Table 5-24 Integration time setting (In case of FRM_LENGTH_CTL=1)” Revise: “Table 5-25 Range of Gains” Revise: “Table 5-26 ADC mode register setting” Revise: “5.5.1. Analog gain settings” Revise: “Table 5-36 Output pixel level according to binning modes(Normal) ” Revise: “5.7.1. 2x1 OCL pixels and PDAF ” Add: “5.7.2.L/R Sensitivity Correction (LRC)” Add: “Figure 5-16 Block diagram for White balance gain”

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		Revise: “Table 7-4 List of corrupted frame causing registers” Revise: “Table 7-10 Gain setting register” Revise: “Table 8-7 Setting for long exposure” Revise: “Table 8-29 XVS control registers” Revise: “Table 9-1 QBC HDR control registers” Revise: “Table 9-2 LSC(Y) control registers” Revise: “Table 9-3 AE control registers” Revise: “Figure 9-8 Example of exposure setting” Revise: “9.5.1.Block diagram for STATS” Revise: “Table A-1 Summary of mode transition related issues”
1.0.1	2018.11.16	Revise: “Table 5-14 Support list of operation mode of capture mode” Revise: “Table 5-25 Range of Gains” Revise: “Table 5-26 ADC mode register setting”
1.0.2	2018.12.10	Add: the description of “3. Image Signal Interface” is added Update: minor editorial error is updated.
1.0.3	2019.03.14	Revise: “Table 4-1 Imaging area determining registers” Revise: “5.1.1 Clock system diagram” Revise: “Table 5-1 PLL pre divider & multiplier setting (PLL single mode)” Revise: “Table 5-2 PLL pre divider & multiplier setting (PLL dual mode)” Revise: “Table 5-5 PLL frequency table IVTCK (include PLL single mode)” Revise: “Table 5-6 PLL frequency table IOPCK” Revise: “Table 5-9 IOPSYCK frequency table” Add: “5.5.4.Bayer weighting function” Revise: Table 8-16 CDS clock cycle Revise: “Table 10-1 LSC configuration registers” Revise: “Table 10-2 LSC Knot point RAM tables” Add “Figure 10-2 LSC knot point table setting” Revise: “10.4. Control scheme -- overall setting scheme” Revise: “Figure 10-8 LSC control flow chart” Update: minor editorial error is updated.

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Note

In this specification, it covers the major F/W or H/W registers with functionality explanation, however the default value which set by Sony is not written. Because H/W or F/W defaults value is sometimes updated faster than document update timing. See "Register setting" (Excel sheet) for the latest default value shown as "Global Setting".

There are two types of “default value”.

- H/W default value: is a default value that is embedded with metal wiring into the silicon. It is decided in very early stage of device design and cannot be changed afterward. These values can be found in the Register Map.
- S/W default value: is a default valued that is recommended value and/or bug fix values to be externally over writing H/W default values after evaluation of engineering samples. These values are shown in “Register Setting”. Sony recommended register setting table or customer unique register setting table will be supplied upon request.

SRM-230-00010-01-258-00011-01

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Glossary

- **Contrast Value**: the value of strength of high-frequency component
- **CRA**: Chief Ray Angle
- **DAC value**: the value of digital analog converter of lens actuator(= the distance from image sensor to lens)
- **DPC**: Defect Pixel Correction
- **Image Height**: the distance from optical light axis on image sensor
- **I^2C**: Inter-Integrated Circuit (= serial bus description)
- **LSC**: Lens Shading Correction
- **MIPI**: Mobile Industry Processor Interface
- **OTP**: One Time Programmable ROM
- **OIS**: Optical Image Stabilization(= the lens shifting function for avoiding blurring the image by hand shake)
- **PDAF**: Phase Detection Auto Focus
- **Register:** a small amount of storage in digital processor
- **QBC HDR:** Quad Bayer Cording HDR
- **QSC:** Quad Bayer Coding Sensitivity Correction
- **OCL:** On Chip Lens

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1. System Outline

This sensor is a diagonal 8.000 mm (Type1/2) 48 Mega-pixel CMOS active pixel type stacked image sensor with a square pixel array. It adopts Sony's back-illuminated and stacked CMOS image sensor to achieve high speed image capturing by column parallel ADC circuits and high sensitivity and low noise image.

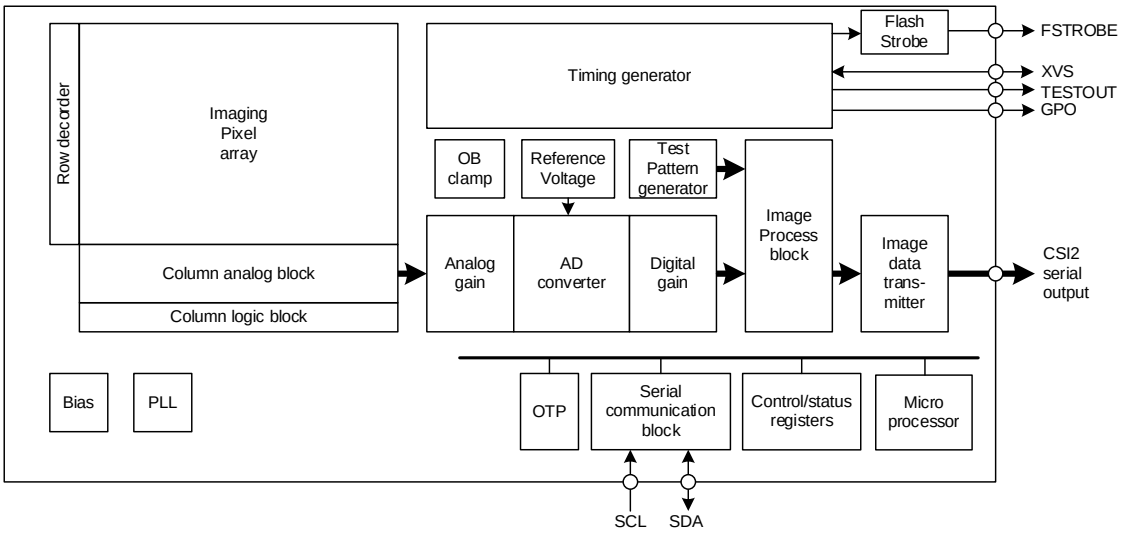


Figure 1-1 System block diagram

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2. Control Register Settings by Serial Communication

This sensor can use the 2-wire serial communication method for sensor control.

In this chapter, specification, method of 2-wire serial communication between the master device and this sensor is described.

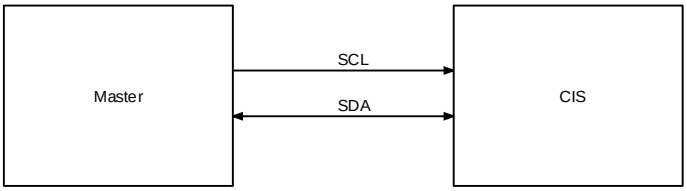


Figure 2-1 2-wire serial communication

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2.1. 2-wire serial communication operation specifications

The 2-wire serial communication method conforms to the camera control interface (CCI). CCI is an I^2C Fast-mode Plus compatible interface, and the data transfer protocol is I^2C standard.

This 2-wire serial communication circuit can be used to access the control-registers and status-registers of this sensor.

Note) only available with INCK ≥ 12.9MHz

Table 2-1 Description of 2-wire serial communication pins

Pin name	Description
SDA	Serial data input/output pin
SCL	Serial clock input pin

The control registers and status registers of this sensor are mapped on the 16-bit address space. The register categories are shown below. Detail register information is shown in each chapter/section in this document and in Register Map.

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Table 2-2 Abstract of register address map for 2-wire serial communication

	Address range	Description	Note
I²C register	0x0000 - 0x0fff	Configuration register Read only and read/write dynamic register	
	0x1000 - 0x1fff	Parameter limit register Read only static register	
	0x2000 - 0x2fff	Reserved	
	0x3000 - 0xffff	Manufacturer specified register	SONY customer register

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2.2. Communication protocol

2-wire serial communication supports a 16-bit register address and 8-bit data message type.

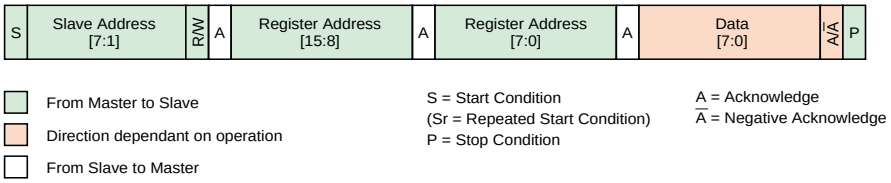


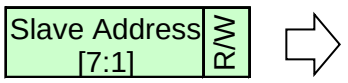
Figure 2-2 2-wire serial communication protocol

SRM-230-00025-03

2.3. Default CCI slave address configuration

2.3.1. First slave address

This sensor has a default Camera Control Interface (CCI) or 1st I²C slave address shown below. When called by this slave address, the serial communication interface is activated. Duplication of the address on the same bus must be prevented. This slave address is controlled by hardware connection of SLASEL pin.



Slave Address (Read or Write) * See Table 2-3 for R/W bit	SLASEL (pin#53)
MSB LSB 0 0 1 1 0 1 0 R/W *1	L or NC
MSB LSB 0 0 1 0 0 0 0 R/W *1	H

Note *1: R/W shows the direction of communication.

Figure 2-3 CCI (I²C) Slave address

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Table 2-3 R/W bit

R/W bit	Direction of communication	Address
0	Write (master → sensor)	0: 0x34 or 0x20
1	Read (sensor → master)	1: 0x35 or 0x21

2.3.2. Secondary slave address

In this sensor, 2nd I2C Slave address is available also. This value is fixed as 0x6c (0x6d in case of read timing). It can be switched On/Off by ”SLAVE_ADD_EN_2ND” and “SLAVE_ADD_ACKEN_2ND” for ACK to 2nd CCI. Refer to the table below.

Table 2-4 2nd CCI ACK control register

	Address	Bit	Name	Description
I2C register	0x3010	[0]	SLAVE_ADD_EN_2ND	2nd Slave Address Enable 0: disable 1: enable
	0x3011	[0]	SLAVE_ADD_ACKEN_2ND	2nd Slave Address Ack Enable 0: disable 1: enable

2.3.3. Second CCI slave address for synchronous dual sensor operation

Two sensors are capable to be differently controlled by one host. In this case two sensors could be operated in parallel with even different integration time and gain while keeping synchronization each other. This kind of dual camera application requires having different first slave address and the same second address at the same time between the two sensors.

See “8.8 Dual camera operation” for more detail.

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2.4. Specification of communication bus state

2.4.1. Idle state

Idle state is specified as follows; neither master nor slave device drives the SDA or SCL, and these bus lines are pulled up to V_{dd} via resistor.

2.4.2. Issue “Start condition”

While the 2-wire communication bus is in idle state, master device (ex. subsequent image processing LSI, etc.) issues the communication-start: Start condition S by driving SDA from “High” to “Low” level. Serial data are transmitted in 8-bit-unit MSB first format. For every 8-bit data transmission, the slave device issues acknowledge or negative acknowledge (explained later). A (acknowledge)/Ā (negative acknowledge).

Data (SDA) is transmitted in sync with the SCL cycle. SDA toggles while SCL is “Low” and holds the value while SCL is “High”.

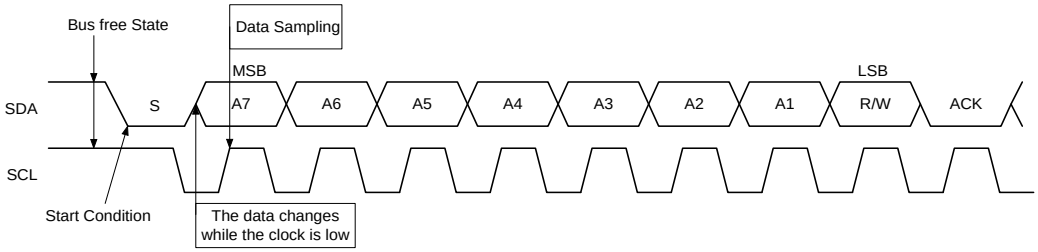


Figure 2-4 Start condition

2.4.3. Issue “Stop condition”

After A (acknowledge)/Ā (negative acknowledge) and while SCL is High, master device issues communication-stop condition: “Stop condition” P by driving SDA from a low to high level. After issuing a “Stop condition”, master release 2-wire serial bus enters into an idle state.

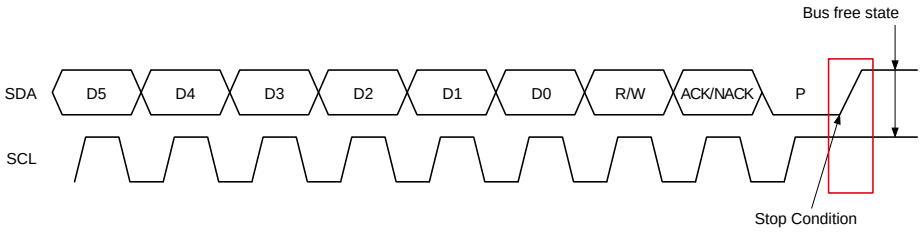


Figure 2-5 Stop condition

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2.4.4. Issue “Repeated start condition”

Master device can issue a start condition after previous transaction without issuing a stop condition.

In this case this start condition is recognized as the “Repeated started condition” Sr.

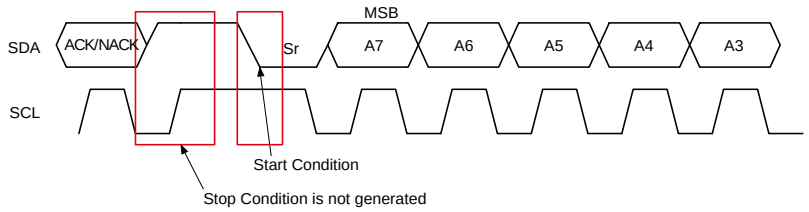


Figure 2-6 Repeated start condition

2.4.5. Issue “Acknowledge/negative acknowledge”

After transmitting each byte, a master or slave device issues an acknowledgement or negative acknowledgement and can release the bus to the idle state. When negative acknowledgement is issued, the master must issue the stop and terminate the communication immediately.

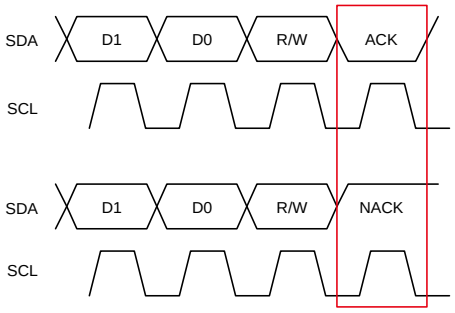


Figure 2-7 Acknowledge and negative acknowledge

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2.5. Read/write operation of 2-wire serial communication

This sensor has an index function that indicates which address is to be accessed. When reading/writing the value from/to a requested address, the master must set the address value to the index. Index value is designated by 2 bytes of dummy write operation after the slave address transmission. The index value is automatically incremented by “one” with the “Acknowledge/Negative Acknowledge” for the following data transfer.

This sensor supports four read modes and two write modes being compliant to Camera Control Interface (CCI).

Table 2-5 Read/Write operations supported by 2-wire serial communication

Access mode	1	CCI single read from random location (Single read from an arbitrary address)
	2	CCI single read from current location (Single read from the held address)
	3	CCI sequential read starting from random location (Sequential read starting from an arbitrary address)
	4	CCI sequential read starting from current location (Sequential read starting from the held address)
	5	CCI single write to random location (Single write to an arbitrary address)
	6	CCI sequential write starting from random location (Sequential write starting from an arbitrary address)

2.5.1. CCI single read from random location

The upper part of the below figure below shows the sensor internal index value, and the lower part shows the SDA data flow. The master sets the sensor index value of M by designating the sensor slave address with a write request. Then the Master generates the Start condition. The Start condition is generated without generating the Stop condition, so it becomes the Repeated Start condition. Next, when the Master sends the slave address with a read request, the sensor outputs an Acknowledge followed immediately by the data at the index address on SDA. After the Master receives the data, it generates a Negative Acknowledge and the Stop condition to end the communication.

When reading single datum from the requested address register, the master device starts write-operation with the slave address of this sensor and by making 2 bytes of dummy write master sets the address value (M) to the index. After that master issues the “start condition” again instead of issuing a “stop condition”. This “start condition” is recognized as “repeated start condition”. Then

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transmitting the read request with the slave address, this sensor issues the “Acknowledge” and starts transmitting the register value from indexed address (M). Master issues the “Negative Acknowledge” and “stop condition” after receiving the transmission.

The figure below indicates the transition of index value and data on SDA line.

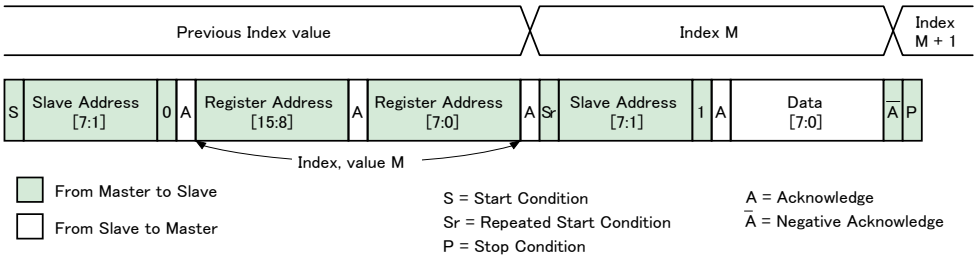


Figure 2-8 CCI single read from random location

2.5.2. CCI single read from current location

When master transmits slave address, but does not designate an index, the previous value is held. And the index value is incremented at the “Acknowledge/Negative Acknowledge” after reading/writing the register value. When master knows the current index value is set to the requested address, master transmits the slave address and read request, then the value in the register appears on SDA line right after the “Acknowledge” issued by this sensor. Master issues the “Negative Acknowledge” and “Stop” and terminates the communication. Since the index value is incremented by “one” with this “Negative Acknowledge”, master can read the register value of the next address with the same procedure.

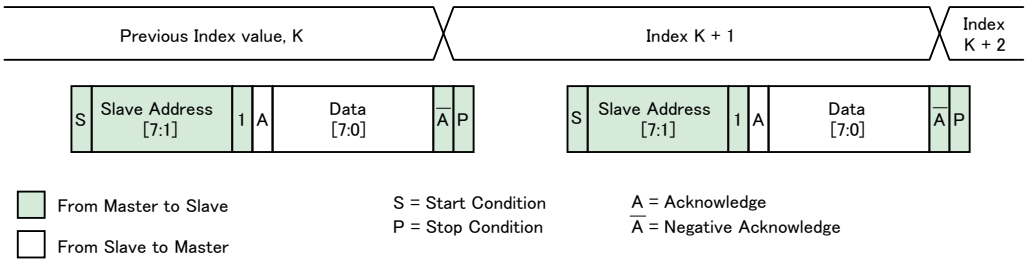


Figure 2-9 CCI single read from current location

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2.5.3. CCI sequential read starting from random location

When reading the data from an arbitrary address sequentially, master reads the first data by the similar procedure to “CCI single read from random location” but issues the “Acknowledge” instead of “Negative Acknowledge”, the index is incremented by “one” with this “Acknowledge” then master can repeat the read operation. This operation is terminated when master issues the “Negative Acknowledge”, “Stop condition”, and the communication is terminated.

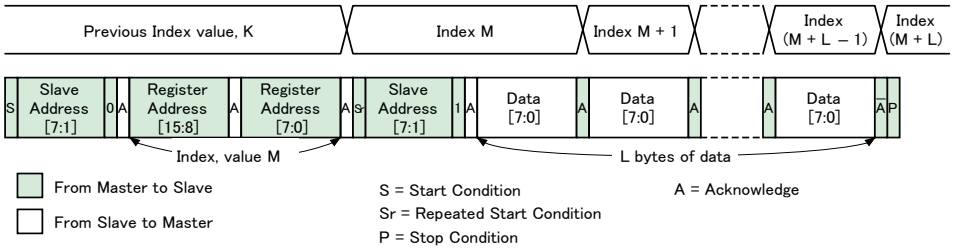


Figure 2-10 CCI sequential read starting from random location

2.5.4. CCI sequential read starting from current location

When master knows the current index value is set to the requested address, master transmits the slave address and read request, then the value from the register appears on SDA line right after the “Acknowledge” issued by this sensor. Master issues the “Acknowledge” after receiving 1-byte of the data and this sensor continuously transmits the data from the next address of register since the index value is incremented by “one” with this “Acknowledge”. By repeating issue of the “Acknowledge” for every 1-byte reading, master can read the data sequentially. After reading necessary bytes of data, master issues “Negative Acknowledge” and “stop condition” and then terminates the communication.

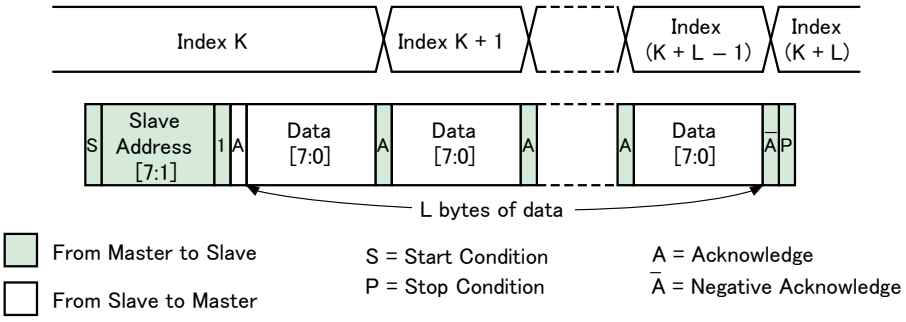


Figure 2-11 CCI sequential read starting from current location

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2.5.5. CCI single write to random location

When writing single datum to a register of target address, the master device starts write-operation with the slave address of this sensor and by making 2-bytes of dummy write master sets the address value (M) to the index. And then master transmits the data to be written to the register addressed by index value. Master issues “stop condition” after it transmits the data and terminates the communication

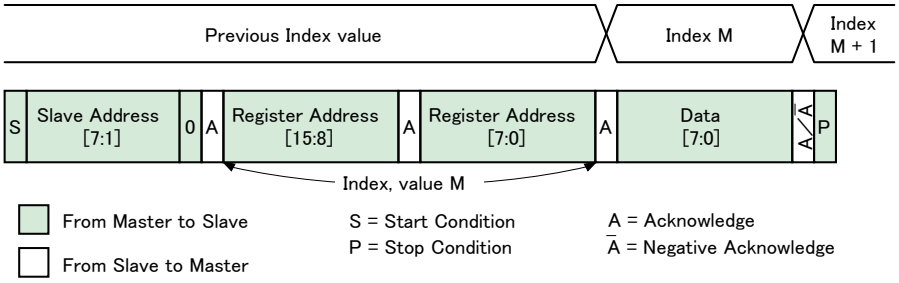


Figure 2-12 CCI single write to random location

2.5.6. CCI sequential write starting from random location

When master writes the data sequentially from the requested address, master uses a similar procedure to do “CCI single write to random location” and without issuing a “stop condition” and continuously transmits the data after each “Acknowledge” issued. This sensor issues “Acknowledge” for each 1-byte write operation. After transmitting necessary data, master issues a “stop condition” and terminates the communication.

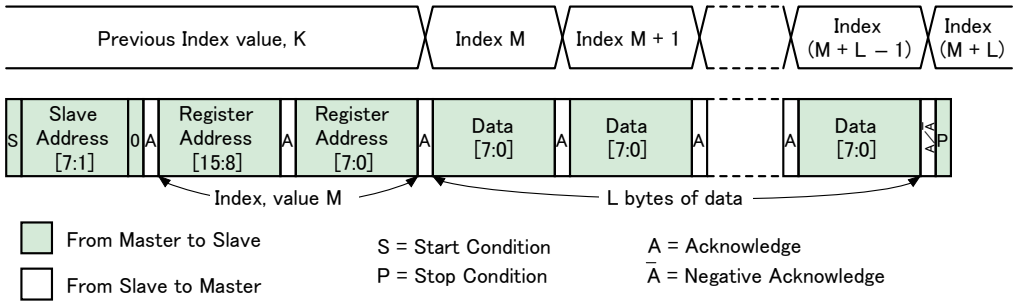


Figure 2-13 CCI sequential write starting from random location

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2.6. 2-wire serial communication register update timing

There are two types of registers in terms of the update timing of the transmitted data: the immediate type and the double buffered type. For immediate type registers, the transmitted data will be written to the registers immediately. As for the v-latched or double buffered type registers, the actual update timing of the register contents is controlled to the proper timing and become valid in the +1frame or +2frame depending on register function. Users can transmit the commands regardless of the internal update timing of this sensor. The registers of double buffered types are indicated with “v” mark at the “reflection” column of Register Map of this sensor.

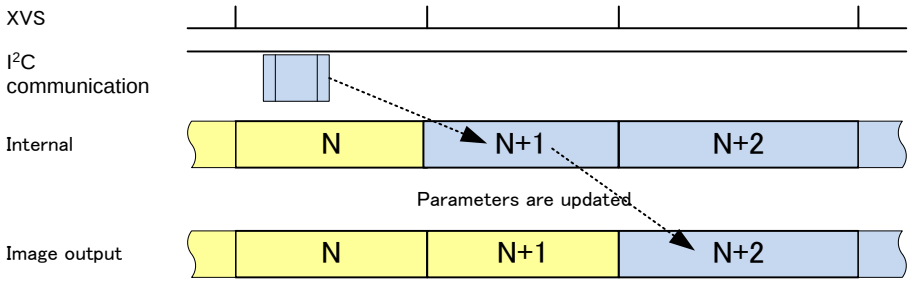


Figure 2-14 2-wire serial communication register update timing diagram (v-latched or double buffered type registers)

Regarding the use case based CCI communications and reflection timings, refer to each section as follows:

For reflection timing of capture mode transition, see 7.2 Streaming mode transition.

For reflection timing of exposure time and gain change, see 7.3 Exposure and gain change sequence and AE bracketing.

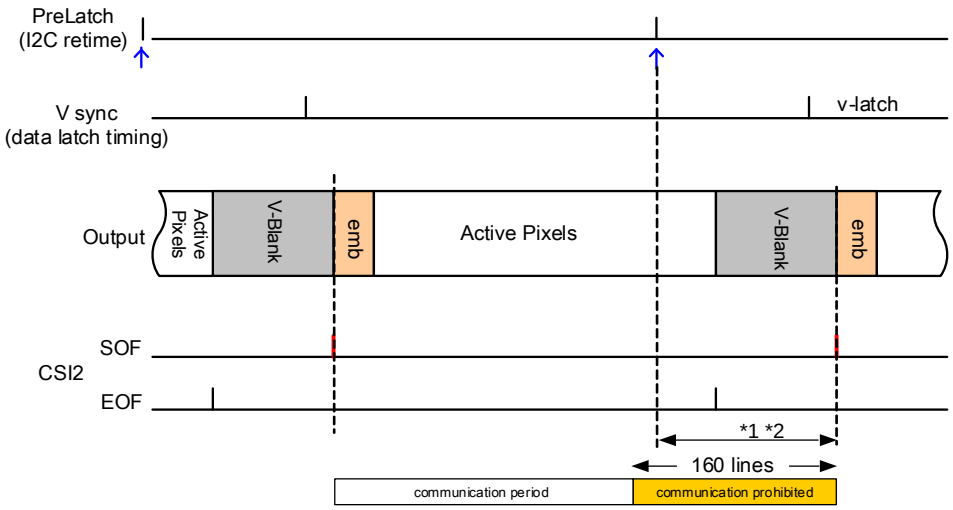
Cautions on pre-latch timing of double buffered registers

Retimed registers are pre-latched before V-sync for internal transaction.

Pre-latch timing exist several lines ahead of the embedded data (FS), and the timing changes depending on the operating condition settings. Registers written ahead of the pre-latch timing are updated internally during the next frame, however registers afterwards are updated internally but may be delayed a frame in addition to the normal update timings. The data transmission to be updated shall be completed within the communication period outlined in Figure2-15 in order to internally update the settings for the target frame; first frame for non-GPH registers and second frame for GPH registers after register settings.

The same timing rule applies for GPH release setting too.

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*1 This period is that number of lines may vary depending on the conditions (less than 160 lines always)
*2 Setting within this period may cause an additional frame delay in addition to the normal update timing of the next frame.

Figure 2-15 Pre-latch timing

2.7. Grouped parameter hold function

2.7.1. Purpose and functional description of grouped parameter hold function

The image shooting parameters are assigned with many registers and they need to be changed within one frame period of the image. However the communication speed is limited and setting all of necessary registers might not be completed. So the double buffered type registers have the “grouped parameter hold” function to behave to be updated at once. While “grouped parameter hold” register is set to “1” the transmitted data are held in the buffer registers and after resets “grouped parameter hold” register to “0”, imaging parameter register values are updated as if they are transmitted at the same time and realize a smooth transition for changing the imaging condition.

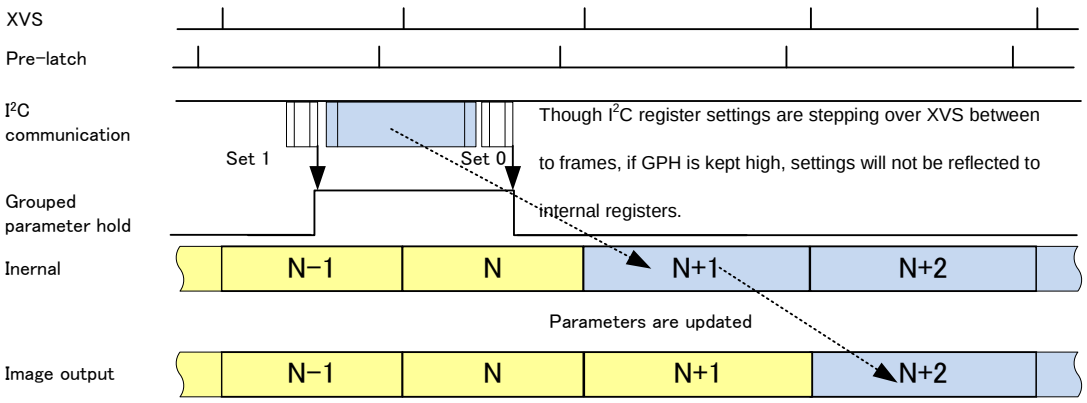
In this section, basic concept of the functionality and general ideas of application are explained. It is required to refer to individual sections for complete understanding on application. In such cases pointers to reference sections are also shown.

The register update timing using the “Group parameter hold function” is shown in Figure2-16. If the timing to set "Group parameter hold" from 1 to 0 is before the pre-latch timing, the register is updated internally during the next frame, however if it is afterwards, the update is delayed by 1 frame. It is recommended to release GPH within the "communication period" outlined in Figure 2-15 in order to update the register with the target timing.

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Table 2-6 Grouped parameter hold function

I²C register	Address	Bit	Name	Description
	0x0104	[0]	GRP_PARAM_HOLD	This register is a hold control register for updating multiple parameters within the same frame. 0 : hold release 1 : hold

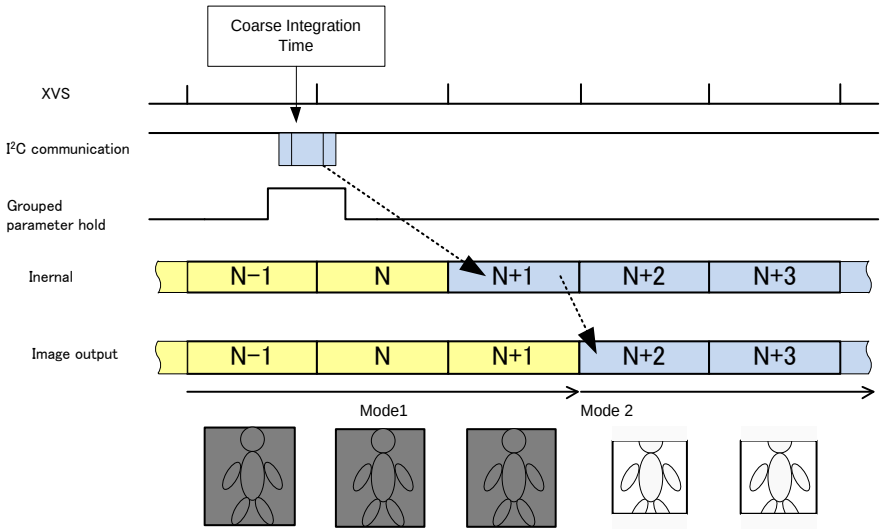


* Example of integration time change.

Figure 2-16 GRP_PARAM_HOLD function timing diagram

2.7.2. Example of mode transition using grouped parameter hold

An example of “On the fly” mode transition using "grouped parameter hold" is shown in the following chart. For more detail of “On the fly” mode transition, see section “2.7 Grouped parameter hold function” and Appendix “A-1.2”.



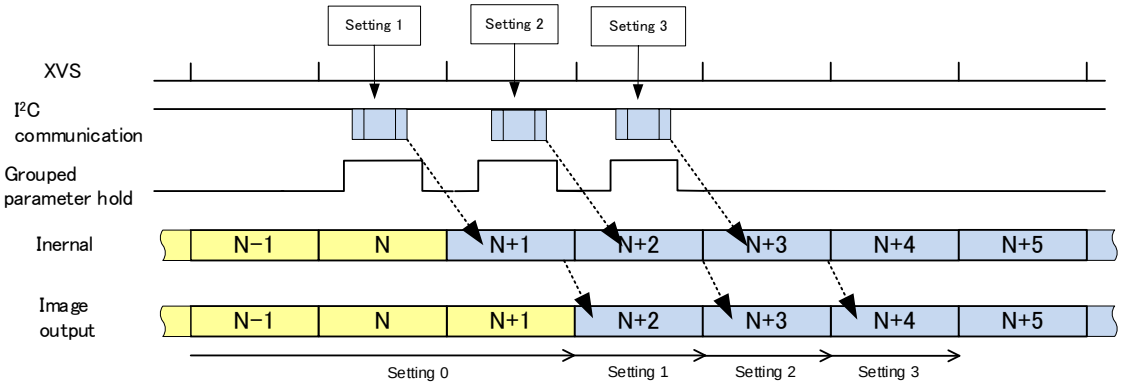
* Example of integration time change.

Figure 2-17 Mode transition using GRP_PARAM_HOLD

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2.7.3. Example of using grouped parameter hold continuously

With this sensor, GPH can be used to update registers in continuous frames, unlikely from older product design. This is outlined in the following timing chart. Note that settings for both setting and resetting GPH must be completed within 1 frame ("communication period" shown in Figure2-15).



* Example of integration time change.

Figure 2-18 GRP_PARM_HOLD continuous frame register update

In “7.2 Streaming mode transition”, you can find more about the grouped parameter function.

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3. Image Signal Interface

This sensor is compliant with MIPI Alliance Standard for Camera Serial Interface 2 (CSI-2) version 1.3 regarding image signal output standards and compliant with D-PHY version 1.2 and C-PHY version 1.0 regarding physical layer standards. With this sensor, PHY settings can be switched with the CSI_SIG_MODE register as listed in the following table. PHY changes must be performed during the SW-STB process.

Table 3-1 CSI sig mode register

I2C Address	Bit	Name	Description
0x0111	[1:0]	CSI_SIG_MODE	The signaling method control 0 : not supported 1 : not supported 2 : CSI-2 Signaling with D-PHY 3 : CSI-2 Signaling with C-PHY

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3.1. MIPI D-PHY transmitter

This sensor outputs image signals by CSI2 high speed serial interface consisting of one pair of clock lines and four pairs of data lines. Refer to MIPI Alliance Standard for Camera Serial Interface 2 (CSI-2) version 1.3 and MIPI Alliance Specification for D-PHY version 1.2 for details. Because signals are transmitted in differential pairs, impedance (generally 100Ω) between differential pairs near the receiver side during HS mode is required.

Otherwise, select a receiver with built-in impedance between differential pairs. Different delay times of differential pairs may reduce the input timing margin of ISP devices, which leads to malfunctions. Therefore, delay time within and among differential pairs must be as similar as possible in layout.

3.1.1. CSI lane mode

Table 3-2 CSI lane mode register

	Address	Bit	Name	Description
I^2C register	0x0114	[1:0]	CSI_LANE_MODE	Number of lanes for CSI 0 : not supported* 1 : 2-lane 2 : not supported * 3 : 4-lane

* This sensor supports only 2Lane and 4Lane.

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MIPI pins function as shown in the following table according to CSI_LANE_MODE.

Table 3-3 MIPI transmitter for D-PHY

	Pin name	2Lane	4Lane
MIPI transmitter	CKP1B/CKN1C	Clock lane	Clock lane
	D1P0C/D1N1A	Data lane 1	Data lane 1
	D2P2A/D2N2B	Data lane 2	Data lane 2
	D3P0A/D3N0B	Not used (LP only)	Data lane 3
	D4P2C/D4N		Data lane 4

3.1.2. CSI data format for D-PHY

The following register table specifies data format of the image output.

It sets data format from the options of output format as shown in the table below.

Table 3-4 CSI data format registers

	Address	Bit	Name	Description
I^2C register	0x0112	[7:0]	CSI_DT_FMT_H	The output data format for CSI CSI_DT_FMT_H : Uncompressed Data Bit Width CSI_DT_FMT_L : Compressed Data Bit Width 0x0808 : RAW8 (top 8bit of internal data) 0x0A08 : COMP8 (10bit to 8bit compression) 0x0A0A : RAW10 (top 10bit of internal data) *Setup other than the above is forbidden.
	0x0113	[7:0]	CSI_DT_FMT_L	

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3.1.3. CSI-2 bus

3.1.3.1. RAW8/COMP8

The 8-bit compression data transmission is performed by transmitting the pixel data over a CSI-2 bus. The following table specifies the packet size constraints for RAW8/COMP8 packets. The length of each packet must be a multiple of the values in the table.

Table 3-5 Packet size constraints for RAW8/COMP8 packet

Pixels	Bytes	Bits
1	1	8

Bit order in transmission follows the general CSI-2 rule, LSB first.

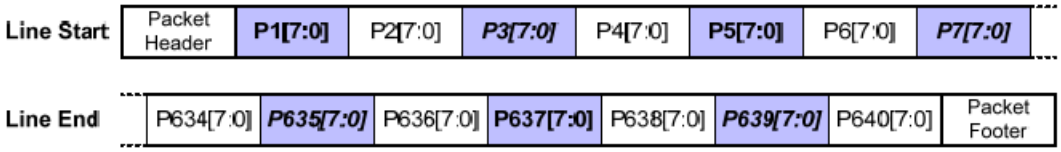


Figure 3-1 RAW8/COMP8 transmission

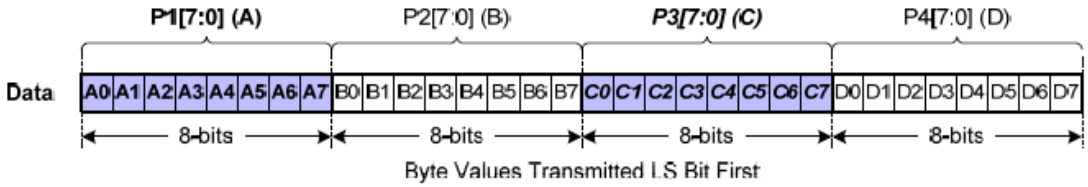


Figure 3-2 RAW8/COMP8 data transmission on CSI-2 bus bitwise illustration

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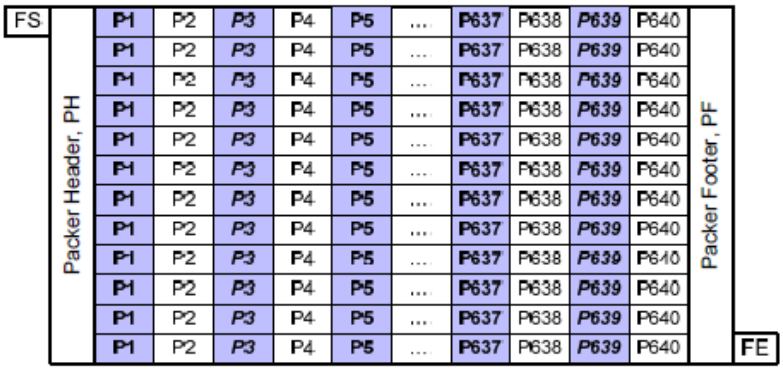


Figure 3-3 RAW8/COMP8 frame format

3.1.3.2. RAW10

The transmission of 10-bit Raw data is accomplished by packing the 10-bit pixel data to look like 8-bit data format. The following table specifies the packet size constraints for RAW10 packets. The length of each packet must be a multiple of the values in the table.

Table 3-6 Packet size constraint for RAW10 packets

Pixels	Bytes	Bits
4	5	40

Bit order in transmission follows the general CSI-2 rule, LSB first.

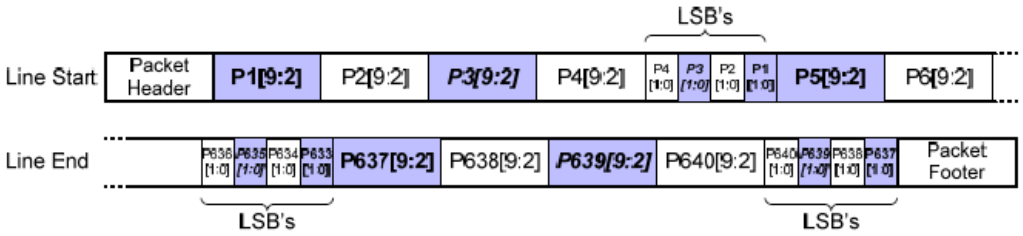


Figure 3-4 RAW10 transmission

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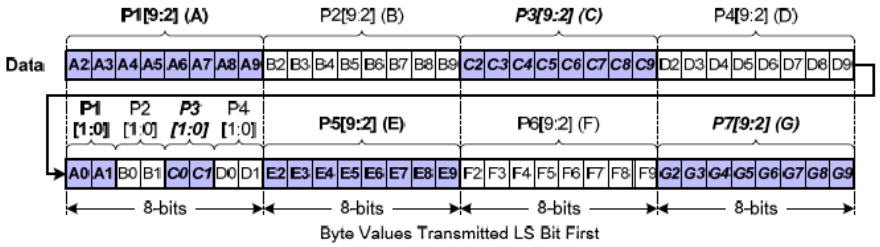


Figure 3-5 RAW10 data transmission on CSI-2 bus bitwise illustration

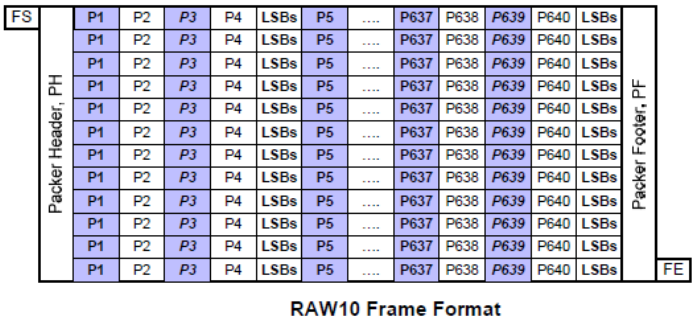


Figure 3-6 RAW10 frame format

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3.1.4. MIPI Global Timing setting for D-PHY

MIPI Global Timing parameters are set with the registers below.

The setting method of MIPI Global Timing can be selected with the PHY_CTRL register.

C-PHY and D-PHY use this register in common.

PHY_CTRL=0

Use automatic control (Automatic Mode)

MIPI Global timing automatically set the appropriate value which is calculated by the below bitrate setting.

Bitrate = (PLL output frequency) / (OP Sys Clock Divider)

(PLL output frequency) = INCK ×IOP_PLL_ MPY / IOP_PREPLLCK_ DIV

PHY_CTRL=1

MIPI Global Timing is automatically set using the values of REQ_LINK_BIT_RATE_MBPS and CSI_LANE_MODE.

Global Timing Setting is automatically set every time the setting of REQ_LINK_BIT_RATE_MBPS is changed.

The value of the REQ_LINK_BIT_RATE_MBPS can be obtained from the following relational equation.

REQ_LINK_BIT_RATE_MBPS = Output bitrate * (CSI_LANE_MODE +1)

Where Output bitrate = OPPXCK * CSI_DT_FMT_L[7:0]

* REQ_LINK_BIT_RATE [31:0] consist of integer part (REQ_LINK_BIT_RATE_MBPS [31:16]) and decimal part (REQ_LINK_BIT_RATE_MBS [15:0]).

* For OPPXCK, refer to “5.1.4 IOPSYCK Clock”.

PHY_CTRL=2

Global Timing must be set with registers from 0x080a to 0x0819 manually.

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Table 3-7 MIPI D-PHY Global timing setting

	Address	Bit	Name	Description
I²C register	0x0808	[1:0]	PHY_CTRL	MIPI Global Timing control selection 0 : automatic control 1 : UI control 2 : register control 3 : reserved C-PHY and D-PHY use this register in common.
	0x080a	[1:0]	TCLK_POST_EX[9:8]	MIPI Global Timing (Tclk)
	0x080b	[7:0]	TCLK_POST_EX[7:0]	
	0x080c	[1:0]	THS_PREPARE_EX[9:8]	MIPI Global Timing (Ths_prepare)
	0x080d	[7:0]	THS_PREPARE_EX[7:0]	
	0x080e	[1:0]	THS_ZERO_MIN_EX[9:8]	MIPI Global Timing (Ths_zero_min)
	0x080f	[7:0]	THS_ZERO_MIN_EX[7:0]	
	0x0810	[1:0]	THS_TRAIL_EX[9:8]	MIPI Global Timing (Ths_trail)
	0x0811	[7:0]	THS_TRAIL_EX[7:0]	
	0x0812	[1:0]	TCLK_TRAIL_MIN_EX[9:8]	MIPI Global Timing (Tclk_trail_min)
	0x0813	[7:0]	TCLK_TRAIL_MIN_EX[7:0]	
	0x0814	[1:0]	TCLK_PREPARE_EX[9:8]	MIPI Global Timing (Tclk_prepare)
	0x0815	[7:0]	TCLK_PREPARE_EX[7:0]	
	0x0816	[1:0]	TCLK_ZERO_EX[9:8]	MIPI Global Timing (Tclk_zero)
	0x0817	[7:0]	TCLK_ZERO_EX[7:0]	
	0x0818	[1:0]	TLPX_EX[9:8]	MIPI Global Timing (Tlpx)
	0x0819	[7:0]	TLPX_EX[7:0]	
	0x0820	[7:0]	REQ_LINK_BIT_RATE_MBPS[31:24]	Output Data Rate [Mbps] Bit[32:16] integer Bit[15:0] decimal
	0x0821	[7:0]	REQ_LINK_BIT_RATE_MBPS[23:16]	
	0x0822	[7:0]	REQ_LINK_BIT_RATE_MBPS[15:8]	
	0x0823	[7:0]	REQ_LINK_BIT_RATE_MBPS[7:0]	

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3.1.5. CLK mode during Frame blanking

This is a function to stop the MIPI clock during Frame Blanking.

The details of the register and timing are shown below.

Table 3-8 CLK mode setting register during Frame Blanking

	Address	Bit	Name	Description
I²C register	0xe000	[0]	FRAME_BLANKSTOP_CL	FRAME_BLANKSTOP_CL Control RUN/STOP of CSI2 during Frame Blanking 0 : Stay in HS mode (Clock/Strobe output) during CLK Frame Blanking 1 : Transition to LP11 (Clock/Strobe stop) during CLK Frame Blanking Details are as follows: - When a valid frame starts, transition to HS-Clock status. - When the frame ends (after EOF output), transition from HS-Clock to LP11.

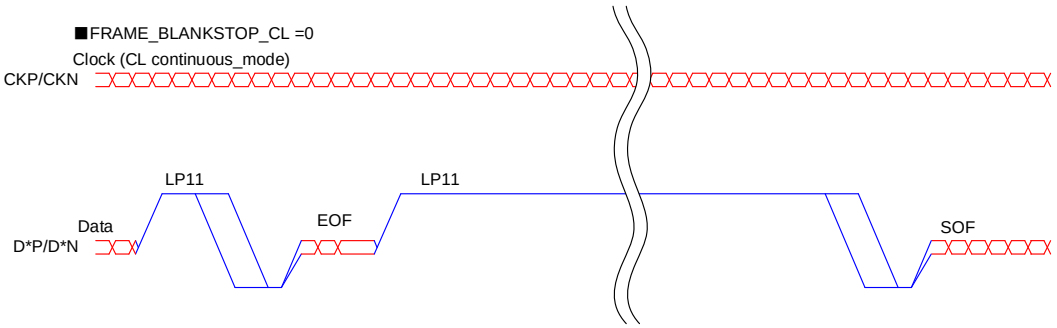


Figure 3-7 CLK mode during Frame Blanking (FRAME_BLANKSTOP_CL =0)

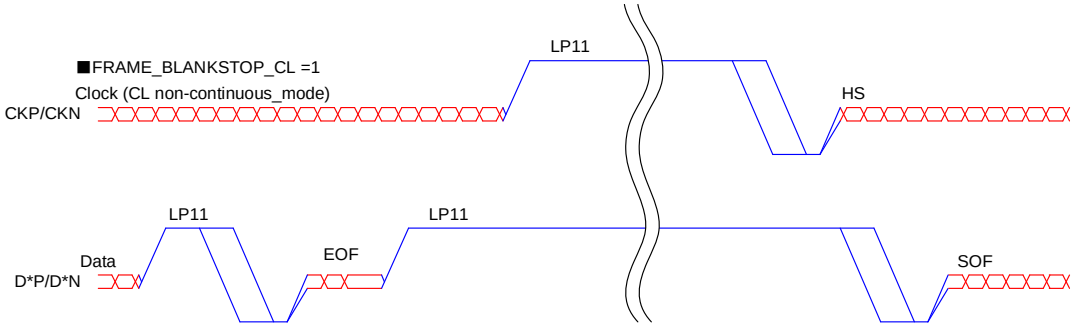


Figure 3-8 CLK mode during Frame Blanking (FRAME_BLANKSTOP_CL =1)

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3.1.6. MIPI Skew Calibration Data Output

This sensor supports skew calibration data output compliant with MIPI Alliance Specification for D-PHY version 1.2. When outputting the data with the bit rate higher than 1.5 Gbps/lane, initial and periodic skew calibration can be transmitted before High-Speed Data Transmission. In this case, the initial skew calibration output can select by SCAL_INIT_EN and the periodic skew calibration output can select by SCAL_PERIOD_EN register. Initial and periodic skew calibrations are composed of a combination of a sync pattern consisting only of "1s" and a clock pattern (010101…) as shown in the following figure. The initial skew calibration transmits a clock pattern after transmitting the 16UI sync pattern. The periodic skew calibration transmits a clock pattern after transmitting the 16UI sync pattern.

Table 3-9 Skew calibration control register

I^2C Address	Bit	Name	Description
0x0830	[0]	SCAL_PERIOD_EN	For period skew calibration control This register can be set only during SW Standby. 0: disable calibration during frame blanking(not auto clear) 1: enable calibration during frame blanking(not auto clear)
0x0832	[0]	SCAL_INIT_EN	For initial skew calibration control This register can be set only during SW Standby. 0: disable calibration in start streaming(not auto clear) 1: enable calibration in start streaming(not auto clear)
0x3400	[0]	SCAL_TSKEWCAL_UNIT	Unit for specifying time that the transmitter drives the skew calibration pattern in the skew calibration mode. This register can be set only during SW Standby. 0: µsec 1: MDBCK (Bit rate [Mbps/lane] / 8)
0x3408	[7:0]	SCAL_TSKEWCAL_INIT[15:8]	Time that the transmitter drives the skew-calibration pattern in the initial skew calibration mode. Unit is specified by SCAL_TSKEWCAL_UNIT. This register can be set only during SW Standby.
0x3409	[7:0]	SCAL_TSKEWCAL_INIT[7:0]	
0x340a	[7:0]	SCAL_TSKEWCAL_PERIOD[15:8]	Time that the transmitter drives the skew- calibration pattern in the periodic skew calibration mode. Unit is specified by SCAL_TSKEWCAL_UNIT. This register can be set only during SW Standby.
0x340b	[7:0]	SCAL_TSKEWCAL_PERIOD[7:0]	

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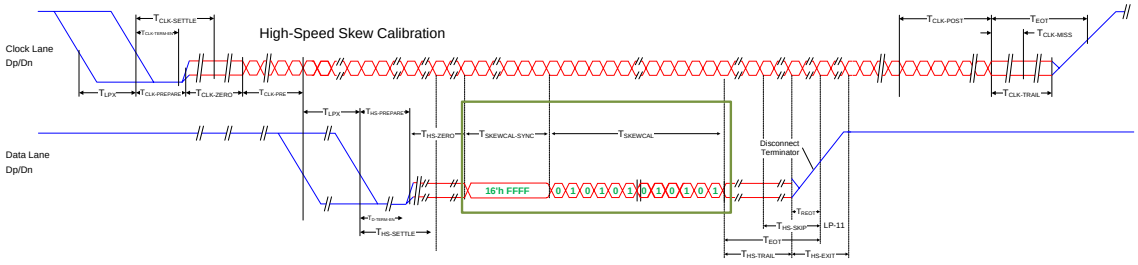


Figure 3-9 Skew calibration data output

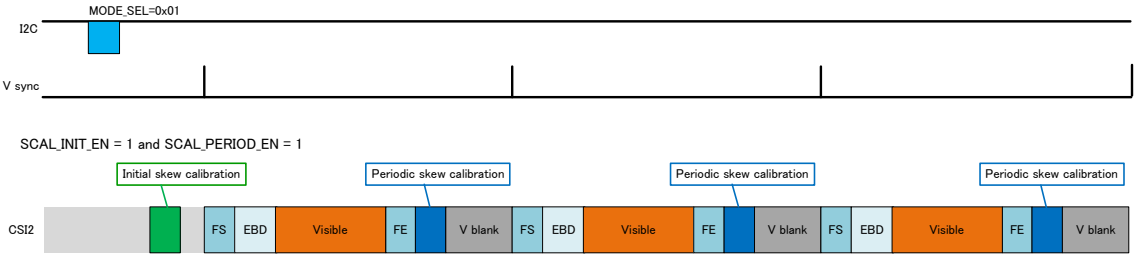


Figure 3-10 Initial and Periodic skew calibration data output

3.2. MIPI C-PHY transmitter

C-PHY can be used, which is one physical layer data transfer standard developed by the MIPI Alliance.

The C-PHY is based on 3-Phase symbol encoding technology delivering 2.28 bits per symbol over three-wire Trios. C-PHY has many characteristics in-common with D-PHY. C-PHY was designed to be able to coexist on the same IC pins D-PHY so that dual-mode devices can be developed.

C-PHY high-speed data coding differs substantially from the D-PHY clock-forwarding system. Clock timing is encoded into each symbol. This is accomplished by requiring that combination of voltages driven onto the wires must change at symbol boundary. This simplifies clock recovery.

As C-PHY signals are output over 3 lines as a single connection, the recommended signal impedance for a single line is 50 ohms. Make sure all 3 lines wired to the receiving device are the same length for the C-PHY output signals. If all lines are not the same length, the differences in wire length cause jitter at the receiving device, which reduces the timing margin. The C-PHY global timing control register is different from that for D-PHY.

Refer to MIPI Alliance C-PHY Standard v1.0 for more detailed information on C-PHY standards.

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3.2.1. CSI lane mode

Table 3-10 CSI lane mode register

I^2C Address	Bit	Name	Description
0x0114	[1:0]	CSI_LANE_MODE	Number of lanes for CSI 0 : 1-Trio 1 : 2-Trio 2 : 3-Trio 3 : not supports *

* In the case of C-PHY, This sensor supports only 1Trio, 2Trio, and 3Trio.

Table 3-11 MIPI transmitter for C-PHY

<table>
<tr><td rowspan="14">MIPI transmitter</td><td>Pin name</td><td colspan="2">1-Trio</td><td colspan="2">2-Trio</td><td colspan="2">3-Trio</td></tr>
<tr><td>D3P0A</td><td rowspan="3">Trio1</td><td>Trio1-A</td><td rowspan="3">Trio1</td><td>Trio1-A</td><td rowspan="3">Trio1</td><td>Trio1-A</td></tr>
<tr><td>D3N0B</td><td>Trio1-B</td><td>Trio1-B</td><td>Trio1-B</td></tr>
<tr><td>D1P0C</td><td>Trio1-C</td><td>Trio1-C</td><td>Trio1-C</td></tr>
<tr><td>D1N1A</td><td colspan="2" rowspan="7">Not used(LP only)</td><td rowspan="3">Trio2</td><td>Trio2-A</td><td rowspan="3">Trio2</td><td>Trio2-A</td></tr>
<tr><td>CKP1B</td><td>Trio2-B</td><td>Trio2-B</td></tr>
<tr><td>CKN1C</td><td>Trio2-C</td><td>Trio2-C</td></tr>
<tr><td>D2P2A</td><td colspan="2" rowspan="4">Not used(LP only)</td><td rowspan="3">Trio3</td><td>Trio3-A</td></tr>
<tr><td>D2N2B</td><td>Trio3-B</td></tr>
<tr><td>D4P2C</td><td>Trio3-C</td></tr>
<tr><td>D4N</td><td colspan="2">Not used(LP only)</td></tr>
</table>

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3.2.2. CSI data format for C-PHY

The following register table specifies data format of the image output.

It sets data format from the options of output format as shown in the table below.

Table 3-12 CSI data format registers

I^2C Address	Bit	Name	Description
0x0112	[7:0]	CSI_DT_FMT_H	The output data format for CSI CSI_DT_FMT_H : Uncompressed Data Bit Width CSI_DT_FMT_L : compressed Data Bit Width 0x0808 : RAW8 (top 8bit of internal data) 0x0A08 : COMP8 (10bit to 8bit compression) 0x0A0A : RAW10 (top 10bit of internal data) *Setup other than the above is forbidden.
0x0113	[7:0]	CSI_DT_FMT_L	

3.2.3. CSI-2 bus

3.2.3.1. RAW8/ COMP8

The 8-bit compression data transmission is performed by transmitting the pixel data over a CSI-2 bus. The following table specifies the packet size constraints for RAW8/COMP8 packets. The length of each packet must be a multiple of the values in the table.

Table 3-13 Packet size constraints for RAW8/COMP8 packet

Pixels	Bytes	Bits
1	1	8

Bit order in transmission follows the general CSI-2 rule, LSB first

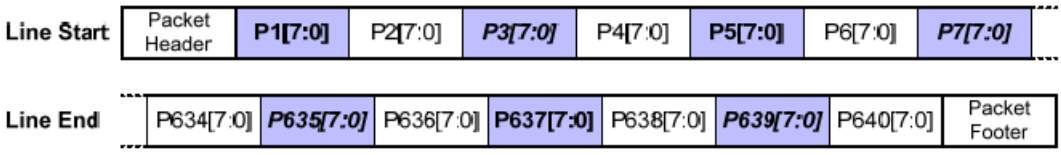


Figure 3-11 RAW8/COMP8 transmission

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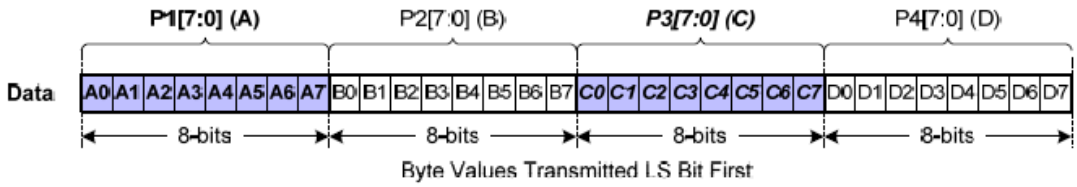


Figure 3-12 RAW8/COMP8 data transmission on CSI-2 bus bitwise illustration

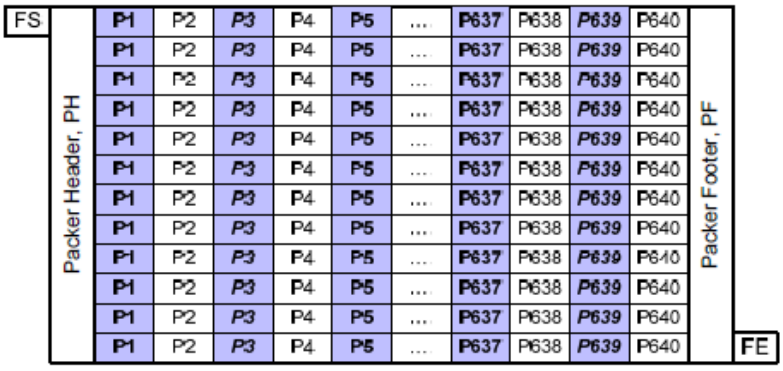


Figure 3-13 RAW8/COMP8 frame format

3.2.3.2. RAW10

The transmission of 10-bit RAW data is accomplished by packing the 10-bit pixel data to look like 8-bit data format. The following table specifies the packet size constraints for RAW10 packets. The length of each packet must be a multiple of the values in the table.

Table 3-14 Packet size constraints for RAW10 packets

Pixels	Bytes	Bits
4	5	40

Bit order in transmission follows the general CSI-2 rule, LSB first.

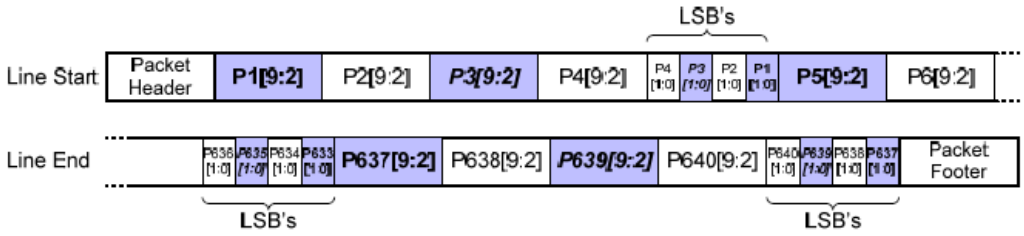


Figure 3-14 RAW10 transmission

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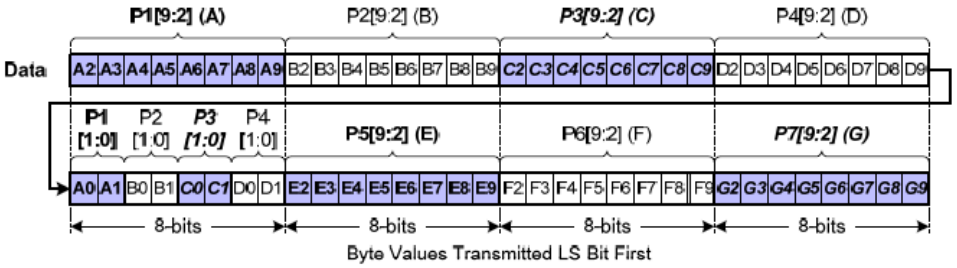


Figure 3-15 RAW10 data transmission on CSI-2 bus bitwise illustration

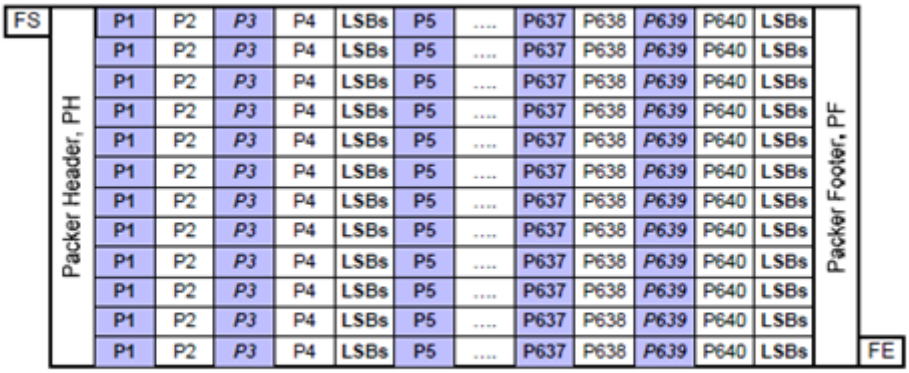


Figure 3-16 RAW10 frame format

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3.2.4. MIPI Global Timing setting for C-PHY

MIPI Global Timing parameters are set with the registers below.

The setting method of MIPI Global Timing can be selected with the PHY_CTRL register.

C-PHY and D-PHY use this register in common.

PHY_CTRL=0

Use automatic control (Automatic Mode)

MIPI Global timing automatically set the appropriate value which is calculated by the below bitrate setting.

Bitrate (symbol per second) = (PLL output frequency) / (OP Sys Clock Divider)

(PLL output frequency) = INCK ×IOP_PLL_MPY / IOP_ PREPLLCK_DIV

PHY_CTRL=1

MIPI Global Timing is automatically set using the values of REQ_LINK_BIT_RATE_MBPS and CSI_LANE_MODE.

Global Timing Setting is automatically set every time the setting of REQ_LINK_BIT_RATE_MBPS is changed.

The value of the REQ_LINK_BIT_RATE_MBPS can be obtained from the following relational equation.

REQ_LINK_BIT_RATE_MBPS = Output bitrate* × (CSI_LANE_MODE +1)

* REQ_LINK_BIT_RATE[31:0] consist of integer part (REQ_LINK_BIT_RATE_MBPS[31:16]) and decimal part (REQ_LINK_BIT_RATE_MBS[15:0]).

Use the value defined by the following expression for the output bit rate described here instead of the symbol bit rate.

Output bitrate = I*OPCK* × *IOPSYCK clock division ratio* *

* See 5.1.4 IOPSYCK Clock

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PHY_CTRL=2

Global Timing must be set with registers from 0x843 to 0x859 manually.

Table 3-15 MIPI C-PHY Global timing setting

I^2C Address	Bit	Name	Description
0x0808	[1:0]	PHY_CTRL	MIPI Global Timing control selection 0 : automatic control 1 : UI control 2 : register control 3 : reserved C-PHY and D-PHY use this register in common.
0x0820	[7:0]	REQ_LINK_BIT_RATE_MBPS[31:24]	Output Data Rate [Mbps] Bit[32:16] integer Bit[15:0] decimal Bit rate [Mbps/lane] * Number of lanes [lane]
0x0821	[7:0]	REQ_LINK_BIT_RATE_MBPS[23:16]	
0x0822	[7:0]	REQ_LINK_BIT_RATE_MBPS[15:8]	
0x0823	[7:0]	REQ_LINK_BIT_RATE_MBPS[7:0]	
0x0843	[5:3]	TGR_PPRG_SEQ_01	Symbol 1 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0843	[2:0]	TGR_PPRG_SEQ_00	Symbol 0 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0844	[5:3]	TGR_PPRG_SEQ_03	Symbol 3 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0844	[2:0]	TGR_PPRG_SEQ_02	Symbol 2 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0845	[5:3]	TGR_PPRG_SEQ_05	Symbol 5 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0845	[2:0]	TGR_PPRG_SEQ_04	Symbol 4 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0846	[5:3]	TGR_PPRG_SEQ_07	Symbol 7 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0846	[2:0]	TGR_PPRG_SEQ_06	Symbol 6 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0847	[5:3]	TGR_PPRG_SEQ_09	Symbol 9 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.

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0x0847	[2:0]	TGR_PPRG_SEQ_08	Symbol 8 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0848	[5:3]	TGR_PPRG_SEQ_11	Symbol 11 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0848	[2:0]	TGR_PPRG_SEQ_10	Symbol 10 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0849	[5:3]	TGR_PPRG_SEQ_13	Symbol 13 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x0849	[2:0]	TGR_PPRG_SEQ_12	Symbol 12 of the Preamble Programmable Sequence. =0 to 4 - symbol value =5 to 7 - invalid value, not a valid symbol value.
0x084E	[1:0]	T3PREPARE[9:8]	C-PHY Global Timing Parameter T3PREPARE [MDBCK]
0x084F	[7:0]	T3PREPARE[7:0]	
0x0850	[1:0]	T3LPX[9:8]	C-PHY Global Timing Parameter TLPX [MDBCK]
0x0851	[7:0]	T3LPX[7:0]	
0x0852	[1:0]	T3HSEXIT[9:8]	C-PHY Global Timing Parameter T3HSEXIT [MDBCK]
0x0853	[7:0]	T3HSEXIT[7:0]	
0x0854	[1:0]	T3PREBEGIN[9:8]	C-PHY Global Timing Parameter T3PREBEGIN [MDBCK]
0x0855	[7:0]	T3PREBEGIN[7:0]	
0x0856	[0]	T3PROGSEQ_EN	MIPI Global Timing for C-PHY control (t3-PROGSEQ enable) 0: t3-PROGSEQ = 0UI 1: t3-PROGSEQ = 14UI
0x0858	[1:0]	T3POST[9:8]	C-PHY Global Timing Parameter T3POST [MDBCK]
0x0859	[7:0]	T3POST[7:0]	

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4. Image Readout Operation

In this chapter, you can find relations among imaging area, pixel readout orientation/readout and transmission frame/signal structure.

By setting the parameters of PLL, image size, start/end position of the imaging area, direction of reading image, binning, shutter mode, integration time, gain, and output format via 2-wire serial communication, this sensor outputs the image data.

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4.1. Physical alignment of imaging pixel array

The figure below shows the physical alignment of the imaging pixel array with pin #1 located at the upper left corner.

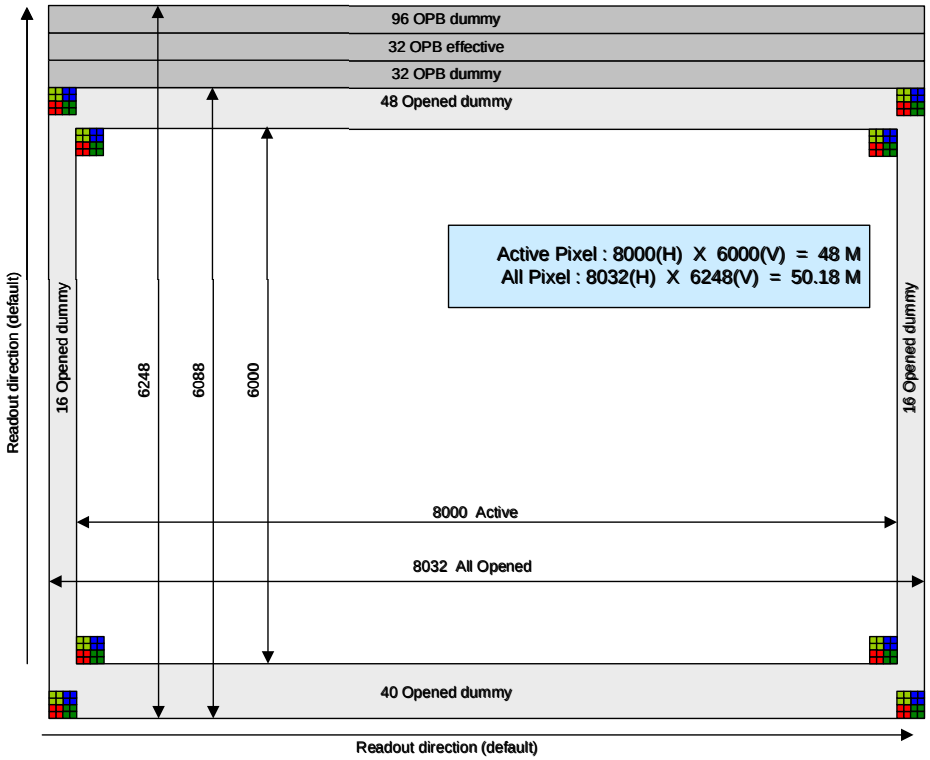


Figure 4-1 Physical alignment of imaging pixel array

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4.2. Color coding and order of reading image data

The original color filter arrangement of the sensor is shown in the figure below. Gr and Gb are the G signals shown at the same line as R signals and B signals respectively. The line with R & Gr signals and the line with Gb & B signals are output alternating one after the other.

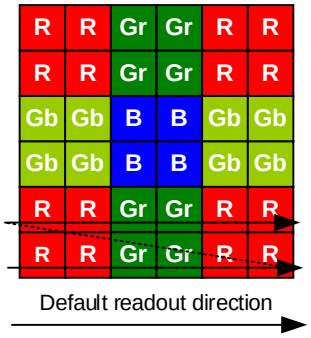


Figure 4-2 Color coding alignment

4.3. Imaging area determination

Imaging area is specified on the output signal with physical coordinate with lower left as (0, 0) and determined by registers shown as below.

Table 4-1 Imaging area determining registers

	Address	Bit	Name	Description
I²C register	0x0344	[4:0]	X_ADD_STA[12:8]	Horizontal direction analog cropping start position within the active pixels area Unit : pixels Format : 13-bit unsigned integer Should be set to 0. * Note that this is the cropping end position when mirroring (IMG_ORIENTATION_H=1)
	0x0345	[7:0]	X_ADD_STA[7:0]	
	0x0346	[4:0]	Y_ADD_STA[12:8]	Vertical direction analog cropping start position within the active pixels area Unit : pixels Format : 13-bit unsigned integer * Note that this is the cropping end position when flipping (IMG_ORIENTATION_V=1)
	0x0347	[7:0]	Y_ADD_STA[7:0]	
	0x0348	[4:0]	X_ADD_END[12:8]	Horizontal direction analog cropping end position within the active pixels area Unit : pixels Format : 13-bit unsigned integer Should be set to 7999. * Note that this is the cropping start position when mirroring (IMG_ORIENTATION_H=1)
	0x0349	[7:0]	X_ADD_END[7:0]	

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	0x034a	[4:0]	Y_ADD_END[12:8]	Vertical direction analog cropping end position within the active pixels area Unit : pixels Format : 13-bit unsigned integer * Note that this is the cropping start position when flipping (IMG_ORIENTATION_V=1)
	0x034b	[7:0]	Y_ADD_END[7:0]	

* Above settings shall be always set based on full pixel basis even in binning mode.

There are capture mode and/or binning mode dependent restrictions in setting Imaging area as shown in the following table.

Table 4-2 Register setting Restrictions list for each mode

	Mode			
	Full-pixel	2x2 Adjacent Pixel Binning	2x2 Adjacent Pixel Binning V2H2	QBC HDR
Y_ADD_STA	multiples of 8	multiples of 8	multiples of 8	multiples of 8
Y_ADD_END – Y_ADD_STA+ 1	multiples of 16	multiples of 8	multiples of 16	multiples of 16
X_ADD_STA	multiples of 4	multiples of 8	multiples of 16	multiples of 4
X_ADD_END – X_ADD_STA + 1	multiples of 4	multiples of 8	multiples of 16	multiples of 4

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4.4. Readout start position

Default readout position of this sensor starts from the lower left when PIN1 is placed at the upper left corner. Because the lens will invert the image both vertically and horizontally, the proper image can be archived when PIN1 is placed at the upper left corner.

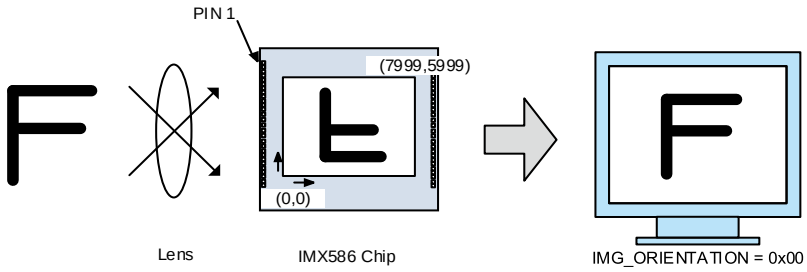


Figure 4-3 Readout start position

Vertical flip and horizontal mirror readout modes can be specified by the register below. And when readout start and end positions are matching the readout size, the same area is displayed when flipping/mirroring the image.

When changing the readout direction, the color of first readout pixel (R/Gr/Gb/B) also changes with it.

Table 4-3 Vertical flip and horizontal mirror

	Address	bit	Name	Description
I²C register	0x0101	[1]	IMG_ORIENTATION_V	Image orientation for Vertical direction 0 : normal 1 : reverse
		[0]	IMG_ORIENTATION_H	Image orientation for Horizontal direction 0 : normal 1 : reverse

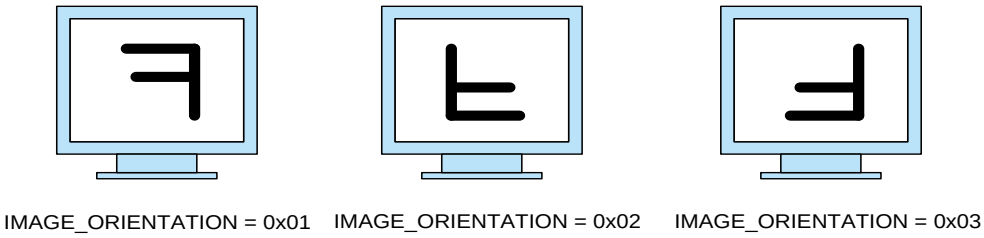


Figure 4-4 Read out image for each combination of flip and mirror

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4.5. Frame structure

Each line of each image frame will be output according to the General Frame Format of CSI2 specification. A period of time from the line end sync code (Packet Footer (PF)) to the line start sync code (Packet Header (PH)) of the next line is called “line blanking”.

Similarly, a period of time from the frame end sync code (Frame End (FE)) to the frame start sync code (Frame Start (FS)) of the next frame is called “frame blanking”.

Frame size is determined by “FRM_LENGTH_LINES” in the vertical direction and “LINE_LENGTH_PCK” in the horizontal direction.

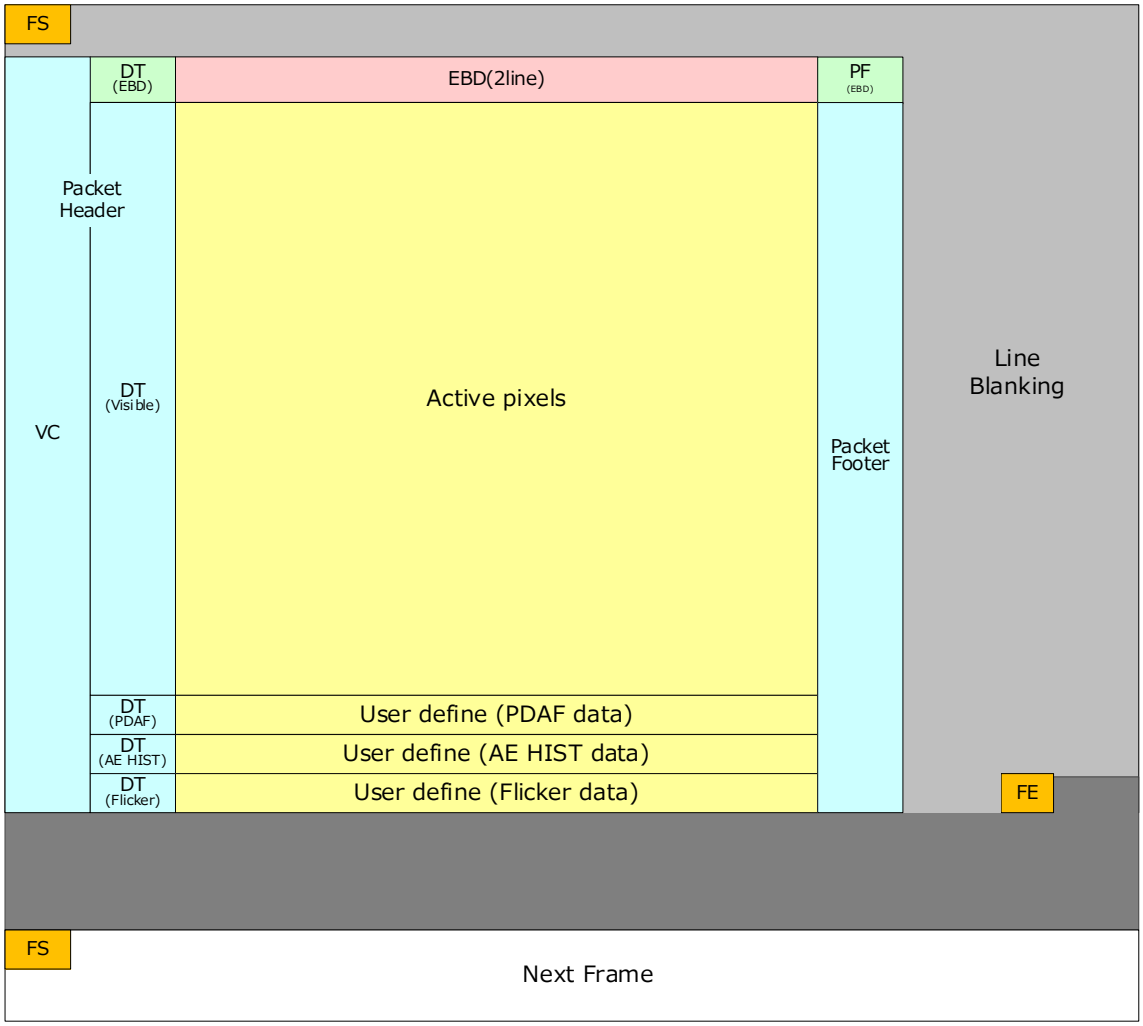


Figure 4-5 Frame structure (Normal)

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4.5.1. Registers that determine frame size

Table 4-4 Registers that determine frame size

	address	bit	name	description
I²C register	0x0340	[7:0]	FRM_LENGTH_LINES[15:8]	The length of frame Unit : lines Format : 16-bit unsigned integer *set value for given capture mode ≦ 65535d
	0x0341	[7:0]	FRM_LENGTH_LINES[7:0]	
	0x0342	[7:0]	LINE_LENGTH_PCK[15:8]	The length of line Unit : pixels Format : 16-bit unsigned integer *LINE_LENGTH_PCK has been constrained. * Must be set fixed value as Table4-5.
	0x0343	[7:0]	LINE_LENGTH_PCK[7:0]	

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4.5.1.1. Constraints of LINE_LENGTH_PCK

Setting value that can be set as LINE_LENGTH_PCK is constrained. It must be set fixed value as follows for each mode and horizontal analog crop setting.

Setting other than these values is prohibited.

H-sync timing is fixed in order to avoid the sensor internal interference (FPN) in this sensor by LINE_LENGTH_PCK.

Table 4-5 Constraints of LINE_LENGTH_PCK

	0x0900 [0]	0x0901 [7:0]	0x3247 [7:0]	0x3F0C [0]	0x3F14 [0]	0x3F15 [1:0]	0x0342, 0x0343 [7:0], [7:0]	
Mode	BINNING_MODE [0]	BINNING_TYPE_H [7:4] BINNING_TYPE_V [3:0]	BINNING_PRIORITY_V [7:0]	BINNING_PDAF_EN [0]	FULL_QHDR_EN [0]	ADC_MODE [1:0]	LINE_LENGTH_PCK [15:0] Default is bold	
							D-PHY	C-PHY
Full-pixel	0x0	x	x	0x0	0x1	0x0	**14704**, 29408	**9440,** 18880
QBC HDR							**9440**, 18880	**9440**, 18880
2x2 Adjacent Pixel Binning with PDAF	0x1	0x22	0x81	0x1	0x0	0x0	**7872**, 15744	**7872**, 15744
2x2 Adjacent Pixel Binning without PDAF				0x0			**7808**, 15616	**5376**, 10752
2x2 Adjacent Pixel Binning V2H2	0x1	0x44	0x89	0x0	0x0	0x0	**5376**, 10752	**5376**, 10752
2x2 Adjacent Pixel Binning V2H2 720p						0x1	**4496**, 8992	**4496**, 8992

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4.6. Output image format

This is the output image diagram of full pixel output mode, Image data is output from the upper left corner of the diagram.

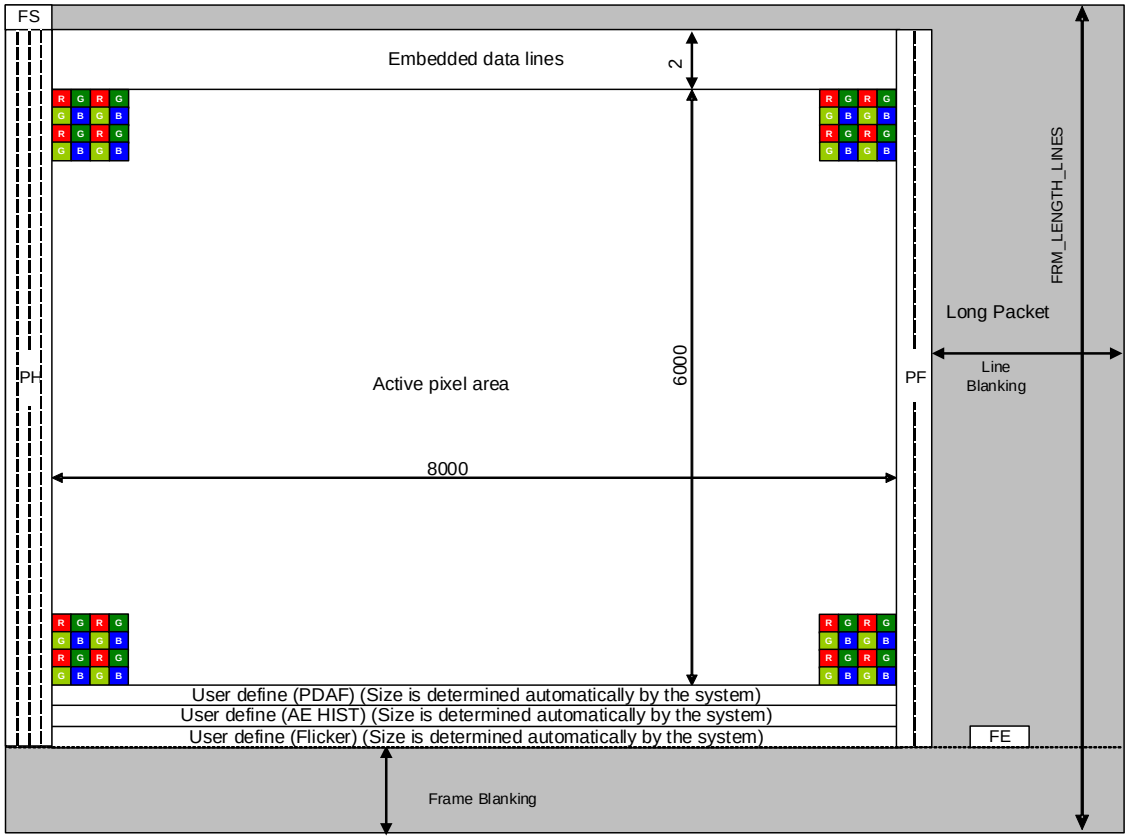


Figure 4-6 Full pixel output mode image data structure

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4.6.1. PD data control

PD (Phase Difference) based on PDAF image data can be output as a packet different from normal image data during the frame blanking period. The function is set to turn ON/OFF with PDAF_CTRL1 [0]. PD data X/Y size is calculated automatically.

Table 4-6 PD data output control

I^2C Address	Bit	Name	Description
0x3e37	[0]	PDAF_CTRL1[0]	PD ON/OFF control 1:Enable 0:Disable

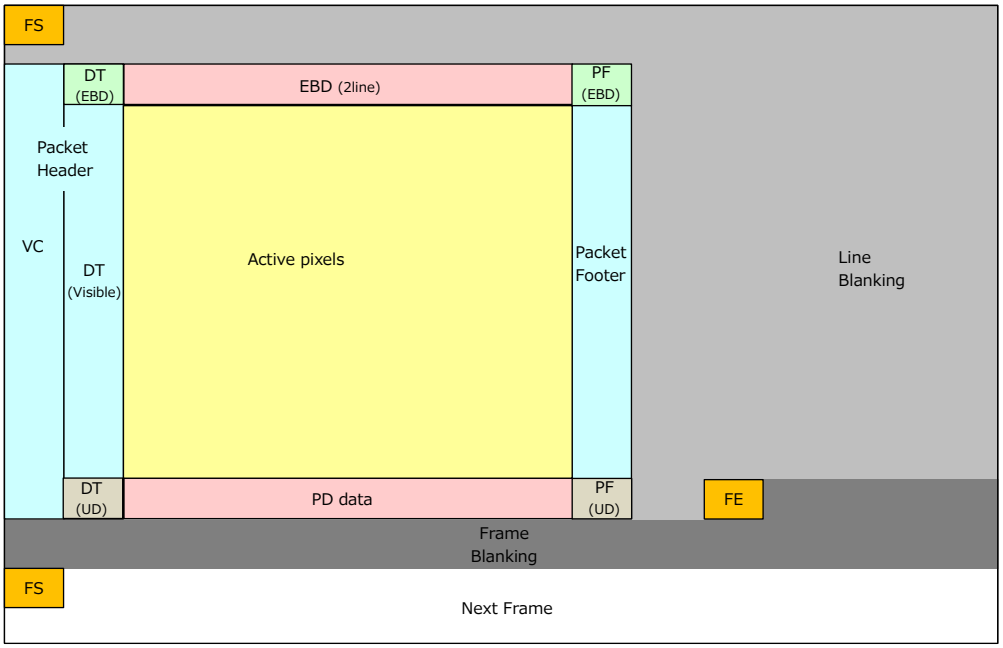


Figure 4-7 Frame Structure (PD output)

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4.6.2. STATS data control

Two kinds of STATS (statistic) data are supported.

AE HIST : Histogram of Y signal in crop area for AE control

Flicker : Histogram of Y signal for flicker detection.

STATS data can be output as a packet different from normal image data during the frame blanking period. The function is set to turn ON/OFF with registers shown in Table 4-7. Refer to the section "9 Settings related to QBC HDR capture mode picture quality".

Table 4-7 STATS data output control

I^2C Address	Bit	Name	Description
0x323b	[0]	AEHIST_OUT_EN	AE HIST data output via MIPI disable/enable 0: disable (default) 1: enable
0x323c	[0]	FLK_OUT_EN	Flicker data output via MIPI disable/enable 0: disable (default) 1: enable

4.6.3. Contents of packet header

The contents of the first byte in the packet header (data identifier) and the corresponding register settings are described in the table below.

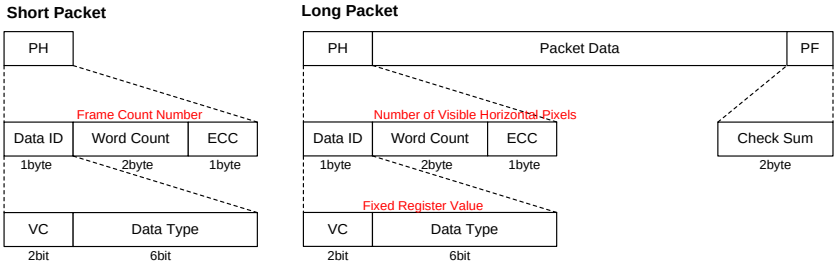


Figure 4-8 Short packet & long packet structures

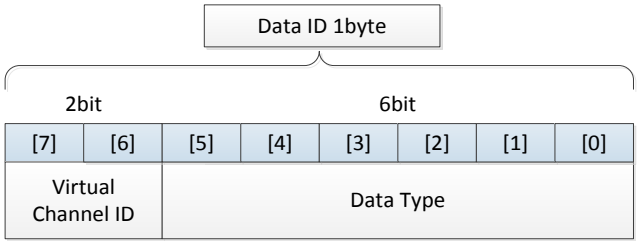


Figure 4-9 Configuration of Data ID

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Table 4-8 Data Identifier

	Bit assignment	Value	Name	Corresponding registers (I^2C)	Description
Packet header	[7:6] Virtual Channel ID*1	2'h0 (Default)	-		Range 0 Others are forbidden.
	[5:0] Data types*2	6'h00	Frame Start Code	NA	
		6'h01	Frame End Code	NA	
		6'h12	Embedded Data	NA	For embedded data line
		6'h2A	RAW8	NA	0x0112:CSI_DT_FMT_H=0x08 0x0113:CSI_DT_FMT_L=0x08
		6'h2B	RAW10	NA	0x0112:CSI_DT_FMT_H=0x0a 0x0113:CSI_DT_FMT_L=0x0a
		6'h30	COMP8 (10bit to 8bit)	NA	0x0112:CSI_DT_FMT_H=0x0a 0x0113:CSI_DT_FMT_L=0x08
		6'h32	AE HIST	0xE2A6 AEHIST_DT[5:0]	
		6'h33	Flicker	0xE2C6 FLK_DT[5:0]	
		6'h36	PDAF	0xE246 CH10_DT[5:0]	

Note 1: Prohibit changing the other Virtual Channel ID from the default value.

Note 2: Register for data types should be updated during SW-standby.

4.6.4. Data type

Types of data in each line are shown below.

Table 4-9 Image pixel area and data type

Image pixel area	Data type
Embedded data lines	Embedded data
Active pixels	RAW10, COMP8(10bit to 8bit), RAW8
User define data	AE HIST data, Flicker data, PDAF data,

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4.6.5. Embedded data line control

It is possible to output certain 2-wire serial register contents on the 2 lines just after the FS sync code of the frame. Regarding actual contents of the embedded data line are listed in Appendix.

Table 4-10 Embedded data line control

	Address	Bit	Name	Description
I²C register	0xBCF1	[7:0]	EBD_SIZE_V	0 is no EBD 2 is EBD 2 line others are forbidden

The sequence of EBD in each output format is as shown in the figures below.

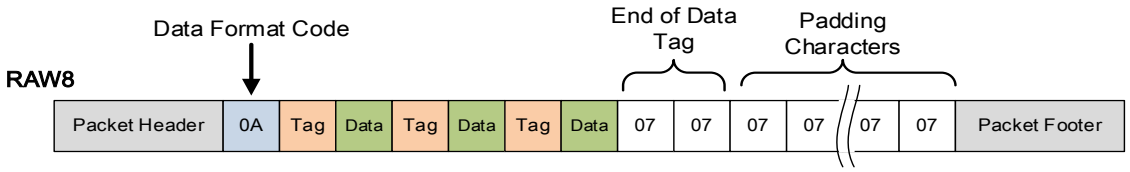


Figure 4-10 Embedded data lines alignment in RAW8/COMP8 mode

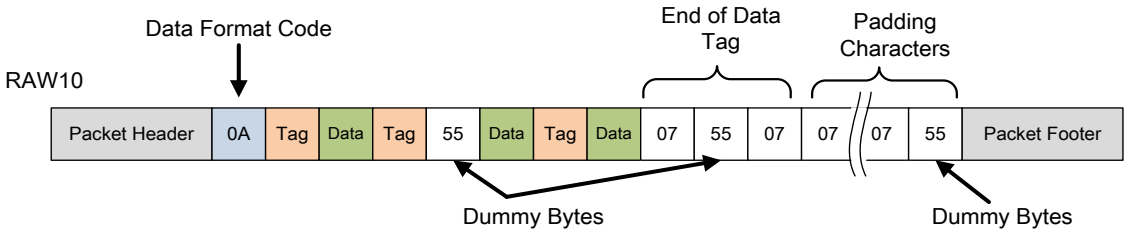


Figure 4-11 Embedded data lines alignment in RAW10 mode

Addresses and the end of register values are distinguished by “Tags” embedded in the data sequence.

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Table 4-11 Embedded data line tag

Tag	Data byte description
00h	Illegal Tag. If found treat as end of Data
07h	End of Data (Data Byte Value = 07H)
aah	CCI Register Index MSB [15:8]
a5h	CCI Register Index LSB [7:0]
5ah	Auto increment the CCI index after the data byte – valid data Data byte contains valid CCI register data
55h	Auto increment the CCI index after the data byte – null data A CCI register does NOT exist for the current CCI index. The data byte value is the 07H
ffh	Illegal Tag. If found treat as end of Data

The definite data sequence is described in the table in the Appendix.

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5. Operation Mode Setting

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5.1. Clock generation and PLL

This sensor is equipped with embedded oscillator and PLL to generate the necessary internal clocks and CSI2 transmission clocks. Set the related registers according to the operation conditions.

5.1.1. Clock system diagram

This sensor is equipped with 2 PLLs, one to output IVTCK for image processing, and the other to output IOPSYCK for MIPI output. The IVTCK PLL can output at 1050 to 2160 MHz, and the IOPCK PLL can output at 1050 to 2500 MHz(C-PHY/D-PHY), based on a clock input with the 6 to 27 MHz range.

The IVTCK PLL could be configured with divider of up to 1/1 to 1/4 range, and multiply in the 88 to 360 range.

The IOPCK PLL could be configured with divider of up to 1/1 to 1/15 range, and multiply in the 88 to 2500 range.

This sensor normally recommend to make it operate in single PLL mode by driving just one PLL (IOPCK PLL), however can also operate in dual PLL mode by driving both PLLs from parameter setting flexibility point of view.

In PLL single mode, IVT_PREPLLCK _DIV and IVT_PLL _MPY are applied to IOPCK PLL, and IOP_PREPLLCK _DIV and IOP_PLL _MPY are ignored.

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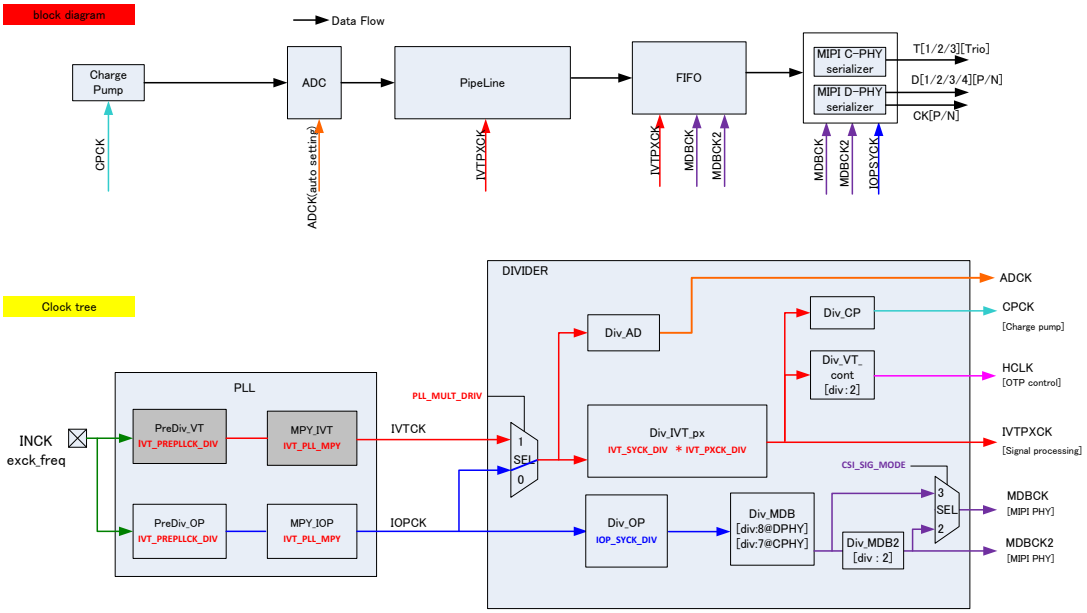


Figure 5-1 Clock system diagram (PLL single mode)

Table 5-1 PLL pre divider & multiplier setting (PLL single mode)

	Address	Bit	Name	Description	Setting constraint
I²C register	0x0305	[3:0]	IVT_PREPLLCK_DIV	The pre-PLL Clock Divider for Internal Video Timing System Clock Range : 1 to 4 Step : 1 Format : 4-bit unsigned integer *Setup other than the above is forbidden.	INCK / IVT_PREPLLCK _DIV = 6 to 12 MHz
	0x0306	[2:0]	IVT_PLL_MPY[10:8]	The PLL multiplier for Internal Video Timing System Clock Range : 88 to 360 Step : 1 Format : 16-bit unsigned integer *Setup other than the above is forbidden.	IVTCK & IOPCK = 1050 to 2160 MHz
	0x0307	[7:0]	IVT_PLL_MPY[7:0]		

*IOP_PREPLLCK _DIV and IOP_PLL _MPY are ignored

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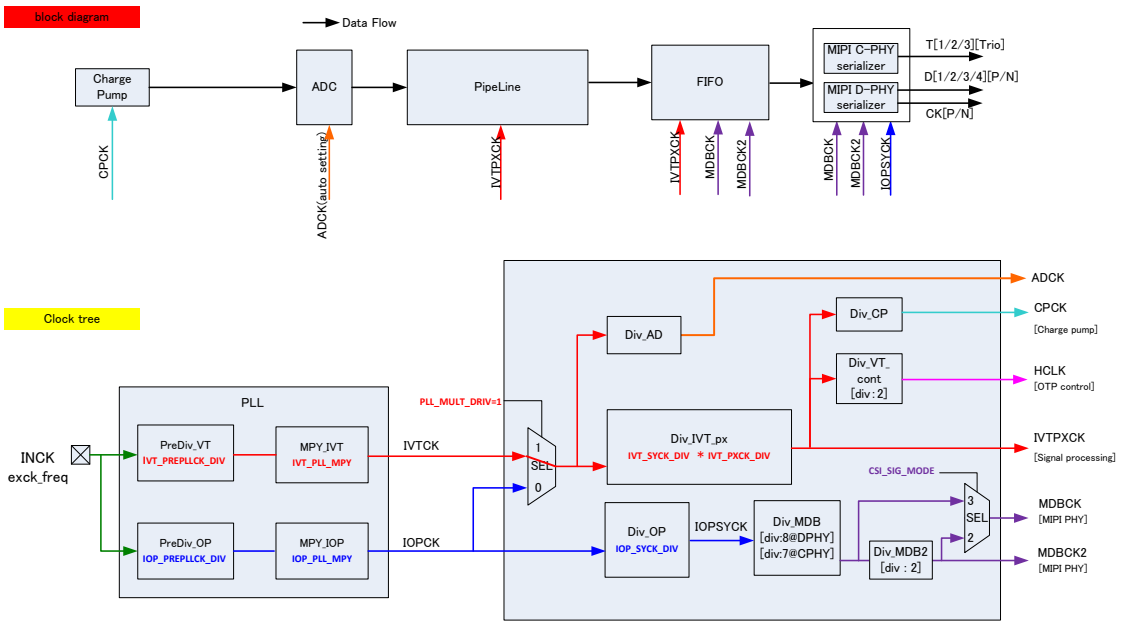


Figure 5-2 Clock system diagram (PLL dual mode)

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Table 5-2 PLL pre divider & multiplier setting (PLL dual mode)

	Address	Bit	Name	Description	Setting constraint
I²C register	0x0305	[3:0]	IVT_PREPLLCK_DIV	The pre-PLL Clock Divider for Internal Video Timing System Clock Range : 1 to 4 Step : 1 Format : 4-bit unsigned integer *Setup other than the above is forbidden.	INCK / IVT_PREPLLCK _DIV = 6 to 12 MHz
	0x030d	[4:0]	IOP_PREPLLCK_DIV	The pre-PLL Clock Divider for Internal Output Pixel System during "Dual PLL mode" (PLL_MULT_DRIV=1). Range : 1 to 27 Step : 1 Format : 4-bit unsigned integer *Setup other than the above is forbidden.	INCK / IOP_PREPLLCK _DIV = 1 to 12 MHz
	0x0306	[2:0]	IVT_PLL_MPY [10:8]	The PLL multiplier for Internal Video Timing System Clock Range : 88 to 360 Step : 1 Format : 16-bit unsigned integer *Setup other than the above is forbidden.	IVTCK = 1050 to 2160 MHz
	0x0307	[7:0]	IVT_PLL_MPY[7:0]		
	0x030e	[3:0]	IOP_PLL_MPY[11:8]	The PLL multiplier for Internal Output Pixel System "Dual PLL mode" (PLL_MULT_DRIV=1). Range : 88 to 2500 Step : 1 Format : 16-bit unsigned integer *Setup other than the above is forbidden.	IOPCK = 1050 to 2500 MHz
	0x030f	[7:0]	IOP_PLL_MPY[7:0]		

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Table 5-3 PLL mode select

I²C register	Address	Bit	Name	Description	Notes
	0x0310	[0]	PLL_MULT_DRIV	PLL mode select 0 : Single PLL mode 1 : Dual PLL mode	

5.1.1.1. INCK

INCK is the base of all clock system which shall be applied by external camera system.

INCK shall be in the range of 6 to 27 MHz

Table 5-4 INCK Frequency setting

I²C register	Address	Bit	Name	Description	Setting constraint
	0x0136	[7:0]	EXCK_FREQ[15:8]	External clock(INCK) frequency [MHz] Bit[15:8] integer Bit[7:0] decimal	=INT(INCK[MHz] * 256)
	0x0137	[7:0]	EXCK_FREQ[7:0]		

5.1.2. Sensor internal operation clocks

These clocks are the root of all the operation clocks in this sensor and it designates the data rate.

5.1.2.1. IVTCK, IOPCK: PLL output(PLL single mode)

IOPCK = INCK frequency × *PreDivider setting* × *PLL multiple setting*

* *PreDivider setting* = 1/ IVT_PREPLLCK _DIV

* *PLL multiple setting* = IVT_PLL _MPY

IVTCK = IOPCK

CKP, CKN = *IOPCK* × *CSI2 divide* × (1/ IOP_SYCK_DIV)

* *CSI2 divide* = 1/2

5.1.2.2. IVTCK, OPCK: PLL output (PLL dual mode)

IVTCK = *INCK frequency* × *PreDivider setting* × *PLL multiple setting*

* *PreDivider setting* = 1/ IVT_PREPLLCK _DIV

* *PLL multiple setting* = IVT_PLL _MPY

IOPCK = *INCK frequency* × *PreDivider setting* × *PLL multiple setting*

* *PreDivider setting* = 1/ IOP_PREPLLCK _DIV

* *PLL multiple setting* = IOP_PLL _MPY

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CKP, CKN = *IOPCK* × *CSI2 divide* × (1/ IOP_SYCK_DIV)

* *CSI2 divide* = 1/2

5.1.2.3. Clock frequency configuration examples

Table 5-5 PLL frequency table IVTCK (include PLL single mode)

CLK	Input Clock (INCK)	Pre division	PLL Input frequency	Multiple	PLL Oscillation frequency
IVTCK	6 MHz	1	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				360	2160.0 MHz
	12 MHz	2	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				360	2160.0 MHz
	13.5 MHz	2	6.75 MHz	156	1053.0 MHz
				266	1795.5 MHz
				320	2160.0 MHz
	18 MHz	3	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				360	2160.0 MHz
	24 MHz	4	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				360	2160.0 MHz
	27 MHz	4	6.75 MHz	156	1053.0 MHz
				266	1795.5 MHz
				320	2160.0 MHz

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Table 5-6 PLL frequency table IOPCK

CLK	Input Clock (INCK)	Pre division	PLL Input frequency	Multiple	PLL Oscillation frequency
IOPCK	6 MHz	1	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				416	2496.0 MHz
	12 MHz	2	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				416	2496.0 MHz
	13.5 MHz	2	6.75 MHz	156	1053.0 MHz
				266	1795.5 MHz
				370	2497.5 MHz
	18 MHz	3	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				416	2496.0 MHz
	24 MHz	4	6.0 MHz	175	1050.0 MHz
				300	1800.0 MHz
				416	2496.0 MHz
	27 MHz	4	6.75 MHz	156	1053.0 MHz
				266	1795.5 MHz
				370	2497.5 MHz

5.1.3. IVTPXCK Clock

The clock for internal image processing is generated by dividing IVTCK into 1/10 or 1/20 frequency. This clock is used as the base of integration time, frame rate, etc.

IVTPXCK *clock frequency* = *IVTCK* × I*VTPXCK clock division ratio*

* *IVTPXCK clock division ratio* = 1 / (IVT_SYCK_DIV * IVT_PXCK_DIV)

IVTPXCK frequency range:

PLL single mode: 52.5MHz ≤ IVTPXCK ≤ 216MHz

PLL dual mode: 52.5MHz ≤ IVTPXCK ≤ 216MHz

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Table 5-7 IVTPXCK divider setting

	Address	Bit	Name	Description	Notes
I²C register	0x0303	[2:0]	IVT_SYCK_DIV	The System Clock Divider for Internal Video Timing System Range : 2, 4 Format : 3-bit unsigned integer *Setup other than the above is forbidden.	
	0x0301	[4:0]	IVT_PXCK_DIV	The Pixel Clock Divider for Internal Video Timing System Range : 5 Format : 5-bit unsigned integer *Setup other than the above is forbidden.	

5.1.4. IOPSYCK Clock

The clock for internal image processing is generated by dividing IOPCK into 1/1, 1/2 or 1/4 frequency according to the word length of the CSI2 interface. This clock designates the pixel rate etc.

IOPSYCK clock frequency = I*OPCK* × *IOPSYCK clock division ratio*

* *IOPSYCK clock division ratio* = 1 / (IOP_SYCK_DIV)

The setting of IOPSYCK frequency shall be used in the range of Table 5-9.

Table 5-8 IOPSYCK divider setting

	Address	Bit	Name	Description	Notes
I²C register	0x030b	[4:0]	IOP_SYCK_DIV	The System Clock Divider for Internal Output Pixel System Range : 1, 2,4 Format : 16-bit unsigned integer *Setup other than the above is forbidden.	

Table 5-9 IOPSYCK frequency table

CLK	IOPCK frequency	IOP_SYCK_DIV	IOPSYCK frequency
IOPSYCK	1050MHz～2500MHz	1	1050.0MHz – 2500.0MHz
		2	525.0MHz - 1250.0MHz
		4	268.0MHz - 625.0MHz(D-PHY) 262.5MHz - 625.0MHz(C-PHY)

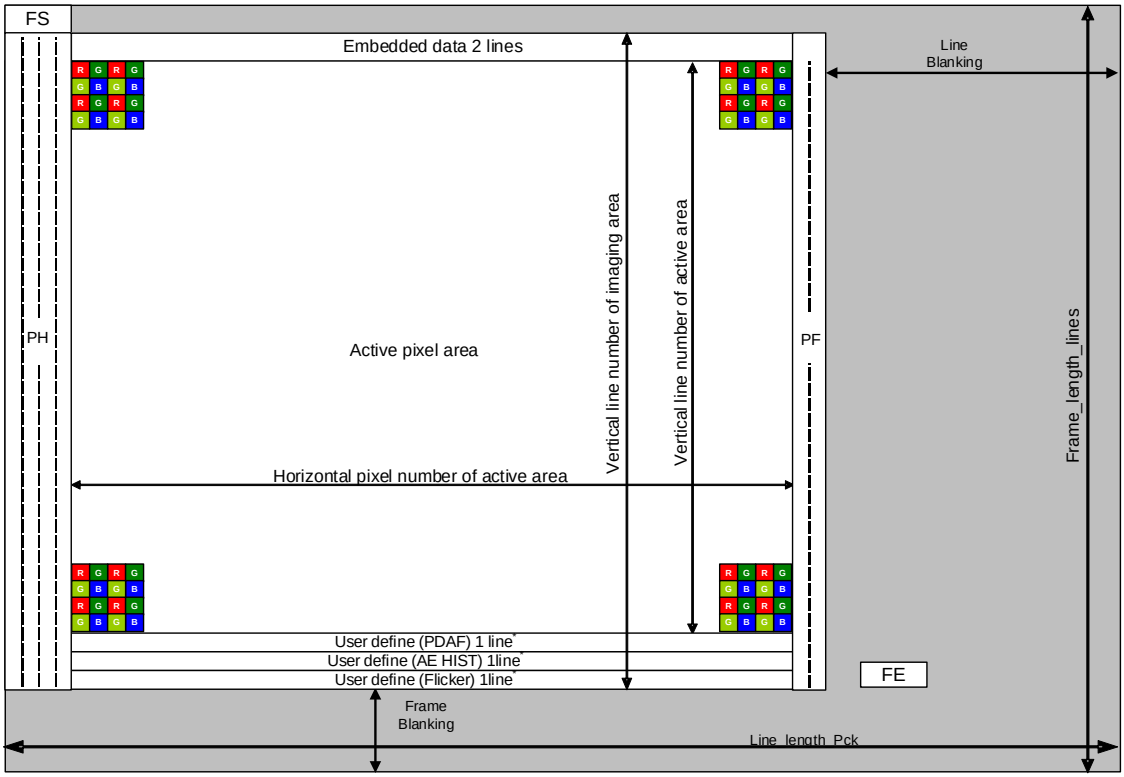
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5.2. Description of mode operation

Main modes of this sensor are Full resolution, 2x2 Adjacent Pixel Binning, 2x2 Adjacent Pixel Binning V2H2 and QBC HDR.

5.2.1. Image size related settings

This sensor can output full size and/or re-sized (shrunk) images. Examples are shown in the table below.



*See 4.6.2 STATS data control

Figure 5-3 Image size parameter definition

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Table 5-10 Typical image output of main capture modes (1)

Operation mode		Mode							
		Full-pixel		2x2 Adjacent Pixel Binning		2x2 Adjacent Pixel Binning V2H2		QBC HDR	
Number of vertical lines in imaging area		6000		3000		1500		3000	
Number of horizontal pixels in active area		8000		4000		2000		4000	
Number of lines and start position		Start position	Number of lines	Start position	Number of lines	Start position	Number of lines	Start position	Number of lines
Name of the areas	Frame start	1	1	1	1	1	1	1	1
	Embedded data lines	2	2	2	2	2	2	2	2
	Number of vertical pixels in active area	4	6000	4	3000	4	1500	4	3000
	PDAF Data	6004	1	3004	1	-	-	3004	1
	AE HIST Data	6005	1	3005	1	1504	1	3005	1
	Flicker Data	6006	1	3006	1	1505	1	3006	1
	Frame end	6006	1	3006	1	1505	1	3006	1

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Table 5-11 Typical image output of main capture modes (2)

	Modes						
	Full-pixel	2x2 Adjacent Pixel Binning	2x2 Adjacent Pixel Binning	2x2 Adjacent Pixel Binning	2x2 Adjacent Pixel Binning V2H2	2x2 Adjacent Pixel Binning V2H2(720p)	QBC HDR
Cropping	Non(4:3)	Non(4:3)	Non(4:3)	Vcrop(16:9)	Vcrop(16:9)	Vcrop(16:9) 720 Crop	Non(4:3)
Binning	Non	H/V	H/V	H/V	H/V	H/V	Non
Scaling	Non	Non	Non	Non	Non	Non	Non
H Pixels	8000	4000	4000	4000	2000	1296	4000*1
V Pixels	6000	3000	3000	2256	1128	736	3000*1
Frame Rate	30fps/19fps*2	100fps/71fps*2	60fps	120fps/94fps*2	240fps	480fps	30fps
PDAF	Support	Not Support	Support	Not Support	Not Support	Not Support	Support
Max Analog Gain	24dB	36dB	36dB	36dB	36dB	30dB	24dB
FOV							
Output							

*1: QBC HDR image is made from 4 same color pixels with different exposure, so the resolution of QBC HDR capture image is half from that of normal capture image.
*2: Notation is frame rate of C-PHY/D-PHY.

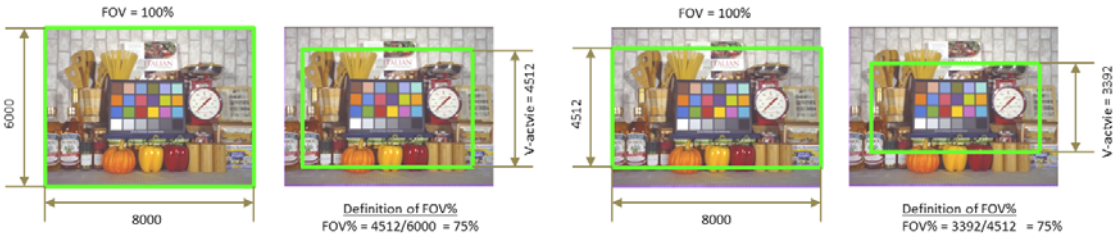
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Table 5-12 FOV% vs FPS (Crop) C-PHY

Skipping	Aspect ratio	FOV %	Actual RAW size	Scaling (V)	Scaling (H)	Max. FPS
No(Full)	4:3	100	8000 x 6000	No	No	30
		75	6000 x 4512	No	No	39
		50	4000 x 3008	No	No	55
		33	2640 x 1984	No	No	85
		20	1600 x 1200	No	No	140
	16:9	100	8000 x 4512	No	No	39
		75	6000 x 3392	No	No	50
		50	4000 x 2256	No	No	75
		33	2640 x 1488	No	No	115
		20	1600 x 912	No	No	180

Table 5-13 FOV% vs FPS (Crop) D-PHY

Skipping	Aspect ratio	FOV %	Actual RAW size	Scaling (V)	Scaling (H)	Max. FPS
No(Full)	4:3	100	8000 x 6000	No	No	19
		75	6000 x 4512	No	No	25
		50	4000 x 3008	No	No	35
		33	2640 x 1984	No	No	55
		20	1600 x 1200	No	No	90
	16:9	100	8000 x 4512	No	No	25
		75	6000 x 3392	No	No	30
		50	4000 x 2256	No	No	50
		33	2640 x 1488	No	No	70
		20	1600 x 912	No	No	115



Note: H-Analog Crop is prohibited.

The following table shows the setting list for the supporting operation modes of capture mode.

Other settings are not supported in this sensor.

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Table 5-14 Support list of operation mode of capture mode

Operation Mode	QBC HDR setting		BINNING setting						ADC setting	QBC Re-mosaic setting
	0x0220 [0]	0x3140 [2:0]	0x0900 [0]	0x0901 [7:0]	0x3247 [7:0]	0x3246 [7:0]	0x3F0C [0]	0x3F14 [0]	0x3F15 [1:0]	0x3620 [0]
	HDR_MODE [0]	HDR_FNCSEL [2:0]	BINNING_MODE [0]	BINNING_TYPE_H [7:4] BINNING_TYPE_V [3:0]	BINNING_PRIORITY_V [7:0]	BINNING_PRIORITY_H [7:0]	BINNING_PDAF_EN [0]	FULL_QHDR_EN [0]	ADC_MODE [1:0]	QBC_RMSC_EN [0]
Full-pixel (QBC RAW)*1	0x0	X	0x0	0x11	0x01	0x01	0x0	0x1	0x0	0x0
Full-pixel (QBC Re-mosaic)										0x1
2x2 Adjacent Pixel Binning without PDAF			0x1	0x22	0x81	0x81	0x0	0x0	0x0	0x0
2x2 Adjacent Pixel Binning with PDAF							0x1			
2x2 Adjacent Pixel Binning V2H2			0x1	0x44	0x89	0x89	0x0	0x0	0x0	0x0
2x2 Adjacent Pixel Binning V2H2 (720p)									0x1	
QBC HDR	0x1	0x4	X	X	X	X	0x0	0x1	0x0	0x0

*1: Use only for QSC calibration

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5.2.2. Normal capture mode settings (full and binning mode)

Register settings for Binning mode are as follows.

Table 5-15 Normal capture mode settings (full and binning mode)

	Address	Bit	Name	Description
I²C register	0x0220	[6:0]	HDR_MODE[6:0]	Setting value: 0x00 (Fix 0x00)
	0x0900	[0]	BINNING_MODE	Binning enable control 0 : disable 1 : enable
	0x0901	[7:4]	BINNING_TYPE_H[3:0]	BINNING_TYPE_H : Binning type selection for Horizontal 1 : no binning 2 : 2x2 Adjacent Pixel Binning 4 : 2x2 Adjacent Pixel Binning V2H2 *Setup other than the above is forbidden.
	0x0901	[3:0]	BINNING_TYPE_V[3:0]	BINNING_TYPE_V : Binning type selection for Vertical 1 : no binning 2 : 2x2 Adjacent Pixel Binning 4 : 2x2 Adjacent Pixel Binning V2H2 *Setup other than the above is forbidden
	0x0902	[7:0]	BINNING_WEIGHTING	Bit[1:0] Binning type selection 0 : Averaged 1 : Summed 2 : Weighting averaged *1 *1 : Only available in 2x2 Adjacent Pixel Binning V2H2 mode *Setup other than the above is forbidden. Bit [7:2] : 6'b000010 Fixed
	0x3247	[7:0]	BINNING_PRIORITY_V	Setting method of Vertical summation 8'h01 : No Binning 8'h81 : 2x2 Adjacent Pixel Binning, QBC HDR 8'h89 : 2x2 Adjacent Pixel Binning V2H2 *Setup other than the above is forbidden
	0x3246	[7:0]	BINNING_PRIORITY_H	Setting method of Horizontal summation 8'h01 : No Binning 8'h81 : 2x2 Adjacent Pixel Binning, QBC HDR 8'h89 : 2x2 Adjacent Pixel Binning V2H2 *Setup other than the above is forbidden

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	0x3F0C	[0]	BINNING_PDAF_EN	2x2 Adjacent Pixel Binning, PDAF enable 0 : disable 1 : enable
	0x3F14	[0]	FULL_QHDR_EN	Full Pixel and QBC HDR mode enable 0 : disable 1 : enable
	0x3620	[0]	QBC_RMSC_EN	QBC Re-mosaic enable 0 : Disable 1 : Enable

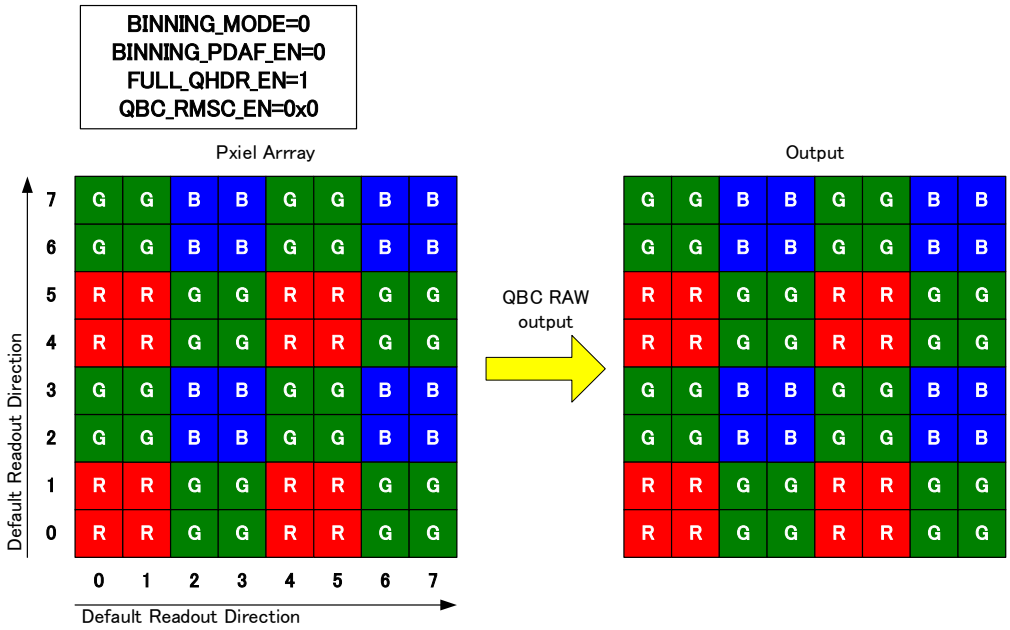


Figure 5-4 Full resolution image (QBC RAW output)

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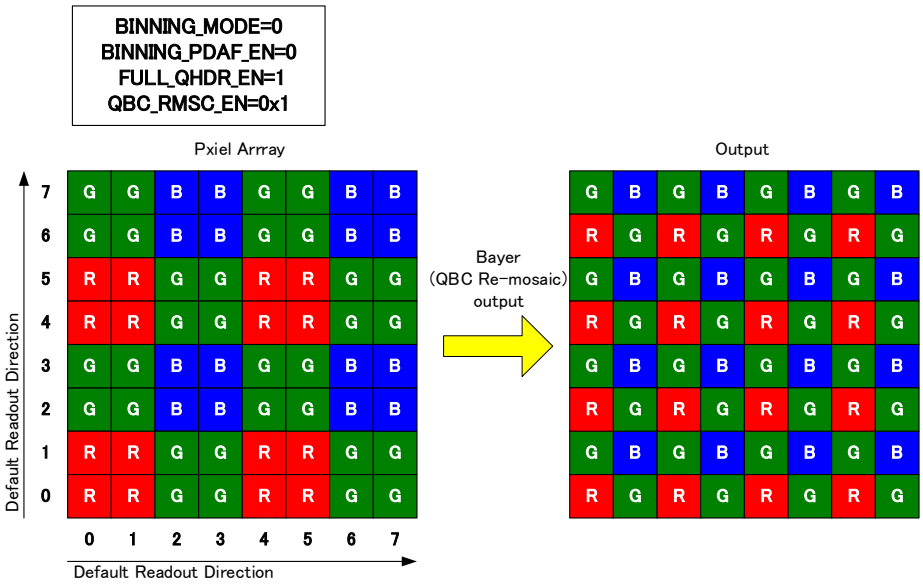


Figure 5-5 Full resolution image (Bayer output)

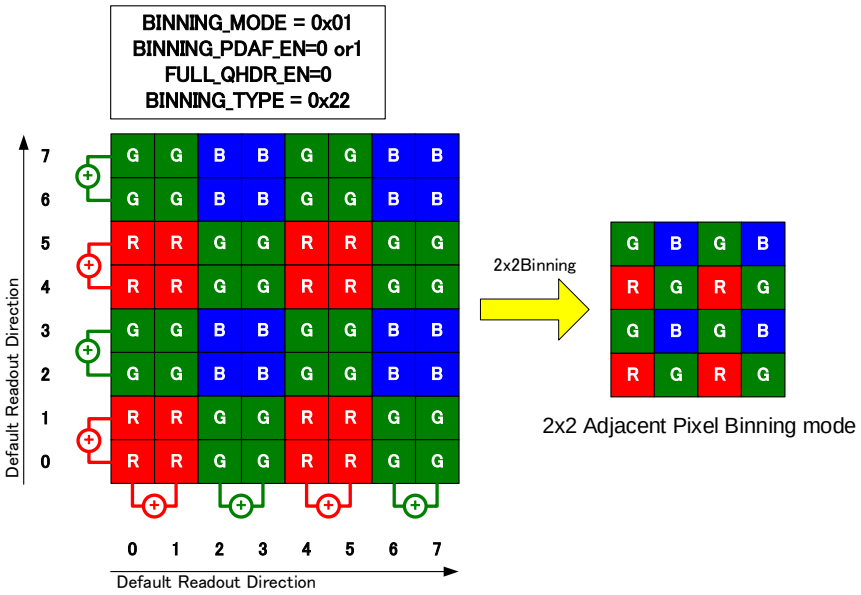


Figure 5-6 2x2 Adjacent Pixel Binning mode image

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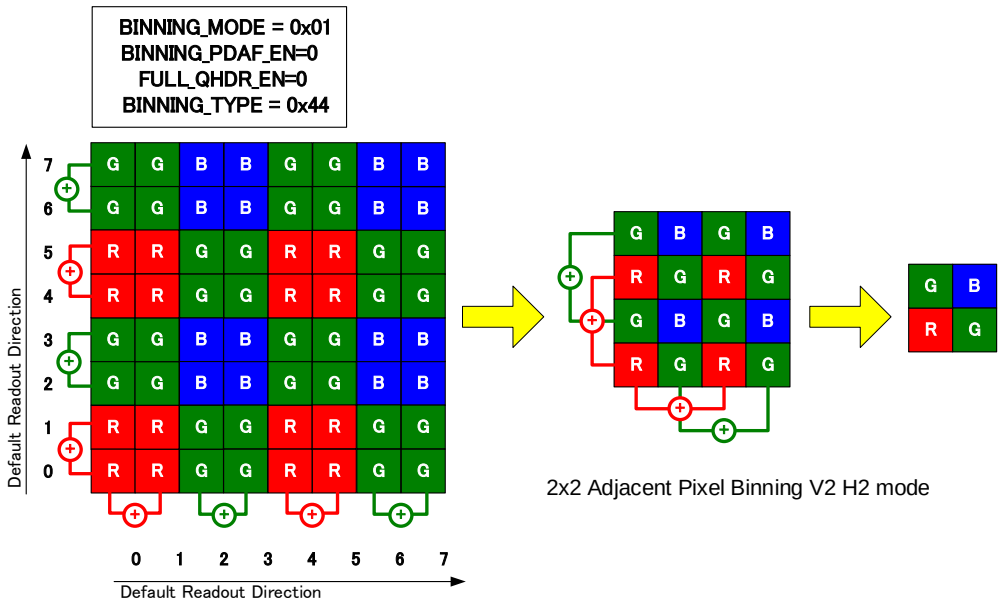


Figure 5-7 2x2 Adjacent Pixel Binning V2H2 mode image

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5.2.3. QBC HDR capture mode settings

Register settings for QBC HDR capture mode are as follows.

QBC HDR image is made from 4 same color pixels with different exposure. The resolution of output image is half from original sensor.

Table 5-16 QBC HDR capture mode setting

	Address	Bit	Name	Description
I²C register	0x0220	[6:0]	HDR_MODE[6:0]	Recommend setting value: 0x63(QBC HDR enable) 0x00(QBC HDR disable) Bit[0] QBC HDR mode enable 0 : QBC HDR disable 1 : QBC HDR enable Bit[1] Analog Gain mode select during QBC HDR 0 : combined gain used 1 : separate gain used Bit[4:2] Reserved Bit[5] exposure mode select during QBC HDR 0 : short exposure determined by ratio (controlled by EXPO_RATIO(0x0222)) 1 : short exposure controlled by direct control Bit[6] Digital Gain mode select during QBC HDR 0 : combined gain used 1 : separate gain used
	0x0222	[4:0]	EXPO_RATIO	Exposure_ratio [4:0] Defines exposure ratio between short and long exposure. In case of QBC HDR, Middle exposure value = coarse_integration_time / Exposure_ratio Short exposure value = coarse_integration_time / Exposure_ratio^2 1, 2, 4, 8(unsigned integer) can be set.
	0x3140	[2:0]	HDR_FNCSEL	QBC HDR function select 4 : QBC HDR Others : reserved

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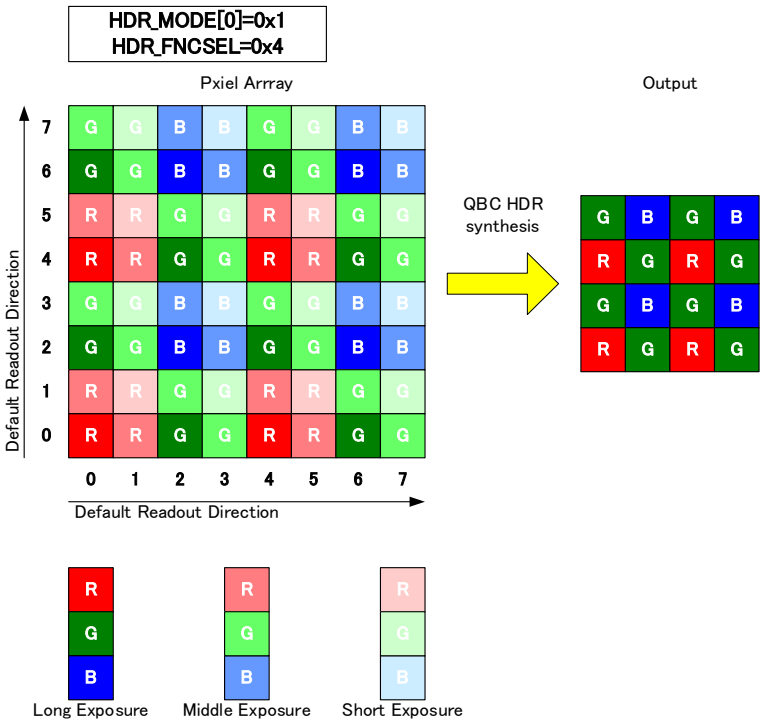


Figure 5-8 QBC HDR image

5.2.4. Image size related functions

The relationships of image output size and the registers are shown below.

In this sensor, horizontal analog cropping is prohibited.

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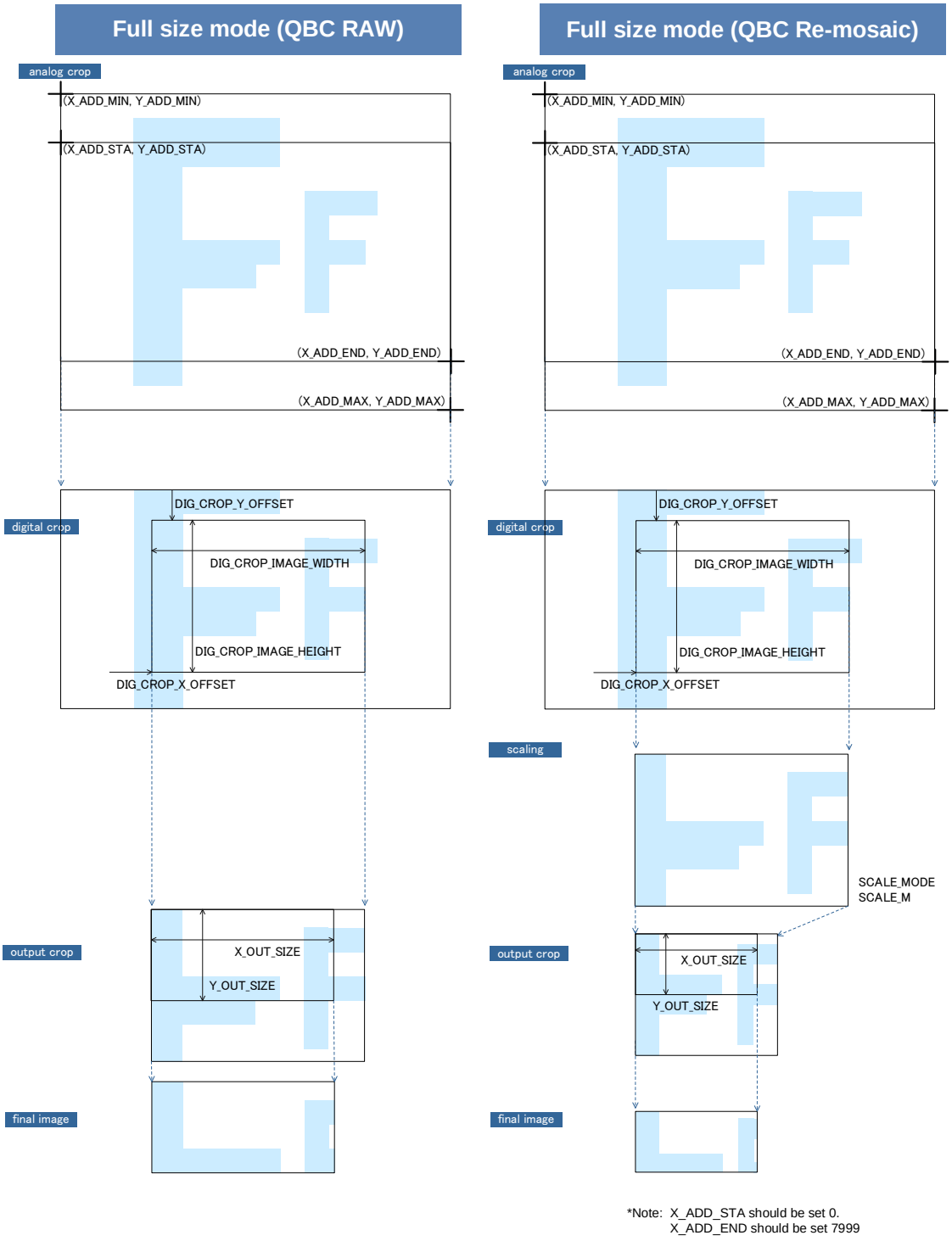


Figure 5-9 Image size related functions for full size mode

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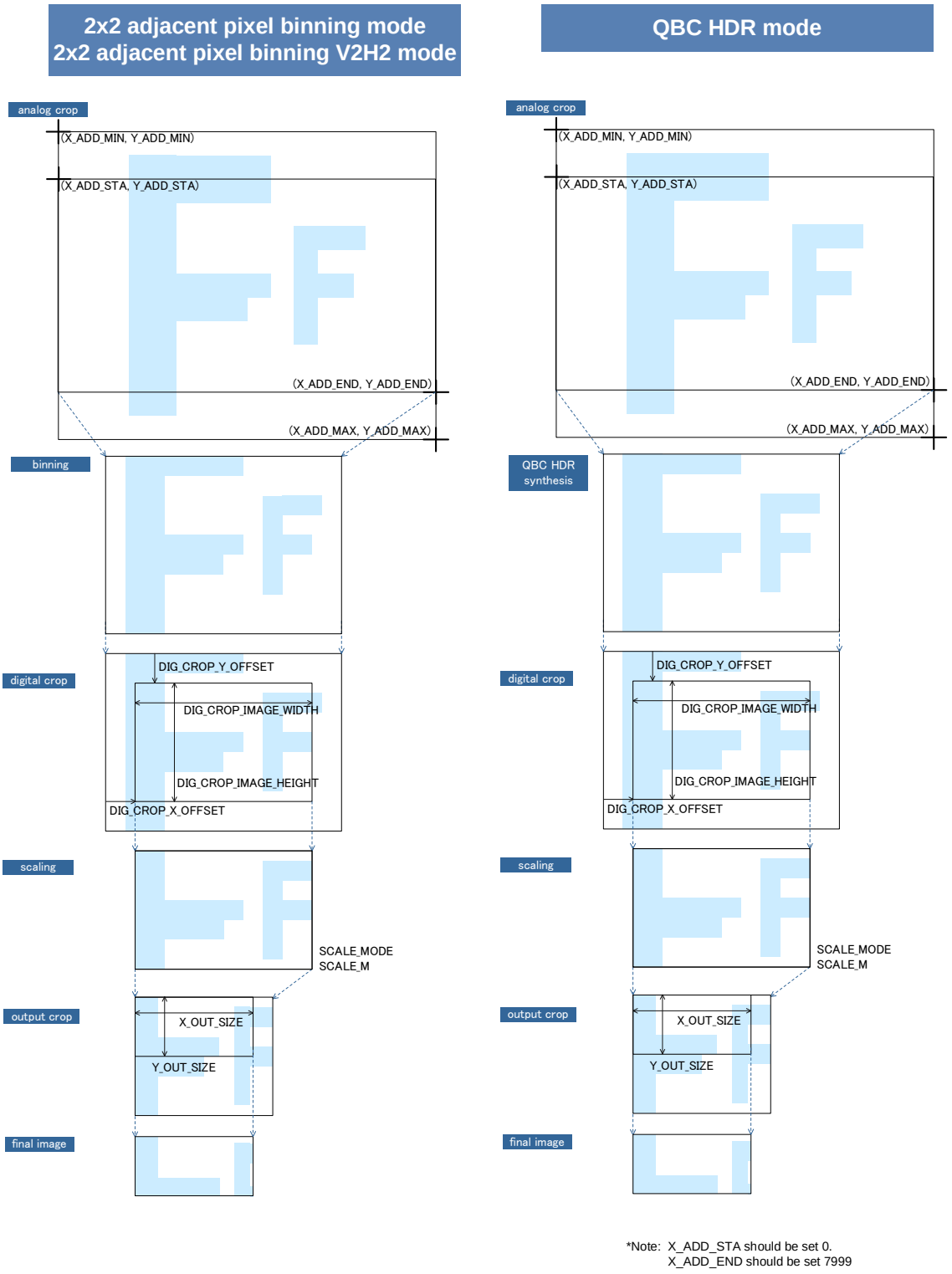


Figure 5-10 Image size related functions for 2x2 adjacent pixel binning mode, 2x2 adjacent pixel binning V2H2 mode and QBC HDR mode

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Analog crop

Please refer to ““4.3 Imaging area determination”.

In this sensor, horizontal analog cropping is prohibited.

Binning

Settings and see 5.2.2 Normal capture mode settings (full and binning mode) for details.

Digital crop

Digital crop is processed after image readout operation or in order of [Analog crop → Binning → Digital crop] and is common for Normal capture mode. The crop size (DIG_CROP_IMAGE_WIDTH and DIG_CROP_IMAGE_HEIGHT) must always be set properly even if cropping is not used.

Table 5-17 Digital crop setting

	Address	Bit	Name	Description
I²C register	0x0408	[4:0]	DIG_CROP_X_OFFSET[12:8]	Offset from X_ADD_STA after binning
	0x0409	[7:0]	DIG_CROP_X_OFFSET[7:0]	
	0x040a	[4:0]	DIG_CROP_Y_OFFSET[12:8]	Offset from Y_ADD_STA after binning
	0x040b	[7:0]	DIG_CROP_Y_OFFSET[7:0]	
	0x040c	[4:0]	DIG_CROP_IMAGE_WIDTH[12:8] *1	Image width after digital cropping
	0x040d	[7:0]	DIG_CROP_IMAGE_WIDTH[7:0] *1	Unit : pixels
	0x040e	[4:0]	DIG_CROP_IMAGE_HEIGHT[12:8] *1	Image height after digital cropping
	0x040f	[7:0]	DIG_CROP_IMAGE_HEIGHT[7:0] *1	Unit : lines

*1 (DIG_CROP_IMAGE_WIDTH and DIG_CROP_IMAGE_HEIGHT) must always be set properly even if cropping is not used.

Scaling

The scaler can’t be used in QBC RAW output mode.

The scaler can be specified by scaling mode and down scale factor registers.

The down scale factor is determined by N/M ratio.

Scaling factor = N/M (N=16 fixed value)

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Table 5-18 Scaling related registers

	Address	Bit	Name	Description
I²C register	0x0401	[1:0]	SCALE_MODE[1:0]	Scaling mode selection 0 : No Scaling 1 : Horizontal Scaling 2 : Horizontal & Vertical Scaling 3 : Reserved
	0x0404	[0]	SCALE_M[8]	Down scale factor M: Available M component: 16 to 511 Format: 9 bit unsigned integer
	0x0405	[7:0]	SCALE_M[7:0]	

An example of picture sizes before and after scaling is as follows.

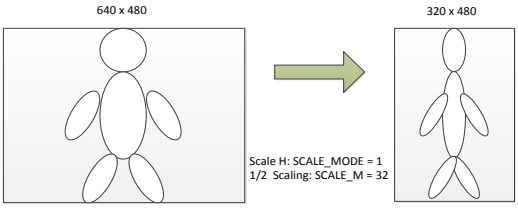


Figure 5-11 Scaling example

Output crop

The width and height of the visible pixel area within the frame of data output from the sensor can be programmed by the “X_OUT_SIZE” and “Y_OUT_SIZE”.

Note that this function does not have “offset” setting just not like as Digital Crop’s Offset capability. So, the picture will not centered any more if the output size is trimmed with Out Crop function.

Table 5-19 Output crop setting

	Address	Bit	Name	Description
I²C register	0x034c	[4:0]	X_OUT_SIZE[12:8]	The sensor output size of horizontal Unit : pixels (multiple of 4)
	0x034d	[7:0]	X_OUT_SIZE[7:0]	
	0x034e	[4:0]	Y_OUT_SIZE[12:8]	The sensor output size of vertical Unit : lines (multiple of 2)
	0x034f	[7:0]	Y_OUT_SIZE[7:0]	

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5.3. Frame rate calculation formula

The frame rate of the output image of this sensor can be calculated by the formula below.

Frame Rate [frame/s] = *Pixel_rate* [pixels/s] / *Total number of pixels* [pixels/frame]

Pixel rate [pixels/s] = *IVTPXCK* [MHz] * 8 (*Total number of IVTPX channel)*

Total number of pixels [pixels/frame]

= FRM_LENGTH_LINES [lines/frame] * LINE_LENGTH_PCK [pixels/line]

See 5.1.3 IVTPXCK Clock for the calculation formula for IVTPXCK.

See 5.1.4 IOPSYCK Clock for the descriptions of registers used for the calculation formula.

5.4. Electronic shutter and integration time settings

5.4.1. Registers related to integration time (electronic shutter setting)

The integration time setting registers are shown below.

Table 5-20 Electronic shutter setting register

	Address	Bit	Name	Description	Notes
I²C register	0x0202	[7:0]	COARSE_INTEG_TIME[15:8]	Coarse storage time Unit : lines Format : 16-bit unsigned integer	
	0x0203	[7:0]	COARSE_INTEG_TIME[7:0]		

Table 5-21 Integration time setting register

	Address	Bit	Name	Description	Notes
I²C register	0x0200	[7:0]	FINE_INTEG_TIME[15:8]	Fine storage time Unit : pixels Format : 16-bit unsigned integer	Read only
	0x0201	[7:0]	FINE_INTEG_TIME[7:0]		
	0x0342	[7:0]	LINE_LENGTH_PCK[15:8]	The length of line Unit : pixels Format : 16-bit unsigned integer	* Must be set fixed value as Table 4-5.
	0x0343	[7:0]	LINE_LENGTH_PCK[7:0]		

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	0x0350	[0]	FRM_LENGTH_CTL	Frame length automatic tracking control Select whether or not the frame length is changed automatically when FRM_LENGTH_LINES < COARSE_INTEG_TIME + α (α = type-specific adjustment parameter, and is 48 for this type. α is the value of CIT_MARGIN.) 0 :no automatic tracking control of frame length 1 :automatic tracking control of frame length In this case, "COARSE_INTEG_TIME + α" operates instead of FRM_LENGTH_LINES	
	0x4bc2	[4:0]	CIT_MARGIN[12:8]	Coarse storage time margin Unit : lines CIT_MARGIN < 48 is prohibited.	
	0x4bc3	[7:0]	CIT_MARGIN[7:0]		

5.4.2. Integration time calculation

The integration time or shutter time (T_{SH}) can be obtained from the following relational equation.

Tline = LINE_LENGTH_PCK [pixels/line] * I*VTPXCK_period* / 8 (*Total number of image pipe lines*)

Tsh = ***Tline*** * (COARSE_INTEG_TIME [lines] + FINE_INTEG_TIME [pixels] / LINE_LENGTH_PCK [pixels/line])

This relationship stands up regardless of the operating mode.

Settings and storage times are shown below.

* FINE_INTEG_TIME is a fixed value.

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COARSE_INTEG_TIME is a variable value. The constraints are as shown below table.

Table 5-22 Constraints of COARSE_INTEG_TIME

Mode	Constraint of COARSE_INTEG_TIME values			
	Min Value (L)	Step value(X)	Max Value	Max Value Margin(S)
Full-pixel (QBC RAW)*1	5 *	1	FRM_LENGTH_LINES – S ※S= Max Value Margin	48
Full-pixel (QBC Re-mosaic)	5 *	1		48
2x2 Adjacent Pixel Binning without PDAF	6	2		48
2x2 Adjacent Pixel Binning with PDAF	6	2		48
2x2 Adjacent Pixel Binning V2H2	6	2		48
2x2 Adjacent Pixel Binning V2H2 (720p)	6	2		48
QBC HDR	5 *	1		48

* CIT must be set odd, if CIT is smaller than 16.

To set exposure time longer than “FRM_LENGTH_LINES”, you can either automatically extend it with the setting FRM_LENGTH_CTL=1 or perform truncation by frame length with the setting FRM_LENGTH_CTL = 0.

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Table 5-23 Integration time setting (In case of FRM_LENGTH_CTL=0)

Parameter Setting		Frame time / Integration Time	
FRM_LENGTH_LINES [15:0]	COARSE_INTEG_TIME [15:0]	Frame time	Integration time (H stands for the duration of one line)
M	L	M x Tline	(L + α)× Tline
	L+ X		(L + X + α) x Tline
	L + 2X		(L + 2X + α) x Tline
	…		…
	M-S-X		((M – S - X) + α) x Tline
	M - S		((M – S) + α) x Tline
	＞M - S	prohibited	

*α=FINE_INTEG_TIME/LINE_LENGTH_PCK

* M: set value for given capture mode <= 65535

*L = Min Value of COARSE_INTEG_TIME. See Table 5-22

*S = Max Value Margin of COARSE_INTEG_TIME. See Table 5-22

*X = Step value of COARSE_INTEG_TIME. See Table 5-22

Table 5-24 Integration time setting (In case of FRM_LENGTH_CTL=1)

Parameter Setting		Frame time / Integration Time	
FRM_LENGTH_LINES [15:0]	COARSE_INTEG_TIME [15:0]	Frame time	Integration time (H stands for the duration of one line)
M	L	M x Tline	(L+α) x Tline
	L + X		(L + X +α) x Tline
	L + 2X		(L + 2X +α) x Tline
	…		…
	M-S		((M-S)+α) x Tline
	(M-S)+X	(M+X) x Tline	((M- S) +X +α) x Tline
	(M-S)+2X	(M+2X) x Tline	((M- S) +2X +α) x Tline
	…	…	…
	N - S	N x Tline	((N-S) +α) x Tline

* α=FINE_INTEG_TIME/LINE_LENGTH_PCK

* M: set value for given capture mode <= 65535

* N: can be set with values greater than M

* L = Min Value of COARSE_INTEG_TIME. See Table 5-22

* S = Max Value Margin of COARSE_INTEG_TIME. See Table 5-22

* X = Step value of COARSE_INTEG_TIME. See Table 5-22

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5.5. Gain setting

This sensor can apply analog gain on photo-electron signal and digital gain on digital signal after ADC.

Range of settable range is as follows.

Table 5-25 Range of Gains

	Min.	Max.	ADC_MODE	Note
Analog Gain	1dB	24dB	0x0	Full-pixel, QBC HDR
	1dB	36dB	0x0	2x2 Adjacent Pixel Binning, 2x2 Adjacent Pixel Binning V2H2
	1dB	30dB	0x1	2x2 Adjacent Pixel Binning V2H2 720p mode
Digital Gain	0dB	24dB	-	Functionally Settable

* For more details of Max Gain, refer to the Table 5-36

Table 5-26 ADC mode register setting

	Address	Bit	Name	Description
I²C register	0x3f15	[1:0]	ADC_MODE	ADC mode 0: Normal ADC mode (analog gain 1- 24dB in Full-pixel and QBC HDR mode) (analog gain 1- 36dB in 2x2 Adjacent Pixel Binning and 2x2 Adjacent Pixel Binning V2H2 mode) 1: High speed ADC mode*1 (analog gain 1-30dB in 2x2 Adjacent Pixel Binning V2H2 720p mode) * Others are forbidden.

*1: High speed ADC Mode: In case that ADC_MODE is set to High speed ADC mode, there are limitation both LINE_LENGTH_PCK and Analog Gain instead of achieving high frame rate. High speed ADC mode can be available only for 2x2 Adjacent Pixel Binning V2H2 720p mode.

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5.5.1. Analog gain settings

The formula for analog gain is shown below.

Analog gain parameters are set with registers. Analog gain and digital gain are set separately.

$$Analog\ Gain = (m0 \times X + c0) / (m1 \times X + c1)$$

The variables of above equation are specified in the table below.

Table 5-27 Variables of analog gain settings

	Address	Bit	Name	Description	Notes
I²C register	0x008c	[7:0]	ANA_GAIN_M0[15:8]	m0: fixed to 0	Read only static
	0x008d	[7:0]	ANA_GAIN_M0[7:0]		
	0x0090	[7:0]	ANA_GAIN_M1[15:8]	m1: fixed to -1	Read only static
	0x0091	[7:0]	ANA_GAIN_M1[7:0]		
	0x008e	[7:0]	ANA_GAIN_C0[15:8]	c0: fixed to 1024	Read only static
	0x008f	[7:0]	ANA_GAIN_C0[7:0]		
	0x0092	[7:0]	ANA_GAIN_C1[15:8]	c1: fixed to 1024	Read only static
	0x0093	[7:0]	ANA_GAIN_C1[7:0]		
	0x0204	[1:0]	ANA_GAIN_GLOBAL[9:8]	X: Analog gain setting value	112 to 1008
	0x0205	[7:0]	ANA_GAIN_GLOBAL[7:0]		

As a result, analog gain is calculated by the formula below.

$$Analog\ Gain = 1024 / (1024 - X)$$

The relationship between the setting values X of ANA_GAIN_GLOBAL and the gain is shown in the following table. The ANA_GAIN_GLOBAL value is normally set in the range from 112 to 1008 [1 dB to 36 dB].

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Table 5-28 Analog gain setting reference 112 to 159]

ana_gain_global	Gain(times)	Gain(dB)
112	1.123	1.006
113	1.124	1.016
114	1.125	1.025
115	1.127	1.035
116	1.128	1.044
117	1.129	1.054
118	1.130	1.063
119	1.131	1.073
120	1.133	1.083
121	1.134	1.092
122	1.135	1.102
123	1.137	1.112
124	1.138	1.121
125	1.139	1.131
126	1.140	1.140
127	1.142	1.150
128	1.143	1.160
129	1.144	1.170
130	1.145	1.179
131	1.147	1.189
132	1.148	1.199
133	1.149	1.208
134	1.151	1.218
135	1.152	1.228
136	1.153	1.238
137	1.154	1.248
138	1.156	1.257
139	1.157	1.267
140	1.158	1.277
141	1.160	1.287
142	1.161	1.297
143	1.162	1.306
144	1.164	1.316
145	1.165	1.326
146	1.166	1.336
147	1.168	1.346
148	1.169	1.356
149	1.170	1.366
150	1.172	1.376
151	1.173	1.386
152	1.174	1.396
153	1.176	1.406
154	1.177	1.416
155	1.178	1.426
156	1.180	1.436
157	1.181	1.446
158	1.182	1.456
159	1.184	1.466

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Table 5-29 Analog gain setting reference [160 to 319]

ana_gain_global	Gain(times)	Gain(dB)
160	1.185	1.476
161	1.187	1.486
162	1.188	1.496
163	1.189	1.506
164	1.191	1.516
165	1.192	1.526
166	1.193	1.536
167	1.195	1.546
168	1.196	1.557
169	1.198	1.567
170	1.199	1.577
171	1.200	1.587
172	1.202	1.597
173	1.203	1.607
174	1.205	1.618
175	1.206	1.628
176	1.208	1.638
177	1.209	1.648
178	1.210	1.659
179	1.212	1.669
180	1.213	1.679
181	1.215	1.689
182	1.216	1.700
183	1.218	1.710
184	1.219	1.720
185	1.221	1.731
186	1.222	1.741
187	1.223	1.751
188	1.225	1.762
189	1.226	1.772
190	1.228	1.783
191	1.229	1.793
192	1.231	1.804
193	1.232	1.814
194	1.234	1.824
195	1.235	1.835
196	1.237	1.845
197	1.238	1.856
198	1.240	1.866
199	1.241	1.877
200	1.243	1.887
201	1.244	1.898
202	1.246	1.909
203	1.247	1.919
204	1.249	1.930
205	1.250	1.940
206	1.252	1.951
207	1.253	1.962
208	1.255	1.972
209	1.256	1.983
210	1.258	1.994
211	1.260	2.004
212	1.261	2.015
213	1.263	2.026
214	1.264	2.036
215	1.266	2.047
216	1.267	2.058
217	1.269	2.069
218	1.270	2.079
219	1.272	2.090
220	1.274	2.101
221	1.275	2.112
222	1.277	2.123
223	1.278	2.133
224	1.280	2.144
225	1.282	2.155
226	1.283	2.166
227	1.285	2.177
228	1.286	2.188
229	1.288	2.199
230	1.290	2.210
231	1.291	2.221
232	1.293	2.231
233	1.295	2.242
234	1.296	2.253
235	1.298	2.264
236	1.299	2.275
237	1.301	2.287
238	1.303	2.298
239	1.304	2.309
240	1.306	2.320
241	1.308	2.331
242	1.309	2.342
243	1.311	2.353
244	1.313	2.364
245	1.315	2.375
246	1.316	2.386
247	1.318	2.398
248	1.320	2.409
249	1.321	2.420
250	1.323	2.431
251	1.325	2.442
252	1.326	2.454
253	1.328	2.465
254	1.330	2.476
255	1.332	2.487
256	1.333	2.499
257	1.335	2.510
258	1.337	2.521
259	1.339	2.533
260	1.340	2.544
261	1.342	2.556
262	1.344	2.567
263	1.346	2.578
264	1.347	2.590
265	1.349	2.601
266	1.351	2.613
267	1.353	2.624
268	1.354	2.636
269	1.356	2.647
270	1.358	2.659
271	1.360	2.670
272	1.362	2.682
273	1.364	2.693
274	1.365	2.705
275	1.367	2.716
276	1.369	2.728
277	1.371	2.740
278	1.373	2.751
279	1.374	2.763
280	1.376	2.775
281	1.378	2.786
282	1.380	2.798
283	1.382	2.810
284	1.384	2.821
285	1.386	2.833
286	1.388	2.845
287	1.389	2.857
288	1.391	2.868
289	1.393	2.880
290	1.395	2.892
291	1.397	2.904
292	1.399	2.916
293	1.401	2.928
294	1.403	2.940
295	1.405	2.951
296	1.407	2.963
297	1.409	2.975
298	1.410	2.987
299	1.412	2.999
300	1.414	3.011
301	1.416	3.023
302	1.418	3.035
303	1.420	3.047
304	1.422	3.059
305	1.424	3.071
306	1.426	3.084
307	1.428	3.096
308	1.430	3.108
309	1.432	3.120
310	1.434	3.132
311	1.436	3.144
312	1.438	3.156
313	1.440	3.169
314	1.442	3.181
315	1.444	3.193
316	1.446	3.205
317	1.448	3.218
318	1.450	3.230
319	1.452	3.242

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Table 5-30 Analog gain setting reference [320 to 479]

ana_gain_global	Gain(times)	Gain(dB)
320	1.455	3.255
321	1.457	3.267
322	1.459	3.279
323	1.461	3.292
324	1.463	3.304
325	1.465	3.316
326	1.467	3.329
327	1.469	3.341
328	1.471	3.354
329	1.473	3.366
330	1.476	3.379
331	1.478	3.391
332	1.480	3.404
333	1.482	3.416
334	1.484	3.429
335	1.486	3.442
336	1.488	3.454
337	1.491	3.467
338	1.493	3.480
339	1.495	3.492
340	1.497	3.505
341	1.499	3.518
342	1.501	3.530
343	1.504	3.543
344	1.506	3.556
345	1.508	3.569
346	1.510	3.581
347	1.513	3.594
348	1.515	3.607
349	1.517	3.620
350	1.519	3.633
351	1.522	3.646
352	1.524	3.659
353	1.526	3.672
354	1.528	3.685
355	1.531	3.697
356	1.533	3.710
357	1.535	3.723
358	1.538	3.737
359	1.540	3.750
360	1.542	3.763
361	1.544	3.776
362	1.547	3.789
363	1.549	3.802
364	1.552	3.815
365	1.554	3.828
366	1.556	3.841
367	1.559	3.855
368	1.561	3.868
369	1.563	3.881
370	1.566	3.894
371	1.568	3.908
372	1.571	3.921
373	1.573	3.934
374	1.575	3.948
375	1.578	3.961
376	1.580	3.974
377	1.583	3.988
378	1.585	4.001
379	1.588	4.015
380	1.590	4.028
381	1.593	4.042
382	1.595	4.055
383	1.598	4.069
384	1.600	4.082
385	1.603	4.096
386	1.605	4.110
387	1.608	4.123
388	1.610	4.137
389	1.613	4.151
390	1.615	4.164
391	1.618	4.178
392	1.620	4.192
393	1.623	4.205
394	1.625	4.219
395	1.628	4.233
396	1.631	4.247
397	1.633	4.261
398	1.636	4.275
399	1.638	4.288
400	1.641	4.302
401	1.644	4.316
402	1.646	4.330
403	1.649	4.344
404	1.652	4.358
405	1.654	4.372
406	1.657	4.386
407	1.660	4.400
408	1.662	4.414
409	1.665	4.428
410	1.668	4.443
411	1.670	4.457
412	1.673	4.471
413	1.676	4.485
414	1.679	4.499
415	1.681	4.514
416	1.684	4.528
417	1.687	4.542
418	1.690	4.557
419	1.693	4.571
420	1.695	4.585
421	1.698	4.600
422	1.701	4.614
423	1.704	4.629
424	1.707	4.643
425	1.710	4.657
426	1.712	4.672
427	1.715	4.687
428	1.718	4.701
429	1.721	4.716
430	1.724	4.730
431	1.727	4.745
432	1.730	4.760
433	1.733	4.774
434	1.736	4.789
435	1.739	4.804
436	1.741	4.818
437	1.744	4.833
438	1.747	4.848
439	1.750	4.863
440	1.753	4.878
441	1.756	4.893
442	1.759	4.908
443	1.762	4.922
444	1.766	4.937
445	1.769	4.952
446	1.772	4.967
447	1.775	4.982
448	1.778	4.998
449	1.781	5.013
450	1.784	5.028
451	1.787	5.043
452	1.790	5.058
453	1.793	5.073
454	1.796	5.089
455	1.800	5.104
456	1.803	5.119
457	1.806	5.134
458	1.809	5.150
459	1.812	5.165
460	1.816	5.180
461	1.819	5.196
462	1.822	5.211
463	1.825	5.227
464	1.829	5.242
465	1.832	5.258
466	1.835	5.273
467	1.838	5.289
468	1.842	5.305
469	1.845	5.320
470	1.848	5.336
471	1.852	5.351
472	1.855	5.367
473	1.858	5.383
474	1.862	5.399
475	1.865	5.415
476	1.869	5.430
477	1.872	5.446
478	1.875	5.462
479	1.879	5.478

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Table 5-31 Analog gain setting reference [480 to 639]

ana_gain_global	Gain(times)	Gain(dB)
480	1.882	5.494
481	1.886	5.510
482	1.889	5.526
483	1.893	5.542
484	1.896	5.558
485	1.900	5.574
486	1.903	5.590
487	1.907	5.607
488	1.910	5.623
489	1.914	5.639
490	1.918	5.655
491	1.921	5.671
492	1.925	5.688
493	1.928	5.704
494	1.932	5.720
495	1.936	5.737
496	1.939	5.753
497	1.943	5.770
498	1.947	5.786
499	1.950	5.803
500	1.954	5.819
501	1.958	5.836
502	1.962	5.853
503	1.965	5.869
504	1.969	5.886
505	1.973	5.903
506	1.977	5.919
507	1.981	5.936
508	1.984	5.953
509	1.988	5.970
510	1.992	5.987
511	1.996	6.004
512	2.000	6.021
513	2.004	6.038
514	2.008	6.055
515	2.012	6.072
516	2.016	6.089
517	2.020	6.106
518	2.024	6.123
519	2.028	6.140
520	2.032	6.157
521	2.036	6.175
522	2.040	6.192
523	2.044	6.209
524	2.048	6.227
525	2.052	6.244
526	2.056	6.261
527	2.060	6.279
528	2.065	6.296
529	2.069	6.314
530	2.073	6.331
531	2.077	6.349
532	2.081	6.367
533	2.086	6.384
534	2.090	6.402
535	2.094	6.420
536	2.098	6.438
537	2.103	6.455
538	2.107	6.473
539	2.111	6.491
540	2.116	6.509
541	2.120	6.527
542	2.124	6.545
543	2.129	6.563
544	2.133	6.581
545	2.138	6.599
546	2.142	6.617
547	2.147	6.636
548	2.151	6.654
549	2.156	6.672
550	2.160	6.690
551	2.165	6.709
552	2.169	6.727
553	2.174	6.746
554	2.179	6.764
555	2.183	6.783
556	2.188	6.801
557	2.193	6.820
558	2.197	6.838
559	2.202	6.857
560	2.207	6.876
561	2.212	6.894
562	2.216	6.913
563	2.221	6.932
564	2.226	6.951
565	2.231	6.970
566	2.236	6.989
567	2.241	7.008
568	2.246	7.027
569	2.251	7.046
570	2.256	7.065
571	2.260	7.084
572	2.265	7.103
573	2.271	7.122
574	2.276	7.142
575	2.281	7.161
576	2.286	7.180
577	2.291	7.200
578	2.296	7.219
579	2.301	7.239
580	2.306	7.258
581	2.312	7.278
582	2.317	7.298
583	2.322	7.317
584	2.327	7.337
585	2.333	7.357
586	2.338	7.377
587	2.343	7.396
588	2.349	7.416
589	2.354	7.436
590	2.359	7.456
591	2.365	7.476
592	2.370	7.496
593	2.376	7.516
594	2.381	7.537
595	2.387	7.557
596	2.393	7.577
597	2.398	7.597
598	2.404	7.618
599	2.409	7.638
600	2.415	7.659
601	2.421	7.679
602	2.427	7.700
603	2.432	7.720
604	2.438	7.741
605	2.444	7.762
606	2.450	7.782
607	2.456	7.803
608	2.462	7.824
609	2.467	7.845
610	2.473	7.866
611	2.479	7.887
612	2.485	7.908
613	2.491	7.929
614	2.498	7.950
615	2.504	7.972
616	2.510	7.993
617	2.516	8.014
618	2.522	8.035
619	2.528	8.057
620	2.535	8.078
621	2.541	8.100
622	2.547	8.121
623	2.554	8.143
624	2.560	8.165
625	2.566	8.187
626	2.573	8.208
627	2.579	8.230
628	2.586	8.252
629	2.592	8.274
630	2.599	8.296
631	2.606	8.318
632	2.612	8.340
633	2.619	8.362
634	2.626	8.385
635	2.632	8.407
636	2.639	8.429
637	2.646	8.452
638	2.653	8.474
639	2.660	8.497

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Table 5-32 Analog gain setting reference [640 to 799]

ana_gain_global	Gain(times)	Gain(dB)
640	2.667	8.519
641	2.674	8.542
642	2.681	8.565
643	2.688	8.587
644	2.695	8.610
645	2.702	8.633
646	2.709	8.656
647	2.716	8.679
648	2.723	8.702
649	2.731	8.725
650	2.738	8.749
651	2.745	8.772
652	2.753	8.795
653	2.760	8.819
654	2.768	8.842
655	2.775	8.865
656	2.783	8.889
657	2.790	8.913
658	2.798	8.936
659	2.805	8.960
660	2.813	8.984
661	2.821	9.008
662	2.829	9.032
663	2.837	9.056
664	2.844	9.080
665	2.852	9.104
666	2.860	9.128
667	2.868	9.153
668	2.876	9.177
669	2.885	9.201
670	2.893	9.226
671	2.901	9.251
672	2.909	9.275
673	2.917	9.300
674	2.926	9.325
675	2.934	9.349
676	2.943	9.374
677	2.951	9.399
678	2.960	9.424
679	2.968	9.450
680	2.977	9.475
681	2.985	9.500
682	2.994	9.525
683	3.003	9.551
684	3.012	9.576
685	3.021	9.602
686	3.030	9.628
687	3.039	9.653
688	3.048	9.679
689	3.057	9.705
690	3.066	9.731
691	3.075	9.757
692	3.084	9.783
693	3.094	9.809
694	3.103	9.836
695	3.112	9.862
696	3.122	9.889
697	3.131	9.915
698	3.141	9.942
699	3.151	9.968
700	3.160	9.995
701	3.170	10.022
702	3.180	10.049
703	3.190	10.076
704	3.200	10.103
705	3.210	10.130
706	3.220	10.157
707	3.230	10.185
708	3.241	10.212
709	3.251	10.240
710	3.261	10.267
711	3.272	10.295
712	3.282	10.323
713	3.293	10.351
714	3.303	10.379
715	3.314	10.407
716	3.325	10.435
717	3.336	10.463
718	3.346	10.492
719	3.357	10.520
720	3.368	10.549
721	3.380	10.577
722	3.391	10.606
723	3.402	10.635
724	3.413	10.664
725	3.425	10.693
726	3.436	10.722
727	3.448	10.751
728	3.459	10.780
729	3.471	10.810
730	3.483	10.839
731	3.495	10.869
732	3.507	10.898
733	3.519	10.928
734	3.531	10.958
735	3.543	10.988
736	3.556	11.018
737	3.568	11.048
738	3.580	11.079
739	3.593	11.109
740	3.606	11.140
741	3.618	11.170
742	3.631	11.201
743	3.644	11.232
744	3.657	11.263
745	3.670	11.294
746	3.683	11.325
747	3.697	11.356
748	3.710	11.388
749	3.724	11.419
750	3.737	11.451
751	3.751	11.483
752	3.765	11.515
753	3.779	11.547
754	3.793	11.579
755	3.807	11.611
756	3.821	11.643
757	3.835	11.676
758	3.850	11.708
759	3.864	11.741
760	3.879	11.774
761	3.894	11.807
762	3.908	11.840
763	3.923	11.873
764	3.938	11.907
765	3.954	11.940
766	3.969	11.974
767	3.984	12.007
768	4.000	12.041
769	4.016	12.075
770	4.031	12.109
771	4.047	12.144
772	4.063	12.178
773	4.080	12.213
774	4.096	12.247
775	4.112	12.282
776	4.129	12.317
777	4.146	12.352
778	4.163	12.387
779	4.180	12.423
780	4.197	12.458
781	4.214	12.494
782	4.231	12.530
783	4.249	12.566
784	4.267	12.602
785	4.285	12.638
786	4.303	12.674
787	4.321	12.711
788	4.339	12.748
789	4.357	12.785
790	4.376	12.822
791	4.395	12.859
792	4.414	12.896
793	4.433	12.934
794	4.452	12.971
795	4.472	13.009
796	4.491	13.047
797	4.511	13.085
798	4.531	13.124
799	4.551	13.162

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Table 5-33 Analog gain setting reference [800 to 1008]

ana_gain_global	Gain(times)	Gain(dB)	ana_gain_global	Gain(times)	Gain(dB)	ana_gain_global	Gain(times)	Gain(dB)	ana_gain_global	Gain(times)	Gain(dB)
800	4.571	13.201	853	5.988	15.546	906	8.678	18.768	959	15.754	23.948
801	4.592	13.240	854	6.024	15.597	907	8.752	18.842	960	16.000	24.082
802	4.613	13.279	855	6.059	15.648	908	8.828	18.917	961	16.254	24.219
803	4.633	13.318	856	6.095	15.700	909	8.904	18.992	962	16.516	24.358
804	4.655	13.358	857	6.132	15.752	910	8.982	19.068	963	16.787	24.499
805	4.676	13.397	858	6.169	15.804	911	9.062	19.144	964	17.067	24.643
806	4.697	13.437	859	6.206	15.856	912	9.143	19.222	965	17.356	24.789
807	4.719	13.477	860	6.244	15.909	913	9.225	19.300	966	17.655	24.937
808	4.741	13.517	861	6.282	15.962	914	9.309	19.378	967	17.965	25.089
809	4.763	13.557	862	6.321	16.016	915	9.394	19.457	968	18.286	25.242
810	4.785	13.598	863	6.360	16.069	916	9.481	19.538	969	18.618	25.399
811	4.808	13.638	864	6.400	16.124	917	9.570	19.618	970	18.963	25.558
812	4.830	13.679	865	6.440	16.178	918	9.660	19.700	971	19.321	25.720
813	4.853	13.720	866	6.481	16.233	919	9.752	19.782	972	19.692	25.886
814	4.876	13.762	867	6.522	16.288	920	9.846	19.865	973	20.078	26.055
815	4.900	13.803	868	6.564	16.344	921	9.942	19.949	974	20.480	26.227
816	4.923	13.845	869	6.606	16.399	922	10.039	20.034	975	20.898	26.402
817	4.947	13.887	870	6.649	16.456	923	10.139	20.120	976	21.333	26.581
818	4.971	13.929	871	6.693	16.512	924	10.240	20.206	977	21.787	26.764
819	4.995	13.971	872	6.737	16.569	925	10.343	20.293	978	22.261	26.951
820	5.020	14.013	873	6.781	16.626	926	10.449	20.381	979	22.756	27.142
821	5.044	14.056	874	6.827	16.684	927	10.557	20.471	980	23.273	27.337
822	5.069	14.099	875	6.872	16.742	928	10.667	20.561	981	23.814	27.537
823	5.095	14.142	876	6.919	16.801	929	10.779	20.652	982	24.381	27.741
824	5.120	14.185	877	6.966	16.860	930	10.894	20.743	983	24.976	27.950
825	5.146	14.229	878	7.014	16.919	931	11.011	20.836	984	25.600	28.165
826	5.172	14.273	879	7.062	16.979	932	11.130	20.930	985	26.256	28.385
827	5.198	14.317	880	7.111	17.039	933	11.253	21.025	986	26.947	28.610
828	5.224	14.361	881	7.161	17.099	934	11.378	21.121	987	27.676	28.842
829	5.251	14.405	882	7.211	17.160	935	11.506	21.218	988	28.444	29.080
830	5.278	14.450	883	7.262	17.222	936	11.636	21.316	989	29.257	29.325
831	5.306	14.495	884	7.314	17.283	937	11.770	21.416	990	30.118	29.576
832	5.333	14.540	885	7.367	17.346	938	11.907	21.516	991	31.030	29.836
833	5.361	14.585	886	7.420	17.408	939	12.047	21.618	992	32.000	30.103
834	5.389	14.631	887	7.474	17.472	940	12.190	21.720	993	33.032	30.379
835	5.418	14.677	888	7.529	17.535	941	12.337	21.824	994	34.133	30.664
836	5.447	14.723	889	7.585	17.599	942	12.488	21.930	995	35.310	30.958
837	5.476	14.769	890	7.642	17.664	943	12.642	22.036	996	36.571	31.263
838	5.505	14.816	891	7.699	17.729	944	12.800	22.144	997	37.926	31.579
839	5.535	14.863	892	7.758	17.795	945	12.962	22.253	998	39.385	31.907
840	5.565	14.910	893	7.817	17.861	946	13.128	22.364	999	40.960	32.247
841	5.596	14.957	894	7.877	17.927	947	13.299	22.476	1000	42.667	32.602
842	5.626	15.005	895	7.938	17.994	948	13.474	22.590	1001	44.522	32.971
843	5.657	15.052	896	8.000	18.062	949	13.653	22.705	1002	46.545	33.358
844	5.689	15.101	897	8.063	18.130	950	13.838	22.821	1003	48.762	33.762
845	5.721	15.149	898	8.127	18.199	951	14.027	22.940	1004	51.200	34.185
846	5.753	15.198	899	8.192	18.268	952	14.222	23.059	1005	53.895	34.631
847	5.785	15.247	900	8.258	18.338	953	14.423	23.181	1006	56.889	35.101
848	5.818	15.296	901	8.325	18.408	954	14.629	23.304	1007	60.235	35.597
849	5.851	15.345	902	8.393	18.479	955	14.841	23.429	1008	64.000	36.124
850	5.885	15.395	903	8.463	18.550	956	15.059	23.556			
851	5.919	15.445	904	8.533	18.622	957	15.284	23.685			
852	5.953	15.495	905	8.605	18.695	958	15.515	23.815			

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5.5.2. Digital gain settings

Digital gain of this sensor can be set by color. The registers for digital gain settings are shown in the table below.

Table 5-34 Digital gain setting

	Address	Bit	Name	Description	Notes
I²C register	0x3130	[0]	DIG_GAIN_MODE	Digital gain control mode select 0 : all color(DIG_GAIN_GLOBAL setting is used for all color) 1 : by color	
	0x020e	[7:0]	DIG_GAIN_GLOBAL[15:8]	Digital gain control code for GR in case of individual control. When global control is selected, treated with a code for all colors. Format : 16-bit unsigned real Range : 0x0100 to 0x0FFF *Setup other than the above is forbidden.	range: 1 to 15
	0x020f	[7:0]	DIG_GAIN_GLOBAL[7:0]		range: 0 to 255
	0x0210	[7:0]	DIG_GAIN_R[15:8]	Digital gain control code for R Format : 16-bit unsigned real	range: 1 to 15
	0x0211	[7:0]	DIG_GAIN_R[7:0]		range: 0 to 255
	0x0212	[7:0]	DIG_GAIN_B[15:8]	Digital gain control code for B Format : 16-bit unsigned real	range: 1 to 15
	0x0213	[7:0]	DIG_GAIN_B[7:0]		range: 0 to 255
	0x0214	[7:0]	DIG_GAIN_GB[15:8]	Digital gain control code for GB Format : 16-bit unsigned real	range: 1 to 15
	0x0215	[7:0]	DIG_GAIN_GB[7:0]		range: 0 to 255

5.5.2.1. Global Digital Gain Control

Global Digital Gain Control function can be used.

5.5.2.2. Individual Digital Gain Control

Each register is comprised of 2 bytes with the *upper_byte* [15:8] setting the integer portion and the *lower_byte* [7:0] setting the decimal portion of the gain. The gain for each color is obtained by the following formula.

Digital gain = upper_byte + lower_byte / 256 [times]

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The unit of Digital gain is times. The upper byte can be set to a value ranging from 1 to 15 and the lower byte to a value ranging from 0 to 255. Therefore, the range of digital gain is shown as follows.

1 + 0/256 [times] (0 dB) ≤ *Digital gain* ≤ 15 + 255/256 [times] (≒24 dB)

When representing the gain by log-linear scale [dB], lower gain takes coarse steps and high gain takes fine steps for the incrimination of the register value. The table below indicates the register values in 0.1 dB steps for reference.

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Table 5-35 Digital gain setting reference (0 to 23.9[dB])

Upper		Lower		Gain [times]	Gain [dB]
dec	hex	dec	hex		
1	1	0	0	1	0
1	1	3	3	1.01	0.1
1	1	6	6	1.02	0.2
1	1	9	9	1.04	0.3
1	1	12	C	1.05	0.4
1	1	15	F	1.06	0.49
1	1	18	12	1.07	0.59
1	1	21	15	1.08	0.68
1	1	25	19	1.1	0.81
1	1	28	1C	1.11	0.9
1	1	31	1F	1.12	0.99
1	1	35	23	1.14	1.11
1	1	38	26	1.15	1.2
1	1	41	29	1.16	1.29
1	1	45	2D	1.18	1.41
1	1	48	30	1.19	1.49
1	1	52	34	1.2	1.61
1	1	55	37	1.21	1.69
1	1	59	3B	1.23	1.8
1	1	63	3F	1.25	1.91
1	1	66	42	1.26	1.99
1	1	70	46	1.27	2.1
1	1	74	4A	1.29	2.21
1	1	78	4E	1.3	2.31
1	1	81	51	1.32	2.39
1	1	85	55	1.33	2.49
1	1	89	59	1.35	2.59
1	1	93	5D	1.36	2.69
1	1	97	61	1.38	2.79
1	1	101	65	1.39	2.89
1	1	106	6A	1.41	3.01
1	1	110	6E	1.43	3.1
1	1	114	72	1.45	3.2
1	1	118	76	1.46	3.29
1	1	123	7B	1.48	3.41
1	1	127	7F	1.5	3.5
1	1	131	83	1.51	3.59
1	1	136	88	1.53	3.7
1	1	140	8C	1.55	3.79
1	1	145	91	1.57	3.90
1	1	150	96	1.59	4.01
1	1	154	9A	1.6	4.09
1	1	159	9F	1.62	4.2
1	1	164	A4	1.64	4.3
1	1	169	A9	1.66	4.4
1	1	174	AE	1.68	4.5
1	1	179	B3	1.7	4.6
1	1	184	B8	1.72	4.7
1	1	189	BD	1.74	4.8
1	1	194	C2	1.76	4.9
1	1	199	C7	1.78	5
1	1	205	CD	1.8	5.11
1	1	210	D2	1.82	5.2
1	1	215	D7	1.84	5.3
1	1	221	DD	1.86	5.41
1	1	226	E2	1.88	5.5
1	1	232	E8	1.91	5.6
1	1	237	ED	1.93	5.69
1	1	243	F3	1.95	5.8
1	1	249	F9	1.97	5.9
1	1	255	FF	2	6
2	2	5	5	2.02	6.11
2	2	11	B	2.04	6.21
2	2	17	11	2.07	6.3
2	2	23	17	2.09	6.4
2	2	29	1D	2.11	6.5
2	2	35	23	2.14	6.59
2	2	42	2A	2.16	6.71
2	2	48	30	2.19	6.8
2	2	55	37	2.21	6.91
2	2	61	3D	2.24	7
2	2	68	44	2.27	7.1
2	2	74	4A	2.29	7.19
2	2	81	51	2.32	7.3
2	2	88	58	2.34	7.4
2	2	95	5F	2.37	7.5
2	2	102	66	2.4	7.6
2	2	109	6D	2.43	7.7
2	2	116	74	2.45	7.79
2	2	124	7C	2.48	7.9
2	2	131	83	2.51	8
2	2	138	8A	2.54	8.09
2	2	146	92	2.57	8.2
2	2	154	9A	2.6	8.3
2	2	161	A1	2.63	8.4
2	2	169	A9	2.66	8.5
2	2	177	B1	2.69	8.6
2	2	185	B9	2.72	8.7
2	2	193	C1	2.75	8.8
2	2	201	C9	2.79	8.9
2	2	210	D2	2.82	9.01
2	2	218	DA	2.85	9.10
2	2	226	E2	2.88	9.2
2	2	235	EB	2.92	9.3
2	2	244	F4	2.95	9.41
2	2	252	FC	2.98	9.5
3	3	5	5	3.02	9.6
3	3	14	E	3.05	9.7
3	3	23	17	3.09	9.8
3	3	32	20	3.13	9.9
3	3	42	2A	3.16	10
3	3	51	33	3	10.1
3	3	60	3C	3.23	10.2
3	3	70	46	3.27	10.3
3	3	80	50	3.31	10.4
3	3	90	5A	3.35	10.5
3	3	99	63	3.39	10.6
3	3	109	6D	3.43	10.7
3	3	120	78	3.47	10.8
3	3	130	82	3.51	10.9
3	3	140	8C	3.55	11
3	3	151	97	3.59	11.1
3	3	161	A1	3.63	11.2
3	3	172	AC	3.67	11.3
3	3	183	B7	3.71	11.4
3	3	194	C2	3.76	11.5
3	3	205	CD	3.8	11.6
3	3	217	D9	3.85	11.7
3	3	228	E4	3.89	11.8
3	3	239	EF	3.93	11.9
3	3	251	FB	3.98	12
4	4	7	7	4.03	12.1
4	4	19	13	4.07	12.2
4	4	31	1F	4.12	12.3
4	4	43	2B	4.17	12.4
4	4	56	38	4.22	12.5
4	4	68	44	4.27	12.6
4	4	81	51	4.32	12.7
4	4	93	5D	4.36	12.8
4	4	106	6A	4.41	12.9
4	4	120	78	4.47	13
4	4	133	85	4.52	13.1
4	4	146	92	4.57	13.2
4	4	160	A0	4.63	13.3
4	4	173	AD	4.68	13.4
4	4	187	BB	4.73	13.5
4	4	201	C9	4.79	13.6
4	4	215	D7	4.84	13.7
4	4	230	E6	4.9	13.8
4	4	244	F4	4.95	13.9
5	5	3	3	5.01	14
5	5	18	12	5.07	14.1
5	5	33	21	5.13	14.2
5	5	48	30	5.19	14.3
5	5	64	40	5.25	14.4
5	5	79	4F	5.31	14.5
5	5	95	5F	5.37	14.6
5	5	111	6F	5.43	14.7
5	5	127	7F	5.5	14.8
5	5	143	8F	5.56	14.9
5	5	160	A0	5.63	15
5	5	176	B0	5.69	15.1
5	5	193	C1	5.75	15.2
5	5	210	D2	5.82	15.3
5	5	227	E3	5.89	15.4
5	5	245	F5	5.96	15.5
6	6	7	7	6.03	15.6
6	6	24	18	6.09	15.7
6	6	42	2A	6.16	15.8
6	6	61	3D	6.24	15.9
6	6	79	4F	6.31	16
6	6	98	62	6.38	16.1
6	6	117	75	6.46	16.2
6	6	136	88	6.53	16.3
6	6	155	9B	6.61	16.4
6	6	175	AF	6.68	16.5
6	6	195	C3	6.76	16.6
6	6	215	D7	6.84	16.7
6	6	235	EB	6.92	16.8
7	7	0	00	7.00	16.9
7	7	20	14	7.08	17
7	7	41	29	7.16	17.1
7	7	63	3F	7.25	17.2
7	7	84	54	7.33	17.3
7	7	106	6A	7.41	17.4
7	7	128	80	7.50	17.5
7	7	150	96	7.59	17.6
7	7	172	AC	7.67	17.7
7	7	195	C3	7.76	17.8
7	7	218	DA	7.85	17.9
7	7	241	F1	7.94	18
8	8	9	9	8.04	18.1
8	8	33	21	8.13	18.2
8	8	57	39	8.22	18.3
8	8	81	51	8.32	18.4
8	8	106	6A	8.41	18.5
8	8	131	83	8.51	18.6
8	8	156	9C	8.61	18.7
8	8	182	B6	8.71	18.8
8	8	207	CF	8.81	18.9
8	8	234	EA	8.91	19
9	9	4	4	9.02	19.1
9	9	31	1F	9.12	19.2
9	9	58	3A	9.23	19.3
9	9	85	55	9.33	19.4
9	9	113	71	9.44	19.5
9	9	141	8D	9.55	19.6
9	9	169	A9	9.66	19.7
9	9	198	C6	9.77	19.8
9	9	227	E3	9.89	19.9
10	A	0	00	10	20
10	A	30	1E	10.12	20.1
10	A	60	3C	10.23	20.2
10	A	90	5A	10.35	20.3
10	A	121	79	10.47	20.4
10	A	152	98	10.59	20.5
10	A	183	B7	10.71	20.6
10	A	215	D7	10.84	20.7
10	A	247	F7	10.96	20.8
11	B	23	17	11.09	20.9
11	B	56	38	11.22	21
11	B	90	5A	11.35	21.1
11	B	123	7B	11.48	21.2
11	B	157	9D	11.61	21.3
11	B	192	C0	11.75	21.4
11	B	227	E3	11.89	21.5
12	C	6	6	12.02	21.6
12	C	41	29	12.16	21.7
12	C	77	4D	12.3	21.8
12	C	114	72	12.45	21.9
12	C	151	97	12.59	22
12	C	188	BC	12.73	22.1
12	C	226	E2	12.88	22.2
13	D	8	8	13.03	22.3
13	D	47	2F	13.18	22.4
13	D	86	56	13.34	22.5
13	D	125	7D	13.49	22.6
13	D	165	A5	13.64	22.7
13	D	206	CE	13.8	22.8
13	D	247	F7	13.96	22.9
14	E	32	20	14.13	23
14	E	74	4A	14.29	23.1
14	E	116	74	14.45	23.2
14	E	159	9F	14.62	23.3
14	E	203	CB	14.79	23.4
14	E	246	F6	14.96	23.5
15	F	35	23	15.14	23.6
15	F	80	50	15.31	23.7
15	F	125	7D	15.49	23.8
15	F	171	AB	15.67	23.9
15	F	217	D9	15.85	24

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5.5.3. Change in output pixel level depending on binning mode

The output pixel level in Binning mode changes as shown in the table.

Table 5-36 Output pixel level according to binning modes(Normal)

Operation Mode	0x0220 [0] HDR_MODE [0]	0x3140 [2:0] HDR_FNCSEL [2:0]	0x0900 [0] BINNING_MODE [0]	0x0901 [7:0] BINNING_TYPE_H [7:4] BINNING_TYPE_V [3:0]	0x3247 [7:0] BINNING_PRIORITY_V [7:0]	0x3246 [7:0] BINNING_PRIORITY_H [7:0]	0x0902 [7:0] BINNING_WEIGHTING [7:0]	0x3F0C [0] BINNING_PDAF_EN [0]	0x3F14 [0] FULL_QHDR_EN [0]	0x3F15 [0] ADC_MODE [1:0]	Output Pixel Level Ratio	Min Analog Gain[dB]	Max Analog Gain[dB]
Full-pixel	0x0	X	0x0	0x11	0x01	0x01	X	0x0	0x1	0x0	1	1	24
2x2 Adjacent Pixel Binning without PDAF			0x1	0x22	0x81	0x81	0x08	0x0	0x0		1	1	36
							0x09				4	1	24
2x2 Adjacent Pixel Binning with PDAF							0x08	0x1	0x0		1	1	36
							0x09				4	1	24
2x2 Adjacent Pixel Binning V2H2			0x1	0x44	0x89	0x89	0x08/0x0A	0x0	0x0		1	1	36
							0x09				4	1	24
2x2 Adjacent Pixel Binning V2H2 720p			0x1	0x44	0x89	0x89	0x08/0x0A	0x0	0x0	0x1	1	1	30
							0x09				4	1	24
QBC HDR	0x1	0x4	X	X	X	X	X	0x0	0x1	0x0	1	1	24

Table 5-37 Binning mode determining registers

	Address	Bit	Name	Description
I²C register	0x0902	[7:0]	BINNING_WEIGHTING	Bit[1:0] Binning type selection 0 : Averaged 1 : Summed 2 : Weighting averaged*1 *1 : Only available in 2x2 Adjacent Pixel Binning V2H2 mode *Setup other than the above is forbidden. Bit [7:2] : 6'b000010 Fixed

*For BINNING_MODE and BINNING_TYPE see 5.2.2

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5.5.4. Bayer weighting function

Bayer weighting function optimizes the center of gravity of every color when using 2x2 Adjacent Pixel Binning V2H2 mode.

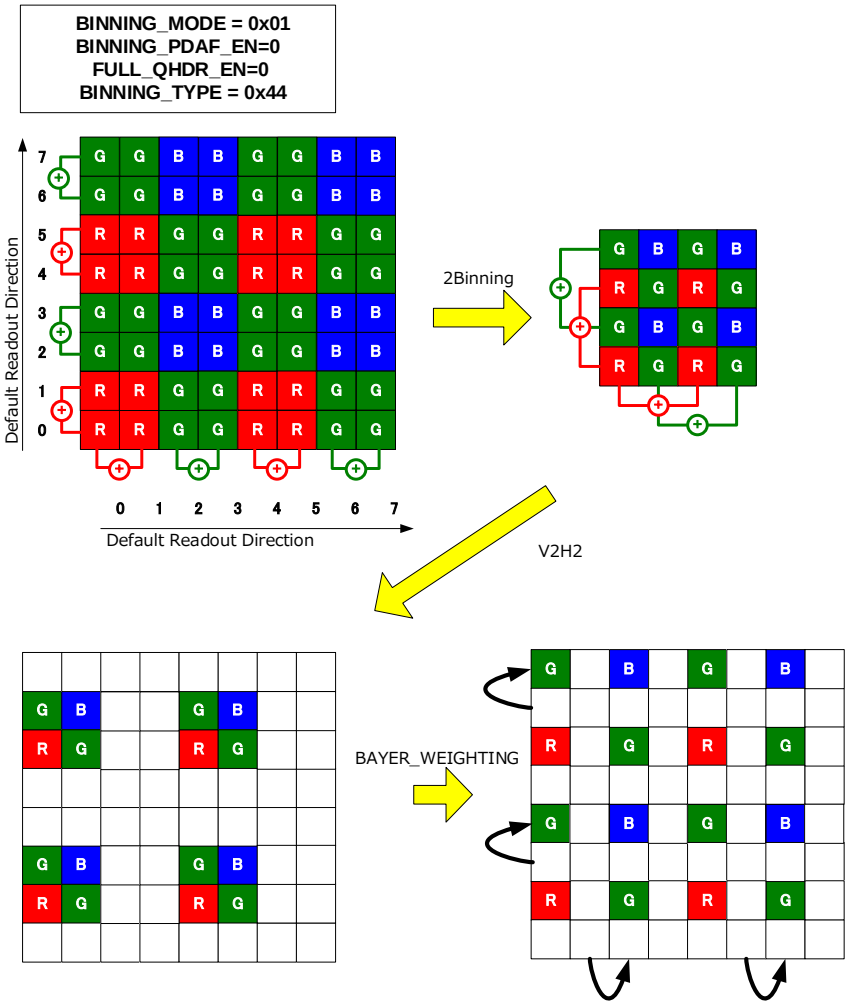


Figure 5-12 Bayer weighting methodology

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5.6. Optical black level clamp

5.6.1. Default setting (normal usage)

This sensor has the optical black level clamping function to make the black level stable against changes in operating conditions. The average value of the black level is adjusted to the pedestal level (0x40).

5.6.2. Test mode setting

Set the registers shown in the table below to control output black level for test purpose.

Table 5-38 Optical black level clamp related registers and descriptions

	Address	Bit	Name	Description
I²C register	0x3032	[1:0]	MANUAL_DATA_PEDESTAL_VALUE[9:8]	Manual setting value for Data Pedestal Default setting value : 0x40
	0x3033	[7:0]	MANUAL_DATA_PEDESTAL_VALUE[7:0]	
	0x3030	[0]	MANUAL_DATA_PEDESTAL_EN	Output black level is controlled with MANUAL_DATA_PEDESTAL_VALUE register value 0x0 : pedestal = 0x40 (fixed value) 0x1 : pedestal = MANUAL_DATA_PEDESTAL_VALUE else : inhibited value

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5.7. Image compensation function setting

There are some other functions in the sensor image pipeline. Use-case may be chosen in terms of trade-off for power consumption and image quality, for example.

5.7.1. 2x1 OCL pixels and PDAF

PDAF capable sensor is equipped with double size OCL pixels called "2x1 OCL pixels". There are two types of 2x1 OCL pixels in this sensor, one is a Left 2x1 OCL pixel and the other is a Right 2x1 OCL pixel. And both 2x1 OCL pixels are located in green pixels. Phase information obtained by both Left and Right 2x1 OCL pixels enables Phase Detection Auto Focus (PDAF) for high-speed auto focus.

For more details of PDAF, refer to the software reference manual for PDAF.

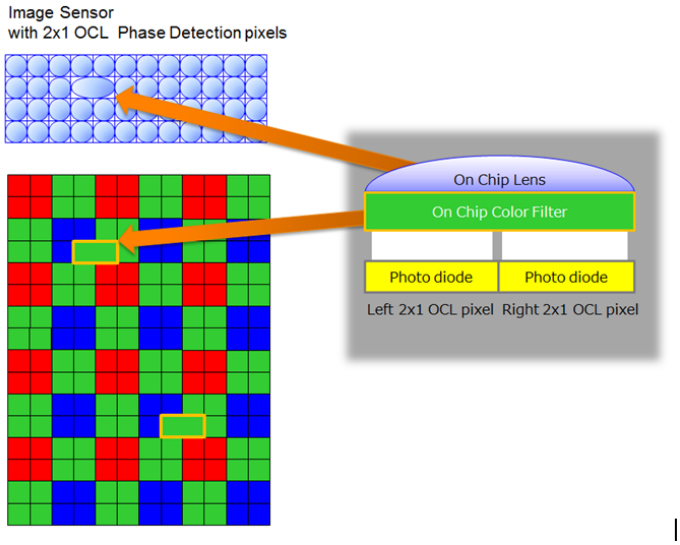


Figure 5-13 2x1 OCL pixels for Phase detection

5.7.2. L/R Sensitivity Correction (LRC)

In this paragraph, abstracts of 2x1 OCL Pixel Correction and an additional defect correction which are both unique function for PDAF capable sensors.

L/R sensitivity correction

Left and right 2x1 OCL pixels have different signal values than normal pixels because of the difference those OCL or Color filter. Left and right 2x1 OCL pixels require having some gain such that the signal level of each color will be equivalent to a normal pixel. This is called “L/R Sensitivity Correction” (LRC).For more details, refer to the software reference manual for PDAF.

Note that, when using LRC, writing the 16 x 12 knot point Gain table to OTP is required. This is called "LRC gain map".This LRC gain map values should be written by the module vendor within the module calibration process.

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5.7.3. Defect Pixel Correction (DPC)

5.7.3.1. Dynamic Defect Correction

This is a function to detect and correct single pixel defects. It performs defect compensation with using pixels surrounding a defect pixel.

* If an adjacent pixel of the same color is defective, this function cannot correct the defect.

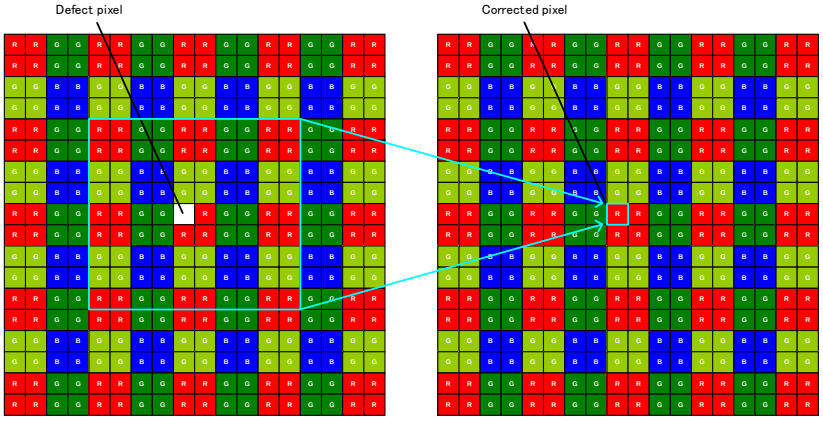


Figure 5-14 Dynamic Defect Correction model figure (Full resolution)

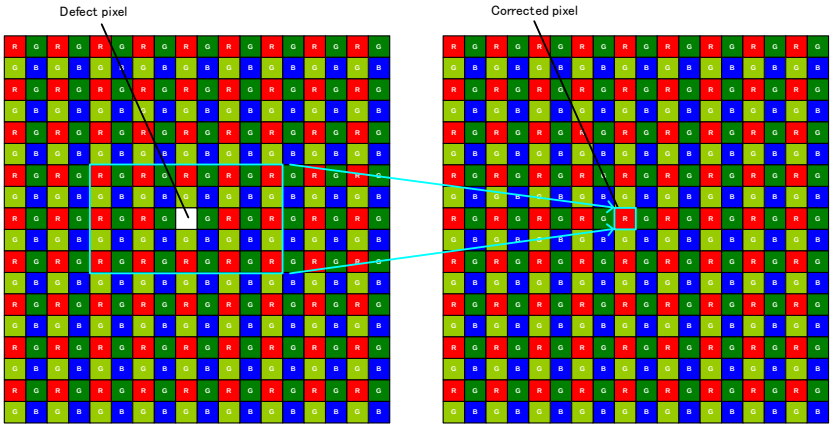


Figure 5-15 Dynamic Defect Correction model figure (binning mode)

Whether to turn each Defect Correction ON or OFF can be set with the following registers.

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Table 5-39 Defect correction register

	Address	Bit	Name	Description
I²C register	0x0b05	[0]	MAP_COUP_CORR_EN	Mapped couplet correction control 0 : disable 1 : enable
	0x0b06	[0]	SING_DEF_CORR_EN	dynamic singlet correction control 0 : disable 1 : enable

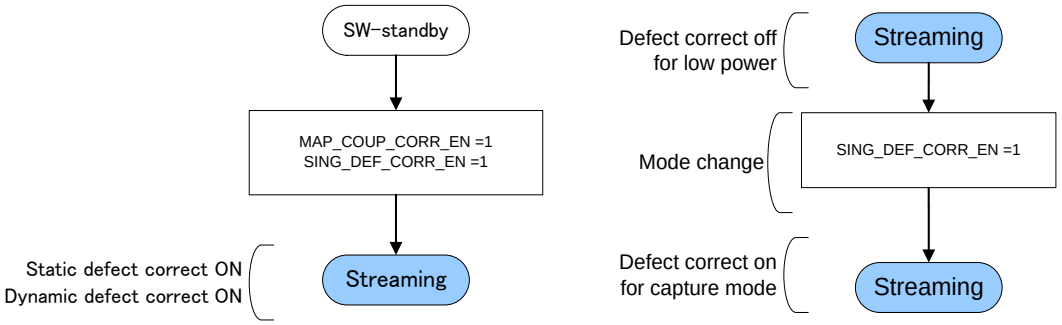


Figure 5-16 Defect correction flow chart

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5.8. White balance gain setting

Sensor expects input white balance information from Host to minimize artifacts. Host needs to feedback values to ABS_GAIN_GR, ABS_GAIN_R, ABS_GAIN_B, and ABS_GAIN_GB. Value of 0 is inhibited.

Usage of GPH (Grouped parameter hold) is mandatory to make the reflection timing synchronous for updating the following registers.

Table 5-40 White balance gain control registers

I^2C Address	Bit	Name	Description
0x0b8e 0x0b8f	[15:8] [7:0]	ABS_GAIN_GR	White balance gain for GR Format : 16-bit iReal Range : 0 to 255.996
0x0b90 0x0b91	[15:8] [7:0]	ABS_GAIN_R	White balance gain for R Format : 16-bit iReal Range : 0 to 255.996
0x0b92 0x0b93	[15:8] [7:0]	ABS_GAIN_B	White balance gain for B Format : 16-bit iReal Range : 0 to 255.996
0x0b94 0x0b95	[15:8] [7:0]	ABS_GAIN_GB	White balance gain for GB Format : 16-bit iReal Range : 0 to 255.996

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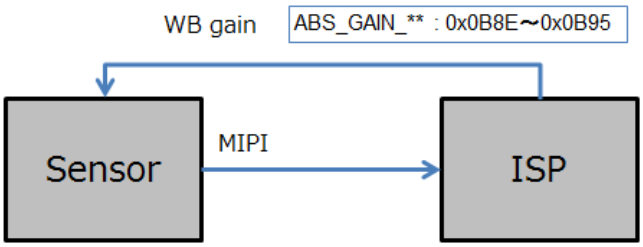


Figure 5-17 Block diagram for White balance gain

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6. Power ON/OFF Sequence

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6.1. Power ON sequence

6.1.1. Power ON reset

This sensor doesn't support a built-in "Power ON Reset" function.

The XCLR pin is set to "LOW" and the power supplies are brought up. Then the XCLR pin should be set to "High" after INCK supplied.

6.1.2. Power ON sequence

6.1.2.1. Startup sequence with 2-wire serial communication (external reset)

Follow the power supply start up sequence below.

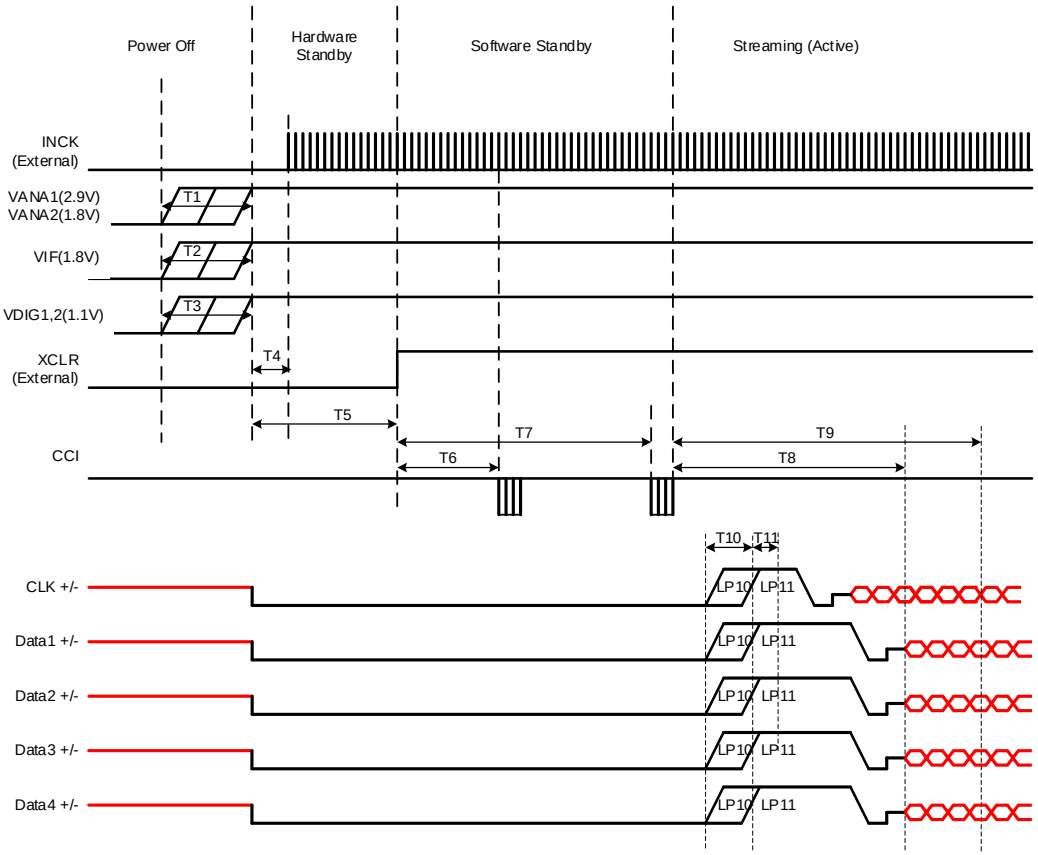


Figure 6-1 Start up sequence with 2-wire serial communication (external reset, D-PHY)

*Presence of INCK during Power Off is acceptable despite of above chart.

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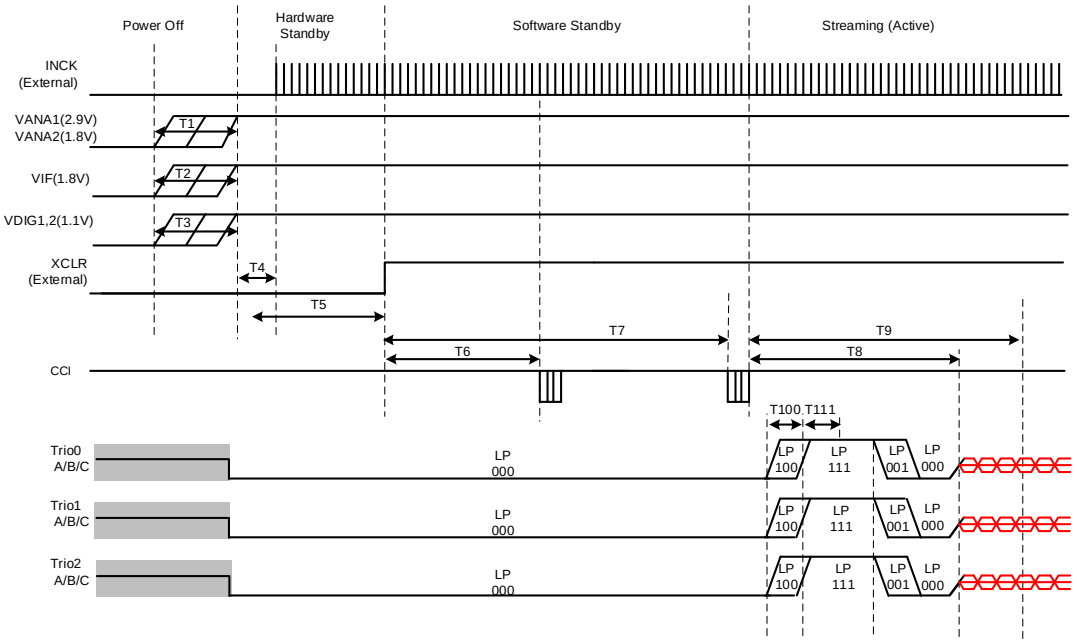


Figure 6-2 Start up sequence with 2-wire serial communication (external reset, C-PHY)

*Presence of INCK during Power Off is acceptable despite of above chart.

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Table 6-1 Startup sequence timing constraints (External reset)

Item	Label	Min.	Max.	Unit	Comment
VANA1,2 rising – VANA1,2 on	T1	VANA1, VANA2, VDIG1, VDIG2 and VIF may rise in any order.		µs	Slew rate of VANA1, VANA2,VDIG1,VDIG2 and VIF(0%-100%) : max 100mV/us
VDIG1,2 rising – VDIG1,2 on	T2			µs	
VIF rising – VIF on	T3			µs	
VANA1, VANA2, VDIG1, VDIG2 and VIF rising – INCK start	T4	0		µs	Later of T1, T2 and T3
VANA1, VANA2, VDIG1, VDIG2 and VIF rising – XCLR rising	T5	0		ms	
XCLR rising till CCI Read version ID register wait time	T6	1		ms	
XCLR rising till Send Streaming Command wait time (To complete reading all parameters from NVM)	T7	8		ms	VRL cap : 2.2µF
Start of first streaming from Sending Streaming Command.	T8		5.0 ms + The delay of the coarse integration time value + (Tline * 90[lines])		iVTPXCK = 216MHz INCK = 6MHz
			8.0 ms + The delay of the coarse integration time value + (Tline * 90[lines])		iVTPXCK = 62.5MHz INCK = 6MHz
Start of first streaming with stable frame from sending streaming command.	T9		T8 + 1Frame	Frames	
DPHY power up	T10	1	1.1	ms	
DPHY init	T11	100	110	µs	
CPHY power up	T100	1	1.1	ms	
CPHY init	T111	100	110	µs	

* XCLR needs to be Low level until all power supplies complete power-ON.

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6.2. Power down sequence

6.2.1. Power down sequence with 2-wire serial communication

Follow the power down sequence below.

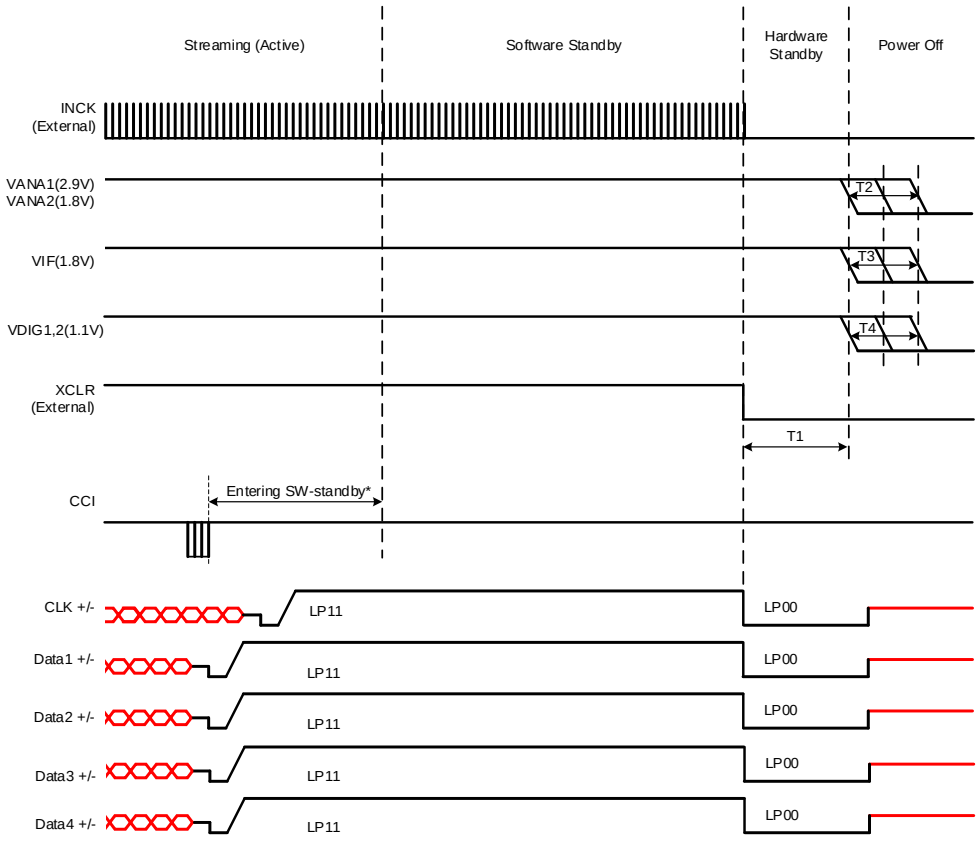


Figure 6-3 Power down sequence with 2-wire serial communication (external reset,D-PHY)

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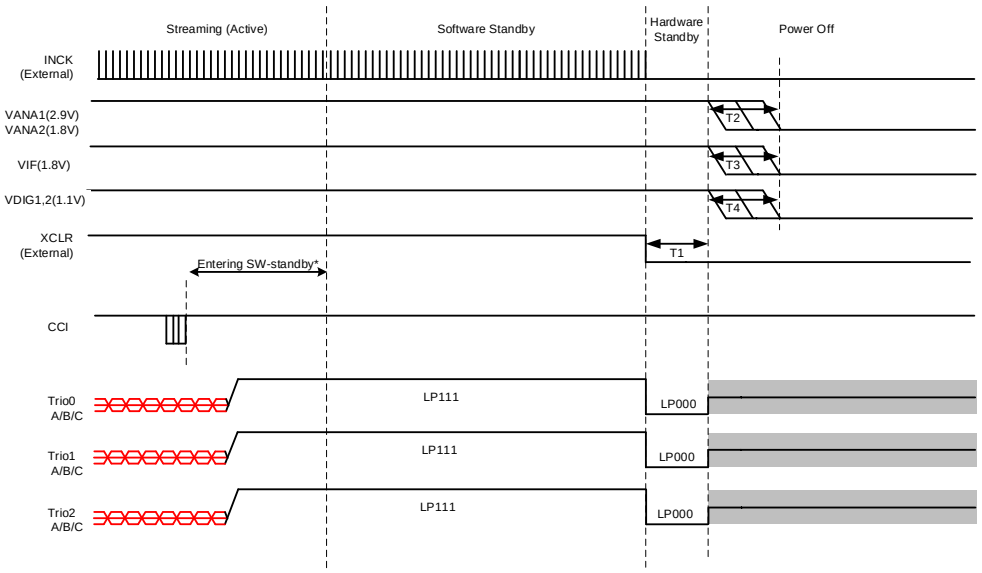


Figure 6-4 Power down sequence with 2-wire serial communication (external reset,C-PHY)

Table 6-2 Power down sequence timing constraints (External reset)

Item	Label	Min.	Max.	Unit	Comment
XCLR Neg-edge – VANA (VDIG or VIF) fall	T1	0		ns	Presence of INCK during Power Off is acceptable.
Sequence free of VANA1,2 falling and VDIG1,2 falling and VIF falling	T2,T3,T4	VANA1, VANA2, VDIG1, VDIG2 and VIF may fall in any order.		ns	

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7. Mode Transition Sequence and Register Update Timing

SRM-230-00088-01

7.1. Transition between SW-standby and Streaming

Sensor can make transition between SW-standby and streaming without using power ON/OFF.

7.1.1. SW-standby to Streaming

The following figure shows how to start streaming at first time after power on sequence and register setting timing restriction

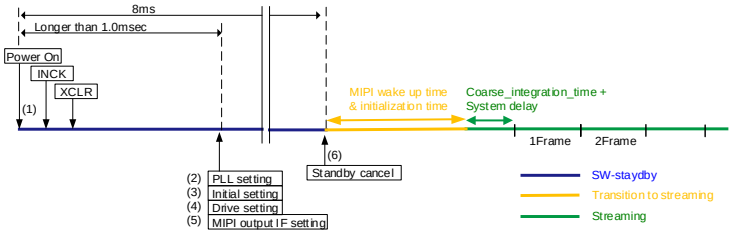


Figure 7-1 Start streaming sequence after power ON

Table 7-1 Start streaming sequence after power ON

(1)	Refer to power up sequence timing diagram	Note: Refer to Power Related sequence of 6.1.2 for procedure of (2) - (6).
(2)	Set PLL parameters	
(3)	Basic settings (operation-critical setting)	
(4)	Set readout mode (start/end position, size, mode, integration time, and gain)	
(5)	Set MIPI interface parameters	
(6)	Start streaming with 0x0100 (MODE_SEL = 1)	
	After “MIPI Wake Up Time” + “Initialize Time”, 1st frame starts and images come out	

The following figure shows how to start a streaming after first streaming.

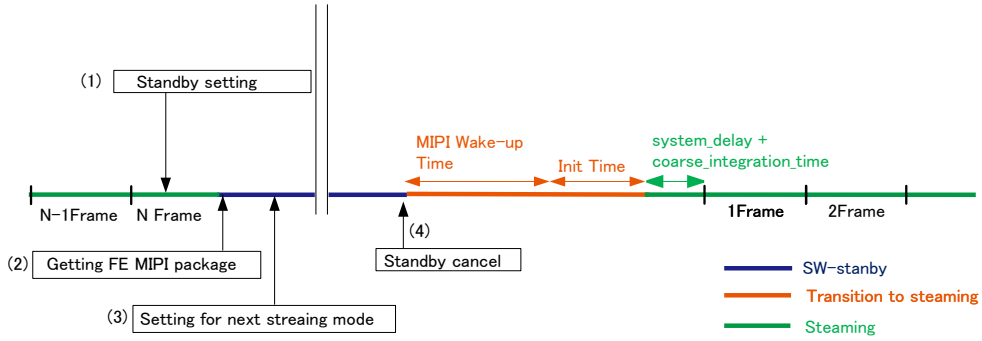


Figure 7-2 Start streaming after first streaming

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Table 7-2 Start streaming after first streaming

(1)	Standby setting in streaming with 0x0100 (MODE_SEL = 0)	Note: Refer to 6.1.2 for procedure of (1) to (4).
(2)	Waiting for MIPI FE package	
(3)	Set register for next streaming mode	
(4)	Start streaming with 0x0100 (MODE_SEL = 1)	
	After “MIPI Wake Up Time” + “Initialize Time”, 1st frame starts and images come out	

7.1.2. Streaming to SW-standby

This section will introduce how to end streaming and exit to SW-standby.

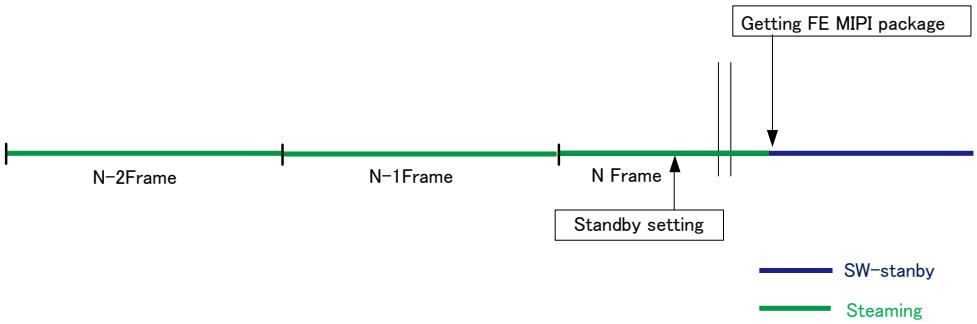


Figure 7-3 transition to SW-standby from streaming

(1) In case a SW-standby command is issued in between “FS” and “FE”, this sensor completes outputting the image data and transits to a standby state.

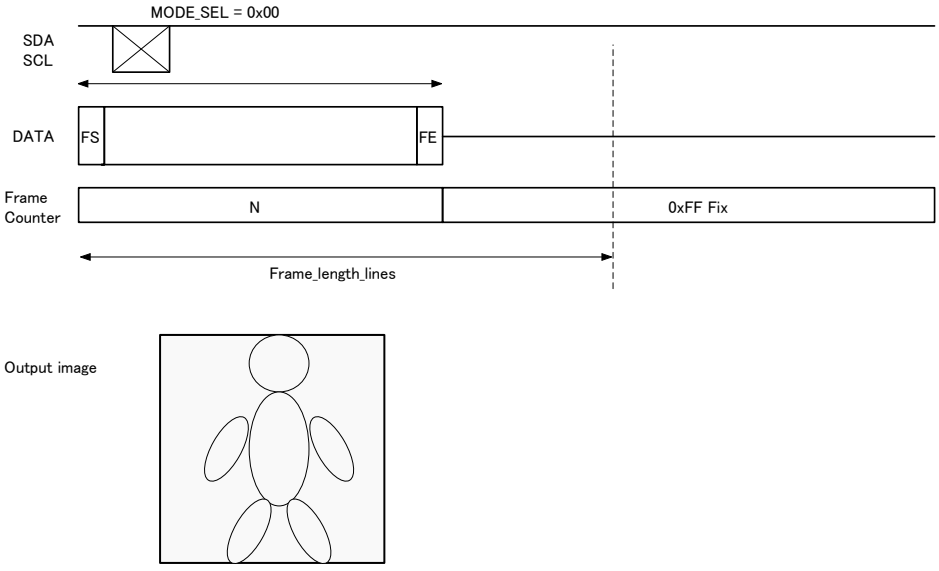


Figure 7-4 SW-standby transition pattern 1

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(2) In case a SW-standby command is issued within a “Frame Blanking” period, this sensor immediately transits to a software standby state.

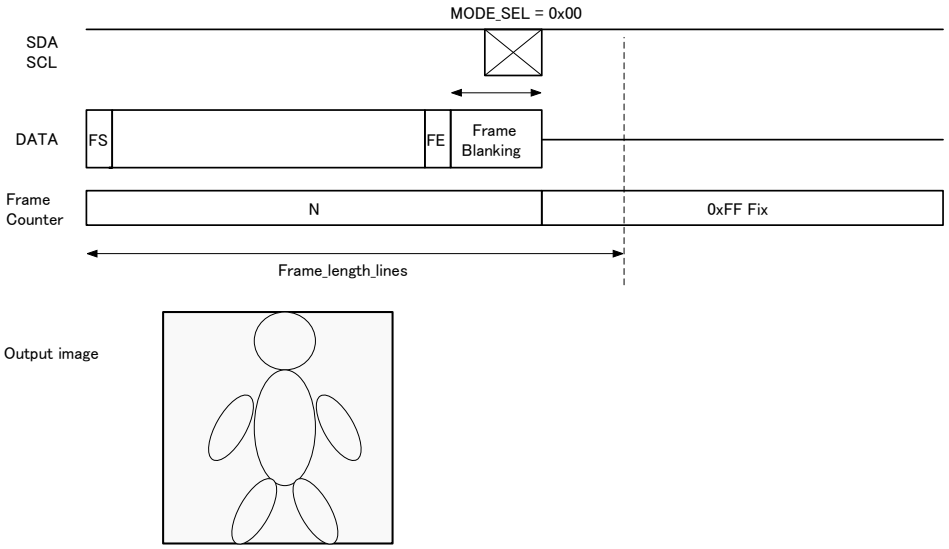


Figure 7-5　SW-standby standby transition pattern 2

(3) In case a SW-standby command is issued in between “FS” and “FE” while the sensor is working in fast SW-standby mode, this sensor completes outputting the line’s image data to where the command belongs and transit to the standby.

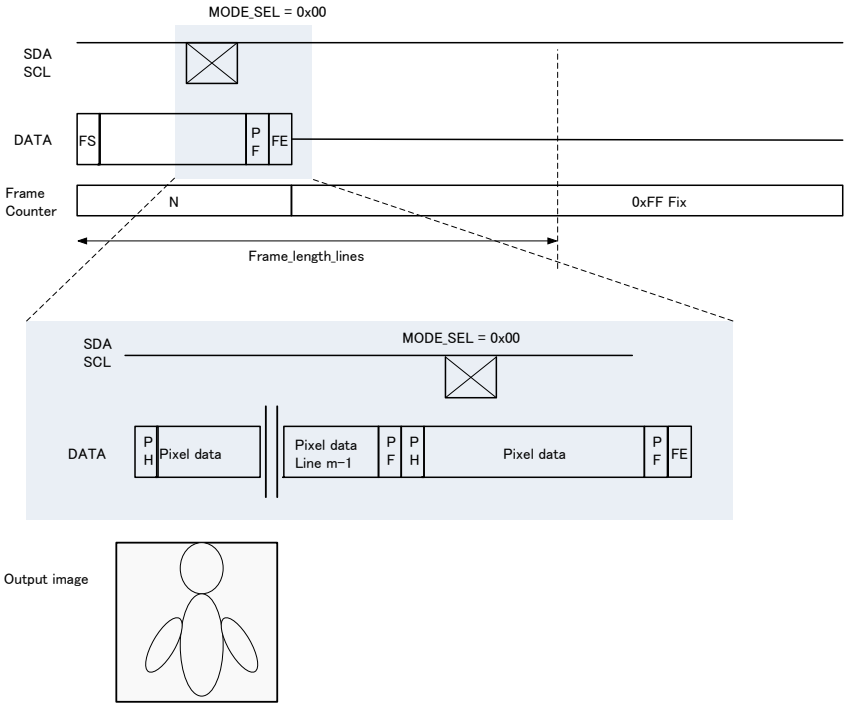


Figure 7-6　Software standby transition pattern 3 (Fast SW standby)

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7.2. Streaming mode transition

In this section, various knowledge on streaming mode change is explained. Basically, mode transition is grouped into two general categories: 1) via software standby transition and 2) on-the-fly transition.

Basic knowledge to know first is about registers which can be updated only in software standby and ones which cause a corrupted frame. Then, based on the knowledge, basic four streaming mode changes are explained.

7.2.1. Registers to be set only in SW-standby as mode transition

In case the following registers are updated between the streaming mode, the streaming mode change must be made via software standby mode.

Table 7-3 List of registers to be set only in SW-standby as mode transition.

	Address	Bit	Name	Description
I²C register	0x0106	[0]	FAST_STANDBY_CTL	Fast standby control 0 : after outputting all current frames, switch to software standby. Effective in Via SW-standby transition 1 : immediately after stopping outputting all current frames, switch to software standby
	0x0350	[0]	FRM_LENGTH_CTL	Frame length automatic tracking control as shutter time exceeding a current operation vertical period Select whether or not the frame length is changed automatically when FRM_LENGTH_LINES < COARSE_INTEG_TIME + α (α = type-specific adjustment parameter, and is 48 for this type. α is the value of CIT_MARGIN.) 0 : no automatic tracking control of frame length 1 : automatic tracking control of frame length In this case, "COARSE_INTEG_TIME + α" operates instead of FRM_LENGTH_LINES
	0x4bc2	[4:0]	CIT_MARGIN[12:8]	Coarse storage time margin Unit : lines CIT_MARGIN < 48 is prohibited.
	0x4bc3	[7:0]	CIT_MARGIN[7:0]	

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	0x3020	[0]	SHORT_FRAME_RS	Fast mode transition. Effective in On-the-fly transition. 0: normal mode change 1: Fast mode change
	0x3021	[0]	COMPLETE_FRAME_TRGS	The type of fast mode transition. 0: truncated frame 1: complete frame
	0x0114	[1:0]	CSI_LANE_MODE	Number of lanes for CSI D-PHY: 0 : not supported 1 : 2-lane 2 : not supported 3 : 4-lane C-PHY: 0 : 1-Trio 1 : 2-Trio 2 : 3-Trio 3 : not supported
	0x0112	[7:0]	CSI_DT_FMT_H	The output data format for CSI CSI_DT_FMT_H : Uncompressed Data Bit Width CSI_DT_FMT_L : Compressed Data Bit Width 0x0808 : RAW8 (top 8bit of internal data) 0x0A08 : COMP8 (10bit to 8bit compression) 0x0A0A : RAW10 (top 10bit of internal data) Setup other than the above is forbidden.
	0x0113	[7:0]	CSI_DT_FMT_L	
	0x0301	[4:0]	IVT_PXCK_DIV	The Pixel Clock Divider for Internal Video Timing System Range : 5 Format : 5-bit unsigned integer *Setup other than the above is forbidden.
	0x0303	[2:0]	IVT_SYCK_DIV	The System Clock Divider for Internal Video Timing System Range : 2, 4 Format : 16-bit unsigned integer *Setup other than the above is forbidden.
	0x0305	[3:0]	IVT_PREPLLCK_DIV	The pre-PLL Clock Divider for Internal Video Timing System Clock, when PLL_MULT_DRIV=1. The pre-PLL Clock Divider for Internal Video Timing System and Internal Output Pixel System, when PLL_MULT_DRIV=0. Range : 1 to 4 Step : 1 Format : 16-bit unsigned integer *Setup other than the above is forbidden.

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	0x0306	[2:0]	IVT_PLL_MPY[10:8]	The PLL multiplier for Internal Video Timing Range : 88 to 360 Step : 1 Format : 16-bit unsigned integer *Setup other than the above is forbidden.
	0x0307	[7:0]	IVT_PLL_MPY[7:0]	
	0x030b	[4:0]	IOP_SYCK_DIV	The System Clock Divider for Internal Output Pixel System Range : 1, 2,4 Format : 16-bit unsigned integer *Setup other than the above is forbidden.
	0x030d	[4:0]	IOP_PREPLLCK_DIV	The pre-PLL Clock Divider for Internal Output Pixel System during "Dual PLL mode" (PLL_MULT_DRIV=1). Range : 1 to 27 Step : 1 Format : 16-bit unsigned integer *Setup other than the above is forbidden.
	0x030e	[3:0]	IOP_PLL_MPY[11:8]	The PLL multiplier for Internal Output Pixel System "Dual PLL mode" (PLL_MULT_DRIV=1). Range : 88 to 2500 Step : 1 Format : 16-bit unsigned integer *Setup other than the above is forbidden.
	0x030f	[7:0]	IOP_PLL_MPY[7:0]	
	0x0310	[0]	PLL_MULT_DRIV	PLL mode select 0 : Single PLL mode 1 : Dual PLL mode
	0x0820	[7:0]	REQ_LINK_BIT_RATE_MBPS[31:24]	Output Data Rate [Mbps] Bit[32:16] integer Bit[15:0] decimal
	0x0821	[7:0]	REQ_LINK_BIT_RATE_MBPS[23:16]	
	0x0822	[7:0]	REQ_LINK_BIT_RATE_MBPS[15:8]	
	0x0823	[7:0]	REQ_LINK_BIT_RATE_MBPS[7:0]	

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7.2.2. Change sequence of corrupted frame related registers

When changing the mode with Corrupted-frame-related registers (vertical direction parameters and some other registers listed in Table 7-3 photo-electron charge integration operation becomes irregular for one frame after updating of the register. This frame shall be treated as invalid frame. Invalid frame should not be output.

Table 7-4 List of corrupted frame causing registers

	Address	Bit	Name	Description
I²C register	0x0101	[1]	IMG_ORIENTATION_V	Image orientation for Vertical direction 0 : normal 1 : reverse
	0x0342	[7:0]	LINE_LENGTH_PCK[15:8]	The length of line Unit : pixels Format : 16-bit unsigned integer * LINE_LENGTH_PCK has been constrained.
	0x0343	[7:0]	LINE_LENGTH_PCK[7:0]	
	0x0346	[4:0]	Y_ADD_STA[12:8]	Vertical direction analog cropping start position within the active pixels area Unit : pixels Format : 16-bit unsigned integer * Note that this is the cropping end position when flipping (IMG_ORIENTATION_V=1)
	0x0347	[7:0]	Y_ADD_STA[7:0]	
	0x034a	[4:0]	Y_ADD_END[12:8]	Vertical direction analog cropping end position within the active pixels area Unit : pixels Format : 16-bit unsigned integer * Note that this is the cropping start position when flipping (IMG_ORIENTATION_V=1)
	0x034b	[7:0]	Y_ADD_END[7:0]	
	0x0385	[3:0]	Y_EVN_INC[3:0]	Y_EVN_INC is fix value(1) in this sensor. Format : 16-bit unsigned integer
	0x0387	[3:0]	Y_ODD_INC[3:0]	Y_ODD_INC is fix value(1) in this sensor. Format : 16-bit unsigned integer
	0x0900	[0]	BINNING_MODE	Binning mode enable 0 : disable 1 : enable
	0x0901	[3:0]	BINNING_TYPE_V[3:0]	Binning type selection for Vertical 1 : no binning 2 : 2x2 Adjacent Pixel Binning 4 : 2x2 Adjacent Pixel Binning V2H2 *Setup other than the above is forbidden.
	0x3247	[7:0]	BINNING_PRIORITY_V[7:0]	Setting method of Vertical summation 8'h01 : No Binning 8'h81 : 2x2 Adjacent Pixel Binning, QBC HDR 8'h89 : 2x2 Adjacent Pixel Binning V2H2 *Setup other than the above is forbidden

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	0x3246	[7:0]	BINNING_PRIORITY_H[7:0]	Setting method of Horizontal summation 8'h01 : No Binning 8'h81 : 2x2 Adjacent Pixel Binning, QBC HDR 8'h89 : 2x2 Adjacent Pixel Binning V2H2
	0x3F0C	[0]	BINNING_PDAF_EN	2x2 Adjacent Pixel Binning, PDAF enable 0 : disable 1 : enable
	0x3F14	[0]	FULL_QHDR_EN	Full Pixel and QBC HDR mode enable 0 : disable 1 : enable
	0x0220	[6:0]	HDR_MODE[6:0]	Recommend setting value: 0x63(QBC HDR enable) 0x00(QBC HDR disable) Bit[0] QBC HDR mode enable 0 : QBC HDR disable 1 : QBC HDR enable Bit[1] Analog Gain mode select during QBC HDR 0 : combined gain used 1 : separate gain used Bit[4:2] Reserved Bit[5] exposure mode select during QBC HDR 0 : short exposure determined by ratio (controlled by EXPO_RATIO(0x0222)) 1 : short exposure controlled by direct control Bit[6] Digital Gain mode select during QBC HDR 0 : combined gain used 1 : separate gain used
	0x3140	[2:0]	HDR_FNCSEL	QBC HDR function select 4 : QBC HDR Others : reserved

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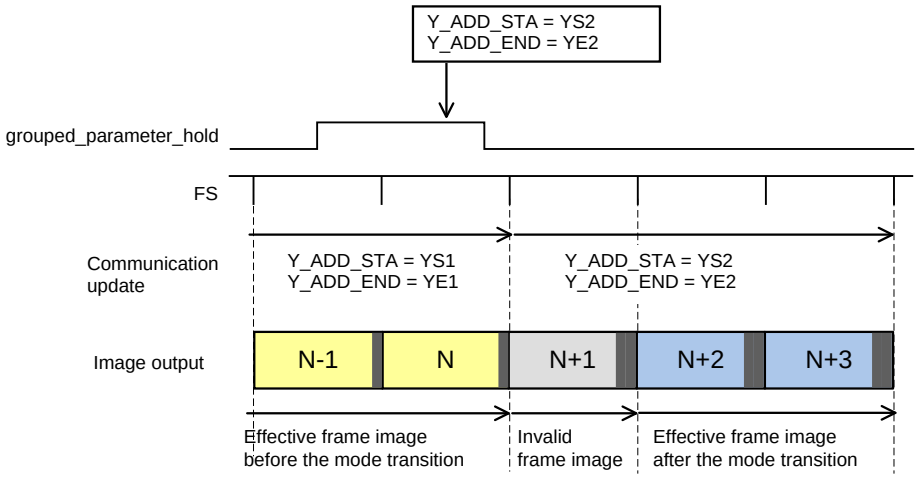


Figure 7-7 Corrupted-frame-related registers change sequence

7.2.3. Summary of Streaming Mode Change

In this paragraph, summary of four available mode transitions in this sensor are explained together with their use case. You can find conditions of using each mode transition method and causing an invalid frame on which details are already explained in previous sections.

Table 7-5 List of available mode transitions (possibility of occurrence of corrupt frames, and other precautions)

Mode transition			Availability of each transition mode	
Mode name		Features and notes/conditions for use	With changes of PLL/MIPI related settings (see Table 7-3)	Without change of PLL/MIPI related settings.
Via SW standby	SW-standby mode transition (default)	- Standard method for any kind of mode transition. - No corrupted frame.	Available	Available
	Fast SW-standby mode transition	- Versatile and fast transition method. - When MODE_SEL[0] falls, picture output stops and mode transition occurs.	Available	Available
On-the-fly	Complete mode transition (default)	- Usable only the case without clock change.	Not available	Available
	Truncate mode transition	- Usable only the case without clock change but fast mode change. - When GRP_PARAM_HOLD falls, picture output stops and mode transition occurs.	Not available	Available.

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7.2.4. SW-standby mode transition

A diagram below illustrates mode transition sequence when clock setting change is required (see Table 7-3). Let the sensor go into SW-standby mode by setting MODE_SEL = 0 before mode change. After setting parameters for the next streaming, set MODE_SEL=1. This mode transition method can also be used for cases which do not require clock change. This mode does not cause any invalid frame both before and after the transition.

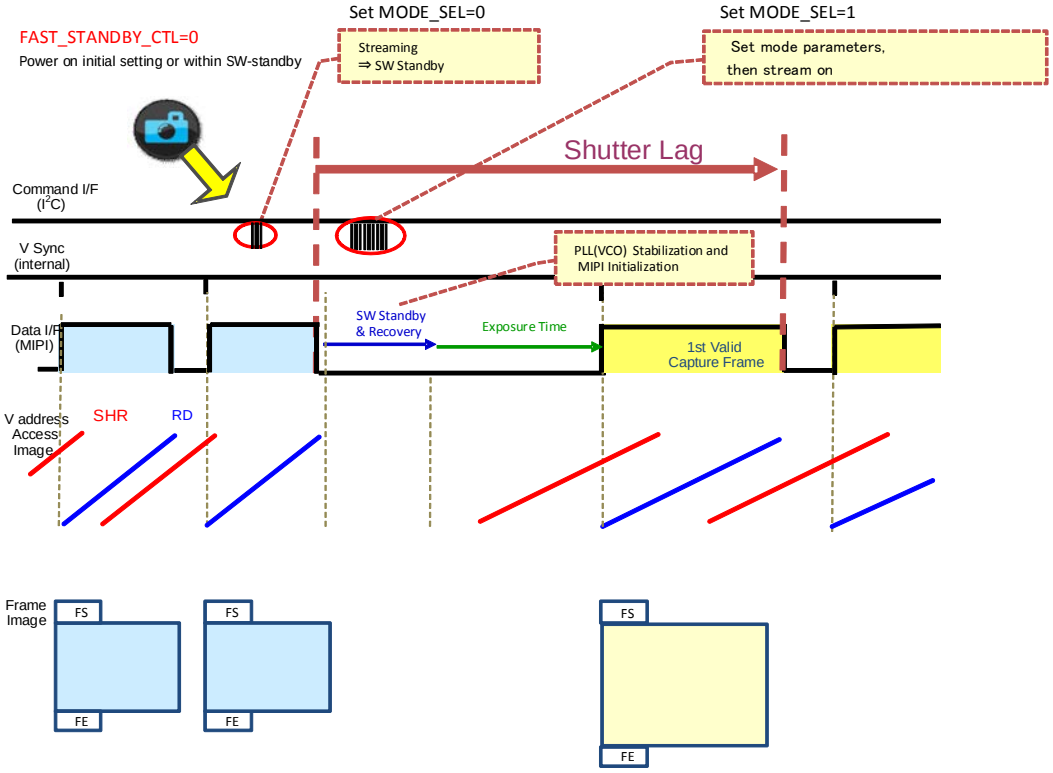


Figure 7-8 SW-standby mode transition (available for with/without clock change)

Table 7-6 SW-standby related register

	Address	Bit	Name	Description
I2C register	0x0100	[4:0]	MODE_SEL[4:0]	Mode Select 0 : Software Standby 1 : Streaming *Setup other than the above is forbidden.

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7.2.5. Fast SW-standby mode transition

A diagram below illustrates mode transition sequence when clock setting change is required. This method of mode transition is shortest shutter lag and is versatile. This mode also go into the SW-standby (MODE_SEL=0) mode once for setting mode parameters for next streaming, then get out by MODE_SEL=1. However differently from the SW-standby mode transition, the last frame before mode transition is stopped during streaming (corrupted frame) by setting MODE_SEL=1 under the condition FAST_STANDBY_CTL = 1 is set once before the mode transition, e.g. in power on sequence or during SW-standby. This mode transition method can be also used for the mode transition without clock change.

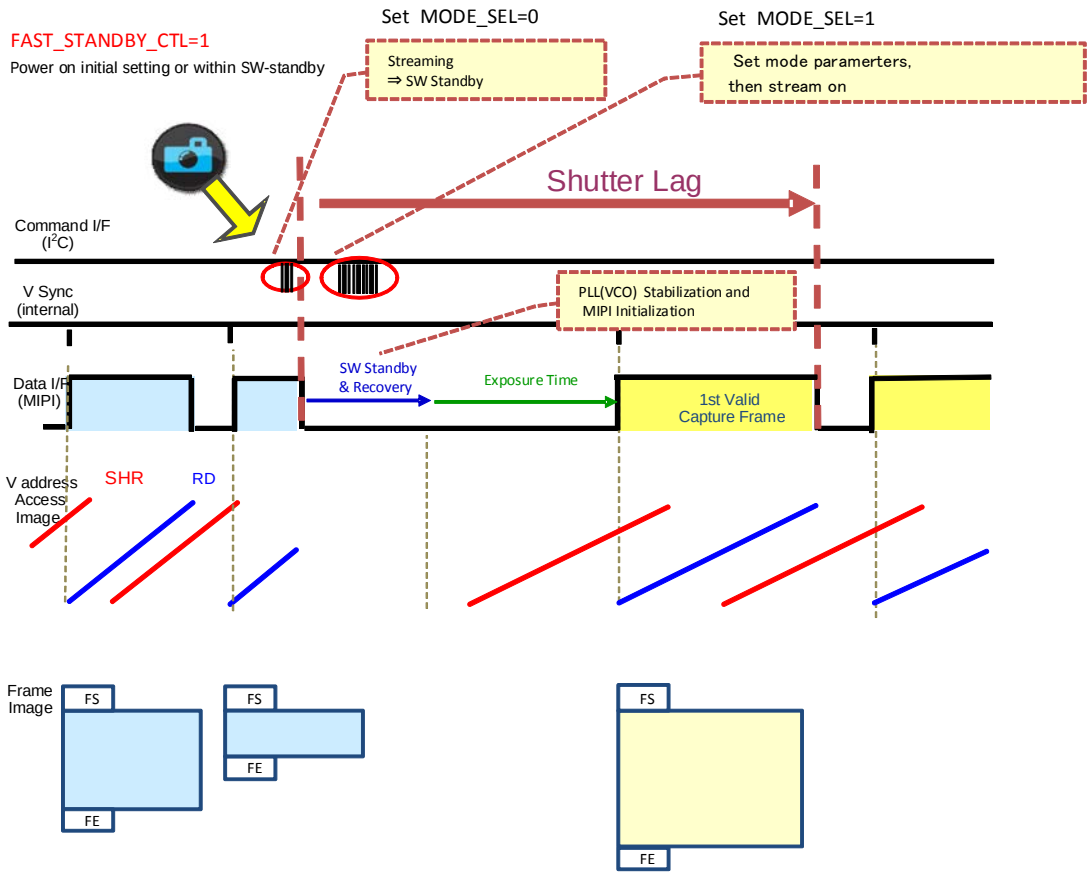


Figure 7-9 Fast SW-standby mode transition

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Table 7-7 Fast SW-standby related registers

	Address	Bit	Name	Description
I²C register	0x0100	[4:0]	MODE_SEL[4:0]	Mode Select 0 : Software Standby 1 : Streaming *Setup other than the above is forbidden.
	0x0106	[0]	FAST_STANDBY_CTL	Fast standby control 0 : after outputting all current frames, switch to software standby 1 : immediately after stopping outputting all current frames, switch to software standby

7.2.6. Complete mode transition

A diagram below illustrates mode transition sequence when clock setting remain the same.

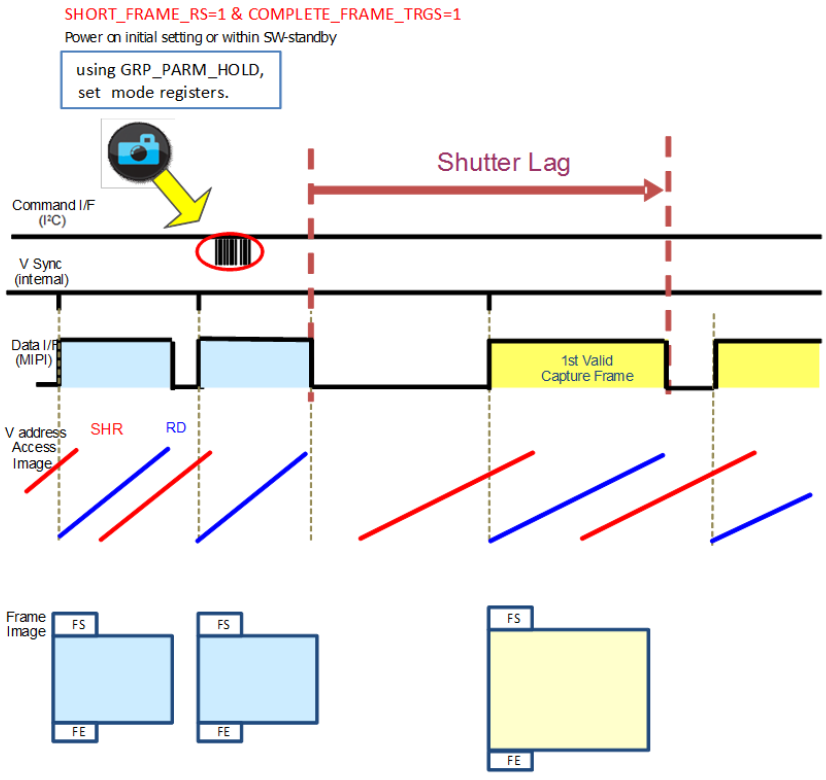


Figure 7-10 Complete mode transition (without clock change)

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7.2.7. Truncate mode transition

A diagram below illustrates mode transition sequence for the sensor when clock setting remain the same (when corrupted frame related registers (see Table 7-4 List of corrupted frame causing registers) and CIT target register are changed at the same time).

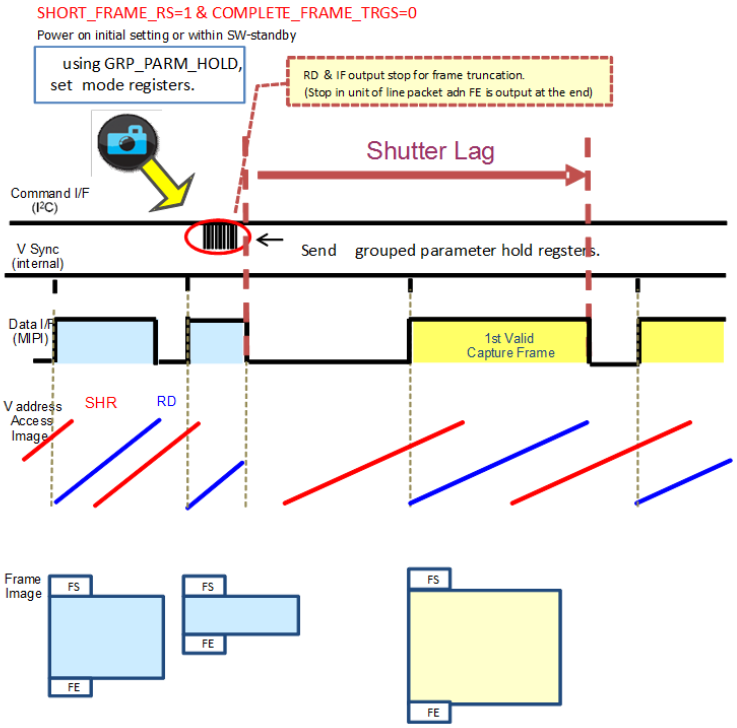


Figure 7-11 Truncate mode transition

Table 7-8 Truncate mode transition related register

	Address	Bit	Name	Description
I²C register	0x3020	[0]	SHORT_FRAME_RS	Mode transition 0: normal mode change 1: Fast mode change
	0x3021	[0]	COMPLETE_FRAME_TRGS	Fast Mode transition (SHORT_FRAME_RS=1) 0: Truncate mode 1: Complete mode
	0x0104	[0]	GRP_PARAM_HOLD	This register is a hold register for updating multiple parameters in the same frame. 0 : consume as normal 1 : hold

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7.3. Exposure and gain change sequence and AE bracketing

This section describes about the timing to update changes in integration time related registers, and changes in gain system registers. In addition, it describes the method to update registers at every vertical period using AE bracketing feature.

7.3.1. Integration time change sequence

Basically, GPH must be used in changing exposure time or shutter time. Set the value “1” to GRP_PARAM_HOLD register and set the integration duration value to COARSE_INTEG_TIME (CIT) register. Then set GRP_PARAM_HOLD back to “0”. The integration time is changed from the next frame and the image shot with the new integration time is output from the second frame after resetting GRP_PARAM_HOLD register.

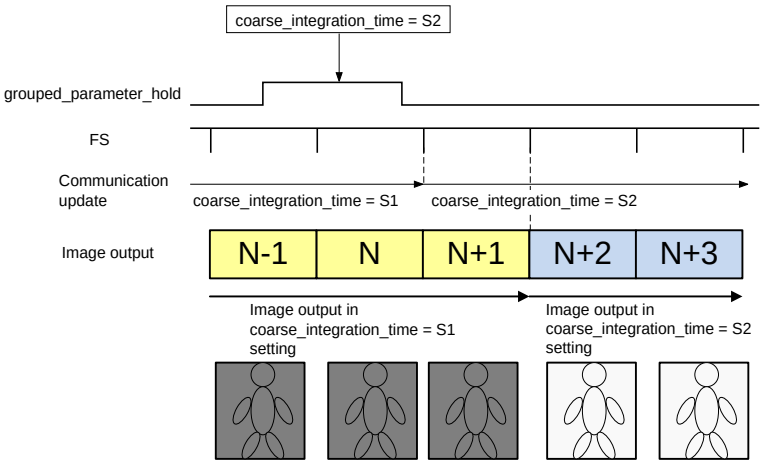


Figure 7-12 Integration time change sequence

Table 7-9 Integration time setting register

	Address	Bit	Name	Description
I²C register	0x0202	[7:0]	COARSE_INTEG_TIME[15:8]	Coarse storage time Unit : lines Format : 16-bit unsigned integer
	0x0203	[7:0]	COARSE_INTEG_TIME[7:0]	
	0x0104	[0]	GRP_PARAM_HOLD	This register is a hold register for updating multiple parameters in the same frame. 0 : consume as normal 1 : hold

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7.3.2. Gain change sequence

With this sensor, register settings for Gain are updated 2 frames after those set with GPH. There are always 2 frames delays for the update timing with this sensor. Gain change sequence also requires a usage of GPH together with integration time settings.

Basic gain change sequence

Set the values to ANA_GAIN_GLOBAL / DIG_GAIN_R / DIG_GAIN_B / DIG_GAIN_GLOBAL / DIG_GAIN_GB registers. Then set GRP_PARAM_HOLD back to “0”. The gain values are changed from the 2nd next frame and the image shot with the new gains is output from the 2nd frame after resetting “GRP_PARAM_HOLD” register. Hence, if the new gain and integration time is changed within the same GPH slot, a capture with both setting in synchronous is output.

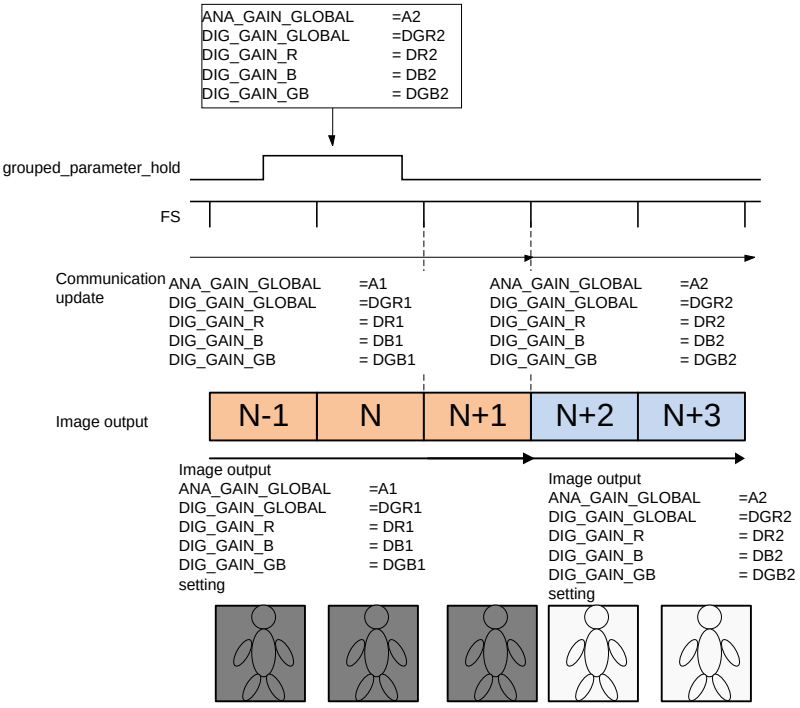


Figure 7-13 Gain change sequence

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Table 7-10 Gain setting register

	Address	Bit	Name	Description
I²C register	0x0204	[1:0]	ANA_GAIN_GLOBAL[9:8]	Analog gain control code for ALL color Format : 16-bit unsigned integer Range : 112 to 1008 *Setup other than the above is forbidden. Gain[times] = 1024 / (1024 - ANA_GAIN_GLOBAL)
	0x0205	[7:0]	ANA_GAIN_GLOBAL[7:0]	
	0x020e	[7:0]	DIG_GAIN_GLOBAL[15:8]	Digital gain control code for ALL color when DIG_GAIN_MODE = 0 Digital gain control code for GR when DIG_GAIN_MODE = 1 Format : 16-bit unsigned real Range : 0x0100 to 0x0FFF *Setup other than the above is forbidden. Gain[times] = DIG_GAIN_GLOBAL[15:8] + (DIG_GAIN_GLOBAL[7:0] / 256)
	0x020f	[7:0]	DIG_GAIN_GLOBAL[7:0]	
	0x0210	[7:0]	DIG_GAIN_R[15:8]	Digital gain control code for R. Valid only DIG_GAIN_MODE = 1. Format : 16-bit unsigned real
	0x0211	[7:0]	DIG_GAIN_R[7:0]	
	0x0212	[7:0]	DIG_GAIN_B[15:8]	Digital gain control code for B. Valid only DIG_GAIN_MODE = 1. Format : 16-bit unsigned real
	0x0213	[7:0]	DIG_GAIN_B[7:0]	
	0x0214	[7:0]	DIG_GAIN_GB[15:8]	Digital gain control code for GB. Valid only DIG_GAIN_MODE = 1. Format : 16-bit unsigned real
	0x0215	[7:0]	DIG_GAIN_GB[7:0]	
	0x3130	[0]	DIG_GAIN_MODE	Digital Gain select 0: all color 1: by color
	0x0104	[0]	GRP_PARAM_HOLD	This register is a hold register for updating multiple parameters in the same frame. 0 : consume as normal 1 : hold

With this sensor, Changing gain and CIT can be updated in every frame from HOST by using GPH.

It is recommended that GPH is used for Gain update setting.

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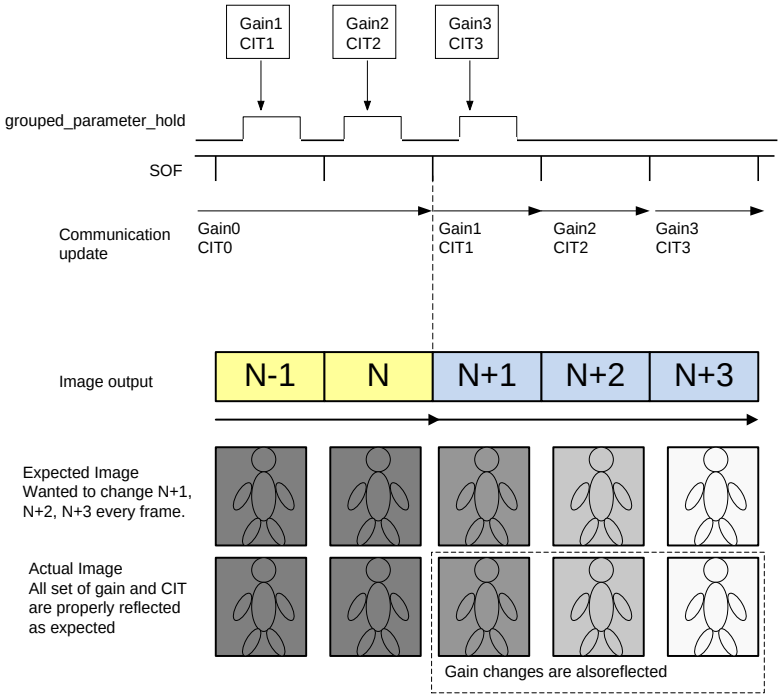


Figure 7-14 Gain change sequence (With CIT change)

Constraints:

1) Register change shall be only CIT, analog gain and digital gain.
2) Host should complete settings by I^2C within 1 frame (see "communication period" shown in Figure 2-15)

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7.3.3. AE bracketing

This function could be used to output the picture for up to five frames continuously, without transmitting coarse integration time, analog/digital gain, and flash settings every frame and setting by setting LUT in advance. This sensor is designed to repeat pre assigned LUT for multiple times.

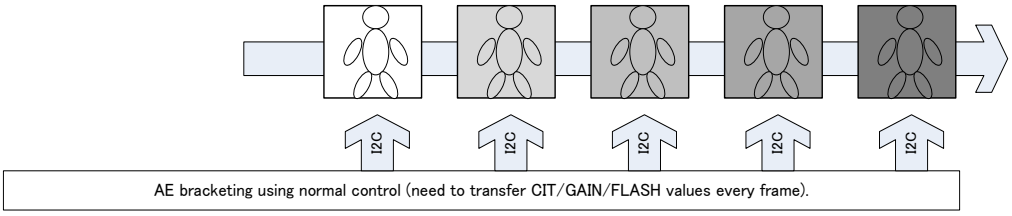


Figure 7-15 AE bracketing - without use of LUT

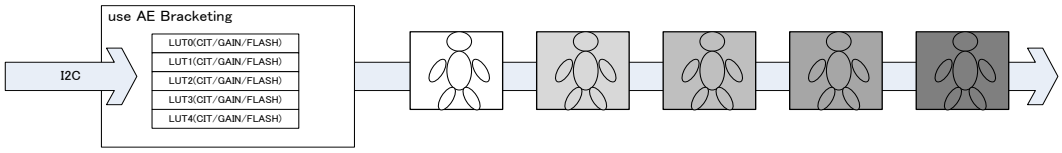


Figure 7-16 AE bracketing - with use of LUT

Table 7-11 AE bracketing related registers

	Address	Bit	Name	Description
I²C register	0x0e00	[7:0]	BRACKETING_LUT_CTL	Bracketing Look Up Table control 0 : disabled 1 to n : bracketing over n frame. (n = type-specific parameter, and is 8(d) for this type)
	0x0e01	[7:0]	BRACKETING_LUT_MODE	Bit[0] Bracketing sequence control 0 : return to SW standby after bracketing 1 : continue in streaming after bracketing Bit[1] Loop control 0 : single bracketing 1 : loop mode Bit[2] Output mask control 0 : no output mask 1 : output mask Bit[7:3] Reserved

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	0x0104	[0]	GRP_PARAM_HOLD	This register is a hold register for updating multiple parameters in the same frame. 0 : consume as normal 1 : hold
	0x0e10	[7:0]	LUT_A_COARSE_INTEG_TIME[15:8]	Coarse integration time for bracketing_LUT_frame_A
	0x0e11	[7:0]	LUT_A_COARSE_INTEG_TIME[7:0]	
	0x0e12	[1:0]	LUT_A_ANA_GAIN_GLOBAL[9:8]	Global analogue gain code for bracketing_LUT_frame_A
	0x0e13	[7:0]	LUT_A_ANA_GAIN_GLOBAL[7:0]	
	0x0e14	[7:0]	LUT_A_DIG_GAIN_GLOBAL[15:8]	Global or green (red row) digital gain value for bracketing_LUT_frame_A
	0x0e15	[7:0]	LUT_A_DIG_GAIN_GLOBAL[7:0]	
	0x0e16	[7:0]	LUT_A_DIG_GAIN_R[15:8]	Red channel digital gain value for bracketing_LUT_frame_A
	0x0e17	[7:0]	LUT_A_DIG_GAIN_R[7:0]	
	0x0e18	[7:0]	LUT_A_DIG_GAIN_B[15:8]	Blue channel digital gain for bracketing_LUT_frame_A
	0x0e19	[7:0]	LUT_A_DIG_GAIN_B[7:0]	
	0x0e1a	[7:0]	LUT_A_DIG_GAIN_GB[15:8]	Green (blue row) channel digital gain value for bracketing_LUT_frame_A
	0x0e1b	[7:0]	LUT_A_DIG_GAIN_GB[7:0]	
	0x0e1d	[4]	LUT_A_FLASH_TRIG_RS	Bit[3:0] Reserved Bit[4] Flash strobe trigger for bracketing_LUT_frame_A 0 : disable 1 : flash strobe is retimed to this frame Bit[7:5] Reserved
	:	:	:	:
	0x0eb8	[7:0]	LUT_H_COARSE_INTEG_TIME[15:8]	Coarse integration time for bracketing_LUT_frame_H
	0x0eb9	[7:0]	LUT_H_COARSE_INTEG_TIME[7:0]	
	:	:	:	:
	0x0eC5	[4]	LUT_H_FLASH_TRIG_RS	Bit[3:0] Reserved Bit[4] Flash strobe trigger for bracketing_LUT_frame_H 0 : disable 1 : flash strobe is retimed to this frame Bit[7:5] Reserved

See this sensor Register Map for complete set of registers for five look-up tables.

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How to control AE bracketing features

AE bracketing sequence always starts during SW-standby, but there are two possible ways to exit; 1) back to SW-standby and 2) move to streaming mode. Additionally, single mode bracketing and loop mode bracketing is available. AE bracketing function continues to repeat until setting BRACKETING_LUT_CTL = 0, See below for details.

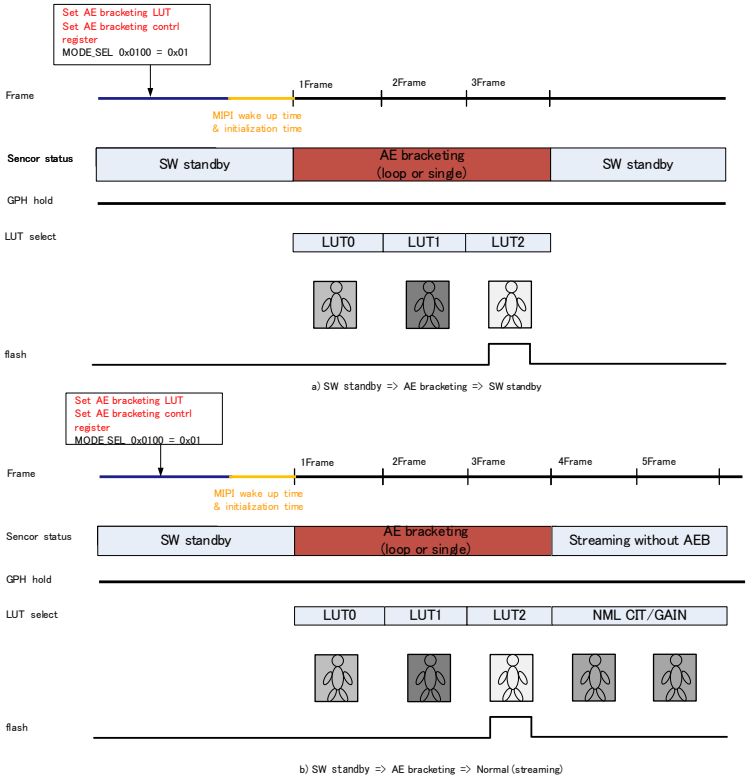


Figure 7-17 AE bracketing sequence timing chart (example of single mode)

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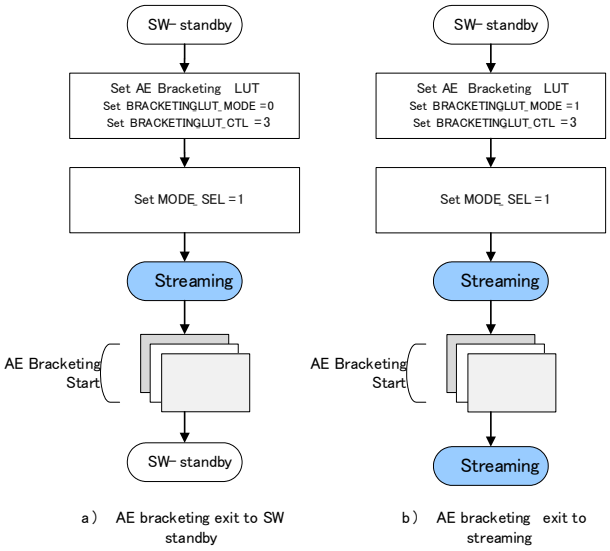


Figure 7-18 Auto bracketing sequence flow chart

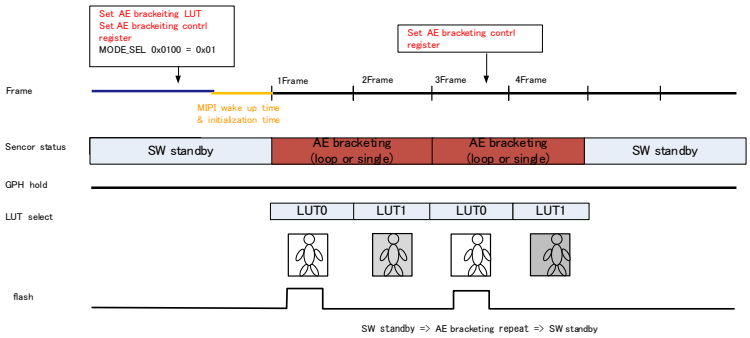


Figure 7-19 AE bracketing sequence (example of 2-repeat mode)

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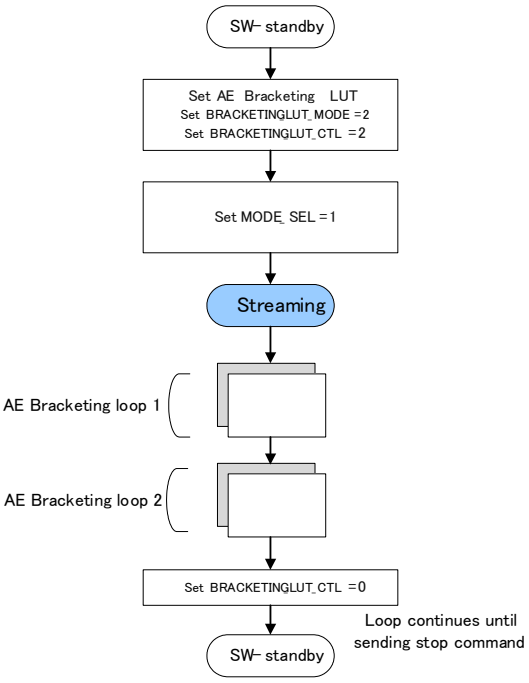


Figure 7-20 AE bracketing sequence (example of 2-repeat mode)

Above example show the limited combination of possible function for AE bracketing. Other combinations of repeat or single mode and either of exit modes are possible with those registers introduced in this paragraph.

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8. Miscellaneous Function

SRM-230-00113-01

8.1. Thermal meter

This function reads the thermal data from an internal sensor and averages it, it can be read as I^2C or EBD data (See A-2)

Thermal meter function only works under the condition with calibration data. The calibration data is optimized by individual sensor and is measured / stored in factory area of OTP.

Also, due to moving average principle, the temperature data is only valid after completion of 8 frame moving average every time after mode transition and power on sequence.

8.1.1. Thermal meter related registers

The following table shows thermal meter related control registers.

Table 8-1 Thermal meter register setting

Name	Address	Bit	Description
TEMP_SEN_CTL	0x0138	[0]	Temperature control enable 0 : disable 1 : enable
TEMP_SEN_OUT	0x013a	[7:0]	Temperature sensor output 0x81~0xEC -20C 0xED -19C … 0x00 0C … 0x4F 79C 0x50~0x7F 80C

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8.1.2. Thermal meter operation sequence

Thermal meter works both during Standby and streaming. While the thermal meter is ON, transition from Standby to Streaming or Streaming to Standby is possible.

Temperature measurement during SW-standby

The sequence for operating the thermal gauge while in Standby is shown as follows.

Valid thermal data is output from the 2nd frame.

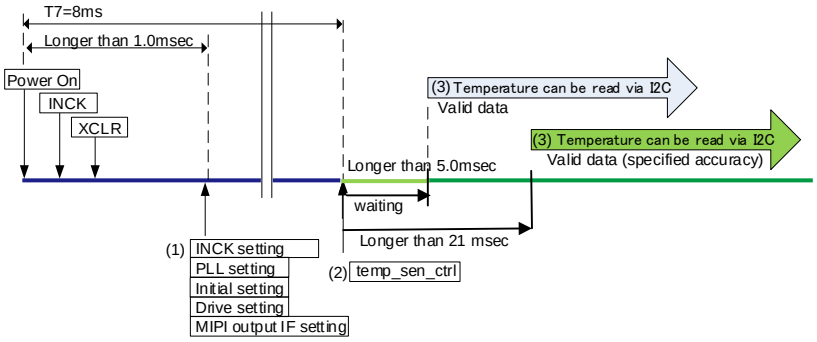


Figure 8-1 Thermal meter working in standby

Table 8-2 Thermal meter working in standby

(1)	Set the INCK frequency (0x0136,0x0137)
(2)	Set the TEMP_SEN_CTL (0x0138)
(3)	Read TEMP_SEN_OUT to get the temperature (0x013a)

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Temperature measurement in streaming mode (1)

The following diagram and table shows about a thermal meter operation sequence working in streaming with TEMP_SEN_CTL being set before MODE_SEL=1 (streaming-1)

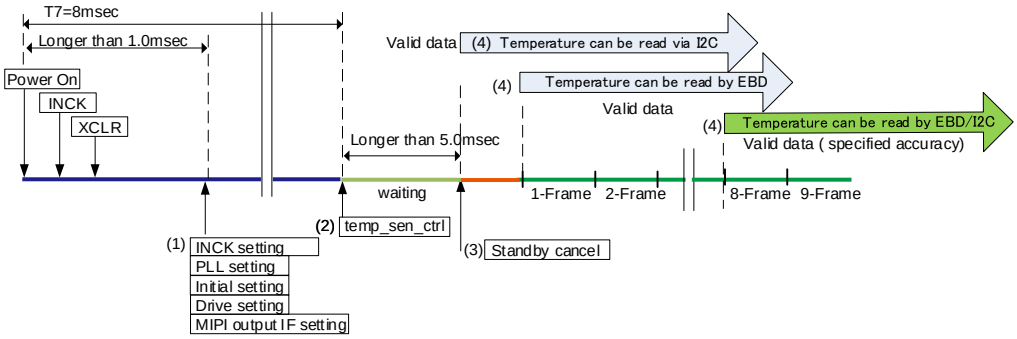


Figure 8-2 Thermal meter working in streaming (1)

Table 8-3 Thermal meter working in streaming (1)

(1)	Set the INCK frequency (0x0136,0x0137)
(2)	Set the TEMP_SEN_CTL (0x0138)
(3)	Set the MODE_SEL(0x0100)
(4)	Read TEMP_SEN_OUT to get the temperature (0x013a)

Temperature measurement in streaming mode (2)

The following diagram and table shows about a thermal meter operation sequence working in streaming with TEMP_SEN_CTL being set after MODE_SEL=1. (streaming-2)

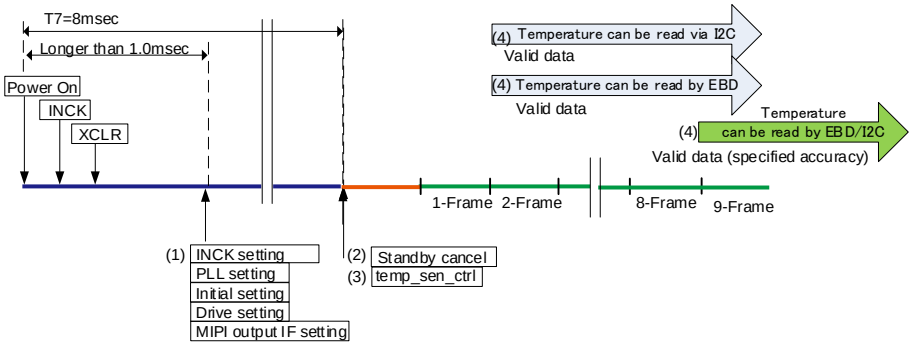


Figure 8-3 Thermal meter working in streaming (2)

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Table 8-4 Thermal meter working in streaming (2)

(1)	Set the INCK frequency (0x0136,0x0137)
(2)	Set the MODE_SEL (0x0100)
(3)	Set the TEMP_SEN_CTL (0x0138)
(4)	Read TEMP_SEN_OUT to get the temperature (0x013a)

8.1.3. Accuracy of temperature measurements

This sensor internally performs 8 frames moving average to output higher accuracy temperature. When it operates in thermal mode during Standby and when it measures temperature during Streaming more than 8 frames after TEMP_SEN_CTL =1, 8 frames averaged temperature data can be obtained after 9 frames.

*** Sensor output is guaranteed for -20-60deg ±3deg (60-80deg ±5deg).**

This specification may be changed without prior notice. And in that case, we will announce the change to customers each time.

*** Thermal mode offers extremely low power consumption.**

*** Sensor shipped with no calibration data in OTP does not operate normally.**

*** If calibration data is not written in OTP, TEMP_SEN_OUT is 0x6c (or other fixed values) regardless of temperature.**

8.2. Test pattern output (types of test patterns)

This sensor can output test patterns by using a built-in pattern generator.

8.2.1. Types of test patterns

This sensor has a function to output fixed video signals from a built-in test pattern generator by setting related registers except for QBC HDR mode.

While related register must be set to output a test pattern, there is no restriction on the sequence for setting the registers. The required test pattern can be output by setting the related registers during image capturing.

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Table 8-5 Test pattern related registers and description

	Address	Bit	Name	Description
I²C register	0x0600 0x0601	[0] [7:0]	TP_MODE [8] TP_MODE [7:0]	Test Pattern selection 0 : No pattern 1 : Solid color 2 : 100% color bars 3 : Fade to grey color bars 4 : PN9 *Setup other than the above is forbidden.
	0x0602 0x0603	[1:0] [7:0]	TD_R [9:8] TD_R [7:0]	Test data for replacing R pixels when using test patterns
	0x0604 0x0605	[1:0] [7:0]	TD_GR [9:8] TD_GR[7:0]	Test data for replacing GR pixels when using test patterns
	0x0606 0x0607	[1:0] [7:0]	TD_B [9:8] TD_B[7:0]	Test data for replacing B pixels when using test patterns
	0x0608 0x0609	[1:0] [7:0]	TD_GB [9:8] TD_GB[7:0]	Test data for replacing GB pixels when using test patterns

Black level is automatically set as 0.

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8.3. Long exposure

Long exposure*1 feature is expected to be an improvement method of sensitivity in dark in some product's use case. In this section, register setting method and some important precautions on image quality impact are explained.

Note1) Long exposure is defined as shutter time of beyond 1/30 [sec] in this document for the convenience of explanation.

8.3.1. Register setting for the long exposure

Basic idea of integration time setting is explained in "5.4.2 Integration time calculation" including shutter time setting beyond the given frame period under constraints of maximum settable range of FRM_LENGTH_LINES and/or LINE_LENGTH_PCK. In this section, Long exposure mode, elongated shutter time setting beyond the aforementioned constraints in much simpler method, is discussed.

Long exposure mode is a function to extend exposure up to approx. 128 times longer than FRM_LENGTH_LINES (0x0340, 0x0341) which is allowed in normal setting. This function enables long exposure by extending only V-blank without changing any other setting such as FRM_LENGTH_LINES and the setting of LINE_LENGTH_PCK (0x0342, 0x0343), while offering low power consumption.

Table 8-6 Long exposure mode related registers

Address	Name	Bit	Description
0x3100	CIT_LSHIFT	[2:0]	Long exposure mode setting. 0 : Long exposure mode OFF Exposure time = COARSE_INTEG_TIME Other : Long exposure mode ON Exposure time = COARSE_INTEG_TIME << LSHIFT ex. 1=2times 2=4times 3=8times … 7=128times (max setting)
0x0202 0x0203	COARSE_INTEG_TIME	[15:0]	Coarse storage time Unit : lines Format : 16-bit unsigned integer

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0x0340 0x0341	FRM_LENGTH_LINES	[15:0]	The length of frame Unit : lines Format : 16-bit unsigned integer
0x0342 0x0343	LINE_LENGTH_PCK	[15:0]	The length of line Unit : pixels Format : 16-bit unsigned integer * LINE_LENGTH_PCK has been constrained.

Long exposure mode setting

Table 8-7 Setting for long exposure

(1)	Set CIT_LSHIFT as you want
	Set COARSE_INTEG_TIME as you want
	Set FRM_LENGTH_LINES as you want
	Select LINE_LENGTH_PCK from Table4-5

The frame rate during long exposure mode is as follows:

Frame Rate [frame/s] = *Pixel_rate* [pixels/s] / (*Total number of pixels* [pixels/frame] x 2^{CIT_LSHIFT})

Pixel rate [pixels/s] = *VTPXCK* [MHz] * 8 (*Total number of VTPX channel)*

Total number of pixels [pixels/frame]

= FRM_LENGTH_LINES [lines/frame] * LINE_LENGTH_PCK [pixels/line]

The setting range of exposure time during long exposure mode is as follows:

CITmin(*1) ≦ COARSE_INTEG_TIME ≦ FRM_LENGTH_LINES – 48(*2)

*1: ***CITmin*** is shown in “5.4.2 Integration time calculation“

*2: 48 is the value of CIT_MARGIN.

Setting example

Table 8-8 Long exposure time related registers (example)

4:3 full resolution output mode VTPXCLK=216MHz	CIT_LSHIFT	3 (x8)
	COARSE_INTEG_TIME	65487
	LINE_LENGTH_PCK	9440 (initial value)
	FRM_LENGTH_LINES	65535

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Exposure time

Exposure time = (65487 * 8 * 9440) / (216[MHz] *8) = 2.86 [sec]

* When this long exposure mode is used, mode transition of SW-standby mode transition or Fast SW-standby mode transition (see 7.2.4 or 7.2.5) is recommended.

8.3.2. Precautions for long exposure application

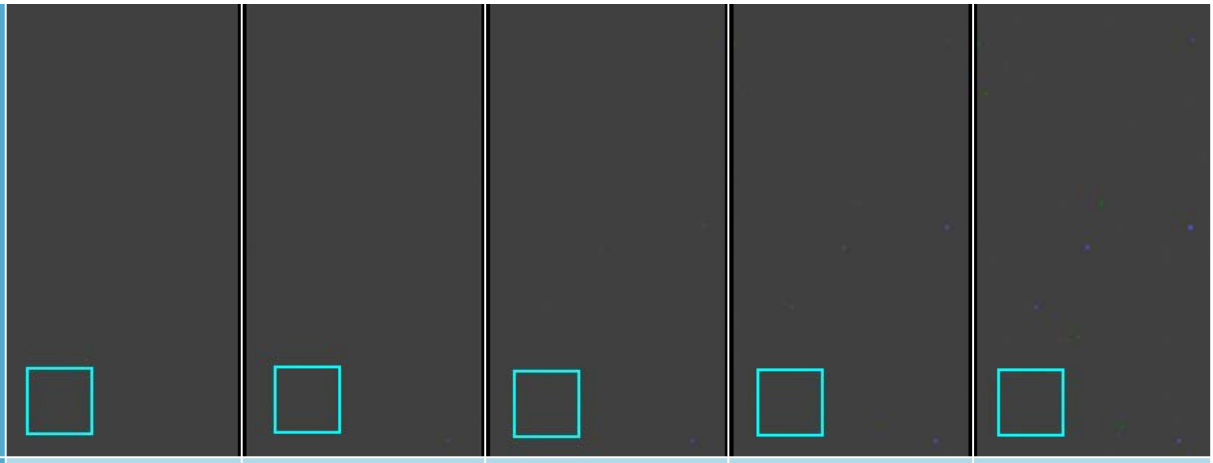
As mentioned beginning, long exposure is an effective mean to improve sensitivity in dark. However, due to and nature of elongated exposure time, there are some known image quality phenomenon of which needed to thought. It is highly recommended to make careful study for the below listed image quality items before making a decision.

1) Growth in level and count of white spots in the dark. (See Table 8-9)

2) Increments of black level attributable to dark signal may happen. (See Table 8-9)

3) Quantization noises of 10bit system become discernable in exchange of improvement in random noise performance. Basically, the phenomenon is more perceptible with a frame addition of multiple frames shot under condition of long exposure and low analog gain.

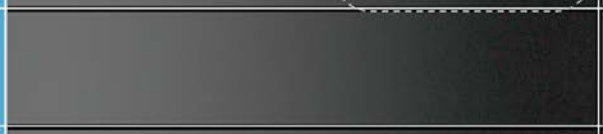
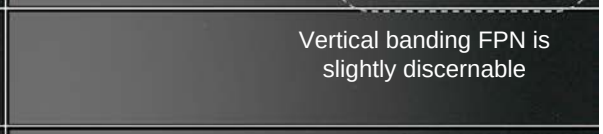
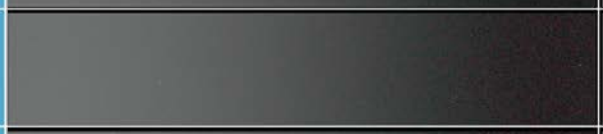
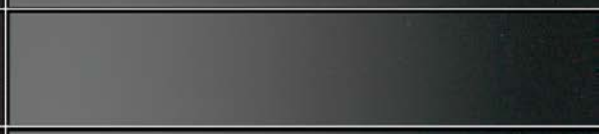
Table 8-9 Example capture of dark image over different exposure time

Exposure Time	1/30s	1/10s	1/3s	1s	3s
Dark Image					
Black level (dec)	63.42	63.41	63.46	63.53	63.76

Note) As setting exposure time longer, white spot count becomes more and its level and the black level goes higher. Note that above captures are obtained with dynamic DPC OFF using typical sensor type only for an explanatory purpose. Turning on the dynamic DPC makes single defect corrected but couplet or cluster defects left uncorrected. Practically, couplet or cluster defects are subjects for evaluation.

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Table 8-10 Quantization error example over an addition of multiple long exposure frames

Gain (dB)	Frame Addition: 0 Exposure time/frame = 2sec	Frame Addition: 32 Exposure time/frame = 2sec
0		
6		
12		 Vertical banding FPN is slightly discernable
18		
24		

Note) As setting exposure time longer and frame addition more in addition to a small analog gain setting, quantization error and/or fixed pattern noise become more perceptible. Note that the above captures are obtained using typical sensor type is enhanced with higher gamma for explanation purpose only.

It is essential that these characteristics that would be attributable to long exposure time shall be evaluated and judged by yourself referring to your product application. Following information may help in understanding the distinctive performances relating to elongated exposure time and then making evaluation plans.

- White spot: Allowable white spots are specified with the signal level and numbers under exposure time of 1/30 [s]. The signal level of white spots is proportional to the exposure time. See “Spot Pixel Specification” in the data sheet for actual level and allowable spot count at the time of shipment.
- Dark signal: Dark signal is specified with the unit of LSB under exposure time of 1/15 [s] in darkness.

The dark signal increases as exposure time become longer and somehow affect the gradation characteristics.

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8.4. Flash light control sequence

This sensor can internally generate the control pulse assuming to trigger the flash light emission and output from the external pins. The flash light trigger pulse width and output timing can be controlled by registers shown in following table. Flash control pulse is output from FSTROBE pins. The setting of strobe pulse is determined by reference point, delay and trigger mode. Details of setting are explained by following sections.

Table 8-11 Flash light control setting registers

	Address	Bit	Name	Description	Notes
I²C register	0x0c12	[7:0]	FLASH_STRB_ADJ [7:0]	Register to control counter for flash pulse width in rolling shutter mode. Flash pulse width (high) = (TFLASH_STRB_WIDTH_H_RS_CTL / EXTCLK frequency) * FLASH_STRB_ADJ Flash pulse width (low) = (TFLASH_STRB_WIDTH_L_RS_CTL / EXTCLK frequency) * FLASH_STRB_ADJ Range : 1 to 255	
	0x0c14	[7:0]	FLASH_STRB_START_POINT [15:8]	Register to select reference point for flash pulse in rolling shutter mode. Step: 1H 2H (Adjacent pixel binning) Range : 0 to 65535	
	0x0c15	[7:0]	FLASH_STRB_START_POINT [7:0]		
	0x0c16	[7:0]	TFLASH_STRB_DLY_RS_CTL [15:8]	Register to control the rising edge point of Flash pulse from FLASH_STRB_START_POINT in rolling shutter mode. Step : 1H 2H (Adjacent pixel binning) Range : 0 to 65535	
	0x0c17	[7:0]	TFLASH_STRB_DLY_RS_CTL [7:0]		
	0x0c18	[7:0]	TFLASH_STRB_WIDTH_H_RS_CTL [15:8]	Register to control the high level width of flash pulse in rolling shutter mode. Range : 1 to 65535 Flash pulse width (high) = (TFLASH_STRB_WIDT_H_RS_CTL / EXTCLK frequency) * FLASH_STRB_ADJ	
	0x0c19	[7:0]	TFLASH_STRB_WIDTH_H_RS_CTL [7:0]		

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	0x0c1a	[7:0]	FLASH_MD_RS [7:0]	Bit[0] flash pulse mode control 0 : single trigger mode. 1 : continuous mode. Bit[1] flash pulse complete or truncated select 0 : complete pulse mode in rolling shutter mode 1 : truncated pulse mode in rolling shutter mode Bit[2] reference start point select 0 : exposure 1 : readout (in rolling shutter mode) Bit[3] flash pulse sync or async select 0 : synchronous 1 : asynchronous Bit[4] Reserved Bit[5] flash pulse single or multi select 0 : single 1 : multiple Bit[7:6] Reserved	
	0x0c1b	[7:0]	FLASH_TRIG_RS [7:0]	Flash trigger 0 : disable 1 : enable *Return to 0 in single mode, automatically *Setup other than the above is forbidden.	
	0x0c1c	[7:0]	FLASH_STAT [7:0]	Bit[0] flash pulse status 0 : Not retimed to this frame 1 : Retimed to this frame When using a flash light control sequence in the AEB mode, this register is fixed to 0. Bit[1] flash status for global reset mode *not supported 0 : Not active in global reset mode 1 : Active in global reset mode Bit[7:2] Reserved	read only
	0x0c26	[7:0]	TFLASH_STRB_WIDTH2_H_RS_CTL [15:8]	Register to control the high level width of 2nd flash pulse in rolling shutter mode when TFLASH_STRB_CNT_RS_CTL=2. Range : 1 to 65535 2nd Flash pulse width (high) = (TFLASH_STRB_WIDT2_H_RS_CTL / EXTCLK frequency) * FLASH_STRB_ADJ	
	0x0c27	[7:0]	TFLASH_STRB_WIDTH2_H_RS_CTL [7:0]		

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	0x0c28	[7:0]	TFLASH_STRB_WIDTH_L_RS_CTL [15:8]	Register to control the low level width of flash pulse in rolling shutter mode. Range : 1 to 65535 Flash pulse width (low) = (TFLASH_STRB_WIDT_L_RS_CTL / EXTCLK frequency) * FLASH_STRB_ADJ	
	0x0c29	[7:0]	TFLASH_STRB_WIDTH_L_RS_CTL [7:0]		
	0x0c2a	[7:0]	TFLASH_STRB_CNT_RS_CTL [7:0]	Register to control flash pulse count in rolling shutter mode. Range : 1 to 255	

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8.4.1. Reference point

Reference point is the base point of delay settings for flash control pulse and can be determined by FLASH_MD_RS[2]. As example, following charts shows single trigger mode. It is also same for other trigger modes.

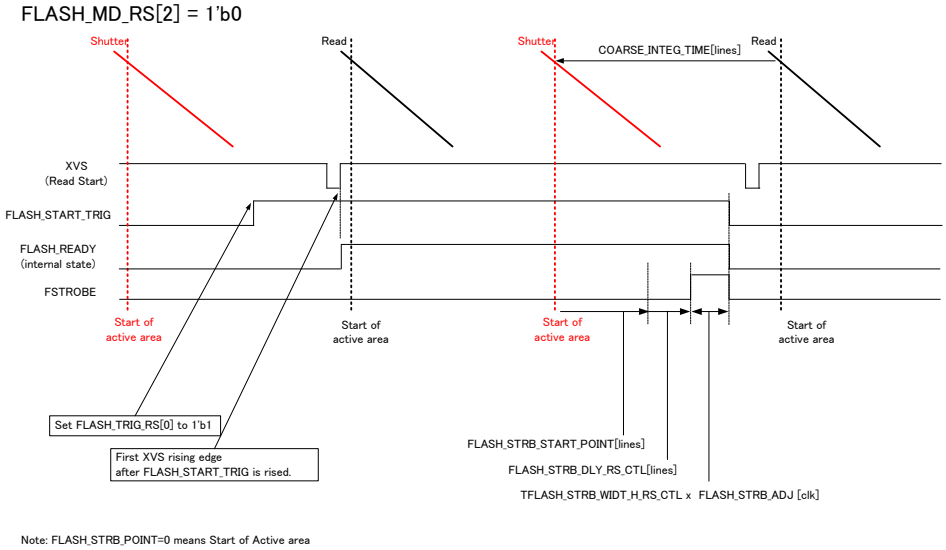


Figure 8-4 Reference point (Start of exposure)

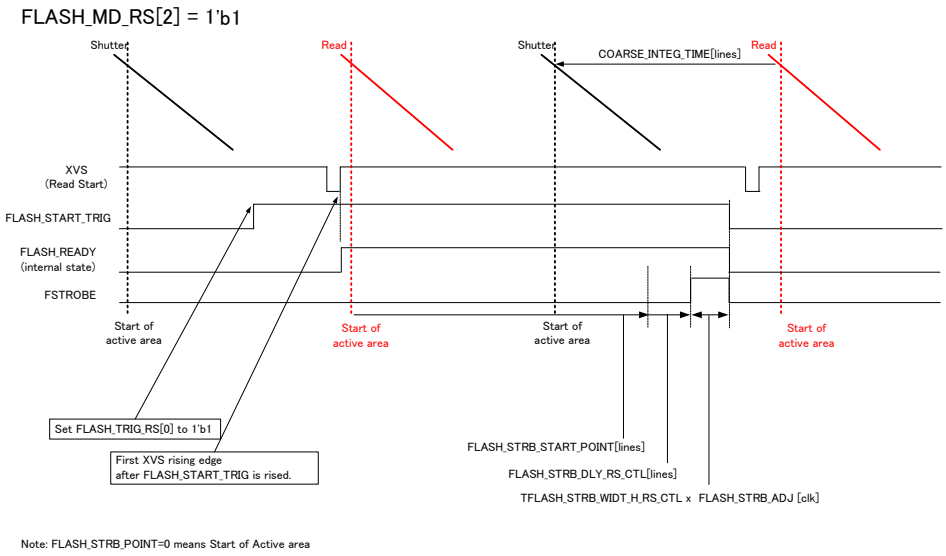


Figure 8-5 Reference point (Start of readout)

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8.4.2. Single trigger mode / Continuous mode

This sensor supports both single trigger strobe mode and continuous mode.

Single trigger mode is enabled by setting FLASH_MD_RS[0] to 1'b0.

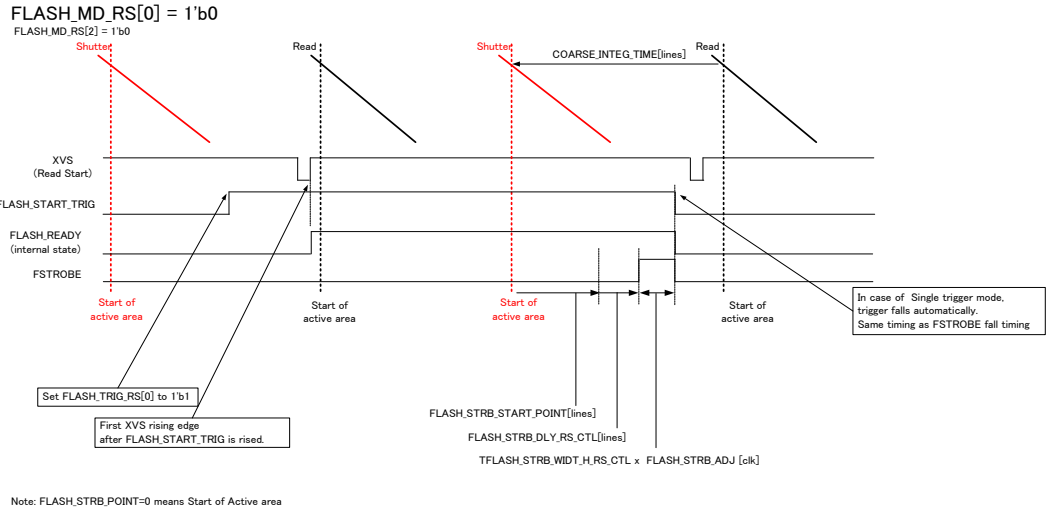


Figure 8-6 Single Trigger mode

Continuous mode is enabled by setting FLASH_MD_RS[0] to 1'b1. In this mode, flash control pulse can be output both continuously and periodically.

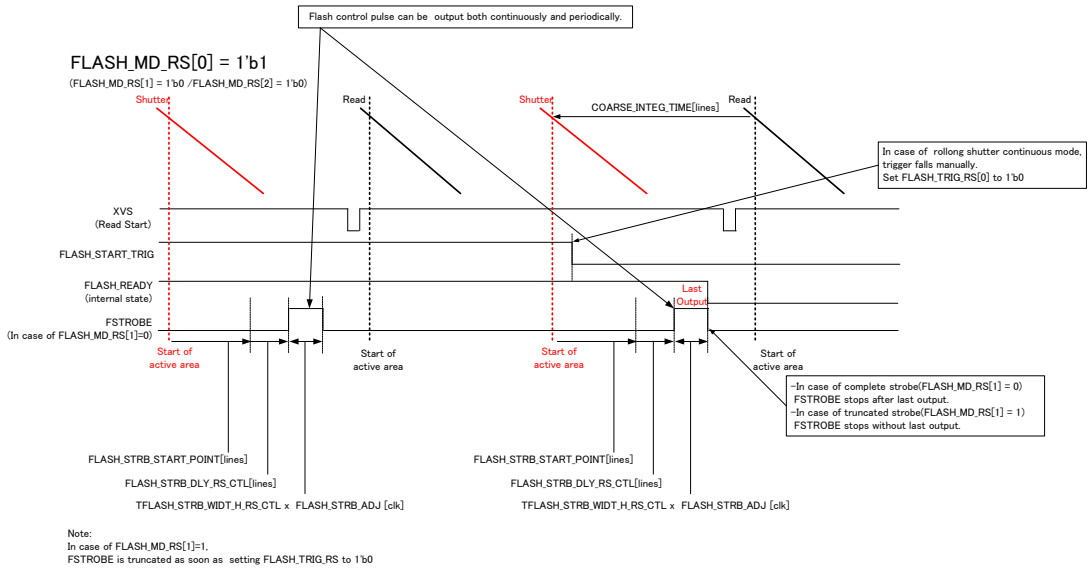


Figure 8-7 Continuous mode

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8.4.3. Multiple strobe

This sensor supports both single flash strobe mode and multiple flash modes. Multiple flash strobe mode is enabled by setting FLASH_MD_RS[5] to 1'b1. In this mode, more than one output of flash control pulse can be controlled. Multiple strobes can set not only in single trigger mode but also in continuous mode. Following chart shows single trigger mode and reference point of shutter as an example. Number of flash control pulse is determined by TFRASH_STRB_CNT_RS_CTL. In case more than 3 strobes are output in 1 frame, width of strobe pulse can't be set separately. On the other hand, dual flash strobe can control the pulse width separately. Following Table shows duty and width period that you can set.

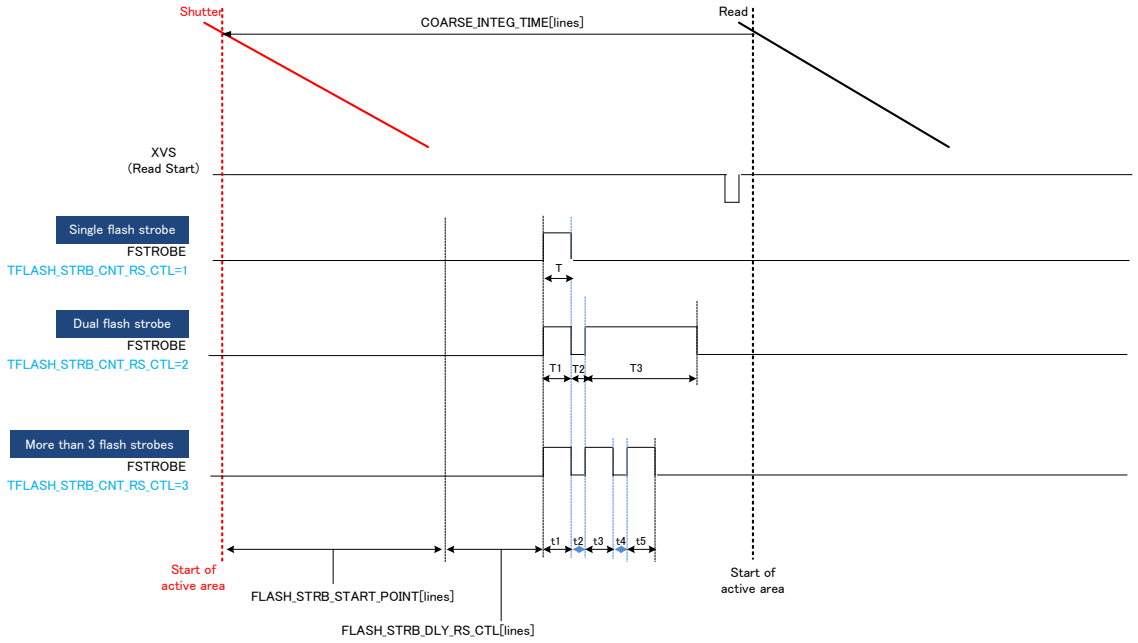


Figure 8-8 Single flash strobe / Multiple flash strobe

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Table 8-12　Single Flash strobe duty control register

Time	Register
T	TFLASH_STRB_WIDTH_H_RS_CTL × FLASH_STRB_ADJ[clk]

Table 8-13　Dual Flash strobe duty control register

Time	Register
T1	TFLASH_STRB_WIDTH_H_RS_CTL × FLASH_STRB_ADJ[clk]
T2	TFLASH_STRB_WIDTH_L_RS_CTL × FLASH_STRB_ADJ[clk]
T3	TFLASH_STRB_WIDTH2_H_RS_CTL × FLASH_STRB_ADJ[clk]

Table 8-14　More than 3 Flash strobe duty control register

Time	Register
t1	TFLASH_STRB_WIDTH_H_RS_CTL × FLASH_STRB_ADJ[clk]
t2	TFLASH_STRB_WIDTH_L_RS_CTL × FLASH_STRB_ADJ[clk]
t3	TFLASH_STRB_WIDTH_H_RS_CTL × FLASH_STRB_ADJ[clk]
t4	TFLASH_STRB_WIDTH_L_RS_CTL × FLASH_STRB_ADJ[clk]
t5	TFLASH_STRB_WIDTH_H_RS_CTL × FLASH_STRB_ADJ[clk]

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8.5. One time programmable (OTP) memory

See “OTP Manual”

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8.6. Pulse for OIS driver

This sensor can output the pulse for OIS driver. This pulse is generated from IVTPXCK.

High period and Low period can be set by registers. The pulse can be reset by H sync.

Refer to 8.7 “Monitor terminal settings” for outputting OIS Pulse from the monitor terminal.

Table 8-15 Pulse output setting registers for OIS driver

	Address	Bit	Name	Description
I²C register	0x3e5a	[7:0]	OIS_H_WIDTH[15:8]	High frequency pulse output for OIS driver is set with IVTPXCK cycle number [1-65535] (def:8)
	0x3e5b	[7:0]	OIS_H_WIDTH[7:0]	
	0x3e5c	[7:0]	OIS_L_WIDTH[15:8]	Low frequency pulse output for OIS driver is set with IVTPXCK cycle number [1-65535] (def:8)
	0x3e5d	[7:0]	OIS_L_WIDTH[7:0]	
	0x4bc6	[0]	OIS_HRESET_EN	Enable H sync reset for pulse output for OIS driver. (def:0) 0: Do not reset 1: Reset with H sync

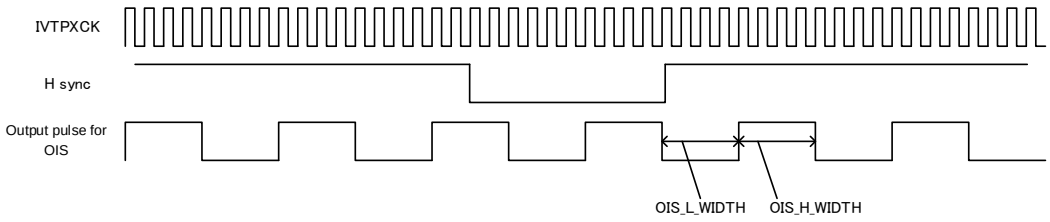


Figure 8-9 OIS_HRESET_EN=0 operation

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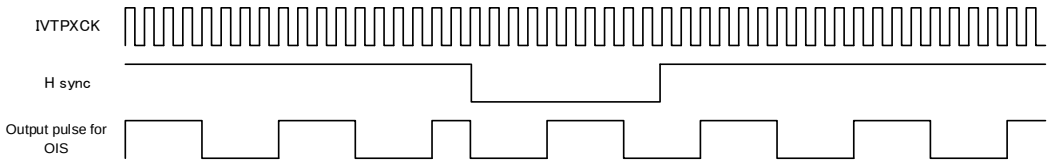


Figure 8-10 OIS_HRESET_EN=1 operation

8.6.1. CDS Frequency

CDS frequency [KHz] = (IVTPXCK [MHz] x 1000) / (CDS Clock)

This sensor's CDS clock cycle are shown below.

Table 8-16 CDS clock cycle

Operation mode	CDS Clock [cycle] @ IVTPXCK
Full resolution (QBC Re-mosaic mode)	623
QBC HDR mode	573
2x2 Adjacent Pixel Binning without PDAF mode	638
2x2 Adjacent Pixel Binning with PDAF mode	1264
2x2 Adjacent Pixel Binning V2H2 mode	641

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8.7. Monitor terminal settings

This sensor can output three internal signals (H Sync/V Sync/OIS pulse) via monitor terminals.

The monitor terminals mean the following two terminals, such as TESTOUT (51 pin) and GPO (54 pin). See the following table for register settings.

Table 8-17 Monitor terminal setting

	Address	Bit	Name	Setting value
I²C register	0x4226	[1:0]	IO_CTRL_GPO	GPO IO Control 0d : Output
	0x4227	[1:0]	IO_CTRL_TESTOUT	TESTOUT IO Control 0d : Output
	0x424A	[2:0]	MON_SEL [10:0]	Monitor select 0d : Output
	0x424B	[7:0]		
	0x4bd5	[7:0]	MNTTEST3_SEL[7:0]	24d: OIS Pulse is selected. 21d: H sync is selected. 22d: V sync is selected.
	0x4bd6	[7:0]	MNTTEST2_SEL[7:0]	
	0x4bd7	[7:0]	MNTTEST1_SEL[7:0]	
	0x423e	[7:0]	MON_GPO	FFh: GPO(54 pin)'s signal is selected by MNTTEST1_SEL[7:0] FEh: GPO(54 pin)'s signal is selected by MNTTEST2_SEL[7:0] FDh: GPO(54 pin)'s signal is selected by MNTTEST3_SEL[7:0]
	0x423f	[7:0]	MON_TESTOUT	FFh: TESTOUT(51 pin)'s signal is selected by MNTTEST1_SEL[7:0] FEh: TESTOUT(51 pin)'s signal is selected by MNTTEST2_SEL[7:0] FDh: TESTOUT(51 pin)'s signal is selected by MNTTEST3_SEL[7:0]
	0x4238	[2:0]	GPIO_SEL[10:0]	00h
	0x4239	[7:0]		

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GPO (54 pin)

V sync / H sync / OIS Pulse can be selected and output via GPO pin.

Signal is selected by MNTTEST1_SEL[7:0] (MON_GPO [7:0] = FFh) or MNTTEST2_SEL [7:0] (MON_GPO [7:0] = FEh) or MNTTEST3_SEL [7:0] (MON_GPO[7:0] = FDh)

TESTOUT (51 pin)

V sync / H sync / OIS Pulse can be selected and output via TESTOUT pin.

Signal is selected by MNTTEST1_SEL[7:0] (MON_TESTOUT [7:0] = FFh) or MNTTEST2_SEL [7:0] (MON_TESTOUT [7:0] = FEh) or MNTTEST3_SEL [7:0] (MON_TESTOUT [7:0] = FDh)

XVS (89 pin)

V Sync Pulse can be selected and output via XVS pin. Refer to Table 8-30.

Setting examples:

Case1: simultaneous output of H sync and OIS Pulse.

Table 8-18 Monitor Setting example case 1

Pin name	Selected Signal	Setting value
GPO (54 pin)	H sync	MNTTEST2_SEL :21d, MNTTEST3_SEL :24d, MON_GPO :FEh, MON_TESTOUT :FDh GPIOSEL :00h
TESTOUT(51 pin)	OIS Pulse	

Case2: simultaneous output of OIS Pulse and V sync.

Table 8-19 Monitor Setting example case 2

Pin name	Selected Signal	Setting value
GPO (54 pin)	OIS Pulse	MNTTEST2_SEL :24d, MNTTEST3_SEL :22d, MON_GPO :FEh, MON_TESTOUT :FDh GPIOSEL :00h
TESTOUT(51 pin)	V sync	

Case3: simultaneous output of V sync and H sync.

Table 8-20 Monitor Setting example case 3

Pin name	Selected Signal	Setting value
GPO (54 pin)	V sync	MNTTEST2_SEL :22d、MNTTEST3_SEL :21d, MON_GPO :FEh, MON_TESTOUT :FDh、GPIOSEL :00h
TESTOUT(51 pin)	H sync	

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8.7.1. V sync / H sync

V sync and H sync are the internal timing of image sensor.

V sync is the timing of the frame unit, and H sync is the line unit ones. Polarity and Pulse width of V sync / H sync can be set.

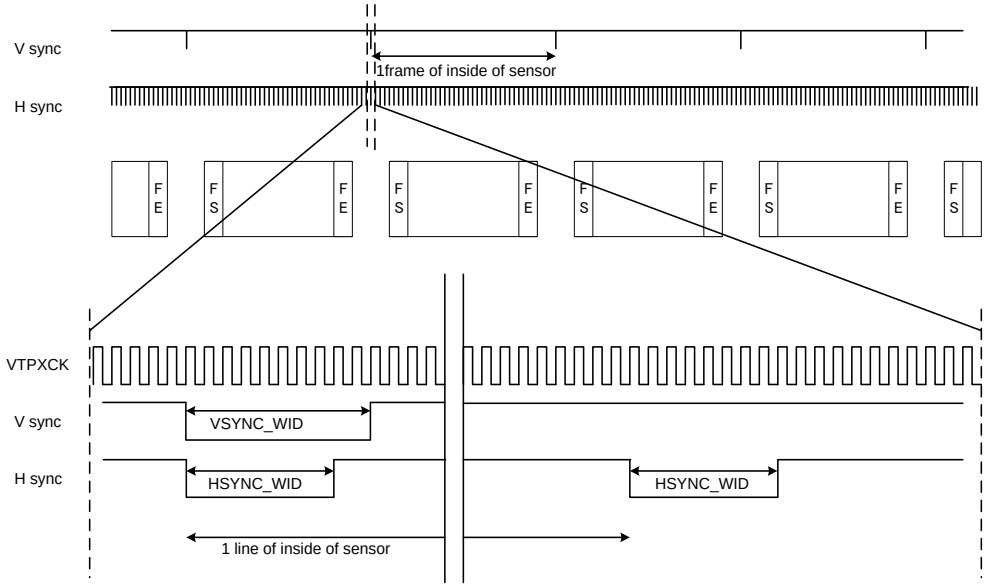


Figure 8-11 Image of V sync / H sync

Table 8-21 V sync / H sync settings

	Address	Bit	Name	comment
I²C register	0x510C	[0]	VSYNC_POL	Polarity of V sync Pulse (def:0) 0:Lo-Active, 1:Hi-Active
	0x510D	[2:0]	VSYNC_WID	Pulse width of V sync Pulse (Unit: IVTPXCK cycle) (def:0) 0d: x10 1d: x20 2d: x50 3d: x100 4d: x200 5d: x500 6d: x1000 7d: x2000
	0x510E	[0]	HSYNC_POL	Polarity of H sync Pulse (def:0) 0:Lo-Active, 1:Hi-Active
	0x510F	[2:0]	HSYNC_WID	Pulse width of H sync Pulse (Unit: IVTPXCK cycle) (def:0) 0d: x10 1d: x20 2d: x50 3d: x100 4d: x200 5d: x500 6d: x1000 7d: x2000

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8.7.2. Setting value output

This sensor can output the setting value(low or high) via monitor terminals. The monitor terminals mean the following three(3) terminals, such as FSTROBE, XVS, and GPO.

Table 8-22 Setting value output from FSTROBE pin

	Address	Bit	Name	comment
I²C register	0x3e50	[0]	GPO_CTRL0[8:0]	FSTROBE pin [8] : Output CTRL 0 Disable : Controlled by sensor (default) 1 Enable : Controlled by user
	0x3e51	[7:0]		[7:1] : Reserved [0] : Output Value 0: low output (default) 1: high output

Table 8-23 Setting value output from XVS pin

	Address	Bit	Name	comment
I²C register	0x3e52	[0]	GPO_CTRL1[8:0]	XVS pin [8] : Output CTRL 0 Disable : Controlled by sensor (default) 1 Enable : Controlled by user
	0x3e53	[7:0]		[7:1] : Reserved [0] : Output Value 0: low output (default) 1: high output
	0x3040	[0]	XVS_IO_CTRL	[0] : XVS IO control 0 : Input (default) 1 : Output

Table 8-24 Setting value output from GPO pin

	Address	Bit	Name	comment
I²C register	0x3e54	[0]	GPO_CTRL2[8:0]	GPO pin [8] : Output CTRL 0 Disable : Controlled by sensor (default) 1 Enable : Controlled by user
	0x3e55	[7:0]		[7:1] : Reserved [0] : Output Value 0: low output (default) 1: high output

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8.8. Dual camera operation

This section describes a method of dual camera operation, which outputs pair of synchronous images from two sensors controlled by one host device. This sensor can be either of a master or a slave device by register configurations. When a sensor operates as a master device, it controls vertical synchronous timings and outputs synchronous signal called V-sync signal or XVS signal for a slave device. When a sensor operates as a slave device, it outputs images synchronizing with V-sync or XVS.

In Sony CIS, there are two dual camera control methods. One is Full-synchronous system and the other is Quasi-synchronous system.

In Full-synchronous system, the master device controls shutter and read timings of the slave device and in Quasi-synchronous system, the master device can only control shutter timings of the slave device and read timings are controlled by slave register settings.

The main feature of Quasi-synchronous system is that Coarse integral time of the slave device is absolutely guaranteed by the register setting whereas in Full synchronous system, Coarse integral time of slave sensor has a possibility of shortened by synchronous signal from master device because the read timing of slave sensor synchronized with master synchronous signal.

This sensor has following functions:

- CCI 1st and 2nd address configuration.
- XVS multiplied, thinned-out and delay function in master/slave mode.
- XVS connectivity check for debugging.
- XVS receive ready-status read function for slave device

In dual camera system, master/slave sensor can be operated with a different operation mode. Host shall be taken care in real time domain [sec], not in COARSE_INTEG_TIME setting because 1H timing may be different in master/slave sensor.

Each sensor is possible to change gain settings individually while keeping synchronous relation between sensors.

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8.8.1. CCI 1st and 2nd slave address for dual camera

The CCI slave 1st and 2nd address is shown in the following Table 8-25, Table 8-26.

See"2.3.1 First slave address" and "2.3.2 Secondary slave address" for detail.

Table 8-25 1st slave address

Slave address (Write or Read)											
Bit [7:0]	7	6	5	4	3	2	1	0*	Type	Value	SLASEL
Bit value	0	0	1	1	0	1	0	0	Write	0x34	L or NC
	0	0	1	1	0	1	0	1	Read	0x35	

Bit [7:0]	7	6	5	4	3	2	1	0*	Type	Value	SLASEL
Bit value	0	0	1	0	0	0	0	0	Write	0x20	H
	0	0	1	0	0	0	0	1	Read	0x21	

* Note : Bit[0] is R/W setting bit.

Table 8-26 2nd slave address

Bit [7:0]	7	6	5	4	3	2	1	0*	Type	Value
Bit value	0	1	1	0	1	1	0	0	Write	0x6C
	0	1	1	0	1	1	0	1	Read	0x6D

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8.8.2. Block diagram

Figure 8-12 shows an example block diagram of the dual camera system using synchronous signal. In this example, accessing two sensors is achieved with shared CCI bus by having two different 1st CCI addresses and two common 2nd CCI addresses between two sensors. Dual camera synchronization is realized by connecting synchronous signal: XVS signal, V-sync signal. Refer to the Figure 8-16 for the definition of XVS signal and V-sync signal.

1st sensor: master mode function

- Outputs the XVS signal from the XVS pin.
- The XVS signal can be replaced with V-sync signal using XVS and GPO pin with some constraints, refer to 8.8.4 for more details.
- If a slave device is not a Sony product, select the XVS signal or V-sync signal according to slave device requirements.

2nd sensor: slave mode function

- Receive the XVS signal if a master device is a Sony product.
- Receive the V-sync signal if a master device is not a Sony product.

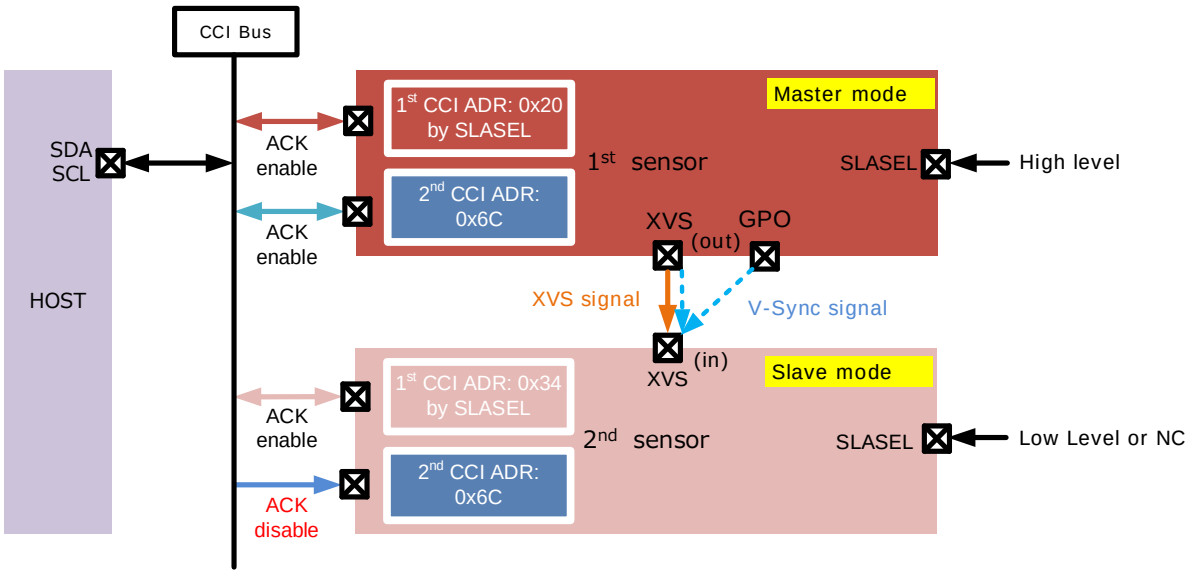


Figure 8-12 Dual camera system block diagram with synchronous signal

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8.8.3. Control sequence

Figure 8-13 shows a control sequence of the dual camera operation with the shared CCI bus.

In Figure 8-13, it is important that a register “MODE_SEL” setting for the slave device shall be set first.

In shared CCI bus connection, CCI address setting should be set as follows,

- Any settings using 1st CCI addresses are independently applied for each sensor.
- Settings with 2nd CCI address are commonly applied for both sensors.

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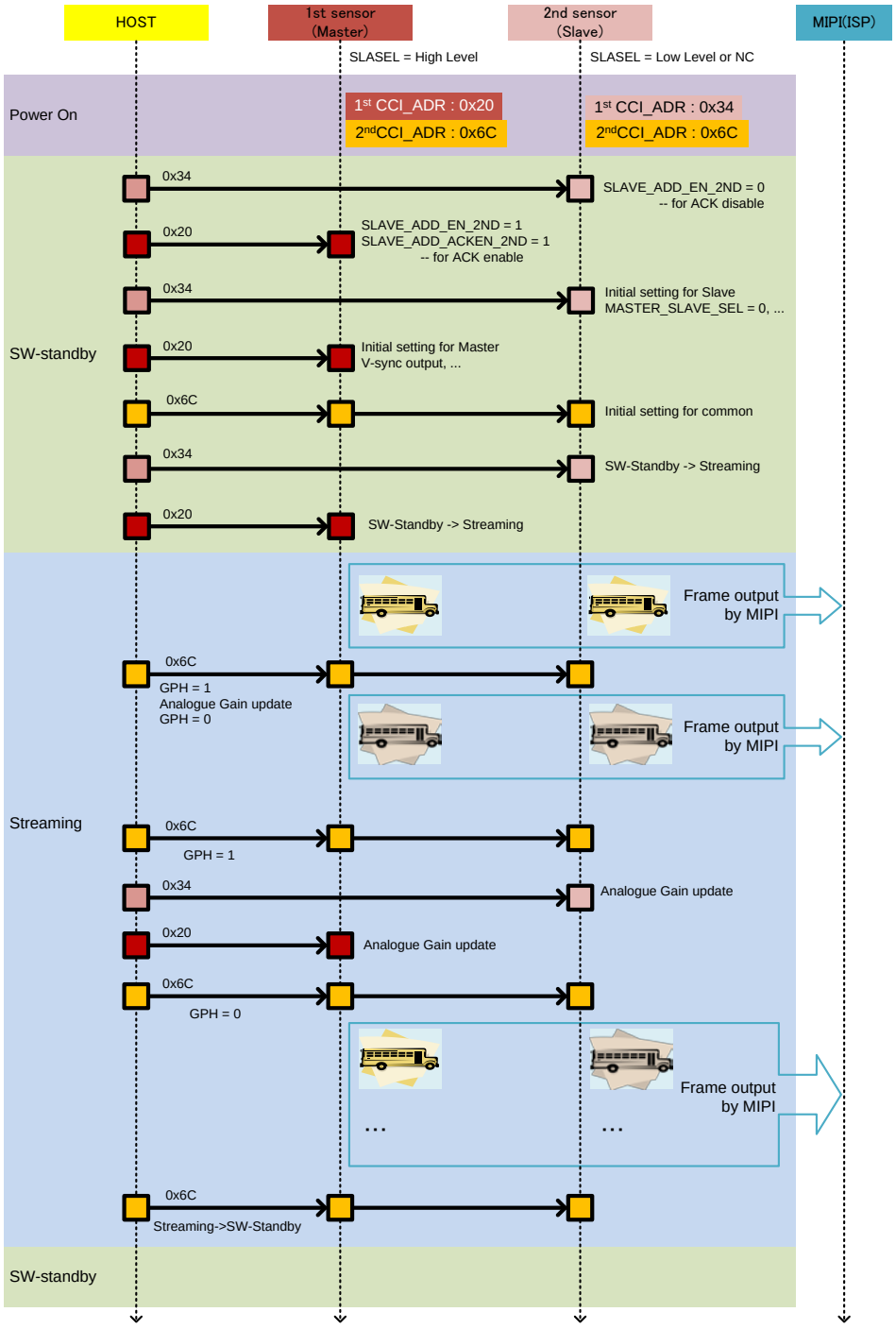


Figure 8-13 Control sequence of the dual camera operation with the shared CCI bus

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8.8.4. Usage in master mode

This chapter describes master mode settings. When the sensor is used in master mode, XVS and GPO pins are available for a synchronous output signal to a slave device.

The relationship between Pin and Signal name is shown in Table 8-27.

Table 8-27 XVS and GPO specification

Pin name	Signal name	Pre-shutter pulse	Delay, Thin-out, Multiple output
XVS(89pin)	XVS signal	Available	Available
	V-Sync signal	Not Available	Not Available
GPO(54pin)	V-Sync signal	Not Available	Not Available

Figure 8-14 shows the difference between the XVS signal and the V-Sync signal. Regarding the XVS signal, a first pulse, which called a Pre-shutter pulse is outputted. The Pre-shutter pulse can be used as triggering the pre-shutter timing at a slave sensor. Regarding the V-Sync signal, Pre-shutter pulse is not outputted.

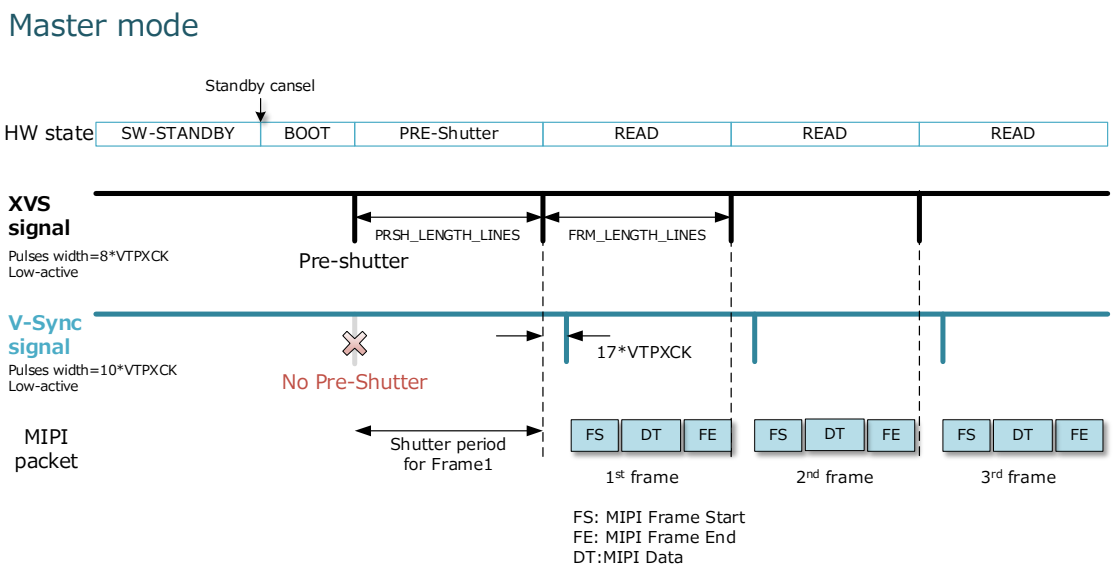


Figure 8-14 Difference between XVS signal and V-Sync signal

V-Sync signal delays four cycles of VTPXCK with respect to XVS because of H/W internal delay. If Sony product is chosen as a slave device, use the XVS signal to output 1st frame simultaneously as master sensor. If a Sony product is not chosen as a slave device, choose XVS signal or V-sync signal according to a slave device requirement.

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Table 8-28 shows the register setting for master.

Table 8-28 Master mode related registers without output signal setting.

Address	Bit	Name	Description	Master Mode Setting (HEX)
0x3F70	[0]	MC_MODE	Multi Camera mode setting 0 : Minimum time mode (default) Pre-shutter period is defined by COARSE_INTEG_TIME 1 : Fixed time mode Pre-shutter period is defined by PRSH_LENGTH_LINES	1'h1
0x3041	[0]	MASTER_SLAVE_SEL	Master or Slave mode setting 0 : Slave mode 1 : Master mode (default)	1'h1 (default)
0x3F79 ~ 0x3F7B	[23:0]	PRSH_LENGTH_LINES	Pre-Shutter period [line] Effective if MC_MODE=1 ≥ COARSE_INTEG_TIME+34 *34 is hardware dependent fixed value	≥ COARSE_INTEG_TIME+34
0x0350	[0]	FRM_LENGTH_CTL	Frame length automatic tracking control Select whether or not the frame length is changed automatically when FRM_LENGTH_LINES < COARSE_INTEG_TIME + α (α = type-specific adjustment parameter, and is 48 for this type. α is the value of CIT_MARGIN.) 0 : no automatic tracking control of frame length In this case, [FRM_LENGTH_LINES - α] operates instead of COARSE_INTEG_TIME 1 : automatic tracking control of frame length In this case, [COARSE_INTEG_TIME + α] operates instead of FRM_LENGTH_LINES	1(default)

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The period of XVS signal and V-Sync signal can be calculated by following equations.

T_period[s] = *Total number of pixel clocks* [pixel clocks/frame] / *Pixel_rate* [pixels/s]

where

Total number of pixel clocks [pixel clocks/frame]

= FRM_LENGTH_LINES [lines/frame] x 2^{CIT_LSHIFT} x LINE_LENGTH_PCK [pixel clocks/line]

Pixel_rate[pixels/s]

= *VTPXCK_clock_frequency* [Hz] x Number of internal pipeline (= 8)

Table 8-29 show XVS output control registers, set this register if XVS signal is used as synchronous signal to slave sensor.

Table 8-29 XVS control registers

Address	Bit	Name	Description	Master Mode Setting (HEX)
0x3040	[0]	XVS_IO_CTRL	XVS IO control 0 : Input (Slave mode) (default) 1 : Output (Master mode)	1'h1
0x3F71	[0]	EXTOUT_EN	XVS output setting 0 : output disable 1 : output enable	1'h1
0x4B82	[2:0]	EXTOUT_XVS_WID	Pulse width = 2^EXTOUT_XVS_WID cycles of VTPXCK	3'h3
0x4B83	[0]	EXTOUT_XVS_POL	XVS output polarity 0 : high active 1: low active	1'h1
0x4B85	[0]	EXTIN_XVS_POL	XVS input polarity 0 : high active 1: low active	1'h1

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If V-Sync signal is chosen as a synchronous signal, refer to Table 8-30.

Table 8-30 GPO/XVS register setting

Pin name	Setting value
GPO	MASTER_SLAVE_SEL(0x3041)=1h MONSEL[10:8] (0x424A)=0h MONSEL[7:0] (0x424B)=0h MON_GPO(0x423E)=FFh MNTTEST1_SEL(0x4BD7) = 16h IO_CTRL_GPO (0x4226)=0h(output)
XVS	MASTER_SLAVE_SEL(0x3041)=1h MONSEL[10:8] (0x424A)=0h MONSEL[7:0] (0x424B)=0h MON_XVS (0x423D)=FFh MNTTEST1_SEL(0x4BD7) = 16h IO_CTRL_XVS (0x4225)=0h(output) XVS_IO_CTRL (0x3040)=1h(output)

Polarity and pulse width of the V-Sync signal is adjustable, refer to Table8-21 VSYNC_POL and VSYNC_WID registers.

This sensor can delay the XVS signal output or internal XVS by setting registers. The register setting shall be made during SW standby.

Table 8-31 Master Control registers to delay XVS output

Address	Bit	Name	Description
0x3F64 ~0x3F65	[15:0]	EXTERNAL_XVS_OFFSET_LINE	Delay setting for XVS output. (set the number of lines) (default : 0)
0x3F66 ~0x3F67	[15:0]	EXTERNAL_XVS_OFFSET_VTPXCK	Delay setting for XVS output. (set the number of clocks) (default : 0)

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Table 8-32 Master Control registers to delay internal XVS

Address	Bit	Name	Description
0x3F68 ~0x3F69	[15:0]	INTERNAL_XVS_OFFSET_LINE	Delay setting for internal XVS. (set the number of lines) (default : 0)
0x3F6A ~0x3F6B	[15:0]	INTERNAL_XVS_OFFSET_VTPXCK	Delay setting for internal XVS (set the number of clocks) (default : 0)

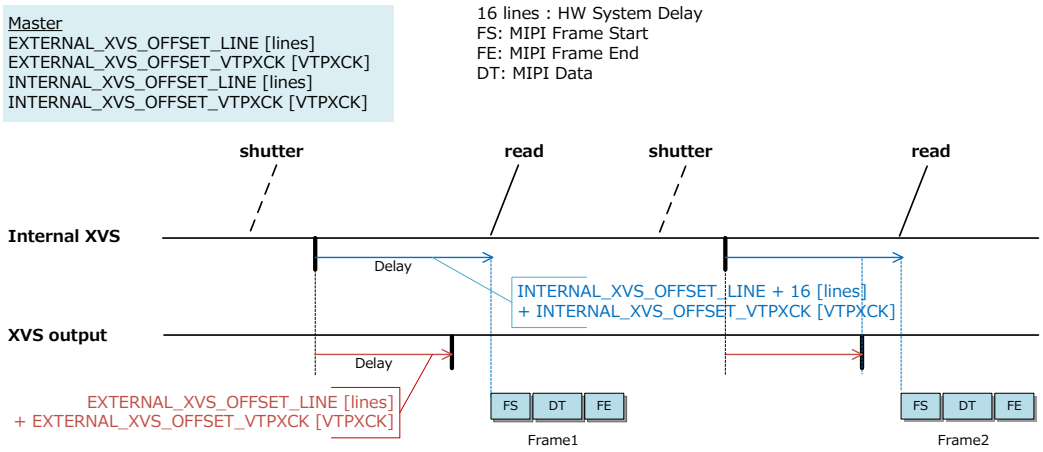


Figure 8-15 Master delay XVS output and internal XVS

This sensor can also thin out or multiply XVS output by setting the registers below. The register setting shall be made during SW standby.

Table 8-33 Master control registers to thin out or multiply XVS output

Address	Bit	Name	Description
0x3F6C	[0]	EXTOUT_XVS_MULT_SEL	0 : Thin out XVS output (default) 1: Multiply XVS output
0x3F6D	[6:0]	EXTOUT_XVS_MULT	Thin out setting for XVS output if EXTOUT_XVS_MULT_SEL=0. 0: not thin out. (output XVS period = Internal XVS period)(default) 1: 1/2 thin out. (output XVS period = Internal XVS period x 2) 2: 1/3 thin out. (output XVS period = Internal XVS period x 3) ... 127: 1/128 thin out. (output XVS period = Internal XVS period x 128) Multiple setting for XVS output if EXTOUT_XVS_MULT_SEL=1 0: not multiply. (output XVS period = Internal XVS period) 1: 2 multiply. (output XVS period x 2 = Internal XVS period) … 127: 128 multiply. (output XVS period x 128 = Internal XVS period)

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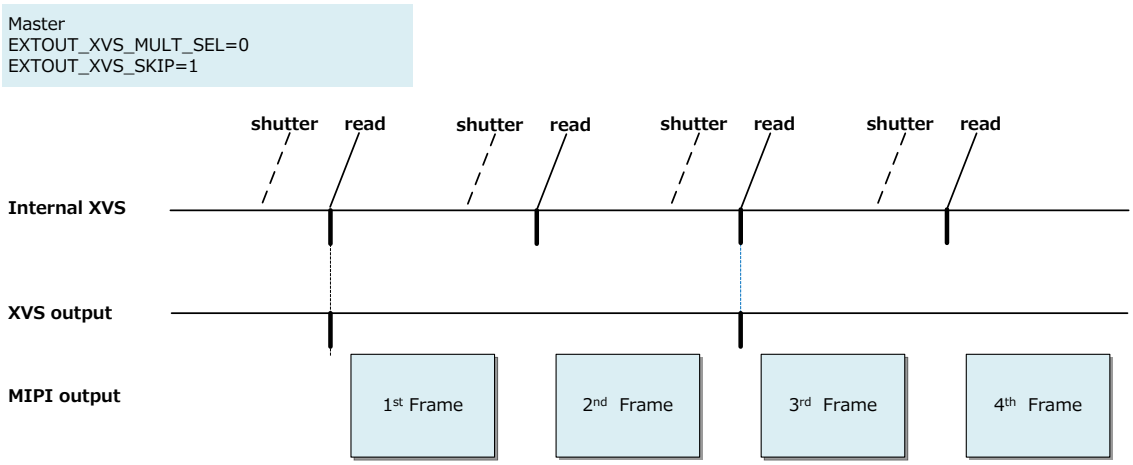


Figure 8-16 Master thin out XVS

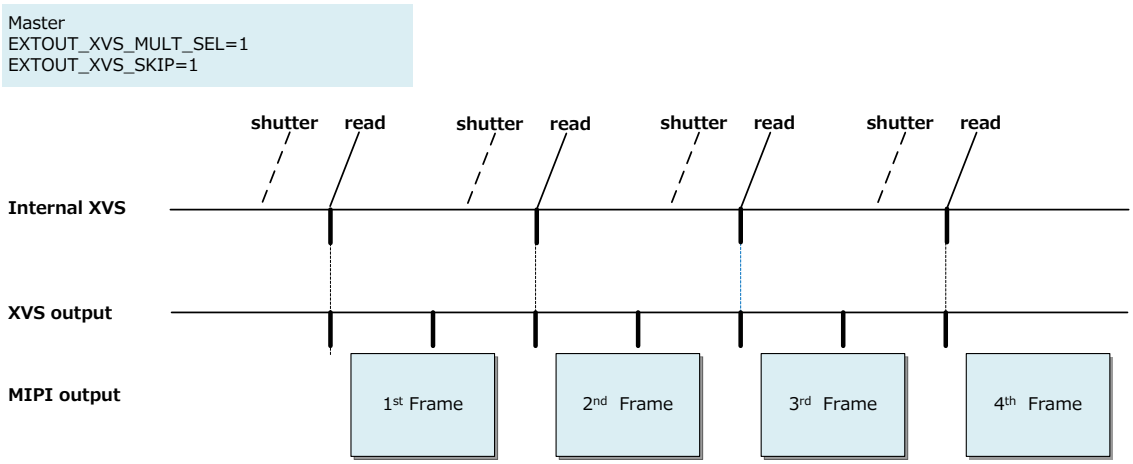


Figure 8-17 Master multiple XVS

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8.8.5. Usage in slave mode

This chapter describes the slave mode setting. When the sensor is used in the slave mode, only XVS pin is available for a synchronous signal. In the slave mode, the 1st image outputs at the 2nd XVS input because 1st XVS input is used as the pre-shutter pulse: start timing of shutter for a 1st image. In the slave mode, after receiving standby cancel, HW state changes from “SW-Standby” to “XVS wait” and keep waiting XVS from a master sensor. When XVS comes from the master, HW state changes to “PRE-Shutter” state.

Figure 8-18 shows how the Pre-Shutter pulse is used in a slave sensor and internal state. The standby cancel shall be set to the slave sensor prior to the master sensor.

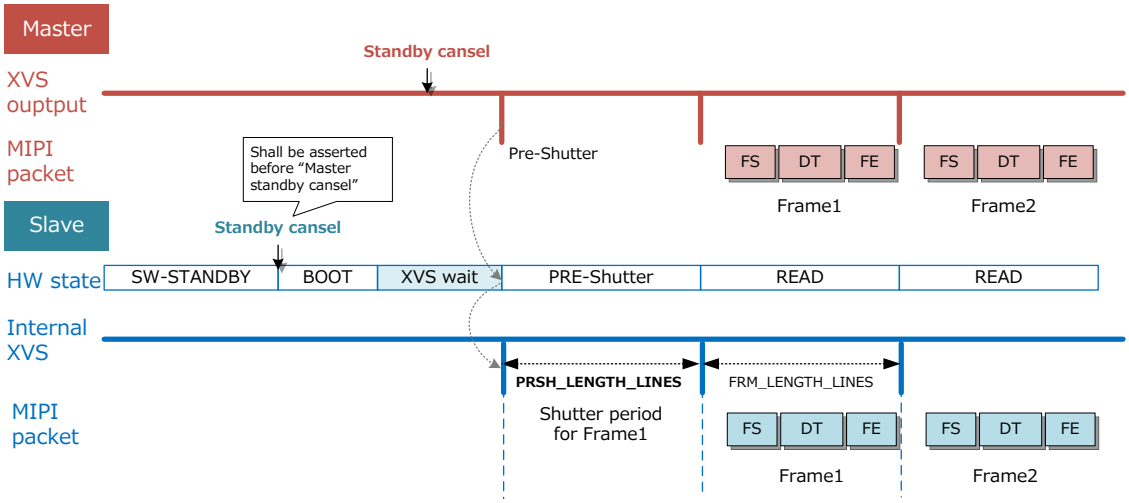


Figure 8-18 Frame output timing and HW state in the slave mode

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The following table shows the register setting for slave.

Table 8-34 Slave mode related registers

Address	Bit	Name	Description	Slave Mode Setting (HEX)
0x3F70	[0]	MC_MODE	Multi Camera mode setting 0 : Minimum time mode (default) Pre-shutter period is defined by COARSE_INTEG_TIME 1 : Fixed time mode Pre-shutter period is defined by PRSH_LENGTH_LINES	1'h1
0x3041	[0]	MASTER_SLAVE_SEL	Master or Slave mode setting 0 : Slave mode 1 : Master mode (default)	1'h0
0x3040	[0]	XVS_IO_CTRL	XVS IO control 0 : Input (Slave mode) (default) 1 : Output (Master mode)	1'h0 (default)
0x3F79 ~ 0x3F7B	[23:0]	PRSH_LENGTH_LINES	Pre-Shutter period [line] Effective if MC_MODE=1 ≥ COARSE_INTEG_TIME+34 *34 is hardware dependent fixed value	≥ COARSE_INTEG_TIME+34
0x3F71	[0]	EXTOUT_EN	XVS output setting 0 : output disable 1 : output enable	1'h0 (default)
0x0350	[0]	FRM_LENGTH_CTL	Frame length automatic tracking control Select whether or not the frame length is changed automatically when FRM_LENGTH_LINES < COARSE_INTEG_TIME + α (α = type-specific adjustment parameter, and is 48 for this type. α is the value of CIT_MARGIN.) 0 : no automatic tracking control of frame length In this case, [FRM_LENGTH_LINES - α] operates instead of COARSE_INTEG_TIME 1 : automatic tracking control of frame length In this case, [COARSE_INTEG_TIME + α] operates instead of FRM_LENGTH_LINES	1(default)

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Internal XVS can be delayed by register.　Register setting shall be issued during SW standby.

Table 8-35　Slave Control registers to delay XVS input

Address	Bit	Name	Description
0x3F68 ~0x3F69	[15:0]	INTERNAL_XVS_OFFSET_LINE	Delay setting for XVS output. (set the number of lines)　(default : 0)
0x3F6A ~0x3F6B	[15:0]	INTERNAL_XVS_OFFSET_VTPXCK	Delay setting for XVS output. (set the number of clocks)　(default : 0)

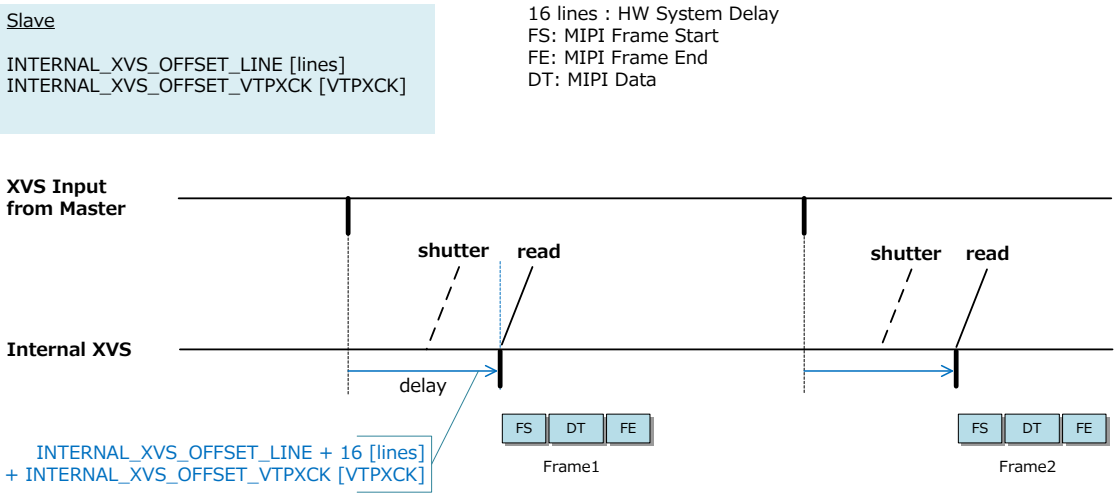


Figure 8-19　Slave delay XVS input

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Table 8-36 Slave control registers to thin out XVS input

Address	Bit	Name	Description
0x3F6E	[0]	EXTIN_XVS_MULT_SEL	0 : thin out internal XVS (default) 1 : multiply internal XVS
0x3F6F	[6:0]	EXTIN_XVS_MULT	Thin out setting for internal XVS if EXTIN_XVS_MULT_SEL=0 0: not thin out. (incoming XVS period = Internal XVS period) (default) 1: 1/2 thin out. (incoming XVS period = Internal XVS period x 2) … 127: 128 thin out. (incoming XVS period = Internal XVS period x 128) Multiply setting for internal XVS if EXTIN_XVS_MULT_SEL=1 0: not multiply. (incoming XVS period = Internal XVS period) 1: 2 multiply. (incoming XVS period = Internal XVS period) … 127: 128 multiply. (incoming XVS period = Internal XVS period)

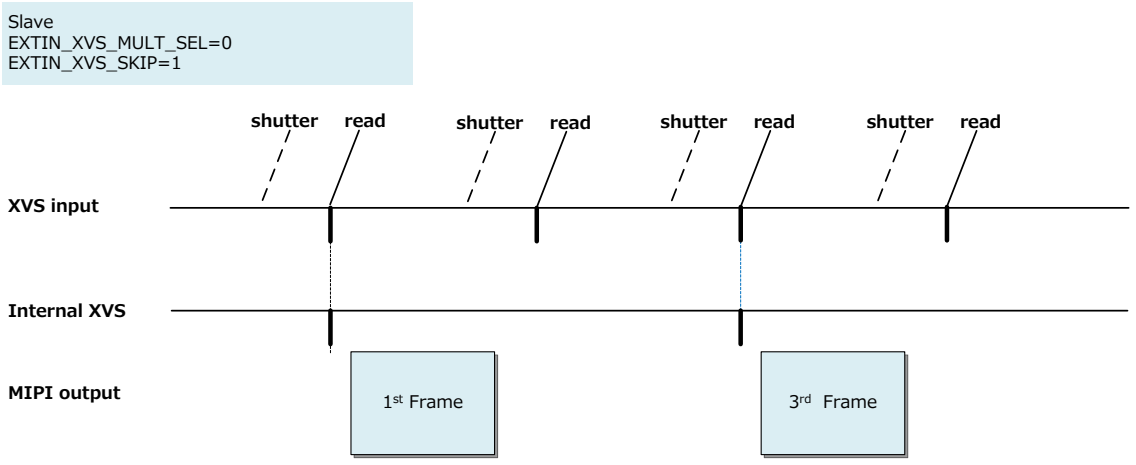


Figure 8-20 Slave thin out XVS input

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Table 8-37 Slave control registers to thin out or multiply XVS input

	Address	Bit	Name	Description
I²C register	0x3F6E	[0]	EXTIN_XVS_MULT_SEL	0 : thin out internal XVS (default) 1 : multiply internal XVS
	0x3F6F	[6:0]	EXTIN_XVS_MULT	Thin out setting for internal XVS if EXTIN_XVS_MULT_SEL=0 0: not thin out. (incoming XVS period = Internal XVS period) (default) 1: 1/2 thin out. (incoming XVS period = Internal XVS period x 2) … 127: 128 thin out. (incoming XVS period = Internal XVS period x 128) Multiply setting for internal XVS if EXTIN_XVS_MULT_SEL=1 0: not multiply. (incoming XVS period = Internal XVS period) 1: 2 multiply. (incoming XVS period = Internal XVS period) … 127: 128 multiply. (incoming XVS period = Internal XVS period)

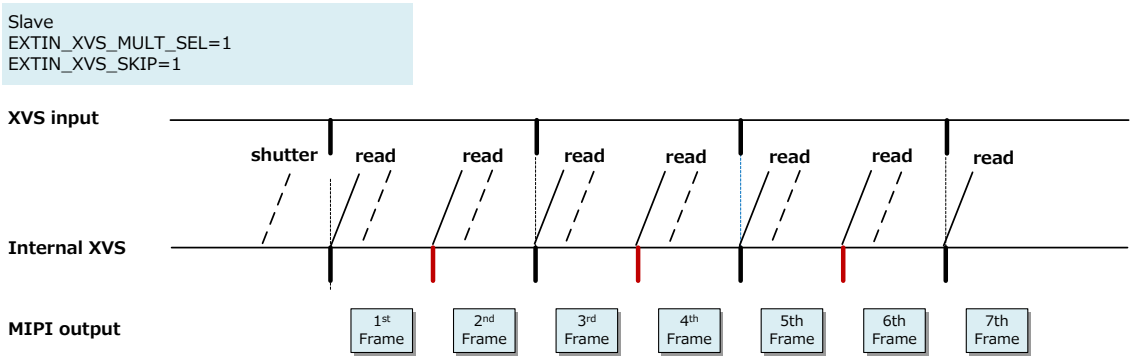


Figure 8-21 Slave multiple XVS input

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8.8.5.1. Restriction in the slave mode operation

1. Keep T_{frame_master}[s] > T_{frame_slave}[s] + $2T_{line_slave}$[s] relationship.
2. If FRM_LENGTH_CTL = 1, slave “COARSE_INTEG_TIME + CIT_MARGIN + 2” shall be less than master FRAME_ LENGTH_LINES. Where default value of CIT_MARGIN is 48.
3. Set T_{frame_master} and T_{frame_slave} as frame length (FRM_LENGTH_LINES x LINE_LENGTH_PCK / *pixel_clock_frequency*) for master and slave sensor. Set n as multiple or thin out number (n = 1,2,4,…,128).
   If XVS is thinned out, n x T_{frame_master} must be greater than $T_{frame_slave} + 2T_{line_slave}$ (see Figure 8-22).
   If XVS is multiplied, T_{frame_master} must be greater than n x $T_{frame_slave} + 2T_{line_slave}$ (see Figure 8-23).
4. In slave mode, the XVS terminal of this sensor inhibits high input signal from a master sensor during power off.
5. The embedded thermal meter of slave sensor in standby mode is not supported. If you want to use embedded thermal meter at standby mode, it is only available in master sensor.
6. Thinned out or multiple function and internal or output XVS delay function must be changed during SW-standby.
7. AEB (AE bracketing) mode is not supported.
8. GPH (grouped parameter hold) must be used for updating gain during streaming.
9. Fast mode transition is not supported
10. Between two sensors, there exists a worst case time difference in MIPI outputs calculated with the following equation. Δtmax = 145 * tINCK =145 / fINCK（Worst case). Where tINCK is external clock period and fINCK is external clock frequency to the sensor.
11. When using thermal meter in master mode, there is restriction on the setting order of TEMP_SEN_CTL, EXTOUT_EN and MODE_SEL; that is, both MODE_SEL and EXTOUT_EN must be 0 or 1 when TEMP_SEN_CTL is 1. (See Figure 8-24).
12. In slave mode, minimum width of XVS input signal is 8 clocks of VTPXCK.
    Maximum width of XVS input signal is “FRM_LENGTH_LINES – 48”.
13. PRSH_LENGTH_LINES must be set to satisfy the following restriction.
    Pre-Shutter period of master[s] > Pre-Shutter period of slave[s] + $2T_{line_slave}$[s]

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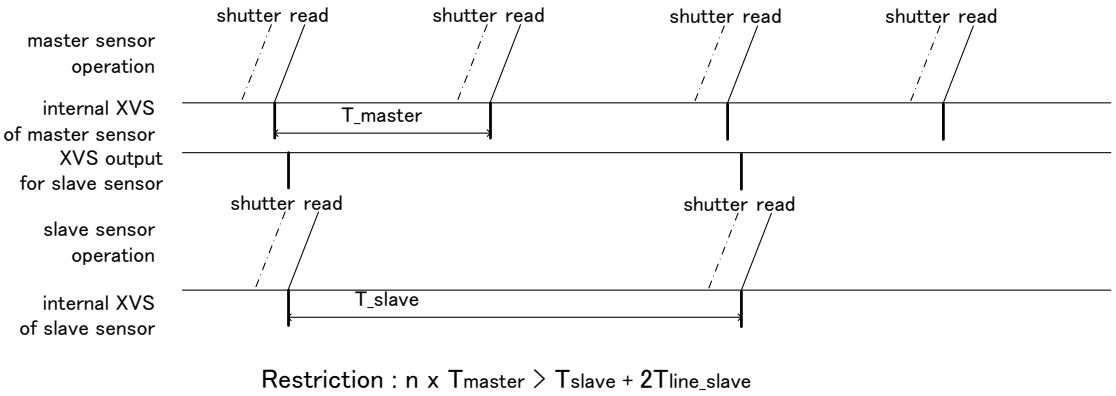


Figure 8-22 Frame length restriction when XVS is thinned out

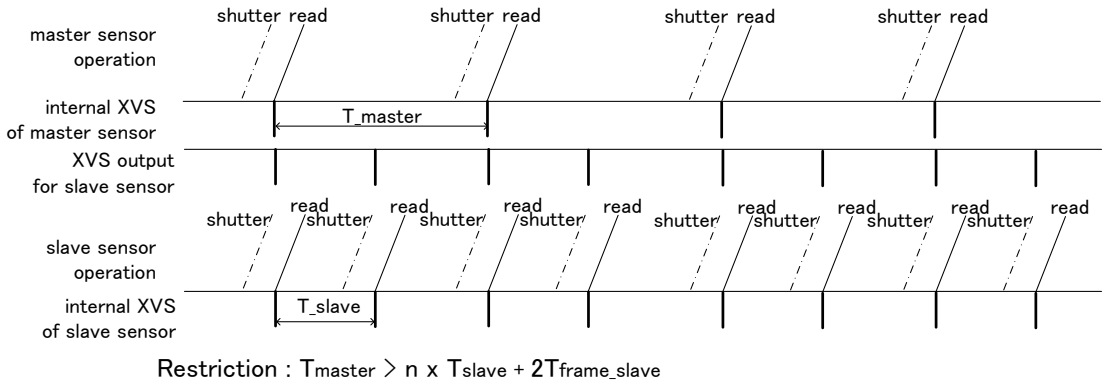


Figure 8-23 Frame length restriction when XVS is multiplied

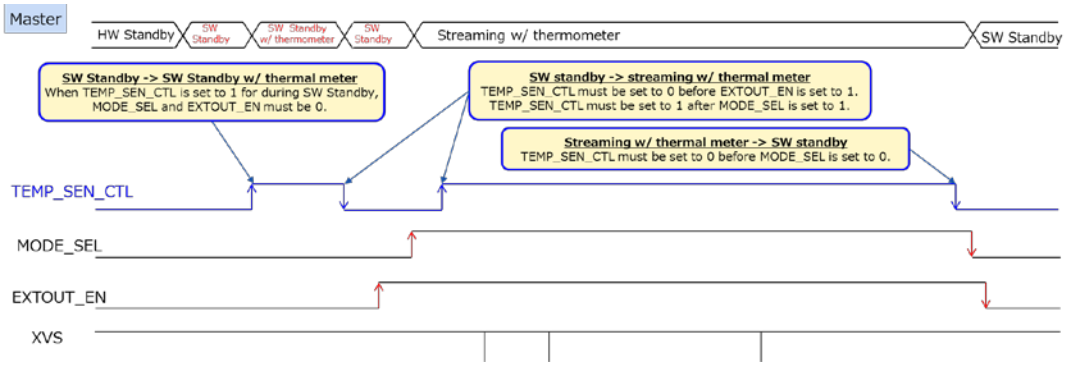


Figure 8-24 Restriction of thermal meter in master mode

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8.8.6. XVS connectivity debug function

This function offers users to check XVS connectivity. Overall XVS check diagram is shown in Figure 8-25. Master sensor can output fixed high/low signal from XVS pin by register control .Slave sensor can monitor XVS input by reading register.

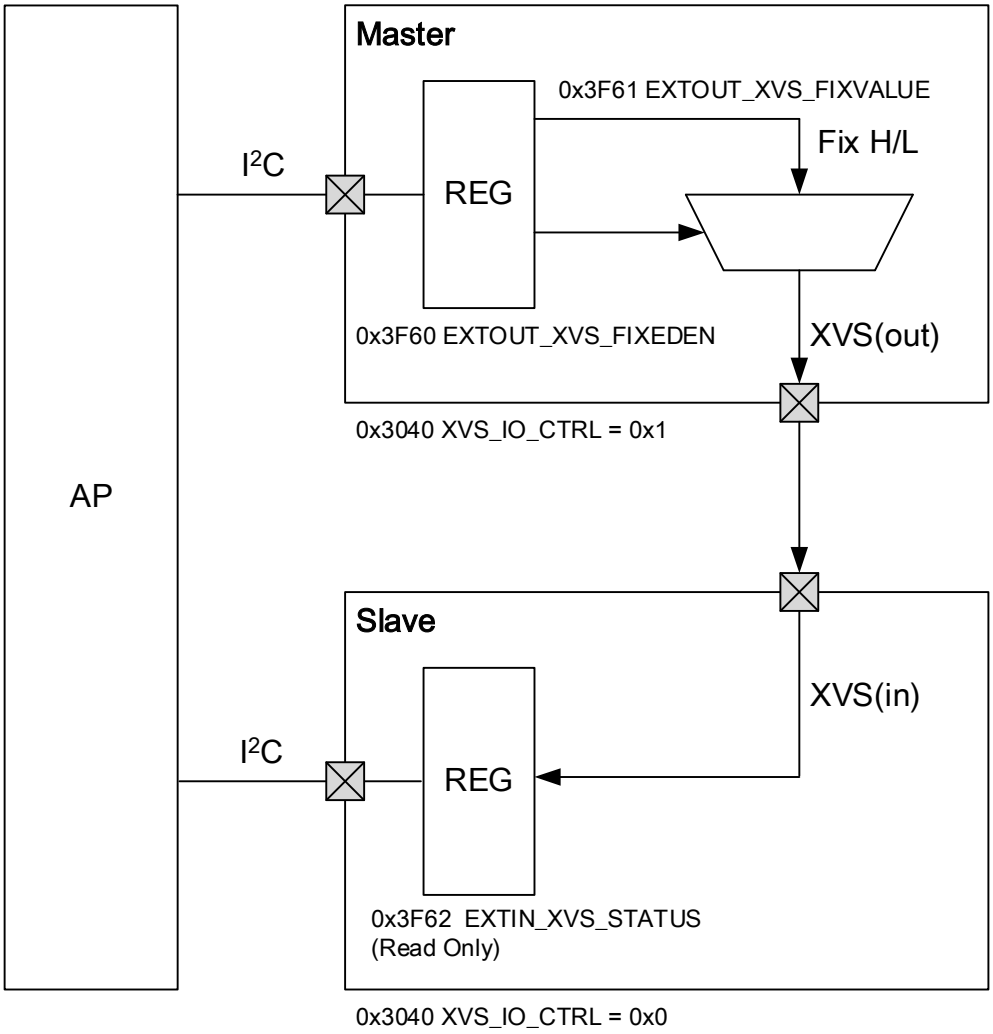


Figure 8-25 XVS connectivity check diagram

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Table 8-38 XVS connectivity check related registers

	Address	Bit	Name	Description	Master mode	Slave mode
I²C register	0x3F61	[0]	EXTOUT_XVS_FIXED_VALUE	Select fixed output level from XVS pin 0: Fixed Low output (default) 1: Fixed High output	0 or 1	-
	0x3F60	[0]	EXTOUT_XVS_FIXED_EN	Fixed output level from XVS pin disable/enable 0: disable (default) 1: enable	1	0
	0x3F62	[0]	EXTIN_XVS_STATUS	Show XVS input level from Master (Read only) 0: Low level 1: High level (default)	-	RO-D
	0x3040	[0]	XVS_IO_CTRL	XVS IO control 0 : Input (Slave mode) (default) 1 : Output (Master mode)	1	0

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8.8.7. Dual camera using shared CCI bus

8.8.7.1. Block diagram

The 2nd slave address shall be activated and the master sensor should be set as ACK responder based on the following table.

Table 8-39 Control register of 2nd slave address

I2C Address	Bit	Name	Description	Initial value (HEX)	Master setting	Slave setting
0x3010	[0]	SLAVE_ADD_EN_2ND	2nd Slave Address Enable 0: disable 1: enable	1'h0	1'h1	1'h1
0x3011	[0]	SLAVE_ADD_ACKEN_2ND	2nd Slave Address Ack Enable 0: disable 1: enable	1'h0	1'h1	1'h0

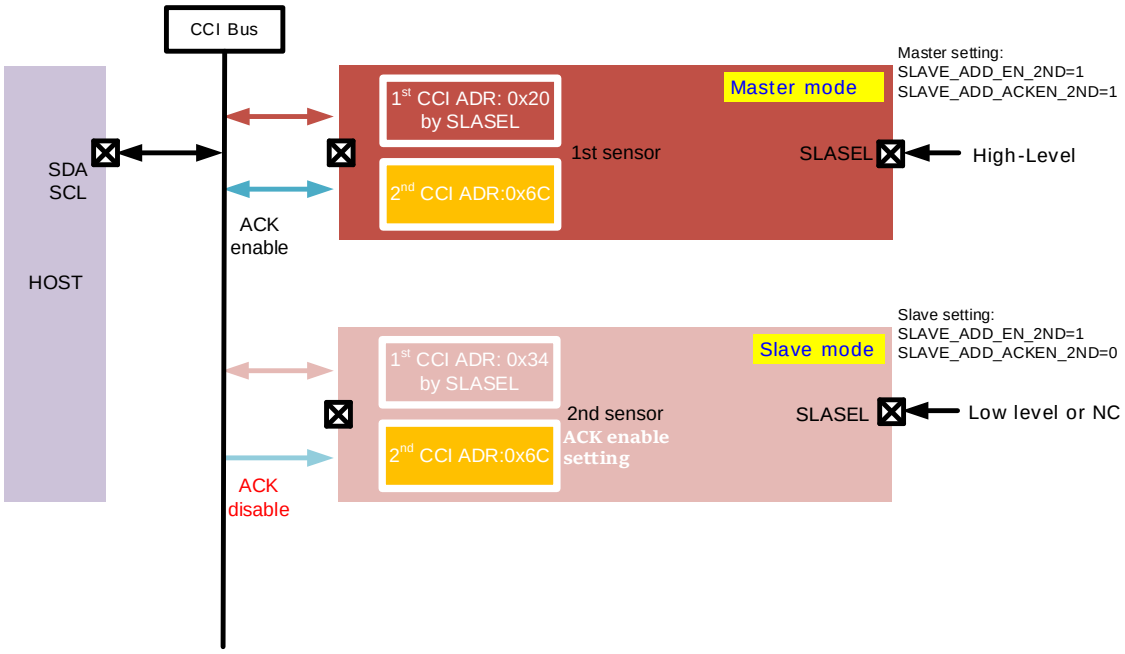


Figure 8-26 Dual camera system block diagram with shared CCI

Figure 8-26 shows the block diagram of the dual camera application using shared CCI bus.

In this sensor, synchronization of two sensors is achieved with shared CCI bus by having different slave address between two sensors and by issuing MODE_SEL=1 at the same time onto the shared bus. With this approach, this sensor can be either of a master device or a slave device.

Besides the above mentioned CCI address, this sensor has a built-in 2nd slave address of 0x6C which is common for all sensors. See "2.3.2 Secondary slave address" about the details of activation of the 2nd slave address.

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As the second CCI address is activated and with different SLASEL setting, each sensor have each CCI addresses; first sensor with 0x20 and 0x6c, second sensor with 0x34 and 0x6c. ACK responder for the 2nd slave address (0x6c) shall be set either one.

The above configuration enables to set different settings on each sensor and talk "start streaming" both together. Consequently each sensor can synchronously capture a same scene with individual exposure time and individual gain setting.

See following paragraph for hardware configuration and control sequence of the dual camera operation.

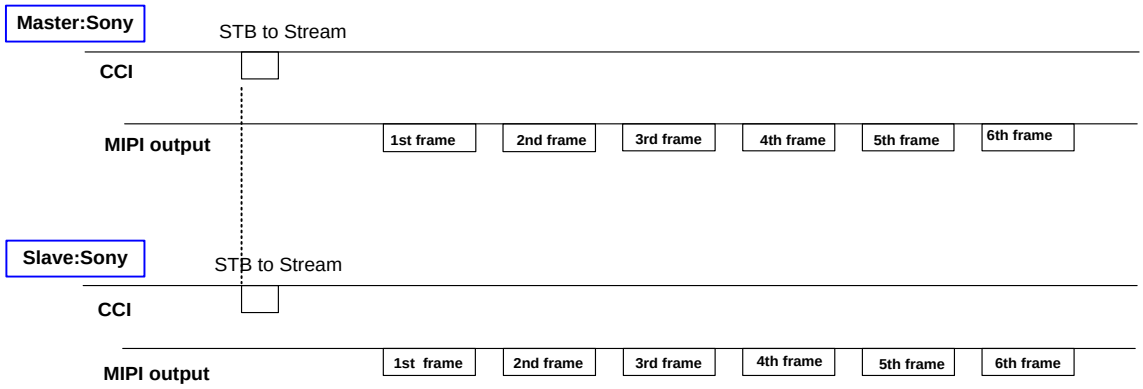


Figure 8-27 Dual camera approach using shared CCI bus

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8.8.7.2. Control sequence

Any settings using 1st CCI addresses are separately applied for each sensor.

Settings with 2nd CCI address are commonly applied for both sensors.

The following figure shows temporal and spatial sequence of dual camera control.

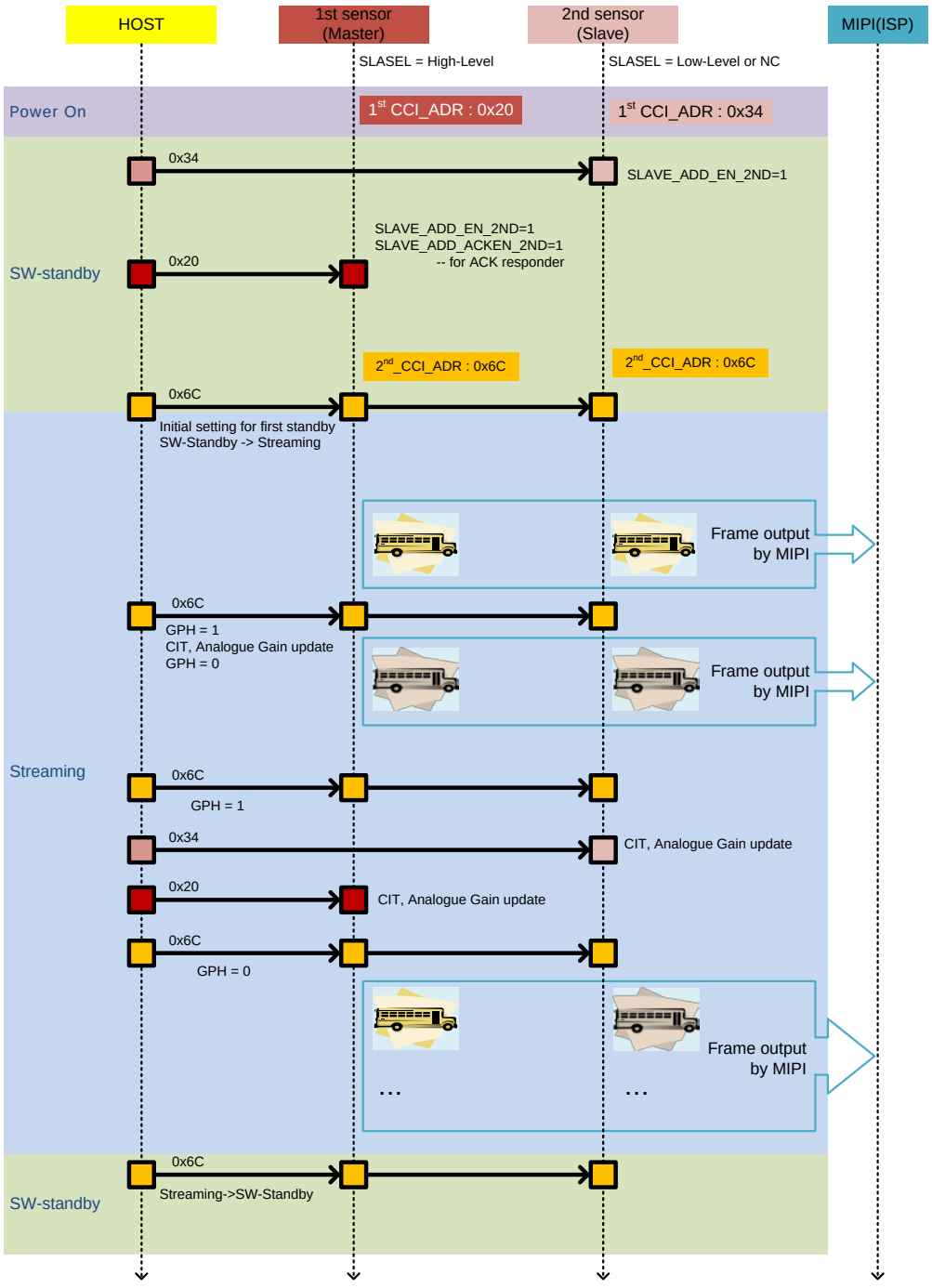


Figure 8-28 Communication control and timing

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8.8.8. Dual camera system with Separated CCI bus (optional)

The following figure shows the block diagram of the Dual camera system using with separated CCI bus. In this use case, one sensor is connected to I^2C Bus (A) and the other is connected to I^2C Bus (B). When the Dual camera system with Separated CCI Bus is operated, both of 1st CCI addresses do not require the different CCI address. It means that the same 1st CCI address is acceptable in this connection. On the other hand, there is no need to setting for 2nd Slave address.

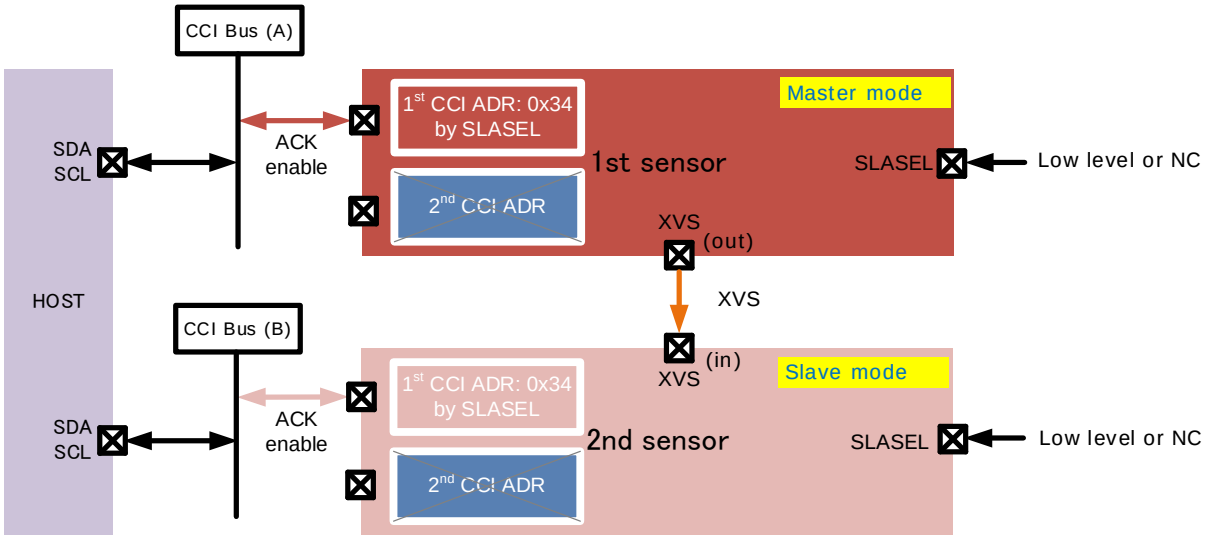


Figure 8-29 Dual camera system block diagram with separated CCI bus

The following figure shows the system control sequence of the Dual camera system using with separated CCI Bus.

First of all, it is necessary to set "MODE_SEL" to the Slave mode sensor before streaming starts. Then, “MODE_SEL” for Master mode sensor should be set.

If you need to change the both sensors setting at same time, it is necessary to pay attention about setting delay time between separated CCI Bus. If there is a large delay time, it will be happen that there is not a synchronized case between master and slave sensors.

The other restrictions are same as Dual camera system with shared CCI Bus.

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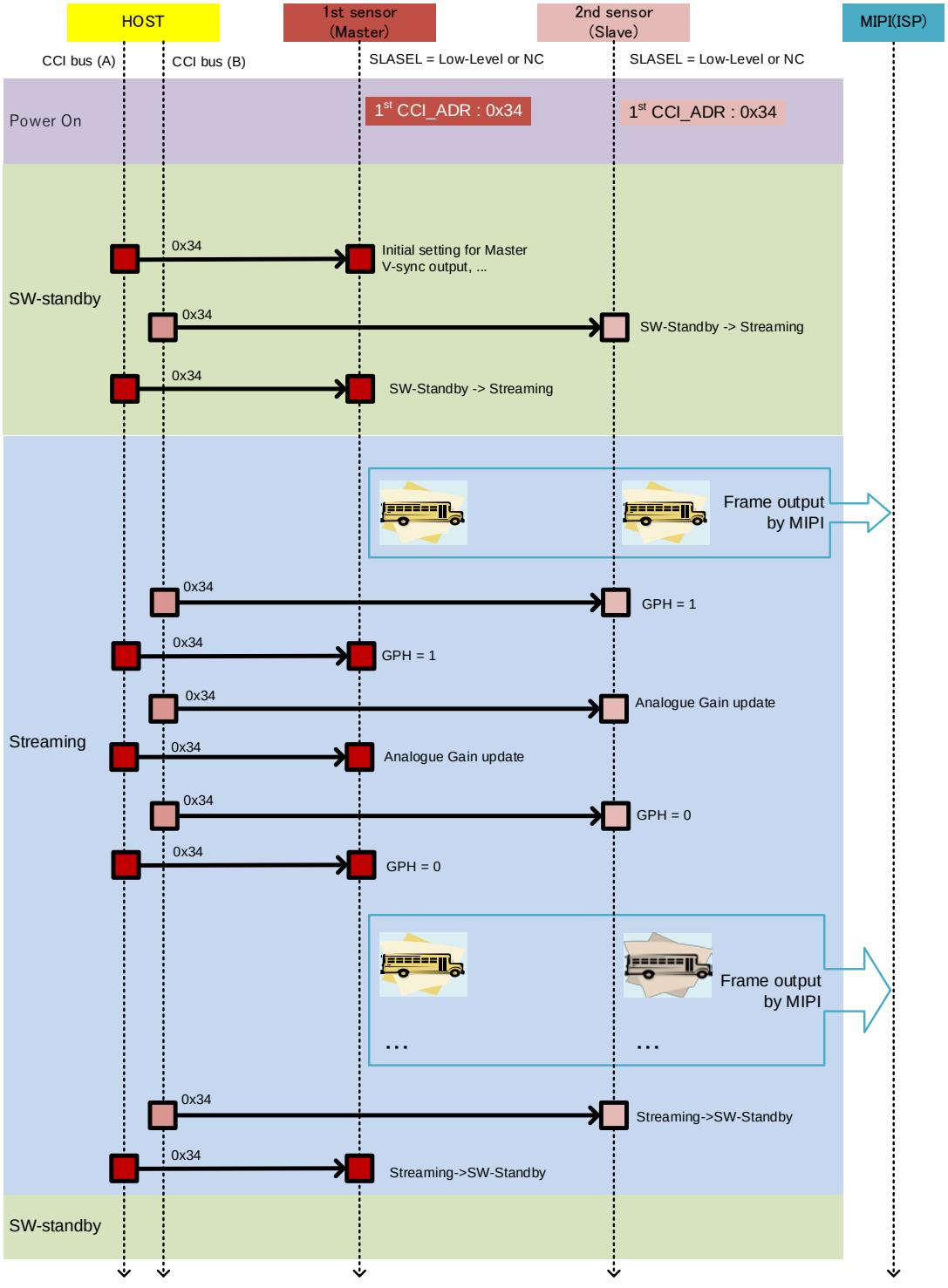


Figure 8-30 Communication control and timing with separated CCI Bus

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8.9. Charge pump setting

This sensor has two charge pump block to generate voltage (VRL, VRLRD) for the pixels array.

The charge pumping clock is common. Internal capacitor value should be set the same value for both charge pumps out.

Setting for CP control mode is below.

Table 8-40 Register setting for CP control mode

	Block	Address	Bit	Name	Description
I²C register	VRL VRLRD	0x521F	[1:0]	CPCKSEL_AUTOEN	Clock auto select 0 : Disable (Manual mode) 2 : Enable (Auto mode, default setting) 1, 3 Reserve

Setting for pumping clock setting (manual mode only) is below.

The charge pumping clock is driven by INCK during SW standby. So this setting is invalid during SW standby.

The pumping frequency shall be in the range of 6MHz～27MHz.

Table 8-41 Register setting for pumping clock

	Block	Address	Bit	Name	Description
I²C register	VRL VRLRD	0x521E	[4:0]	CPCKSEL_NOM	Pumping clock setting 1: 1/1 of IVTPXCK 2: 1/2 of IVTPXCK 3: 1/4 of IVTPXCK 4: 1/6 of IVTPXCK 5: 1/8 of IVTPXCK 6: 1/10 of IVTPXCK 7: 1/12 of IVTPXCK 8: 1/14 of IVTPXCK 9: 1/16 of IVTPXCK 10: 1/18 of IVTPXCK 11: 1/20 of IVTPXCK 12: 1/22 of IVTPXCK 13: 1/24 of IVTPXCK 14: 1/26 of IVTPXCK 15: 1/28 of IVTPXCK Others: reserved

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9. Settings related to QBC HDR capture mode picture quality

This Sensor built-in QBC HDR function consists of the blocks shown in the following block diagram. In This Sensor, most of control functions are handled by built-in firmware. In this chapter, user accessible control registers of each block are explained.

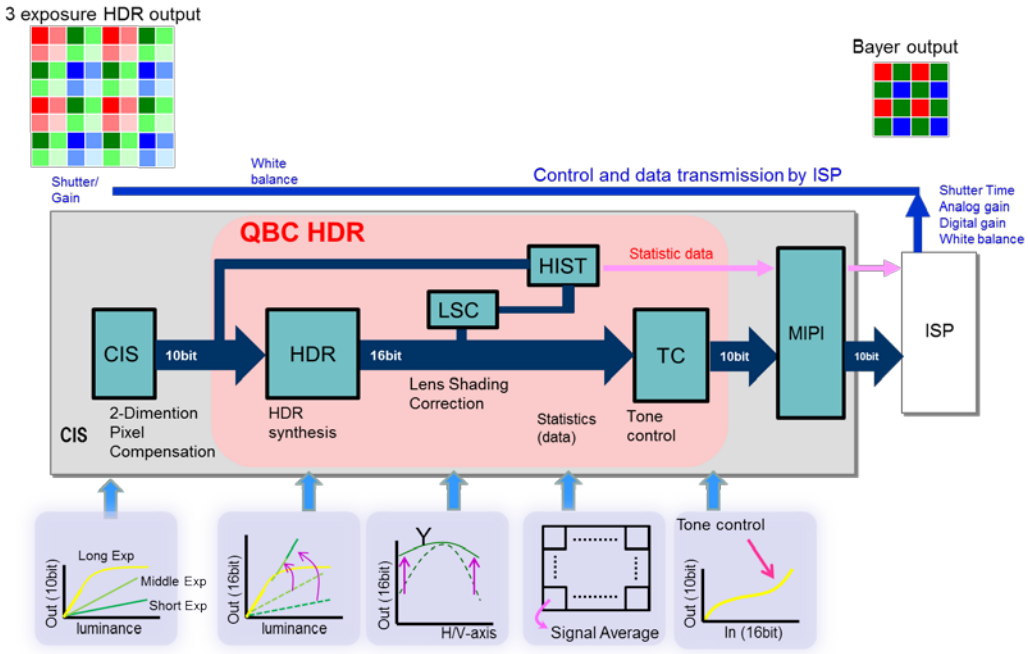


Figure 9-1 Block diagram for QBC HDR

QBC HDR system of This Sensor requires following signal input/feedback from ISP:

- Input from the sensor: Image data and statistic data.
- Feedback from the ISP: Settings of shutter time, analog gain, digital gain and white balance gains.

9.1. Outline of QBC HDR

QBC HDR is a one of HDR system for preview and movie mode. QBC HDR image is made from 4 same color pixels which have different exposure as shown in following figure. The resolution of output image is a half from original sensor. The output image in QBC HDR mode is Bayer coding alignment so that there is no need to apply the re-mosaic processing outside of the sensor.

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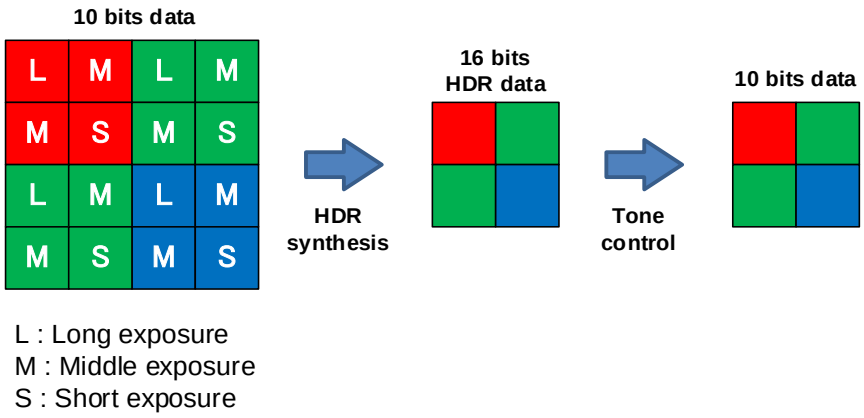


Figure 9-2 Image of QBC HDR synthesis

QBC HDR data is compressed to 10 bits by tone control (TC). In actual processing it is possible to select the settings from three kinds of tone curves;

1. Global tone curve
2. Blend of Global and Local tone curve.
3. Local tone curve

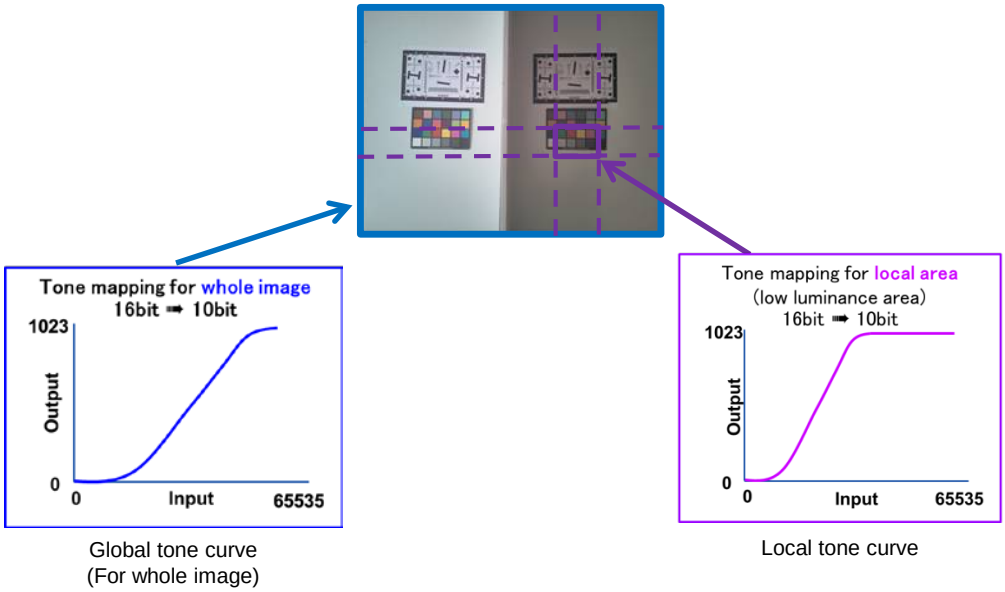


Figure 9-3 Blend of Global and Local tone curve

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9.2. QBC HDR control

The registers for QBC HDR control are shown in the following table. It is recommended that Grouped Parameter Hold (GPH) is used for setting of these registers.

Table 9-1 QBC HDR control registers

	Address	Bit	Name	Description
I²C register	0x0220	[6:0]	HDR_MODE[6:0]	Recommend setting value: 0x63(QBC HDR enable) 0x00(QBC HDR disable) Bit[0] QBC HDR mode enable 0 : QBC HDR disable 1 : QBC HDR enable Bit[1] Analog Gain mode select during QBC HDR 0 : combined gain used 1 : separate gain used Bit[4:2] Reserved Bit[5] exposure mode select during QBC HDR 0 : short exposure determined by ratio (controlled by EXPO_RATIO(0x0222)) 1 : short exposure controlled by direct control Bit[6] Digital Gain mode select during QBC HDR 0 : combined gain used 1 : separate gain used
	0x3140	[2:0]	HDR_FNCSEL	QBC HDR function select 4 : QBC HDR Others : reserved
	0x3C0F	[7:0]	TC_POST_GAIN	Digital Gain after tone control Gain = TC_POST_GAIN / 32 0x10 : Gain = x0.5 0x20 : Gain = x1.0 0x40 : Gain = x2.0 0xFF: Gain = x7.97 (Max)
	0x3C00	[5:0]	TC_LTC_RATIO_1	Ratio to use local tone control (*1) Local ratio = TC_LTC_RATIO_* / 32 0x00 : Global 100%, Local 0% : 0x10 : Global 50%, Local 50% : 0x20 : Global 0%, Local 100%
	0x3C01	[5:0]	TC_LTC_RATIO_2	
	0x3C02	[5:0]	TC_LTC_RATIO_3	

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	0x3C03	[5:0]	TC_LTC_RATIO_4	Others : reserved Same value is necessary to be set in five registers
	0x3C04	[5:0]	TC_LTC_RATIO_5	
	0x0B8E	[7:0]	ABS_GAIN_GR[15:8]	Gain of Gr pixel for white balance. The setting of white balance gain is mandatory.(*2) Format : 16-bit iReal Range : 0 to 255.996
	0x0B8F	[7:0]	ABS_GAIN_GR[7:0]	
	0x0B90	[7:0]	ABS_GAIN_R[15:8]	Gain of R pixel for white balance. The setting of white balance gain is mandatory. (*2) Format : 16-bit iReal Range : 0 to 255.996
	0x0B91	[7:0]	ABS_GAIN_R[7:0]	
	0x0B92	[7:0]	ABS_GAIN_B[15:8]	Gain of B pixel for white balance. The setting of white balance gain is mandatory.(*2) Format : 16-bit iReal Range : 0 to 255.996
	0x0B93	[7:0]	ABS_GAIN_B[7:0]	
	0x0B94	[7:0]	ABS_GAIN_GB[15:8]	Gain of Gb pixel for white balance. The setting of white balance gain is mandatory. (*2) Format : 16-bit iReal Range : 0 to 255.996
	0x0B95	[7:0]	ABS_GAIN_GB[7:0]	

Note 1: This table shows the relationship between TC_LTC_RATIO_* register and local tone curve ratio.

TC_LTC_RATIO_*	Local tone curve ratio
0x00	0.0%
0x01	3.1%
0x02	6.3%
0x03	9.4%
0x04	12.5%
0x05	15.6%
0x06	18.8%
0x07	21.9%
0x08	25.0%
0x09	28.1%
0x0A	31.3%
0x0B	34.4%
0x0C	37.5%
0x0D	40.6%
0x0E	43.8%
0x0F	46.9%
0x10	50.0%

TC_LTC_RATIO_*	Local tone curve ratio
0x11	53.1%
0x12	56.3%
0x13	59.4%
0x14	62.5%
0x15	65.6%
0x16	68.8%
0x17	71.9%
0x18	75.0%
0x19	78.1%
0x1A	81.3%
0x1B	84.4%
0x1C	87.5%
0x1D	90.6%
0x1E	93.8%
0x1F	96.9%
0x20	100.0%

Note 2: Gain for white balance setting value is calculated by the following formula.

Gain = ABS_GAIN_xx / 256 (xx means one of color, GR, R, B, and GB)

For example:

Gain = x1, ABS_GAIN is set to 0x0100

Gain = x1.5, ABS_GAIN is set to 0x0180

Gain = x2, ABS_GAIN is set to 0x0200

Gain = x8; ABS_GAIN is set to 0x0800

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The ratio between global tone curve and local tone curve is set by TC_LTC_RATIO_* register. The example is shown in the following figure.

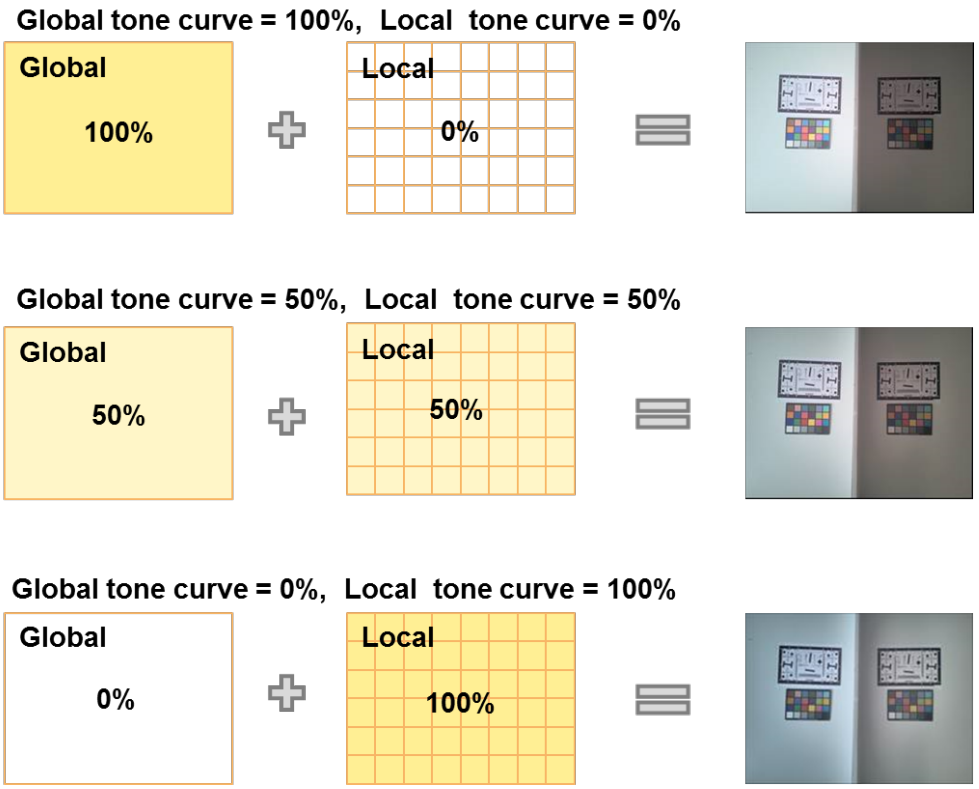


Figure 9-4 Comparison images for Local tone ratio

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9.3. LSC control

9.3.1. LSC(Y) setting for AE-HIST

In order to realize QBC HDR function, Lens Shading Correction (LSC) needs to be done by application processor, but LSC(Y) in this sensor should be enable to output AE-HIST. For this reason, it is necessary to set LSC data to this sensor beforehand in order to output correct AE-HIST from LSC(Y) block. For more details about LSC, refer to the chapter 10 Lens Shading Correction (LSC).

For more details about STATS data (AE-HIST and Flicker), refer to the chapter 9.5. STATS (statistic) data control.

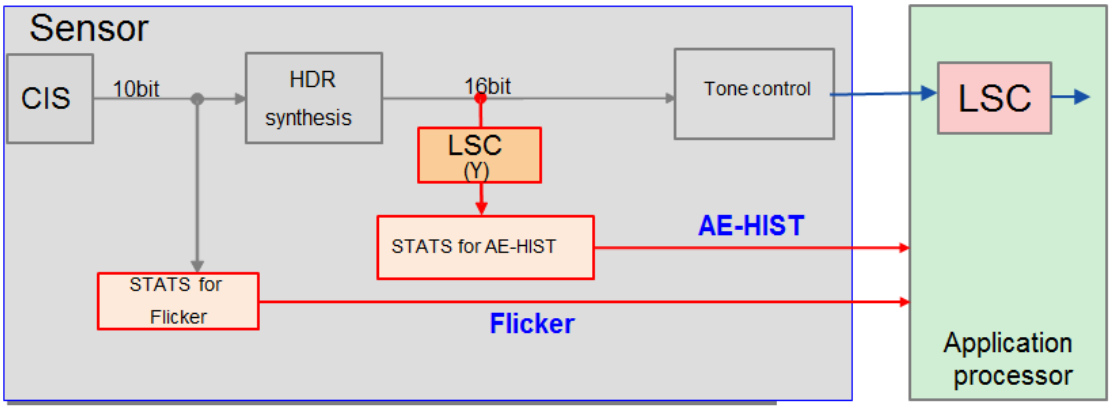


Figure 9-5 LSC location

Table 9-2 LSC(Y) control registers

I^2C Address	Bit	Name	Description
0x0B00	[0]	SHD_CORR_EN	LSC Enable 0:disable, 1:enable Note: LSC enable is necessary for QBC HDR mode
0x380C	[7:0]	LSC_APPLICATION_RATE	Set to 0x00 (For QBC HDR mode) Others : reserved
0x380D	[7:0]	LSC_APPLICATION_RATE_Y	Set to 0x80 (For QBC HDR mode) Others : reserved

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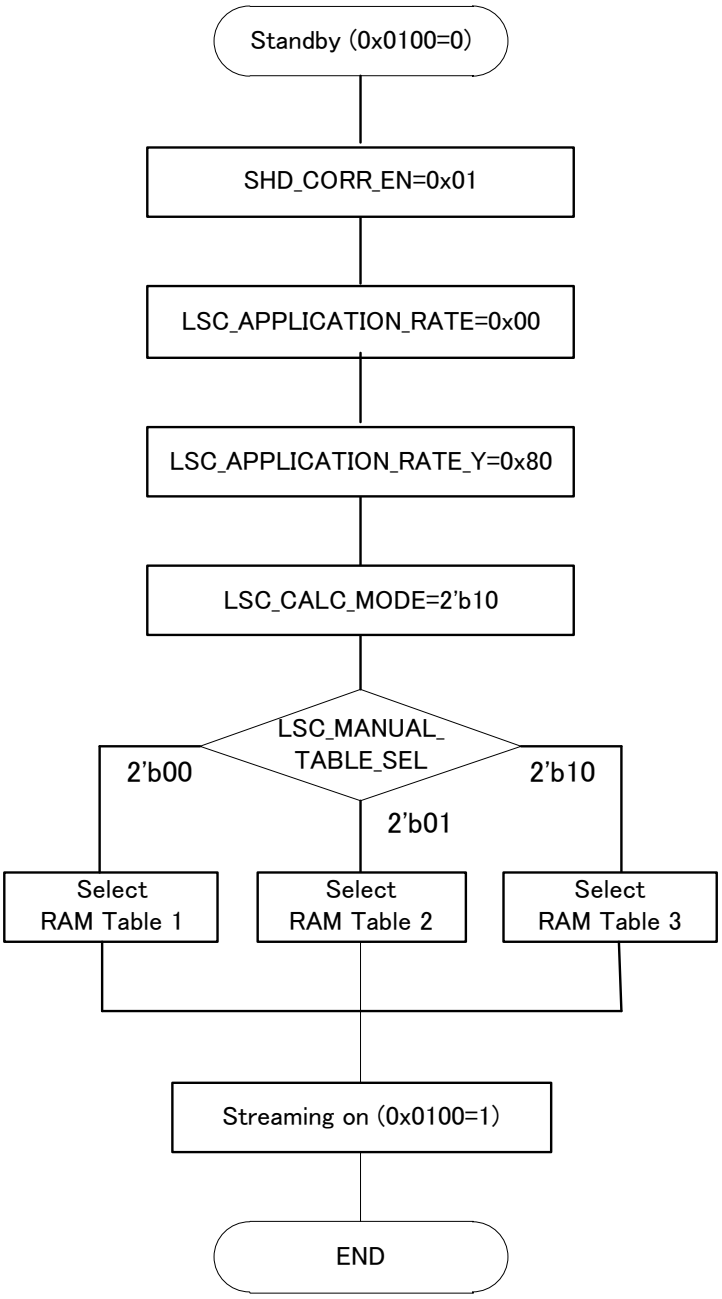


Figure 9-6 LSC related setting flow chart

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9.3.2. Calibration data usage for LSC(Y)

The calibration data to correct the whole lens shading is used for QBC HDR as shown in the following figure. This is for the histogram data without the element of lens shading.

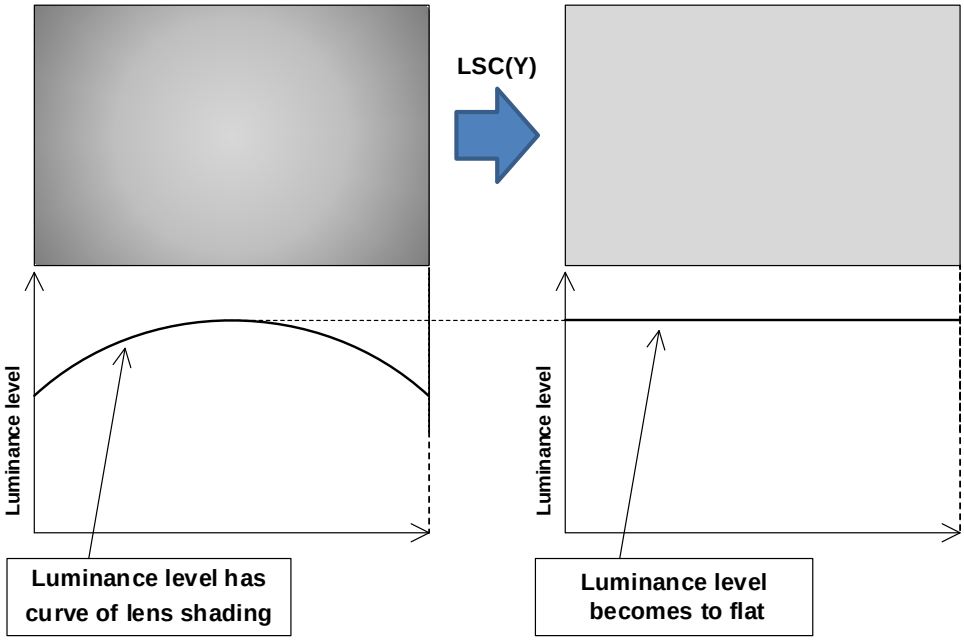


Figure 9-7 Correction level of LSC

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9.4. AE control

For QBC HDR, exposure time, analog gain and digital gain for three different exposures are set individually by the following registers. It is recommended that GPH is used for setting of these registers.

Table 9-3 AE control registers

	Address	Bit	Name	Description
I2C register	0x0202	[7:0]	COARSE_INTEG_TIME[15:8]	Exposure time for long exposure Unit : lines Format : 16-bit unsigned integer
	0x0203	[7:0]	COARSE_INTEG_TIME[7:0]	
	0x0204	[1:0]	ANA_GAIN_GLOBAL[9:8]	Analog gain for long exposure (*1)
	0x0205	[7:0]	ANA_GAIN_GLOBAL[7:0]	
	0x020E	[7:0]	DIG_GAIN_GLOBAL[15:8]	Digital gain for long exposure (*2) (*3)
	0x020F	[7:0]	DIG_GAIN_GLOBAL[7:0]	
	0x3FE0	[7:0]	MID_COARSE_INTEG_TIME[15:8]	Exposure time for middle exposure Unit : lines Format : 16-bit unsigned integer
	0x3FE1	[7:0]	MID_COARSE_INTEG_TIME[7:0]	
	0x3FE2	[1:0]	MID_ANA_GAIN_GLOBAL[9:8]	Analog gain for middle exposure (*1)
	0x3FE3	[7:0]	MID_ANA_GAIN_GLOBAL[7:0]	
	0x3FE4	[7:0]	MID_DIG_GAIN_GLOBAL[15:8]	Digital gain for middle exposure (*2) (*3)
	0x3FE5	[7:0]	MID_DIG_GAIN_GLOBAL[7:0]	
	0x0224	[7:0]	ST_COARSE_INTEG_TIME[15:8]	Exposure time for short exposure Unit : lines Format : 16-bit unsigned integer
	0x0225	[7:0]	ST_COARSE_INTEG_TIME[7:0]	
	0x0216	[1:0]	ST_ANA_GAIN_GLOBAL[9:8]	Analog gain for short exposure (*1)
	0x0217	[7:0]	ST_ANA_GAIN_GLOBAL[7:0]	
	0x0218	[7:0]	ST_DIG_GAIN_GLOBAL[15:8]	Digital gain for short exposure (*2) (*3)
	0x0219	[7:0]	ST_DIG_GAIN_GLOBAL[7:0]	

Note 1: Format: 16 bit unsigned integer.

Range: 112 to 960

Gain [times] = 1024 / (1024 - **ANA_GAIN_GLOBAL)

Note 2: Format: 16 bit unsigned integer

Range: 0x0100 to 0x0FFF

Gain [times] = **DIG_GAIN_GLOBAL / 256

Note 3: All color as DIG_GAIN_MODE is supported for QBC HDR. Individual gain is not available.

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In QBC HDR, ANA_GAIN_GLOBAL, DIG_GAIN_GLOBAL and COARSE_INTEG_TIME settings for three different exposures (long, middle and short) should be always keeping the relationship as shown in the following formula. The ratio Long/Middle is used as QBC HDR ratio. Maximum ratio is eight (8).

Long / Middle = Middle / Short

Long = COARSE_INTEG_TIME
x (1024 / (1024 - ANA_GAIN_GLOBAL)
x ((DIG_GAIN_GLOBAL / 256) + 1)

Middle = MID_COARSE_INTEG_TIME
x (1024 / (1024 – MID_ANA_GAIN_GLOBAL)
x ((MID_DIG_GAIN_GLOBAL / 256) + 1)

Short = ST_COARSE_INTEG_TIME
x (1024 / (1024 – ST_ANA_GAIN_GLOBAL)
x ((ST_DIG_GAIN_GLOBAL / 256) + 1)

Especially, the settings for three different exposures should be always keeping the following relationship.

COARSE_INTEG_TIME ≥ MID_COARSE_INTEG_TIME ≥ ST_COARSE_INTEG_TIME

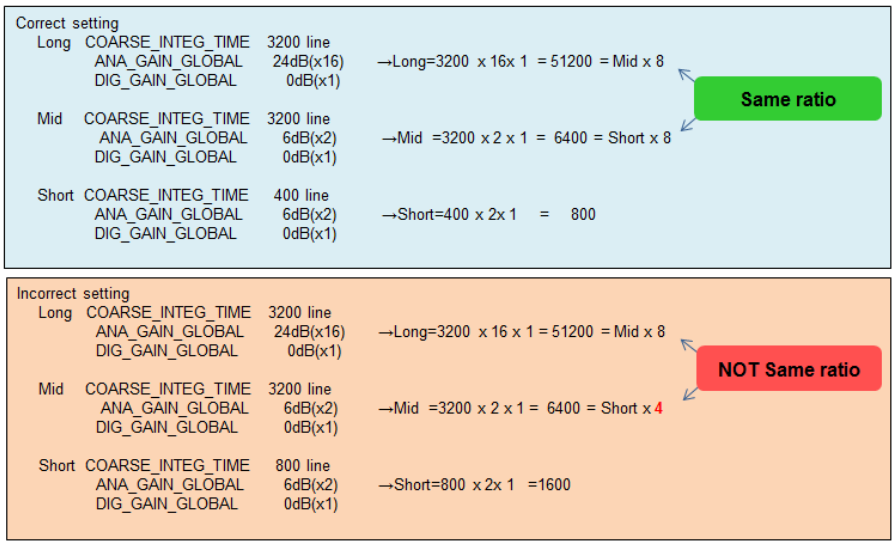


Figure 9-8 Example of exposure setting

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9.5. STATS (statistic) data control

9.5.1. Block diagram for STATS

Block diagram for STATS is shown in the follow figure.

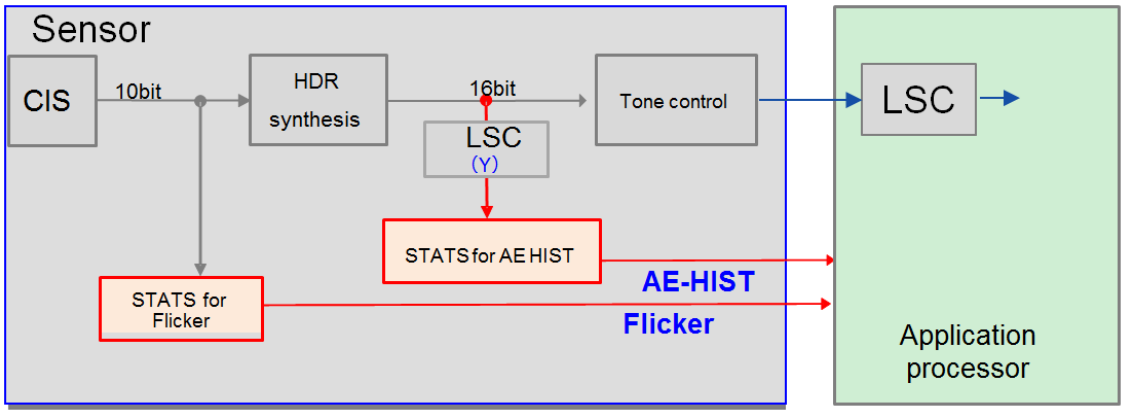


Figure 9-9 Block diagram for STATS

The following two kinds of STATS are supported in this sensor.

AE-HIST	: Histogram of Y (Luminance) signal in AE-HIST area for AE control
Flicker	: Histogram of Y (Luminance) signal in Flicker area for flicker detection.

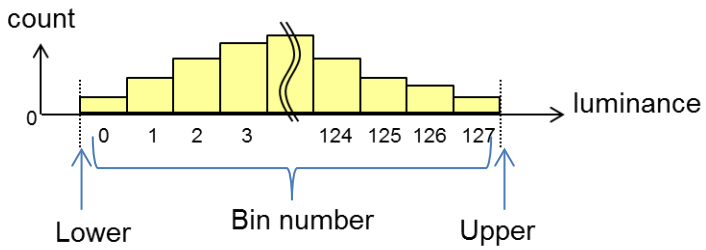
AE-HIST is made from after QBC HDR synthesis signal, and Flicker data is made from before QBC HDR synthesis signal.

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9.5.2. AE-HIST

AE-HIST is the histogram which can output two types of pixel count against the linear and log scale simultaneously. The linear scale of AE-HIST is output a total number of 128 Bin number. The log scale of AE-HIST is output a total number of 16 Bin number. AE-HIST have two of auto and manual mode, and these mode for linear and log scale can be set individually. In auto mode, AE-HIST data covers all luminance level. Lower and upper signal levels are determined by the long and middle exposure ratio, the lower limit is set to zero (0) and the upper limit set to maximum. In manual mode, lower and upper signal level can change manually by the registers shown in Table 9-4 to output partial luminance level. One bin number consists of 24 bit data. It is recommended that GPH is used for setting of these registers.

Linear scale output example



Log scale output example

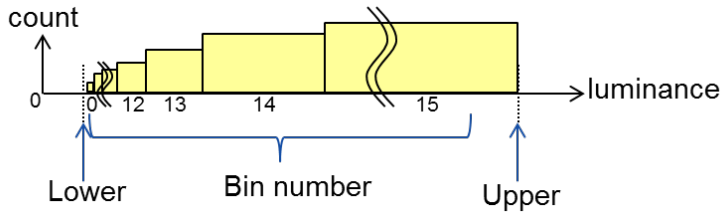


Figure 9-10 AE-HIST data

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Table 9-4 Register for data of AE-HIST

I^2CAddress	Bit	Name	Description
0x3C05	[0]	AEHIST_LINER_AUTO_THRETH	0 : Lower and upper signal level are decided by the register 0x3C06 – 0x3C09 1 : Lower and upper level are decided by AE setting automatically
0x3C06	[7:0]	AEHIST_LINER_LOWER_TH[15:8]	Lower signal level for linear scale
0x3C07	[7:0]	AEHIST_LINER_LOWER_TH [7:0]	
0x3C08	[7:0]	AEHIST_LINER_UPPER_TH [15:8]	Upper signal level for linear scale
0x3C09	[7:0]	AEHIST_LINER_UPPER_TH [7:0]	
0x3C0A	[0]	AEHIST_LOG_AUTO_THRETH	0 : Lower and upper signal level are decided by the register 0x3C0B– 0x3C0E 1 : Lower and upper level are decided by AE setting automatically
0x3C0B	[7:0]	AEHIST_LOG_LOWER_TH [15:8]	Lower signal level for log scale
0x3C0C	[7:0]	AEHIST_LOG_LOWER_TH [7:0]	
0x3C0D	[7:0]	AEHIST_LOG_UPPER_TH [15:8]	Upper signal level for log scale
0x3C0E	[7:0]	AEHIST_LOG_UPPER_TH [7:0]	

AE-HIST area is defined by register as shown in the table.

Table 9-5 Register for area of AE-HIST

	Address	Bit	Name	Description
I2C register	0x37E0	[4:0]	AEHIST_AREA_X_OFFSET[12:8]	For X_OFFSET Range: 0 - (maximum H output size - 1)
	0x37E1	[7:0]	AEHIST_AREA_X_OFFSET[7:0]	
	0x37E2	[4:0]	AEHIST_AREA_Y_OFFSET[12:8]	For Y_OFFSET Range: 0 - (maximum V output size - 1)
	0x37E3	[7:0]	AEHIST_AREA_Y_OFFSET[7:0]	
	0x37E4	[4:0]	AEHIST_AREA_WIDTH[12:8]	For Width Range: 1 - (maximum H output size)
	0x37E5	[7:0]	AEHIST_AREA_WIDTH[7:0]	
	0x37E6	[4:0]	AEHIST_AREA_HEIGHT[12:8]	For Height Range: 1 - (maximum V output size)
	0x37E7	[7:0]	AEHIST_AREA_HEIGHT[7:0]	

Above registers are need to be set based on the output image size.

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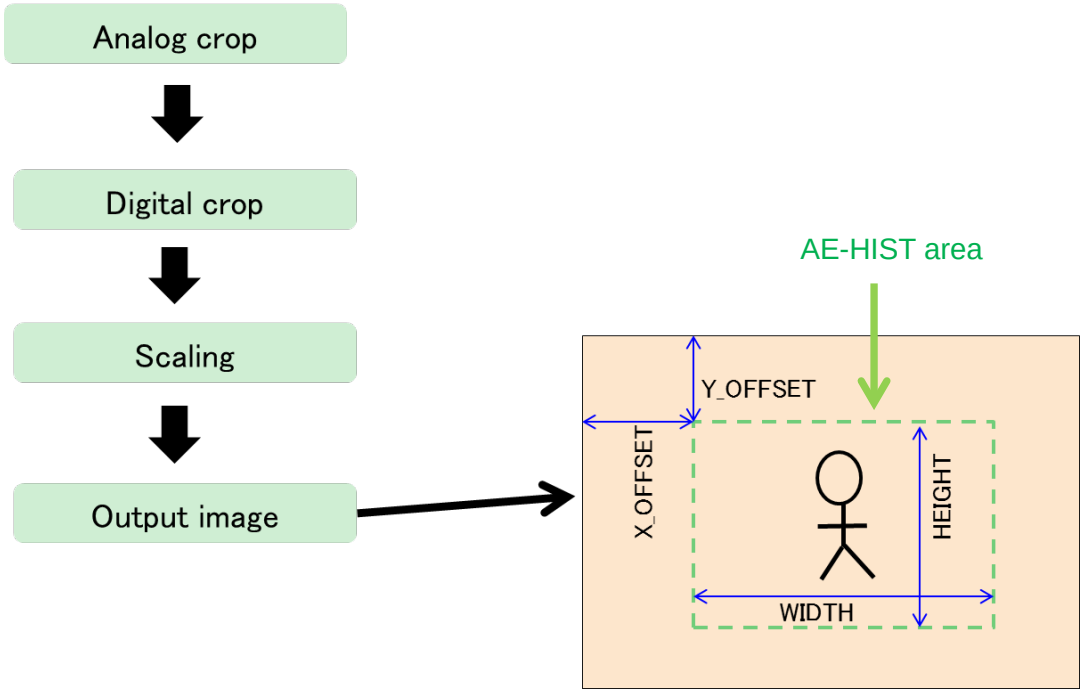


Figure 9-11 AE-HIST area

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The data format in MIPI is shown in the following figure. This is the case of RAW10.

Word count	Bit							
	7	6	5	4	3	2	1	0
Packet header	Data ID							
	16bit word count							
	ECC							
1	Bin0 for linear[23:16]							
2	Bin0 for linear[15:8]							
3	Bin0 for linear[7:0]							
4	0	1	0	1	0	1	0	1
5	0	1	0	1	0	1	0	1
6	Bin1 for linear[23:16]							
7	Bin1 for linear[15:8]							
8	Bin1 for linear[7:0]							
9	0	1	0	1	0	1	0	1
10	0	1	0	1	0	1	0	1
…	…							
636	Bin127 for linear[23:16]							
637	Bin127 for linear[15:8]							
638	Bin127 for linear[7:0]							
639	0	1	0	1	0	1	0	1
640	0	1	0	1	0	1	0	1
641	Bin0 for scale[23:16]							
642	Bin0 for scale[15:8]							
643	Bin0 for scale[7:0]							
644	0	1	0	1	0	1	0	1
645	0	1	0	1	0	1	0	1
…	…							
716	Bin15 for scale[23:16]							
717	Bin15 for scale[15:8]							
718	Bin15 for scale[7:0]							
719	0	1	0	1	0	1	0	1
720	0	1	0	1	0	1	0	1
721	0	1	0	1	0	1	0	1
722	0	1	0	1	0	1	0	1

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723	0	1	0	1	0	1	0	1
724	0	1	0	1	0	1	0	1
725	0	1	0	1	0	1	0	1
…	…							
X_out_size x 1.25	0	1	0	1	0	1	0	1
Packet footer	16bit checksum							

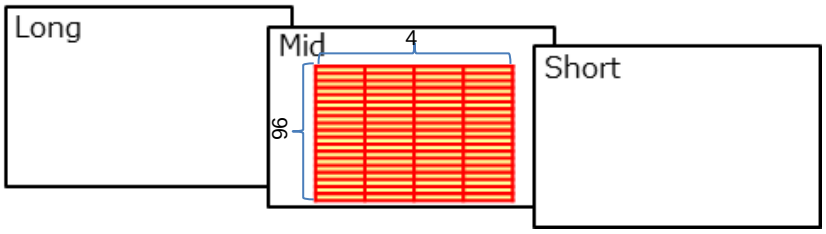
Figure 9-12 MIPI data format for AE HIST

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9.5.3. Flicker

Flicker is the data used for flicker detection. This data shows the averaged luminance level for each block. Flicker has two output modes. One is the histogram of one exposure selected by register setting, and the number of block is 96x4. Another is the histogram of three exposures, and the number of block of each exposure is 32x4. The following figure shows the block image for two modes.

The case of one exposure



The case of three exposures

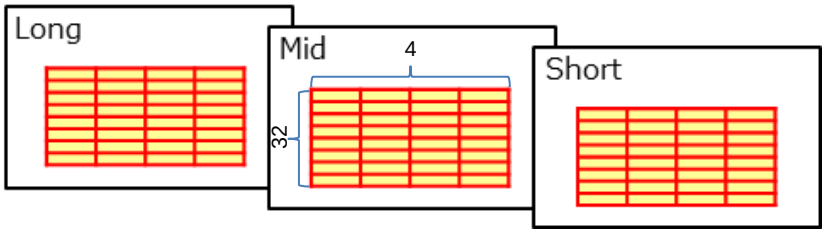


Figure 9-13 STATS data for Flicker

The following table shows the register to select the mode and exposure.

Table 9-6 Flicker data control registers

	Address	Bit	Name	Description
I2C register	0x37F8	[1:0]	FLK_EXP_SEL	Flicker data detection setting 0x0 : Long 0x1 : Middle 0x2 : Short 0x3 : Long/Middle/Short Three exposure

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The area to make the Flicker data is set by register setting, and definition of area is shown in the following figure. The registers are set to output image size.

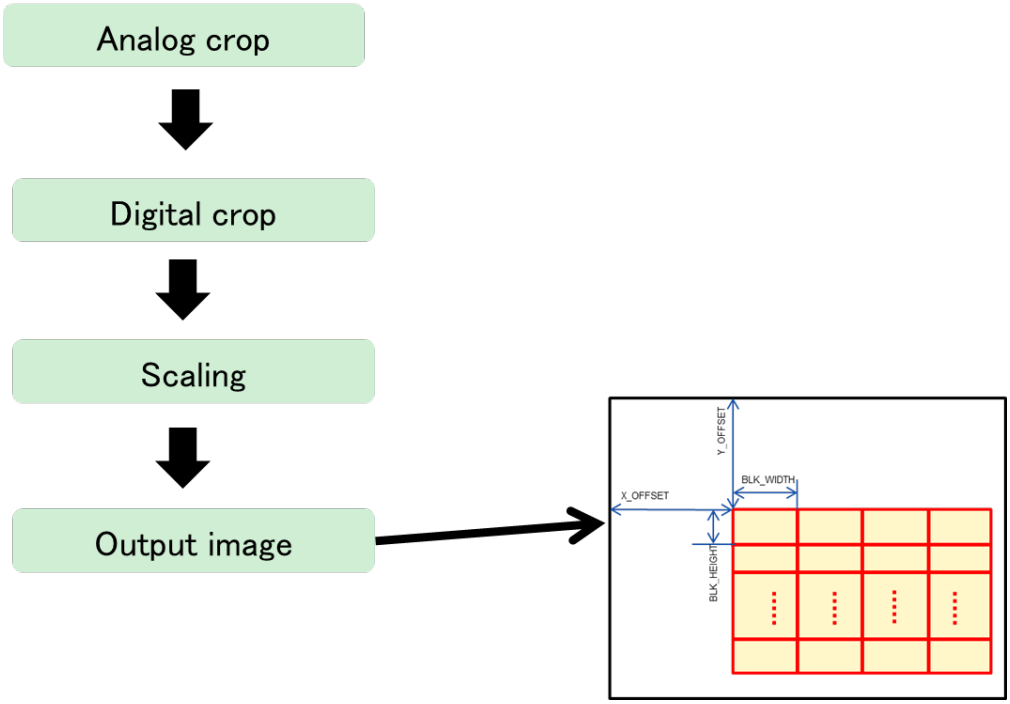


Figure 9-14 Area for Flicker

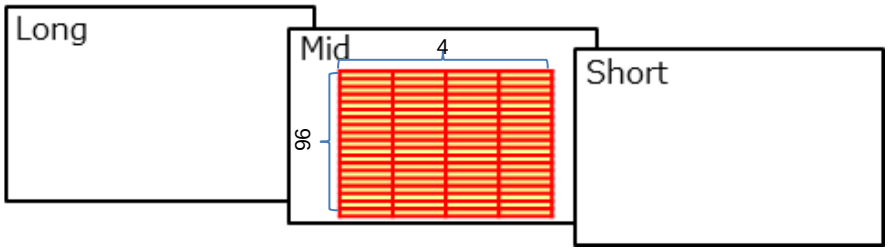
Table 9-7 Register for area of Flicker

I^2C Address	Bit	Name	Description
0x37F0	[4:0]	FLK_AREA_X_OFFSET[12:8]	For X_OFFSET Range : 0 – (maximum H output size -8) Multiple of 2
0x37F1	[7:0]	FLK_AREA_X_OFFSET[7:0]	
0x37F2	[4:0]	FLK_AREA_Y_OFFSET[12:8]	For Y_OFFSET Range : 0 – (maximum V output size -8) Multiple of 2
0x37F3	[7:0]	FLK_AREA_Y_OFFSET[7:0]	
0x37F4	[4:0]	FLK_AREA_BLK_WIDTH[12:8]	For Width of block Range : 2 – (maximum H output size / 4) Multiple of 2
0x37F5	[7:0]	FLK_AREA_BLK_WIDTH[7:0]	
0x37F6	[4:0]	FLK_AREA_BLK_HEIGHT[12:8]	For Height of block Range : 2 – (maximum V output size / 32) Multiple of 2
0x37F7	[7:0]	FLK_AREA_BLK_HEIGHT[7:0]	

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9.5.3.1. One exposure mode

Image area is divided by 4 blocks in X direction, and 96 blocks in Y direction. The following figure shows the case which middle is selected.



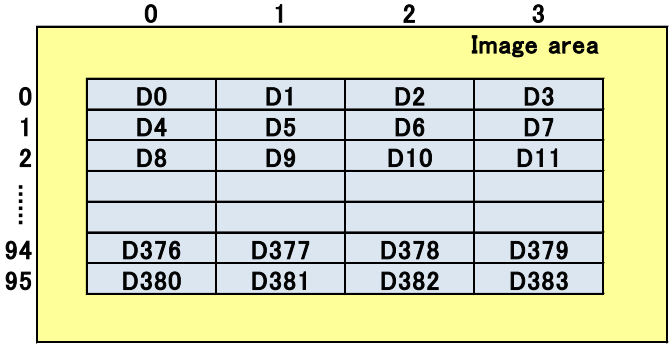


Figure 9-15 STATS data of one exposure mode

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The RAW10 data format in MIPI is shown in following figure.

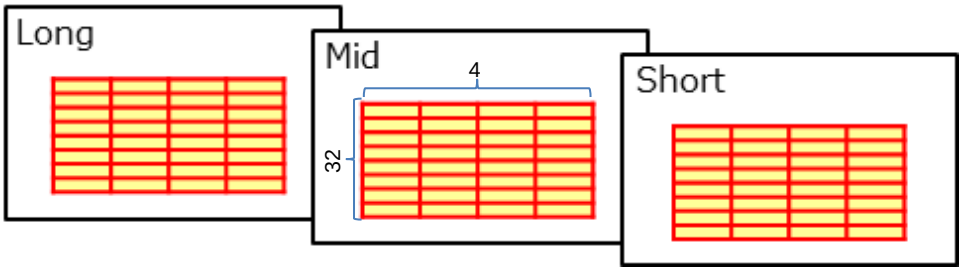
Word count	Bit							
	7	6	5	4	3	2	1	0
Packet header	Data ID							
	16 bit word count							
	ECC							
1	D0 for one exposure [9:2]							
2	D0[1:0]		0	1	0	1	0	1
3	D1 for one exposure [9:2]							
4	D1[1:0]		0	1	0	1	0	1
5	0	1	0	1	0	1	0	1
6	D2 for one exposure [9:2]							
7	D2[1:0]		0	1	0	1	0	1
8	D3 for one exposure [9:2]							
9	D3[1:0]		0	1	0	1	0	1
10	0	1	0	1	0	1	0	1
⋮	⋮							
956	D382 for long exposure [9:2]							
957	D382[1:0]		0	1	0	1	0	1
958	D383 for long exposure [9:2]							
959	D383[1:0]		0	1	0	1	0	1
960	0	1	0	1	0	1	0	1
961	0	1	0	1	0	1	0	1
962	0	1	0	1	0	1	0	1
963	0	1	0	1	0	1	0	1
964	0	1	0	1	0	1	0	1
965	0	1	0	1	0	1	0	1
⋮	⋮							
X_out_size x 1.25	0	1	0	1	0	1	0	1
Packet footer	16 bit checksum							

Figure 9-16 STATS data of one exposure mode

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9.5.3.2. Three exposure mode

Image area is divided by 4 blocks in X direction, and 32 blocks in Y direction.



Long

	0	1	2	3
0	D0	D1	D2	D3
1	D4	D5	D6	D7
2	D8	D9	D10	D11
⋮				
30	D376	D377	D378	D379
31	D124	D125	D126	D127

Middle

0	D128	D129	D130	D131
1	D132	D133	D134	D135
2	D136	D137	D138	D139
⋮				
30	D248	D249	D250	D251
31	D252	D253	D254	D255

Short

0	D256	D257	D258	D259
1	D260	D261	D262	D263
2	D264	D265	D266	D267
⋮				
30	D376	D377	D378	D379
31	D380	D381	D382	D383

Figure 9-17 STATS data for all exposure

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The RAW10 data format in MIPI is shown in the following figure.

Word count	Bit							
	7	6	5	4	3	2	1	0
Packet header	Data ID							
	16 bit word count							
	ECC							
1	D0 for long exposure [9:2]							
2	D0_long[1:0]		0	1	0	1	0	1
3	D1 for long exposure [9:2]							
4	D1_long[1:0]		0	1	0	1	0	1
5	0	1	0	1	0	1	0	1
6	D2 for long exposure [9:2]							
7	D2_long[1:0]		0	1	0	1	0	1
8	D3 for long exposure [9:2]							
9	D3_long[1:0]		0	1	0	1	0	1
10	0	1	0	1	0	1	0	1
11	D128 for middle exposure [9:2]							
12	D128_middle[1:0]		0	1	0	1	0	1
13	D129 for middle exposure [9:2]							
14	D129_middle[1:0]		0	1	0	1	0	1
15	0	1	0	1	0	1	0	1
16	D130 for middle exposure [9:2]							
17	D130_middle[1:0]		0	1	0	1	0	1
18	D131 for middle exposure [9:2]							
19	D131_middle[1:0]		0	1	0	1	0	1
20	0	1	0	1	0	1	0	1
21	D256 for short exposure [9:2]							
22	D256_short[1:0]		0	1	0	1	0	1
23	D257 for short exposure [9:2]							
24	D257_short[1:0]		0	1	0	1	0	1
25	0	1	0	1	0	1	0	1
26	D258 for short exposure [9:2]							
27	D258_short[1:0]		0	1	0	1	0	1
28	D259 for short exposure [9:2]							
29	D259_short[1:0]		0	1	0	1	0	1
30	0	1	0	1	0	1	0	1
31	D4 for long exposure [9:2]							
32	D4_long[1:0]		0	1	0	1	0	1
33	D5 for long exposure [9:2]							
34	D5_long[1:0]		0	1	0	1	0	1
35	0	1	0	1	0	1	0	1
36	D6 for long exposure [9:2]							
37	D6_long[1:0]		0	1	0	1	0	1
38	D7 for long exposure [9:2]							
39	D7_long[1:0]		0	1	0	1	0	1
40	0	1	0	1	0	1	0	1
41	D132 for middle exposure [9:2]							
42	D132_middle[1:0]		0	1	0	1	0	1
43	D133 for middle exposure [9:2]							
44	D133_middle[1:0]		0	1	0	1	0	1
45	0	1	0	1	0	1	0	1
46	D134 for middle exposure [9:2]							
47	D134_middle[1:0]		0	1	0	1	0	1
48	D135 for middle exposure [9:2]							
49	D135_middle[1:0]		0	1	0	1	0	1
50	0	1	0	1	0	1	0	1
51	D260 for short exposure [9:2]							
52	D260_short[1:0]		0	1	0	1	0	1
53	D261 for short exposure [9:2]							
54	D261_short[1:0]		0	1	0	1	0	1
55	0	1	0	1	0	1	0	1
56	D262 for short exposure [9:2]							
57	D262_short[1:0]		0	1	0	1	0	1
58	D263 for short exposure [9:2]							
59	D263_short[1:0]		0	1	0	1	0	1
60	0	1	0	1	0	1	0	1
⋮	⋮							

Word count	Bit							
	7	6	5	4	3	2	1	0
931	D124 for long exposure [9:2]							
932	D124_long[1:0]		0	1	0	1	0	1
933	D125 for long exposure [9:2]							
934	D125_long[1:0]		0	1	0	1	0	1
935	0	1	0	1	0	1	0	1
936	D126 for long exposure [9:2]							
937	D126_long[1:0]		0	1	0	1	0	1
938	D127 for long exposure [9:2]							
939	D127_long[1:0]		0	1	0	1	0	1
940	0	1	0	1	0	1	0	1
941	D252 for middle exposure [9:2]							
942	D252_middle[1:0]		0	1	0	1	0	1
943	D253 for middle exposure [9:2]							
944	D253_middle[1:0]		0	1	0	1	0	1
945	0	1	0	1	0	1	0	1
946	D254 for middle exposure [9:2]							
947	D254_middle[1:0]		0	1	0	1	0	1
948	D255 for middle exposure [9:2]							
949	D255_middle[1:0]		0	1	0	1	0	1
950	0	1	0	1	0	1	0	1
951	D380 for short exposure [9:2]							
952	D380_short[1:0]		0	1	0	1	0	1
953	D381 for short exposure [9:2]							
954	D381_short[1:0]		0	1	0	1	0	1
955	0	1	0	1	0	1	0	1
956	D382 for short exposure [9:2]							
957	D382_short[1:0]		0	1	0	1	0	1
958	D383 for short exposure [9:2]							
959	D383_short[1:0]		0	1	0	1	0	1
960	0	1	0	1	0	1	0	1
⋮	⋮							
X_out_size x 1.25	0	1	0	1	0	1	0	1
Packet footer	16 bit checksum							

Figure 9-18 MIPI data format for Flicker

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9.6. STATS data in image

These STATS data put on end of the frame as shown in the following figure. Each packet for STATS has the different data type in packet header.

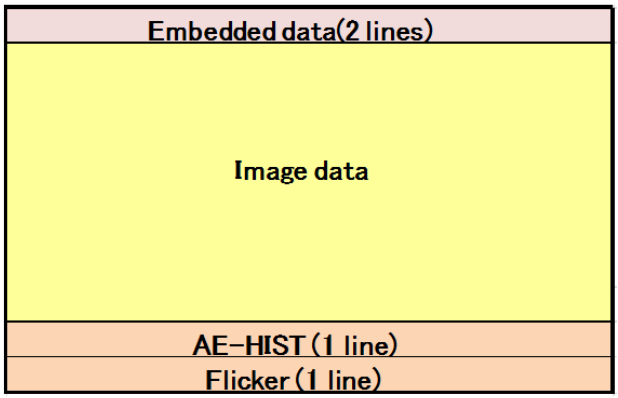


Figure 9-19 Location of STATS data in frame

Table 9-8 Data type

Data	Data type value of MIPI
Embedded data	0x12
Image data (RAW10)	0x2B
AE-HIST	0x32
Flicker	0x33

The output enable/disable of each STATS is set by the following register.

Table 9-9 Enable rgister for STATS

I^2C Address	Bit	Name	Description
0x323B	[0]	AEHIST_OUT_EN	MIPI AE-HIST output enable 0 : Disable 1 : Enable
0x323C	[0]	FLK_OUT_EN	MIPI Flicker output enable 0 : Disable 1 : Enable

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10. Lens Shading Correction (LSC)

This sensor has a built-in lens shading correction (LSC) function for each color (R/Gr/Gb/B) separately.

A purpose of the LSC function in this sensor is primarily for module level shading calibration of four colors. However, it can be also usable as a substitute function of LSC function as usually done in ISP.

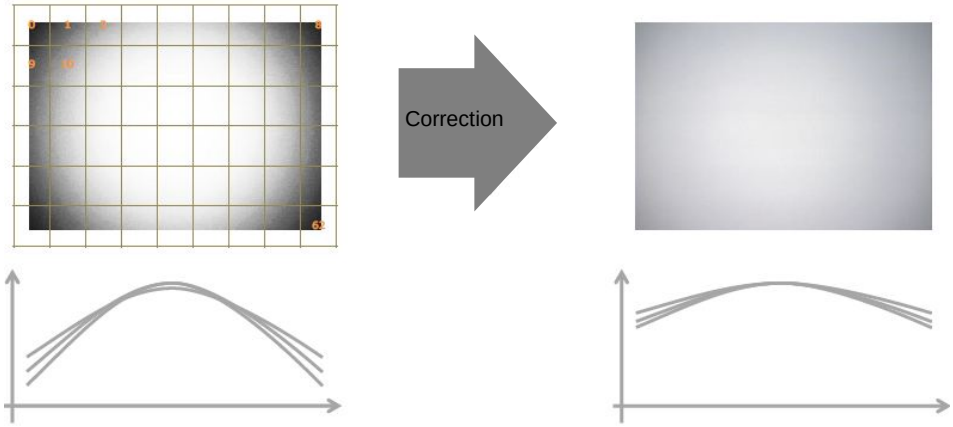


Figure 10-1 Lens Shading Correction

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10.1. LSC control registers

The following table shows control registers of LSC and knot point tables.

Table 10-1 LSC configuration registers

I^2C Address	Bit	Name	default	Description
0x0b00	[0]	SHD_CORR_EN	1'b0	LSC Enable 0:disable, 1:enable
0x380C	[7:0]	LSC_APPLICATION_RATE	0x80	LSC application rate Set to 0x80 (Other than QBC HDR mode) Set to 0x00 (For QBC HDR mode) others : reserved
0x3804	[1:0]	LSC_CALC_MODE	2'b10	Table select mode 2'b00: Sensor adjust (auto) mode 2'b10: Table select (manual) mode others: reserved
0x3805	[1:0]	LSC_MANUAL_TABLE_SEL	2'b00	Table selector for manual mode when LSC_CALC_MODE = 2'b10 2'b00: table1 2'b01: table2 2'b10: table3 2'b11:reserved
0x9750	[1:0]	KNOT_FMT_R	2'b00	define Knot point (R pixel) *1 2'b00: u2.8, 2'b01:u3.7, 2'b10:u1.9, 2'b11:u4.6
0x9751	[1:0]	KNOT_FMT_GR	2'b00	define Knot point (Gr pixel) *1 2'b00: u2.8, 2'b01:u3.7, 2'b10:u1.9, 2'b11:u4.6
0x9752	[1:0]	KNOT_FMT_GB	2'b00	define Knot point (Gb pixel) *1 2'b00: u2.8, 2'b01:u3.7, 2'b10:u1.9, 2'b11:u4.6
0x9753	[1:0]	KNOT_FMT_B	2'b00	define Knot point (B pixel) *1 2'b00: u2.8, 2'b01:u3.7, 2'b10:u1.9, 2'b11:u4.6

Note1: copied from OTP and it should be set as common value.

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Table 10-2 LSC Knot point RAM tables

I^2C Address	Bit	Name	default	Description
0x9B00 - 0x9C7F	[9:0]	KNOT_POINT_TABLE_R	0x100	Knot point parameter R for Table1/2/3 *2, *3 16(H) * 12(V) * 10(resolution)
0x9C80 - 0x9DFF	[9:0]	KNOT_POINT_TABLE_GR	0x100	Knot point parameter Gr for Table1/2/3 *2, *3 16(H) * 12(V) * 10(resolution)
0x9E00 - 0x9F7F	[9:0]	KNOT_POINT_TABLE_GB	0x100	Knot point parameter Gb for Table1/2/3 *2, *3 16(H) * 12(V) * 10(resolution)
0x9F80 - 0xA0FF	[9:0]	KNOT_POINT_TABLE_B	0x100	Knot point parameter B for Table1/2/3 *2, *3 16(H) * 12(V) * 10(resolution)
0x9014	[1:0]	LSC_TABLE_SEL	0x00	Select LSC knot point table number for I^2C access. 0: Prohibited *1 1: Table1 2: Table2 3: Table3
0xB870	[1:0]	LSC_TABLE_STATUS	0x00	Show selected LSC knot point table number for I2C access. 0: reserved 1: Table1 2: Table2 3: Table3
0x4439	[0]	SETTING_LOAD	0x00	LSC knot point table switching trigger *4 0 : disable 1 : enable (automatically return to 0 after settings are done)

Note1: Must be set 1 or 2 or 3 when set the register of SETTING_LOAD to 1.

Note2: Automatically loaded the calibration data after power-on when the calibration data is stored in OTP.

Note3: This sensor has three RAM tables for LSC knot point. The same I2C address is allocated to three RAM tables, so the user needs to switch RAM table and set LSC knot point during SW standby.

Note4: During switching of LSC knot point table, I2C write and read access are prohibit for 50 µs after setting SETTING_LOAD to 1

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The following flowchart describes how to set three RAM tables.

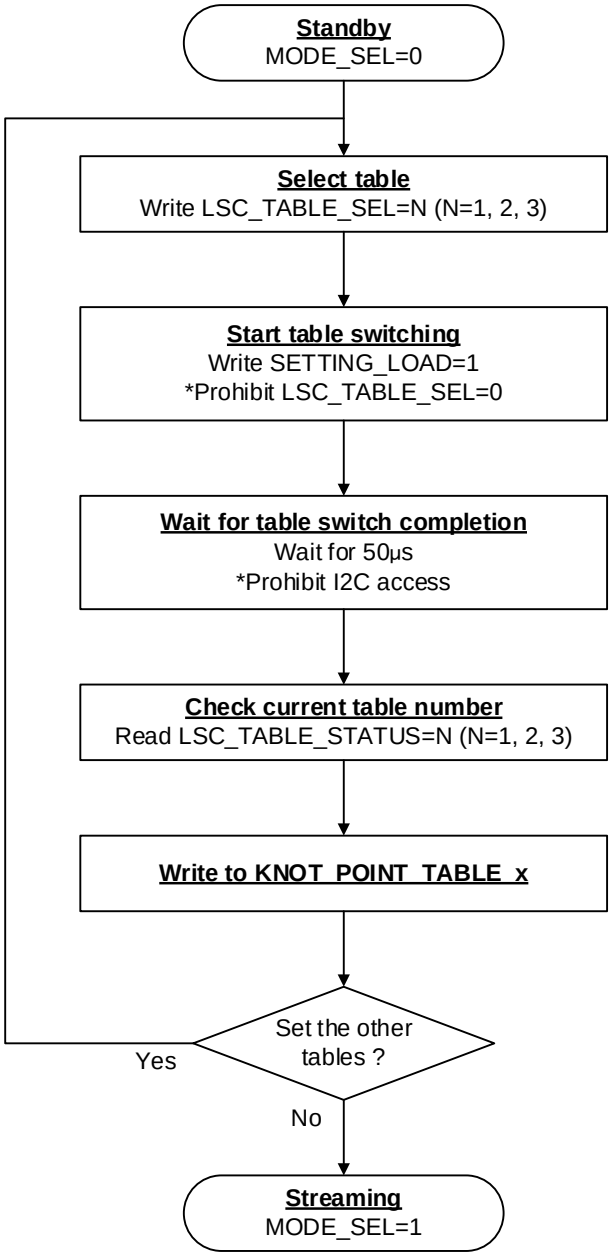


Figure 10-2 LSC knot point table setting

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Table 10-3 Sensor adjust (auto) mode configuration registers

I^2C Address	Bit	Name	default	Description
0x9742 0x9743	[15:8] [7:0]	LSC_TH_COLOR_BLEND_1	16'h0000	threshold level of color temperature boundary 1 >= WBG_R_GR : select table3
0x9744 0x9745	[15:8] [7:0]	LSC_TH_COLOR_BLEND_2	16'h0000	threshold level of color temperature boundary 2 > WBG_R_GR: select blend table(2&3)
0x9746 0x9747	[15:8] [7:0]	LSC_TH_COLOR_BLEND_3	16'h0000	threshold level of color temperature boundary 3 >= WBG_R_GR : select table2
0x9748 0x9749	[15:8] [7:0]	LSC_TH_COLOR_BLEND_4	16'h0000	threshold level of color temperature boundary 4 > WBG_R_GR : select blend table(1&2) <= WBG_R_GR : select table1
0xBAA6 0xBAA7	[15:8] [7:0]	WBG_R_GR	16'h0000	white balance ratio (ABS_GAIN_R / ABS_GAIN_GR) (u8.8) - When LSC_CALC_MODE = 2'b00 (auto mode) is valid *This value vs. color temperature relation shall be defined according to the relation between RAM table and color temperature.
0x974C 0x974D	[15:0] [7:0]	LSC_OUTDOOR_TH_IN	16'h03E3	Need to change by LSC_CALC_MODE setting. LSC_CALC_MODE = 2b'00 (auto mode): 16'h0000 (Fixed value) Others : reserved
0x974E 0x974F	[15:0] [7:0]	LSC_OUTDOOR_TH_OUT	16'h01F0	Need to change by LSC_CALC_MODE setting. LSC_CALC_MODE = 2b'00 (auto mode): 16'h0000 (Fixed value) Others : reserved

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10.2. Constrained condition of LSC

- LSC is not supported in QBC RAW output mode.
- During switching of LSC knot point table, I2C write and read access are prohibited for 50 us after setting SETTING_LOAD to 1.

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10.3. Abstract of LSC system

In this sensor, LSC function supports two modes (manual and auto) and consists of an OTP table, three RAM tables and a blend block. Each table consists of 16(H) x 12(V) Knot point tables for each color.

Table 10-4 LSC function

LSC function		Note
Correction algorithm		B-spline correction
Color correction mode		4 color mode (R / Gr / Gb / B)
Knot point	Horizontal	16
	Vertical	12
	Resolution	10bit
Color temperature correction RAM table		3 Knot point tables (Update by ISP via CCI bus. Only Table1 could be read from OTP. See Figure 10-3)
Correction method		Calibration (manual) mode Sensor adjust (auto) mode
OTP memory size		771Byte (768Byte x 1 table + 3Byte for LSC control)

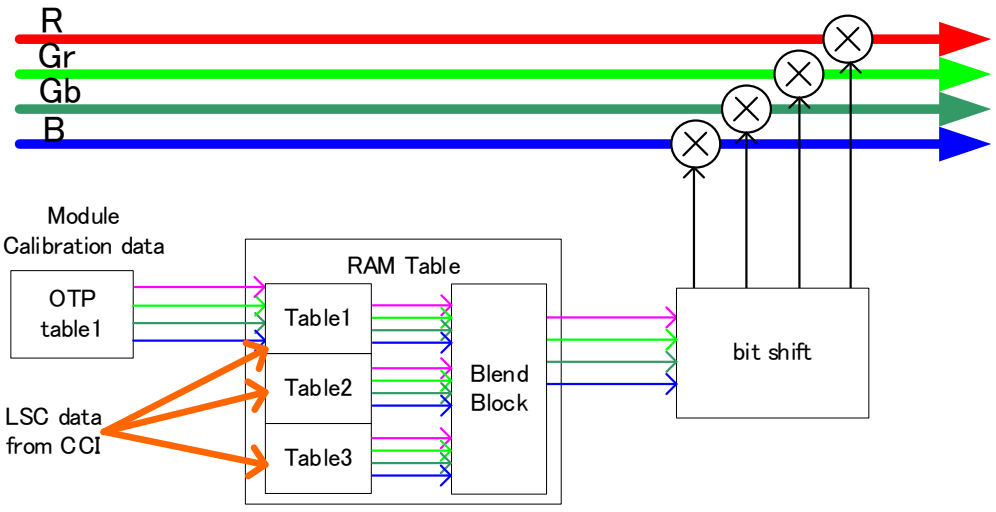


Figure 10-3 Block diagram for LSC knot point table setting

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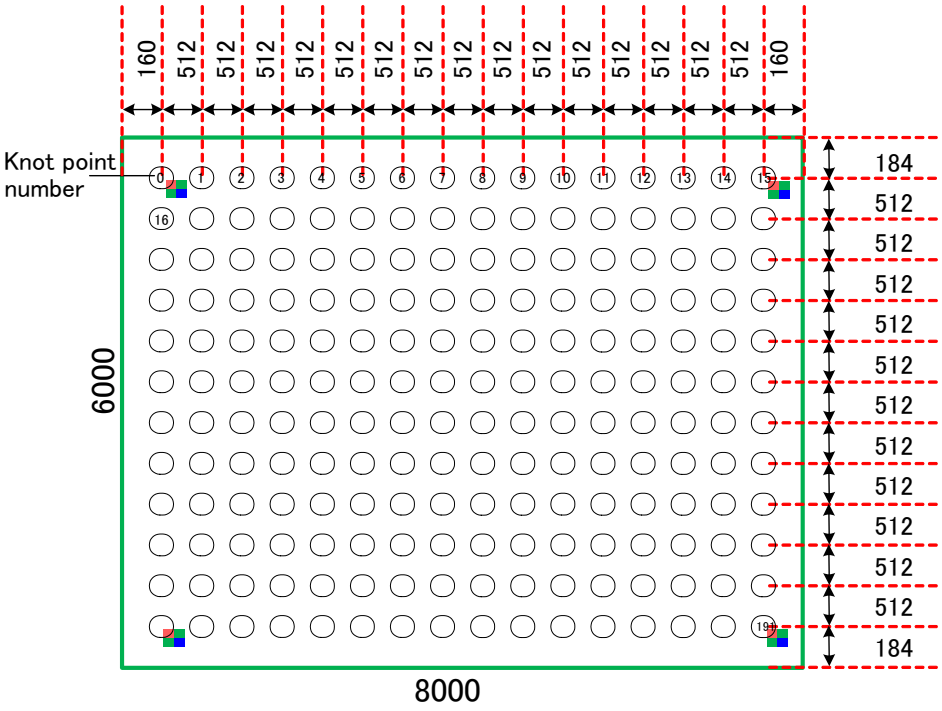


Figure 10-4 Knot point tables (16(H) x 12(V))

Calibration mode (manual mode)

Each data of Knot point tables can be stored in OTP and the data is automatically loaded into RAM Table 1 during initialization (See Figure 10-3). LSC function performs the Lens shading correction by adapting the B-spline interpolation with Knot point data.

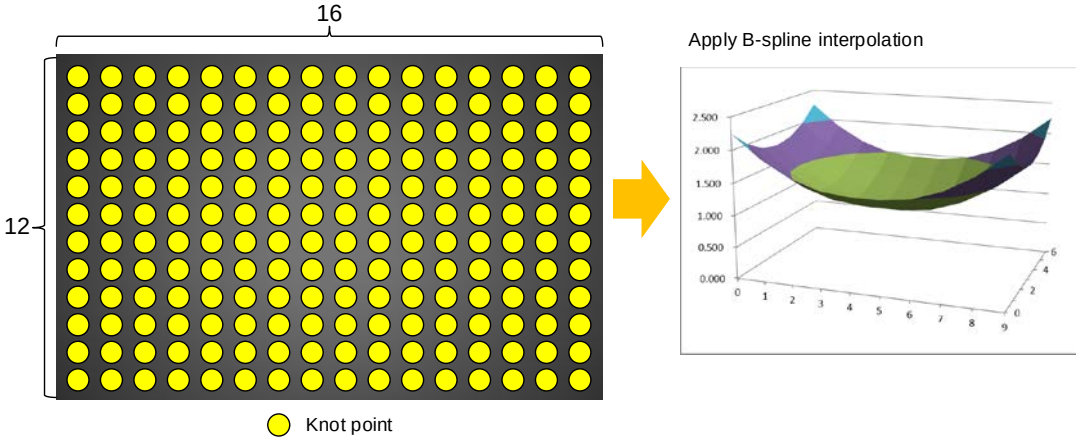
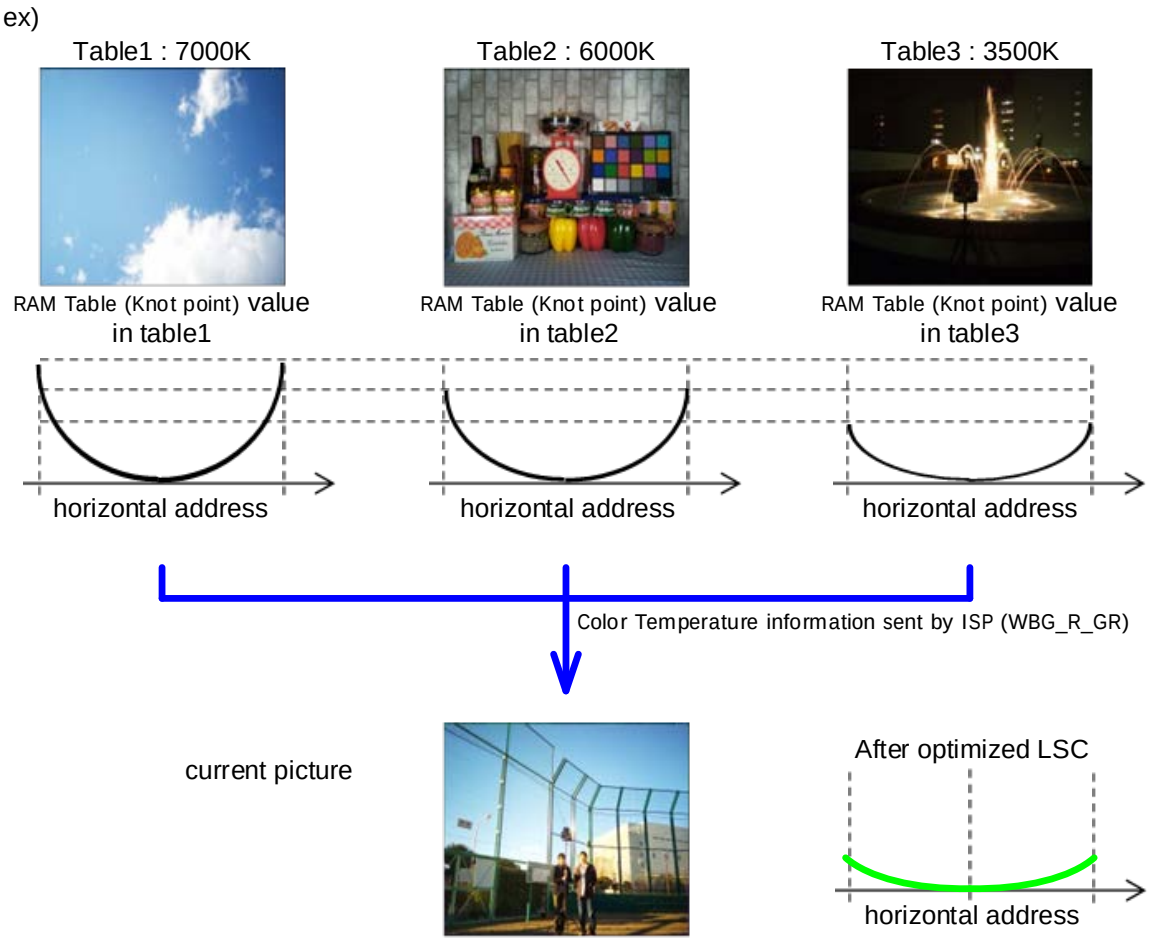


Figure 10-5 B-spline interpolation

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Sensor adjust mode (auto mode)

This mode supports generally called LSC function that is usually done in back-end ISP. Three color shading tables (RAM Tables 1-3) updated via CCI bus can be selected/blended according to color temperature information sent by ISP/AP.



* Sensor calculates optimized LSC parameters by white balance ratio

Figure 10-6 RAM(Knot) table selection(auto mode)

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ISP can get module calibration data which is stored in sensor's OTP area by reading RAM Table1 to activate Sensor adjust mode.

Then LSC data calculated and tuned over color temperature change in ISP should be transferred back to sensor RAM Table1-3.

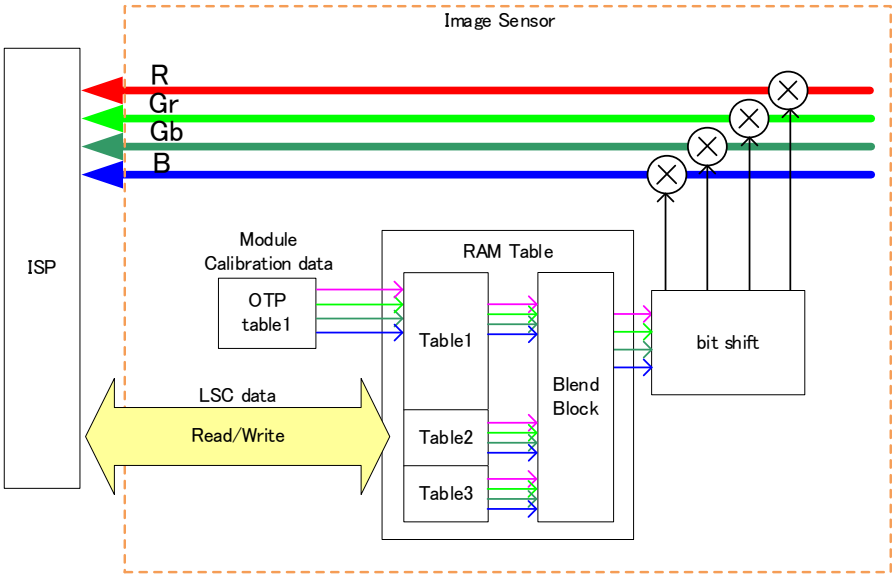


Figure 10-7 Sensor adjust mode (auto mode)

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10.4. Control scheme -- overall setting scheme

This sensor supports two Lens Shading Correction mode, “Calibration (manual) mode” and “Sensor adjust (auto) mode”. In Calibration (manual) mode, select RAM Table 1 to reflect calibration values to main RAW data path (see Figure 10-3, Figure 10-8). In Sensor adjust (auto) mode, the optimized lens shading correction setting which is calculated from the RAM table 1, 2, 3 by this sensor can be used in each scheme.

Calibration (manual) Mode

Refer to Manual mode in Figure 10-8

1: Set Software Standby (0x0100=0)

2: Set LSC function to enable (SHD_CORR_EN=1)

3: Set LSC application rate (LSC_APPLICATION_RATE =0x80)

4: Set LSC calculation mode to manual (LSC_CALC_MODE=2’b10)

5: Select the one of RAM Table (ex. RAM table1: LSC_MANUAL_TABLE_SEL=2’b00)

6: Set Streaming on (0x0100=1)

Sensor adjust (auto) mode

Refer to Auto mode in Figure 10-8

1: Set Software Standby (0x0100=0)

2: Set LSC function to enable (SHD_CORR_EN=1)

3: Set LSC application rate (LSC_APPLICATION_RATE =0x80)

4: Set LSC_OUTDOOR_TH_IN =0x0000 (*)

5: Set LSC_OUTDOOR_TH_OUT =0x0000 (*)

6: Set LSC calculation mode to auto (LSC_CALC_MODE=2’b00)

7: Set the initial settings for table blend threshold (LSC_TH_COLOR_BLEND_1/2/3/4 = user defined value, refer to Figure 10-9　LSC table calculation according to color temperature)

8: Set the initial value of color temperature (WBG_R_GR = initial value as user defined)

9: Set Streaming on (0x0100=1)

10: Update color temperature setting (WBG_R_GR=new value)

Note: The following registers are prohibited to change the value in auto mode function

0x974C、0x974D : LSC_OUTDOOR_TH_IN = 0x0000

0x974E、0x974F : LSC_OUTDOOR_TH_OUT = 0x0000

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The following two charts show overview of LSC control flow

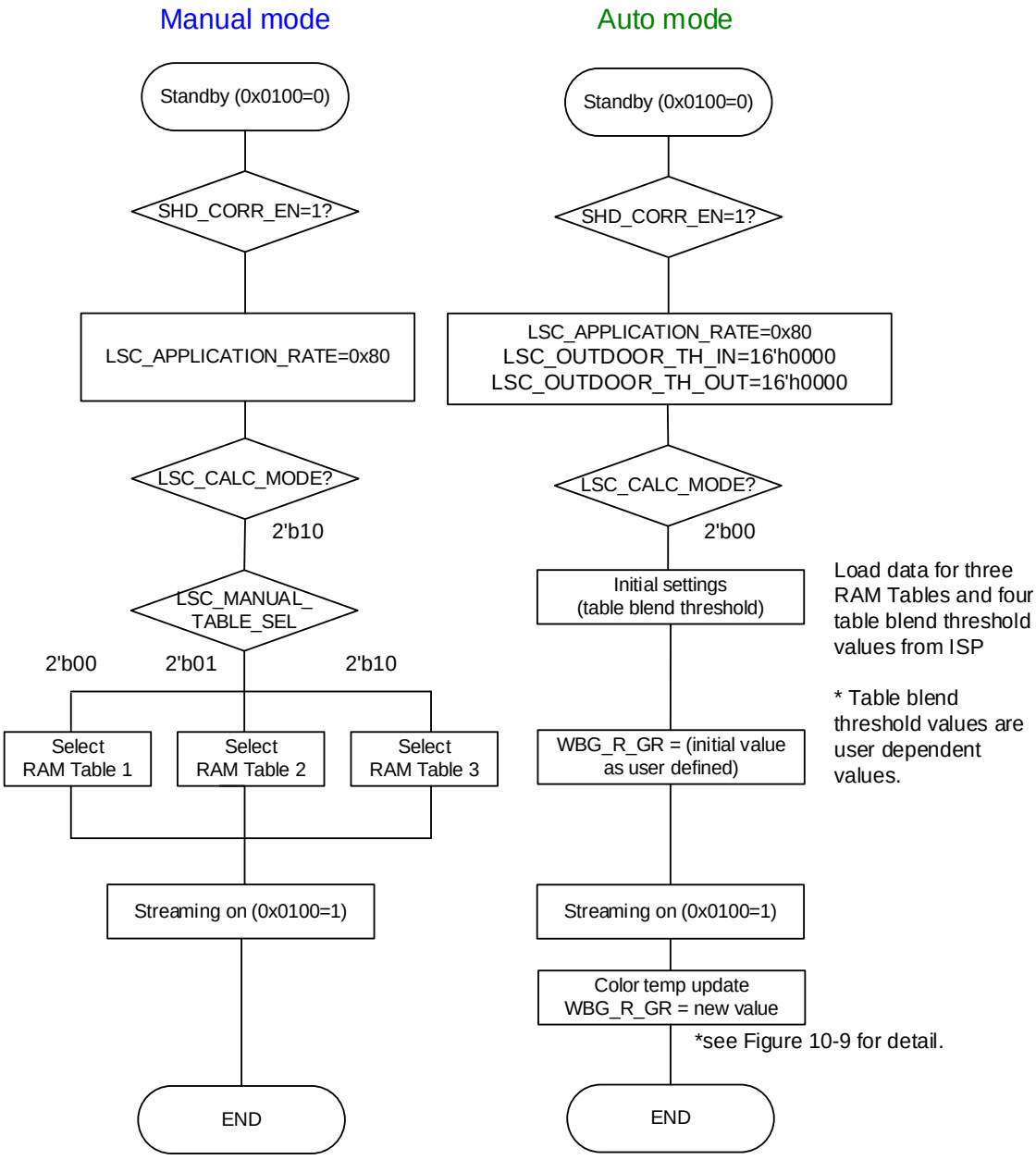


Figure 10-8 LSC control flow chart

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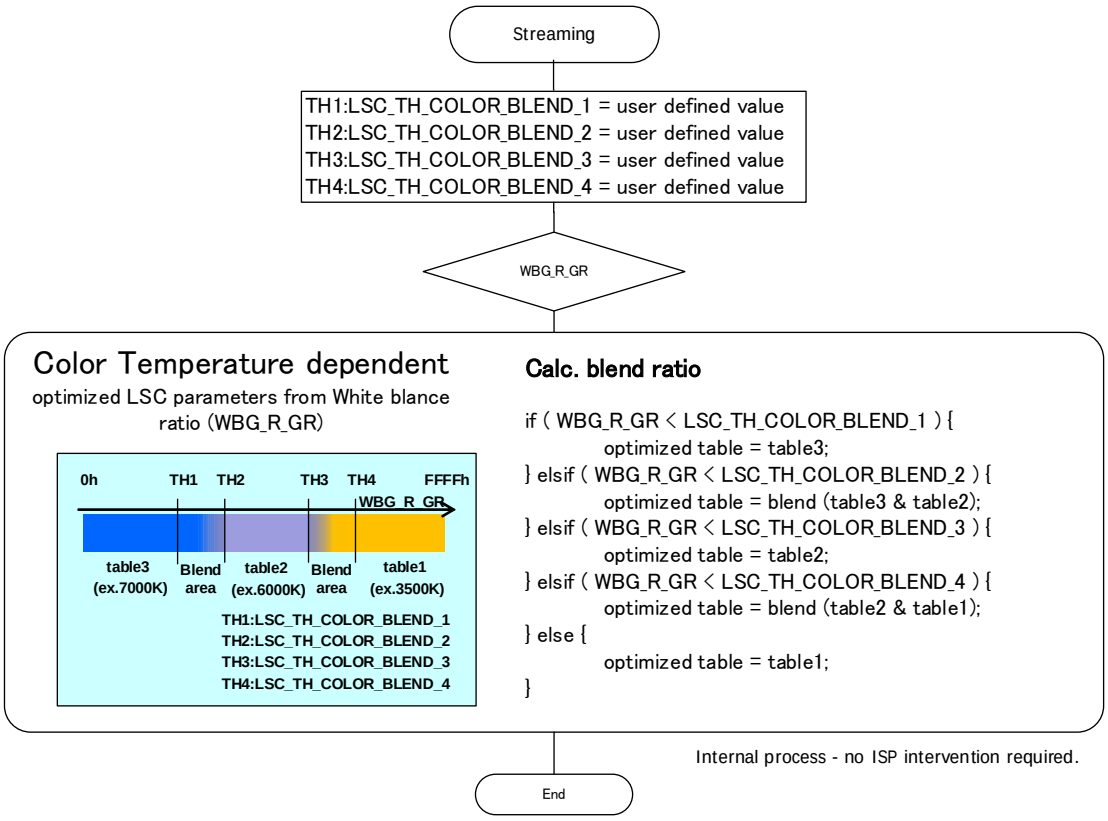


Figure 10-9 LSC table calculation according to color temperature

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10.5. LSC data information in OTP

There are two areas to store the LSC related data in OTP. One is for LSC compressed data storage area which is consists of three tables and the other is LSC control parameters storage area. These LSC data and control parameters are automatically loaded to the sensor after power on.

Relation between the sensor loaded data and OTP data

Each LSC Knot Point data stored in OTP is transformed as follows and common setting value of GLOBAL_OFFSET is used in each color(R/Gr/Gb/B) and each table.

RAM table data (Knot_xx[9:0]) is copied from OTP table with the following relation.

Knot_xx[9:0] = GLOBAL_OFFSET [9:0] + LSC_TABLE_y_zz[7:0]

Note: GLOBAL_OFFSET[9:0] (address 3136 to 3137),

LSC_TABLE_y_zz[7:0] (address 2368 to 3135)

(y=R, Gr, Gb, B; zz=00, 01,..., 191)

Knot point is an u2.8 number. E.g. 1101111111b means 3+127/ 256 in decimal.

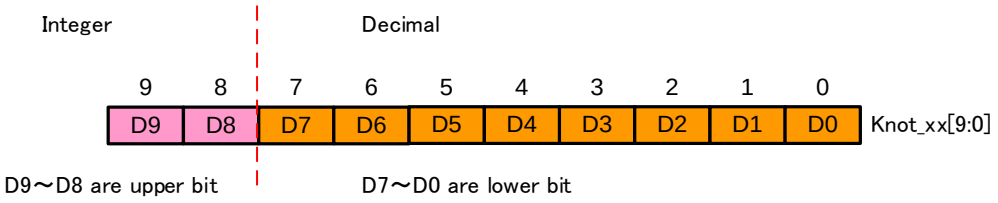


Figure 10-10 Knot point DATA structure (u2.8)

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OTP Map

		Page (dec)	Row (dec)	OTP addr (dec)	addr (hex)	MSB 7	LSB 0
Table1 (768byte)	R(192byte) = 192(knot points) x 1byte	37	148	2368	940	LSC_TABLE1_R_00[7:0]	
		37	148	2369	941	LSC_TABLE1_R_01[7:0]	
		~	~	~	~	~	
		39	159	2559	9FF	LSC_TABLE1_R_191[7:0]	
	GR(192byte) = 192(knot points) x 1byte	40	160	2560	A00	LSC_TABLE1_GR_00[7:0]	
		40	160	2561	A01	LSC_TABLE1_GR_01[7:0]	
		~	~	~	~	~	
		42	171	2751	ABF	LSC_TABLE1_GR_191[7:0]	
	GB(192byte) = 192(knot points) x 1byte	43	172	2752	AC0	LSC_TABLE1_GB_00[7:0]	
		43	172	2753	AC1	LSC_TABLE1_GB_01[7:0]	
		~	~	~	~	~	
		45	183	2943	B7F	LSC_TABLE1_GB_191[7:0]	
	B(192byte) = 192(knot points) x 1byte	46	184	2944	B80	LSC_TABLE1_B_00[7:0]	
		46	184	2945	B81	LSC_TABLE1_B_01[7:0]	
		~	~	~	~	~	
		48	195	3135	C3F	LSC_TABLE1_B_191[7:0]	
Control (3byte)	Control (3byte)	49	196	3136	C40	Reserved	GLOBAL_OFFSET[9:8]
		49	196	3137	C41	GLOBAL_OFFSET[7:0]	
		49	196	3138	C42	GLOBAL_FORMAT	

Total (771byte)

Copy from OTP
"OTP data[7:0] + GLOBAL_OFFSET[9:0]" to SRAM at Power On Initializing

(Ex.) T1_R_00[9:0] = GLOBAL_OFFSET[9:0] + LSC_TABL1_R_00[7:0]

Register Map

		I2C Address	MSB 7	LSB 0
Table1 (1536byte)	R(384byte) = 192(knot points) x 2byte	9B00h	Reserved	T1_R_00[9:8]
		9B01h	T1_R_00[7:0]	
		9B02h	Reserved	T1_R_01[9:8]
		9B03h	T1_R_01[7:0]	
		~	~	
		9C7Eh	Reserved	T1_R_191[9:8]
		9C7Fh	T1_R_191[7:0]	
	GR(384byte) = 192(knot points) x 2byte	9C80h	Reserved	T1_GR_00[9:8]
		9C81h	T1_GR_00[7:0]	
		9C82h	Reserved	T1_GR_01[9:8]
		9C83h	T1_GR_01[7:0]	
		~	~	
		9DFEh	Reserved	T1_GR_191[9:8]
		9DFFh	T1_GR_191[7:0]	
	GB(384byte) = 192(knot points) x 2byte	9E00h	Reserved	T1_GB_00[9:8]
		9E01h	T1_GB_00[7:0]	
		9E02h	Reserved	T1_GB_01[9:8]
		9E03h	T1_GB_01[7:0]	
		~	~	
		9F7Eh	Reserved	T1_GB_191[9:8]
		9F7Fh	T1_GB_191[7:0]	
	B(384byte) = 192(knot points) x 2byte	9F80h	Reserved	T1_B_00[9:8]
		9F81h	T1_B_00[7:0]	
		9F82h	Reserved	T1_B_01[9:8]
		9F83h	T1_B_01[7:0]	
		~	~	
		A0FEh	Reserved	T1_B_191[9:8]
		A0FFh	T1_B_191[7:0]	

Figure 10-11 Knot point data configuration

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11. Quad Bayer Coding Sensitivity Correction (QSC)

This sensor has a Quad Bayer Coding Sensitivity Correction (QSC) function for each color (R/Gr/Gb/B) separately. The purpose of the QSC function of this sensor is the sensitivity difference correction between adjacent 2×2 pixels of the same color.

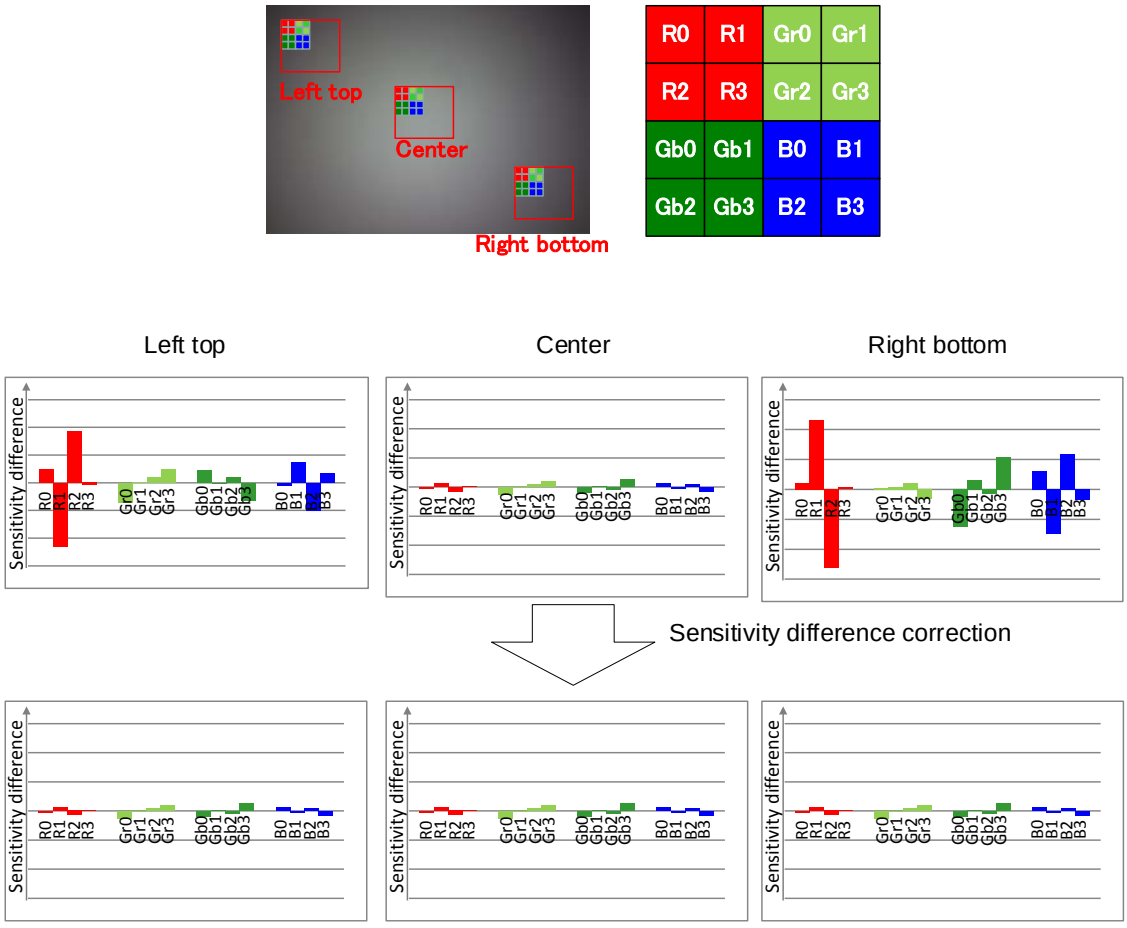


Figure 11-1 Quad Bayer Coding Sensitivity Correction

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11.1. Abstract of QSC system

QSC knot point table consists of 16(H) x 12(V) Knot point tables for each color.

QSC is supported under full-resolution mode, so QSC_EN can be set to 1 when HDR_MODE is 0 and BINNING_MODE is 0.

Table 11-1 QSC function

QSC function		Note
Correction algorithm		bi-linear interpolation
Color correction mode		4 color mode (R / Gr / Gb / B)
Knot point	Horizontal	16
	Vertical	12
	Resolution	6bit
Knot point table		1 knot point table (Updated by ISP via CCI bus).
Total QSC memory size		2304Byte

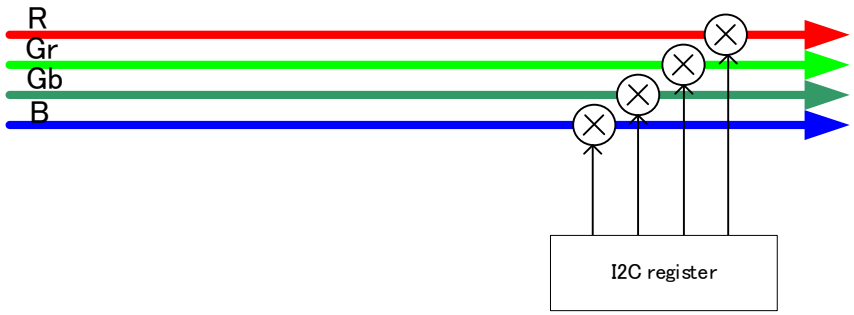


Figure 11-2 Block diagram for QSC knot point table setting

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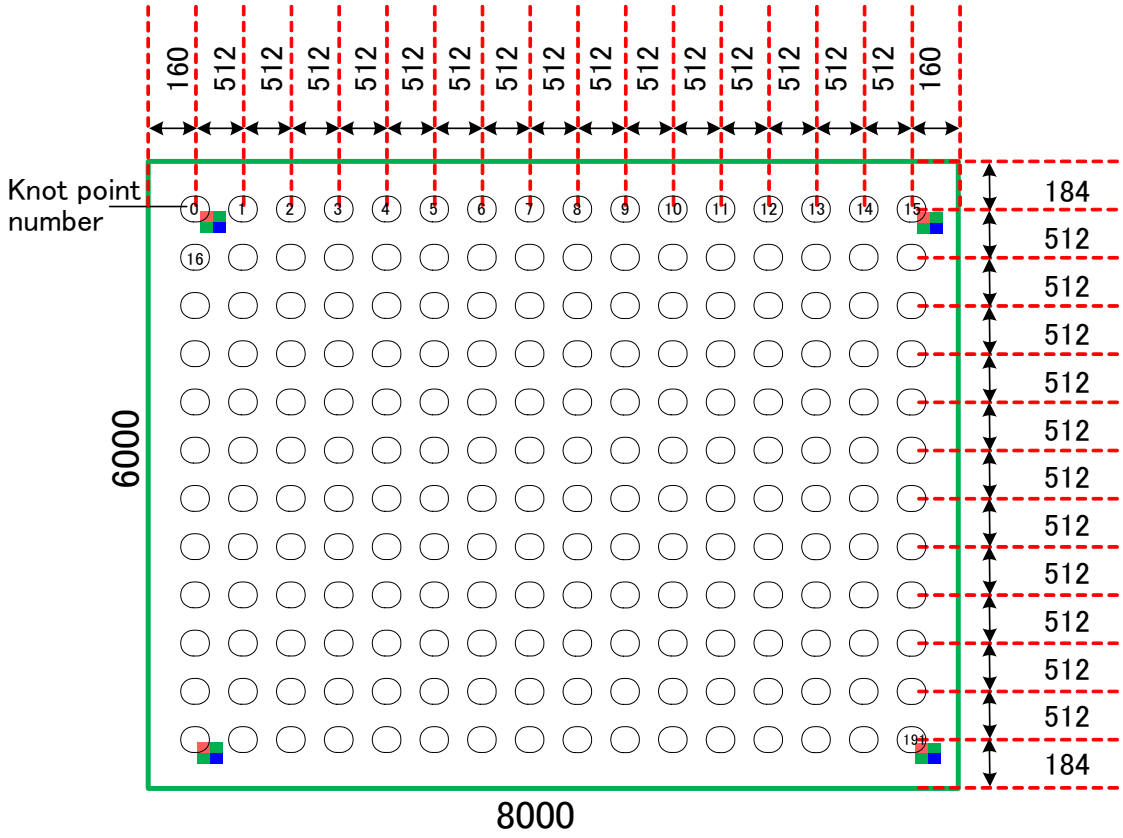


Figure 11-3 QSC knot point tables (16(H) x 12(V))

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11.2. QSC control registers

The following table shows control registers of QSC.

Table 11-2 QSC control register

I^2C Address	Bit	Name	default	Description
0x3621	[0]	QSC_EN	1'b0	QSC Enable 0:disable, 1:enable
0x7F00	[7:0]	QSC_KNOT_POINT_R00[23:16]	0x00	0th knot point of 16x12 knot point tables for R. *1 *2 *3
0x7F01	[7:0]	QSC_KNOT_POINT_R00[15:8]	0x00	
0x7F02	[7:0]	QSC_KNOT_POINT_R00[7:0]	0x00	

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0x7F03	[7:0]	QSC_KNOT_POINT_GR_H00[7:0]	0x00	0th knot point of 16x12 knot point tables for GR. *1 *2 *3
0x7F04	[7:0]	QSC_KNOT_POINT_GR_L00[15:8]	0x00	
0x7F05	[7:0]	QSC_KNOT_POINT_GR_L00[7:0]	0x00	
0x7F06	[7:0]	QSC_KNOT_POINT_GB_H00[15:8]	0x00	0th knot point of 16x12 knot point tables for GB. *1 *2 *3
0x7F07	[7:0]	QSC_KNOT_POINT_GB_H00[7:0]	0x00	
0x7F08	[7:0]	QSC_KNOT_POINT_GB_L00[7:0]	0x00	
0x7F09	[7:0]	QSC_KNOT_POINT _B00[23:16]	0x00	0th knot point of 16x12 knot point tables for B. *1 *2 *3
0x7F0A	[7:0]	QSC_KNOT_POINT _B00[15:8]	0x00	
0x7F0B	[7:0]	QSC_KNOT_POINT _B00[7:0]	0x00	
:	:	:	:	:
0x87F4	[7:0]	QSC_KNOT_POINT_R191[23:16]	0x00	191th knot point of 16x12 knot point tables for R. *1 *2 *3
0x87F5	[7:0]	QSC_KNOT_POINT_R191[15:8]	0x00	
0x87F6	[7:0]	QSC_KNOT_POINT_R191[7:0]	0x00	
0x87F7	[7:0]	QSC_KNOT_POINT_GR_H191[7:0]	0x00	191th knot point of 16x12 knot point tables for GR. *1 *2 *3
0x87F8	[7:0]	QSC_KNOT_POINT_GR_L191[15:8]	0x00	
0x87F9	[7:0]	QSC_KNOT_POINT_GR_L191[7:0]	0x00	
0x87FA	[7:0]	QSC_KNOT_POINT_GB_H191[15:8]	0x00	191th knot point of 16x12 knot point tables for GB. *1 *2 *3
0x87FB	[7:0]	QSC_KNOT_POINT_GB_H191[7:0]	0x00	
0x87FC	[7:0]	QSC_KNOT_POINT_GB_L191[7:0]	0x00	
0x87FD	[7:0]	QSC_KNOT_POINT_B191[23:16]	0x00	191th knot point of 16x12 knot point tables for B. *1 *2 *3
0x87FE	[7:0]	QSC_KNOT_POINT_B191[15:8]	0x00	
0x87FF	[7:0]	QSC_KNOT_POINT_B191[7:0]	0x00	

Note1: bit alignment of QSC knot point is shown Figure11-13.

Note2: copied from OTP table.

Note3: write and read access via CCI bus to knot point table is allowed only during SW standby

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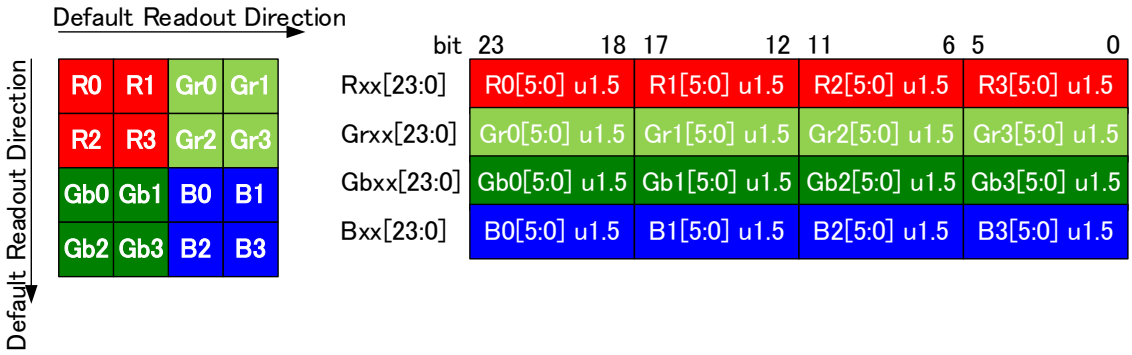


Figure 11-4 Bit alignment of QSC knot points

QSC knot point (u1.5) is converted to gain value (u1.7) by the following equation.

Gain value (u1.7) = (QSC_KNOT_xy[5:0] + 96) / 128 (x=R, GR, GB, B; y=00, 01, .. , 191)

Table 11-3 Setting example of QSC knot point

QSC_KNOT_xy[5:0]	Gain(times)
0	0.75
:	:
16	0.875
:	:
32	1.000
:	:
48	1.125
:	:
63	1.2421875

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11.3. Constrained condition of QSC

- Since QSC is not supported in QBC HDR mode and 2x2 adjacent binning mode, QSC_EN must be set to 0.
- It is allowed only during SW standby when write and read access via CCI bus to QSC knot point table.

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11.4. QSC data information in OTP

OTP has an area that stores QSC knot point gain data (address 64 to 2367). These QSC knot point gain data is automatically loaded to the sensor registers (address 0x7F00 to 0x87FF) after power on. Data loaded from OTP can be updated via CCI bus during SW standby.

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12. Register Map

See “Register Map document”.

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Appendix

A-1. Register update grouping

In 7.1 and 7.2, operation and setting method of each register is mainly discussed. In this Appendix chapter, on the other hand, registers are discussed as groups classified by update timing, setting reason, behavior of each register depending on various mode transition, Practical method of combing and updating various types of registers are main theme of this section.

A-1.1 Procedure for mode transition setting

In this sensor, most of registers are grouped as follows from register update aspect.

Group A (Global setting only in Power On)

A-1: INCK and PLL setting

A-2: Global settings (basic configuration, Analog parameters and IQ parameters)

Group B (Operation mode settings for every mode change)

B-1: must be set in SW-standby (see Table 7-3)

B-2: corrupted frame related registers (see Table 7-4)

B-3: other most registers in address # of 0x0xxx and 0x3xx (except Group A/B-1/B-2/C)

Group C (special purpose registers)

C-2: must be set every time before change in streaming as work around (defined as needed).

C-3: Temperature sensor related registers (for detail see 8.1)

A-1.2 Things to be considered for mode transitions

The following table shows various issues around mode transitions that have to be kept in mind.

As you can see in the following table, there are two methods in the capture mode transitions as follows:

Via SW-standby transition

In this method, all mode change related registers are set during software standby state and is universal method of mode transition. The SW-standby is controlled by MODE_SEL.

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On-the-fly transition

This mode transition is to instantly change the streaming without using SW-standby. This mode does not insert black out of image during mode transition if the condition of usage is matched. On-the-fly transition method usually has some constraints for usage, and so the usable case is limited. On-the-fly transition is controlled by GPH (Grouped parameter hold).

Table A-1 Summary of mode transition related issues

Capture mode transition method		Use of GPH	Shutter lag	Corrupted frame output	Applicable register update Group	Register example *1
Via SW-standby	SW-standby mode transition	No	Long	No	All of A, B, C	IOP_SYS_DIV REQ_LINK_BIT_RATE_MBPS
	Fast SW-standby mode transition	No	Short	Yes	All of A, B, C	
On the fly	Normal mode transition	Yes	Long	Yes (depend on setting)	B-2, B-3	Y_ADD_STA BINNING_TYPE
	Fast mode transition	Yes	Short	Yes	B-2, B-3	

Note *1: To know more about group classification of each register, see Register Map or Register Setting Table.

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A-1.3 Mode transition procedure

In this paragraph, practical method of mode transition sequence and register settings are explained. From mode transition view point, sensor's states are divided into the following four stats; (1) Power ON, (2) Initial SW-standby, (3) Streaming and (4) SW-standby.

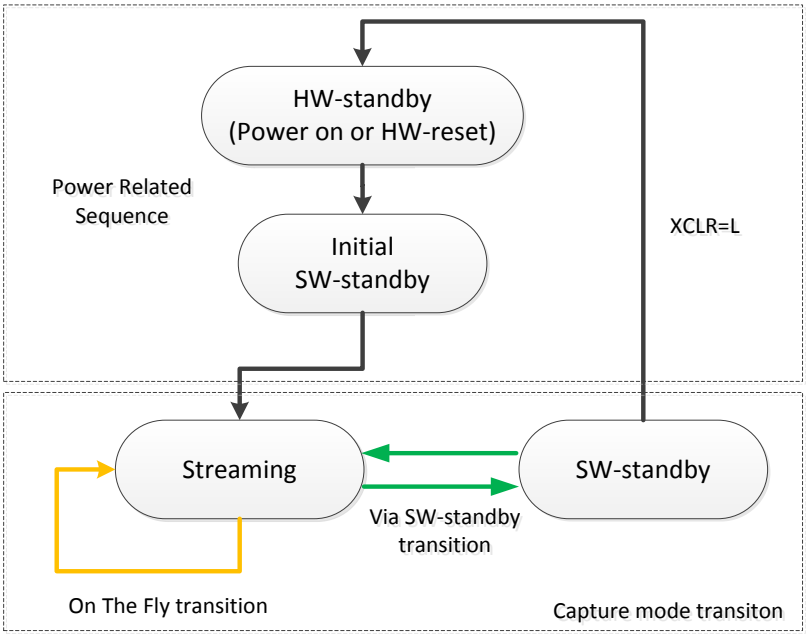


Figure A-1 Mode transitions

Purpose of the above figure is to simply describe the mode transitions from power on to streaming (Power related sequence) and between streaming (Capture mode transitions).

Table A-2 Mode transition procedures

Mode transition	Transition procedures
Power related sequence	Group A-1 → Group A-2, Group B(all) → MODE_SEL=1 Minimum setting*1: Difference from registers' default value of Group B
On The Fly transition	GPH=1 → Group B-2, B-3, Group C-1 → GPH=0 Example: From Normal capture mode of full 2 Binning to Normal capture mode of full resolution. Minimum setting*1: Difference of register value between the transition modes.
Via SW-standby	MODE_SEL=0 → Group B-1, B-2, B-3, Group C-1 → MODE_SEL=1 Example: From Normal capture mode of full resolution to Normal capture mode of 2-binning with MIPI low data rate Minimum setting*1: Difference of register value between the transition modes.

Note *1: Minimum setting can reduce CCI transfer time.

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A-2. Embedded data lines

Contents and output sequence of Embedded Data Lines are shown as below.

Table A-3 EBD LINE0

Pixel#	Index	Register Name	EBD Byte	Meaning of EBD Byte
1			0x0A	EBD Format Code (0x0a = Simplified 2byte Tagged Data Format)
2			0xAA	Tag
3			0x00	register address MSB [15:8]
4			0xA5	Tag
5			0x05	register address LSB [7:0]
6			0x5A	Tag
7	0x0005	FRM_CNT	value	Valid data
8			0x5A	Tag
9	0x0006	PIX_ORDER	value	Valid data
10			0x55	Tag
11			0x07	Dummy data
12			0x5A	Tag
13	0x0008	DT_PEDESTAL[9:8]	value	Valid data
14			0x5A	Tag
15	0x0009	DT_PEDESTAL[7:0]	value	Valid data
16			0xAA	Tag
17			0x01	register address MSB [15:8]
18			0xA5	Tag
19			0x01	register address LSB [7:0]
20			0x5A	Tag
21	0x0101	IMG_ORIENTATION_V/IMG_ORIENTATION_H	value	Valid data
22			0xA5	Tag
23			0x10	register address LSB [7:0]
24			0x5A	Tag
25			value	Invalid data
26			0xA5	Tag
27			0x36	register address LSB [7:0]
28			0x5A	Tag
29	0x0136	EXCK_FREQ[15:8]	value	Valid data

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Pixel#	Index	Register Name	EBD Byte	Meaning of EBD Byte
30			0x5A	Tag
31	0x0137	EXCK_FREQ[7:0]	value	Valid data
32			0x5A	Tag
33	0x0138	TEMP_SEN_CTL	value	Valid data
34			0x55	Tag
35			0x07	Dummy data
36			0x5A	Tag
37	0x013A	TEMP_SEN_OUT	value	Valid data
38			0xAA	Tag
39			0x02	register address MSB [15:8]
40			0xA5	Tag
41			0x00	register address LSB [7:0]
42			0x5A	Tag
43	0x0200	FINE_INTEG_TIME[15:8]	value	Valid data
44			0x5A	Tag
45	0x0201	FINE_INTEG_TIME[7:0]	value	Valid data
46			0x5A	Tag
47	0x0202	COARSE_INTEG_TIME[15:8]	value	Valid data
48			0x5A	Tag
49	0x0203	COARSE_INTEG_TIME[7:0]	value	Valid data
50			0x5A	Tag
51	0x0204	ANA_GAIN_GLOBAL[9:8]	value	Valid data
52			0x5A	Tag
53	0x0205	ANA_GAIN_GLOBAL[7:0]	value	Valid data
54			0xA5	Tag
55			0x0E	register address LSB [7:0]
56			0x5A	Tag
57	0x020E	DIG_GAIN_GLOBAL[15:8]	value	Valid data
58			0x5A	Tag
59	0x020F	DIG_GAIN_GLOBAL[7:0]	value	Valid data
60			0x5A	Tag
61	0x0210	DIG_GAIN_R[15:8]	value	Valid data
62			0x5A	Tag
63	0x0211	DIG_GAIN_R[7:0]	value	Valid data

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Pixel#	Index	Register Name	EBD Byte	Meaning of EBD Byte
64			0x5A	Tag
65	0x0212	DIG_GAIN_B[15:8]	value	Valid data
66			0x5A	Tag
67	0x0213	DIG_GAIN_B[7:0]	value	Valid data
68			0x5A	Tag
69	0x0214	DIG_GAIN_GB[15:8]	value	Valid data
70			0x5A	Tag
71	0x0215	DIG_GAIN_GB[7:0]	value	Valid data
72			0x5A	Tag
73	0x0216	ST_ANA_GAIN_GLOBAL[9:8]	value	Valid data
74			0x5A	Tag
75	0x0217	ST_ANA_GAIN_GLOBAL[7:0]	value	Valid data
76			0x5A	Tag
77	0x0218	ST_DIG_GAIN_GLOBAL[15:8]	value	Valid data
78			0x5A	Tag
79	0x0219	ST_DIG_GAIN_GLOBAL[7:0]	value	Valid data
80			0xA5	Tag
81			0x20	register address LSB [7:0]
82			0x5A	Tag
83	0x0220	HDR_MODE	value	Valid data
84			0x5A	Tag
85	0x0221		value	Invalid data
86			0x5A	Tag
87	0x0222	EXPO_RATIO	value	Valid data
88			0x55	Tag
89			0x07	Dummy data
90			0x5A	Tag
91	0x0224	ST_COARSE_INTEG_TIME[15:8]	value	Valid data
92			0x5A	Tag
93	0x0225	ST_COARSE_INTEG_TIME[7:0]	value	Valid data
94			0xAA	Tag
95			0x03	register address MSB [15:8]
96			0xA5	Tag
97			0x00	register address LSB [7:0]

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Pixel#	Index	Register Name	EBD Byte	Meaning of EBD Byte
98			0x5A	Tag
99			0x00	Reserved data
100			0x5A	Tag
101	0x0301	IVT_PXCK_DIV	value	Valid data
102			0x5A	Tag
103			0x00	Reserved data
104			0x5A	Tag
105	0x0303	IVT_SYCK_DIV	value	Valid data
106			0x5A	Tag
107			0x00	Reserved data
108			0x5A	Tag
109	0x0305	IVT_PREPLLCK_DIV	value	Valid data
110			0x5A	Tag
111	0x0306	IVT_PLL_MPY[10:8]	value	Valid data
112			0x5A	Tag
113	0x0307	IVT_PLL_MPY[7:0]	value	Valid data
114			0x5A	Tag
115			0x00	Reserved data
116			0x5A	Tag
117	0x0309		value	Invalid data
118			0x5A	Tag
119			0x00	Reserved data
120			0x5A	Tag
121	0x030B	IOP_SYCK_DIV	value	Valid data
122			0x5A	Tag
123			0x00	Reserved data
124			0x5A	Tag
125	0x030D	IOP_PREPLLCK_DIV	value	Valid data
126			0x5A	Tag
127	0x030E	IOP_PLL_MPY[1 1 :8]	value	Valid data
128			0x5A	Tag
129	0x030F	IOP_PLL_MPY[7:0]	value	Valid data
130			0x5A	Tag
131	0x0310	PLL_MULT_DRIV	value	Valid data

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Pixel#	Index	Register Name	EBD Byte	Meaning of EBD Byte
132			0xA5	Tag
133			0x40	register address LSB [7:0]
134			0x5A	Tag
135	0x0340	FRM_LENGTH_LINES[15:8]	value	Valid data
136			0x5A	Tag
137	0x0341	FRM_LENGTH_LINES[7:0]	value	Valid data
138			0x5A	Tag
139	0x0342	LINE_LENGTH_PCK[15:8]	value	Valid data
130			0x5A	Tag
141	0x0343	LINE_LENGTH_PCK[7:0]	value	Valid data
142			0x5A	Tag
143	0x0344	X_ADD_STA[12:8]	value	Valid data
144			0x5A	Tag
145	0x0345	X_ADD_STA[7:0]	value	Valid data
146			0x5A	Tag
147	0x0346	Y_ADD_STA[12:8]	value	Valid data
148			0x5A	Tag
149	0x0347	Y_ADD_STA[7:0]	value	Valid data
150			0x5A	Tag
151	0x0348	X_ADD_END[12:8]	value	Valid data
152			0x5A	Tag
153	0x0349	X_ADD_END[7:0]	value	Valid data
154			0x5A	Tag
155	0x034A	Y_ADD_END[12:8]	value	Valid data
156			0x5A	Tag
157	0x034B	Y_ADD_END[7:0]	value	Valid data
158			0x5A	Tag
159	0x034C	X_OUT_SIZE[12:8]	value	Valid data
160			0x5A	Tag
161	0x034D	X_OUT_SIZE[7:0]	value	Valid data
162			0x5A	Tag
163	0x034E	Y_OUT_SIZE[12:8]	value	Valid data
164			0x5A	Tag
165	0x034F	Y_OUT_SIZE[7:0]	value	Valid data

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Pixel#	Index	Register Name	EBD Byte	Meaning of EBD Byte
166			0xA5	Tag
167			0x80	register address LSB [7:0]
168			0x5A	Tag
169			0x00	Reserved data
170			0x5A	Tag
171	0x0381	X_EVN_INC	value	Valid data
172			0x5A	Tag
173			0x00	Reserved data
174			0x5A	Tag
175	0x0383	X_ODD_INC	value	Valid data
176			0x5A	Tag
177			0x00	Reserved data
178			0x5A	Tag
179	0x0385	Y_EVN_INC	value	Valid data
180			0x5A	Tag
181			0x00	Reserved data
182			0x5A	Tag
183	0x0387	Y_ODD_INC	value	Valid data
184			0xAA	Tag
185			0x04	register address MSB [15:8]
186			0xA5	Tag
187			0x01	register address LSB [7:0]
188			0x5A	Tag
189	0x0401	SCALE_MODE	value	Valid data
190			0xA5	Tag
191			0x04	register address LSB [7:0]
192			0x5A	Tag
193	0x0404	SCALE_M[8]	value	Valid data
194			0x5A	Tag
195	0x0405	SCALE_M[7:0]	value	Valid data
196			0xA5	Tag
197			0x08	register address LSB [7:0]
198			0x5A	Tag
199	0x0408	DIG_CROP_X_OFFSET[12:8]	value	Valid data

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Pixel#	Index	Register Name	EBD Byte	Meaning of EBD Byte
200			0x5A	Tag
201	0x0409	DIG_CROP_X_OFFSET[7:0]	value	Valid data
202			0x5A	Tag
203	0x040A	DIG_CROP_Y_OFFSET[12:8]	value	Valid data
204			0x5A	Tag
205	0x040B	DIG_CROP_Y_OFFSET[7:0]	value	Valid data
206			0x5A	Tag
207	0x040C	DIG_CROP_IMAGE_WIDTH[12:8]	value	Valid data
208			0x5A	Tag
209	0x040D	DIG_CROP_IMAGE_WIDTH[7:0]	value	Valid data
210			0x5A	Tag
211	0x040E	DIG_CROP_IMAGE_HEIGHT[12:8]	value	Valid data
212			0x5A	Tag
213	0x040F	DIG_CROP_IMAGE_HEIGHT[7:0]	value	Valid data
214			0x07	End of Data
215			0x07	

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Table A-4 EBD LINE 1

Pixel#	Index	Register Name	EBD Byte	Meaning of EDB Byte
1			0x0A	EBD Format Code (0x0a = Simplified 2byte Tagged Data Format)
2			0xAA	Tag
3			0x09	register address MSB [15:8]
4			0xA5	Tag
5			0x00	register address LSB [7:0]
6			0x5A	Tag
7	0x0900	BINNING_MODE	value	Valid data
8			0x5A	Tag
9	0x0901	BINNING_TYPE_H/BINNING_TYPE_V	value	Valid data
10			0x5A	Tag
11	0x0902	BINNING_WEIGHTING	value	Valid data
12			0xAA	Tag
13			0x0B	register address MSB [15:8]
14			0xA5	Tag
15			0x00	register address LSB [7:0]
16			0x5A	Tag
17	0x0B00	SHD_CORR_EN	value	Valid data
18			0x55	Tag
19			0x07	Dummy data
20			0x5A	Tag
21	0x0B02		value	Invalid data
22			0x5A	Tag
23	0x0B03		value	Invalid data
24			0x5A	Tag
25	0x0B04	BLK_LEV_CORR_EN	value	Valid data
26			0x5A	Tag
27	0x0B05	MAP_COUP_CORR_EN	value	Valid data
28			0x5A	Tag
29	0x0B06	SING_DEF_CORR_EN	value	Valid data
30			0xA5	Tag
31			0x8E	register address LSB [7:0]

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Pixel#	Index	Register Name	EBD Byte	Meaning of EDB Byte
32			0x5A	Tag
33	0x0B8E	ABS_GAIN_GR[15:8]	value	Valid data
34			0x5A	Tag
35	0x0B8F	ABS_GAIN_GR[7:0]	value	Valid data
36			0x5A	Tag
37	0x0B90	ABS_GAIN_R[15:8]	value	Valid data
38			0x5A	Tag
39	0x0B91	ABS_GAIN_R[7:0]	value	Valid data
40			0x5A	Tag
41	0x0B92	ABS_GAIN_B[15:8]	value	Valid data
42			0x5A	Tag
43	0x0B93	ABS_GAIN_B[7:0]	value	Valid data
44			0x5A	Tag
45	0x0B94	ABS_GAIN_GB[15:8]	value	Valid data
46			0x5A	Tag
47	0x0B95	ABS_GAIN_GB[7:0]	value	Valid data
48			0xAA	Tag
49			0x0C	register address MSB [15:8]
50			0xA5	Tag
51			0x12	register address LSB [7:0]
52			0x5A	Tag
53	0x0C12	FLASH_STRB_ADJ	value	Valid data
54			0x55	Tag
55			0x07	Dummy data
56			0x5A	Tag
57	0x0C14	FLASH_STRB_START_POINT[15:8]	value	Valid data
58			0x5A	Tag
59	0x0C15	FLASH_STRB_START_POINT[7:0]	value	Valid data
60			0x5A	Tag
61	0x0C16	TFLASH_STRB_DLY_RS_CTL[15:8]	value	Valid data
62			0x5A	Tag
63	0x0C17	TFLASH_STRB_DLY_RS_CTL[7:0]	value	Valid data
64			0x5A	Tag
65	0x0C18	TFLASH_STRB_WIDTH_H_RS_CTL[15:8]	value	Valid data

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Pixel#	Index	Register Name	EBD Byte	Meaning of EDB Byte
66			0x5A	Tag
67	0x0C19	TFLASH_STRB_WIDTH_H_RS_CTL[7:0]	value	Valid data
68			0x5A	Tag
69	0x0C1A	FLASH_MD_RS	value	Valid data
70			0x55	Tag
71			0x07	Dummy data
72			0x5A	Tag
73	0x0C1C	FLASH_STAT	value	Valid data
74			0xA5	Tag
75			0x26	register address LSB [7:0]
76			0x5A	Tag
77	0x0C26	TFLASH_STRB_WIDTH2_H_RS_CTL[15:8]	value	Valid data
78			0x5A	Tag
79	0x0C27	TFLASH_STRB_WIDTH2_H_RS_CTL[7:0]	value	Valid data
80			0x5A	Tag
81	0x0C28	TFLASH_STRB_WIDTH_L_RS_CTL[15:8]	value	Valid data
82			0x5A	Tag
83	0x0C29	TFLASH_STRB_WIDTH_L_RS_CTL[7:0]	value	Valid data
84			0x5A	Tag
85	0x0C2A	TFLASH_STRB_CNT_RS_CTL	value	Valid data
86			0xAA	Tag
87			0x30	register address MSB [15:8]
88			0xA5	Tag
89			0x03	register address LSB [7:0]
90			0x5A	Tag
91	0x3003		value	Invalid data
92			0xAA	Tag
93			0x31	register address MSB [15:8]
94			0xA5	Tag
95			0x00	register address LSB [7:0]
96			0x5A	Tag
97	0x3100	CIT_LSHIFT	value	Valid data
98			0xA5	Tag
99			0x30	register address LSB [7:0]

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Pixel#	Index	Register Name	EBD Byte	Meaning of EDB Byte
100			0x5A	Tag
101	0x3130	DIG_GAIN_MODE	value	Valid data
102			0xA5	Tag
103			0x68	register address LSB [7:0]
104			0x5A	Tag
105	0x3168		value	Invalid data
106			0x5A	Tag
107	0x3169		value	Invalid data
108			0xAA	Tag
109			0x32	register address MSB [15:8]
110			0xA5	Tag
111			0x43	register address LSB [7:0]
112			0x5A	Tag
113	0x3243		value	Invalid data
114			0xA5	Tag
115			0x46	register address LSB [7:0]
116			0x5A	Tag
117	0x3246	BINNING_PRIORITY_H	value	Valid data
118			0x5A	Tag
119	0x3247	BINNING_PRIORITY_V	value	Valid data
120			0xA5	Tag
121			0x61	register address LSB [7:0]
122			0x5A	Tag
123	0x3261		value	Invalid data
124			0xA5	Tag
125			0x64	register address LSB [7:0]
126			0x5A	Tag
127	0x3264		value	Invalid data
128			0x5A	Tag
129	0x3265		value	Invalid data
130			0xAA	Tag
131			0x38	register address MSB [15:8]
132			0xA5	Tag
133			0x08	register address LSB [7:0]

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Pixel#	Index	Register Name	EBD Byte	Meaning of EDB Byte
134			0x5A	Tag
135	0x3808		value	Invalid data
136			0x5A	Tag
137	0x3809		value	Invalid data
138			0xAA	Tag
139			0x3F	register address MSB [15:8]
140			0xA5	Tag
141			0x50	register address LSB [7:0]
142			0x5A	Tag
143	0x3F50		value	Invalid data
144			0xA5	Tag
145			0x56	register address LSB [7:0]
146			0x5A	Tag
147	0x3F56		value	Invalid data
148			0x5A	Tag
149	0x3F57		value	Invalid data
150			0xA5	Tag
151			0x6C	register address LSB [7:0]
152			0x5A	Tag
153	0x3F6C	EXTOUT_XVS_MULT_SEL	value	Valid data
154			0x5A	Tag
155	0x3F6D	EXTOUT_XVS_MULT	value	Valid data
156			0x5A	Tag
157	0x3F6E	EXTIN_XVS_MULT_SEL	value	Valid data
158			0x5A	Tag
159	0x3F6F	EXTIN_XVS_MULT	value	Valid data
160			0xA5	Tag
161			0xE0	register address LSB [7:0]
162			0x5A	Tag
163	0x3FE0	MID_COARSE_INTEG_TIME[15:8]	value	Valid data
164			0x5A	Tag
165	0x3FE1	MID_COARSE_INTEG_TIME[7:0]	value	Valid data
166			0x5A	Tag
167	0x3FE2	MID_ANA_GAIN_GLOBAL[9:8]	value	Valid data

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Pixel#	Index	Register Name	EBD Byte	Meaning of EDB Byte
168			0x5A	Tag
169	0x3FE3	MID_ANA_GAIN_GLOBAL[7:0]	value	Valid data
170			0x5A	Tag
171	0x3FE4	MID_DIG_GAIN_GLOBAL[15:8]	value	Valid data
172			0x5A	Tag
173	0x3FE5	MID_DIG_GAIN_GLOBAL[7:0]	value	Valid data
174			0x07	End of Data
175			0x07	

SRM-576-00881-03-586-01261-02

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